

Characterization and Optimization of Additively Manufactured Rocket Engine Components

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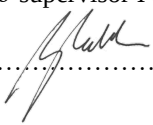
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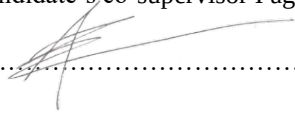
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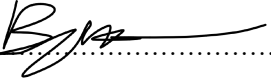
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DECLARATION 2 - PUBLICATIONS

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ABSTRACT

Rocket components manufactured using Additive Manufacturing are becoming increasingly popular in the space industry due to their commercial and performance advantages. The development and maintenance of orbital launch capabilities and allied technologies have historically been expensive, with few countries successfully achieving these goals. However, recent advances in Additive Manufacturing and other technologies have made space-based activities more accessible. This thesis reports both on the characterisation of materials, and the development of components for rocket engine applications for the South African First Integrated Rocket Engine (SAFFIRE) programme. This thesis aimed to gain insight and experience into the potential applications of Additive Manufacturing, with a focus on practical solutions and quality control.

For the characterization of materials, the methodology used comprised of the manufacture of standardized cube samples, overhang samples and wall-thickness specimens to be individually analysed using the non-destructive testing method, microcomputed tomography. Porosity and surface roughness were of particular interest in this investigation to determine the defects present in the Additively Manufactured materials evaluated. Applications of these Additively Manufactured materials were then manufactured and investigated with respect to impeller and volute pump components, and an injector flange. Using microcomputed tomography methods, these applications were evaluated to quantify surface deviation and other Additive Manufacturing process-based defects. Furthermore, the effects of Design for Additive Manufacturing techniques on these applications were investigated.

Both the quantitative and qualitative results for porosity and surface roughness exhibited acceptable quality for all the materials under investigation, except for the 6061-T6 samples. For the 6061-T6 samples, porosity and surface deviation results were of concern, and Additive Manufacturing process parameter modifications were implemented which reduced these defect occurrences. For the impeller, volute and injector flange component applications, various defects were quantified and mitigated primarily through the use of Design for Additive Manufacturing techniques. The results in this thesis provide valuable insights into the specific quality of the Additive Manufacturing processes and materials under investigation. In the context of the material characterization, the results provide clarity on the quality of additively manufactured materials available in the South African supply chain. These standardized sample testing methods can easily be applied to future Additive Manufacturing material investigations, where specific material properties are not well known. The process parameter modification and quality control techniques presented in this thesis can similarly be applied to future Additive Manufacturing material investigations. For the component applications presented, the observed defects provided an understanding of the formation of these defects, and the adopted mitigation strategies showed clear improvements in quality, which may be applicable to future component applications.

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1. INTRODUCTION

1.1. Space Industry

The space industry and its constituents have historically been expensive endeavours with only a handful of countries successfully developing and maintaining national programs for orbital launch capabilities and other related space-based activities. However, recent advances in microelectronics, cost-effective manufacturing, metallurgy, computational design and analyses have made space-based activities more accessible than ever before. The small satellite market, in particular, has seen tremendous growth since 2012 and the market is expected to grow considerably with an estimated 7,000 small satellites likely to be launched between the years 2018 and 2027 [1]. Small satellites typically have a mass range of 1-500 kg and therefore do not require large orbital launch vehicles commonly used in the traditional space industry. This has allowed many nations to embark on the development of national and/or commercial launch capabilities since smaller orbital launch vehicle programmes are considerably cheaper to develop, test and maintain.

1.2. South African First Integrated Rocket Engine (SAFFIRE) Programme Overview

With no current orbital launch capabilities established in Africa, there is a keen interest within South Africa to develop such a capability in order support socioeconomic advancement of the continent. The Aerospace Systems Research Group (ASReG) located at the University of KwaZulu-Natal was formed in 2009 with the purpose of designing, manufacturing and testing rocket propulsion systems. The South African First Integrated Rocket Engine (SAFFIRE) project at ASReG aims to develop liquid rocket engines for a small satellite launch vehicle configured to deliver payloads to orbit from South Africa. With ongoing development work on rocket engine technology, as well as allied systems and infrastructure, the SAFFIRE programme is developing the local skills and knowledge necessary to support the establishment of an indigenous orbital launch capability.

1.3. Additive Manufacturing

The vast majority of orbital launch vehicles employ liquid fuel rocket engines for propulsion. These engines have traditionally been extremely complex and expensive to develop, manufacture and test. In this regard, Additive Manufacturing (AM) techniques present a wide variety of benefits such as light-weighting, part consolidation, cost reduction, design sophistication, and reduced lead times [2]. AM techniques are relatively new, with the first application for a spacecraft being launched on the Juno mission to Jupiter in 2011 [3]. However, AM technology has developed rapidly over the last decade and currently, the majority of orbital launch providers are either developing or using AM for various launch vehicle components. A good example of this is Rocket Lab's Rutherford liquid fuel rocket engine, which has been using AM techniques to manufacture all of their primary engine components such as combustion chambers, injectors, pumps, and main propellant valves since 2013 [4].

1.4. Problem Statement and Research Objectives

Based on previous contributions to the SAFFIRE programme and current market developments, it is clear that utilizing Additive Manufacturing (AM) in the production process for liquid rocket engine components has some distinct advantages. Although some components have been designed and manufactured using AM techniques by ASReG, the expertise required to execute the development of highly accurate and complex components was not available. This lack of practical experience gave rise to this project. In collaboration with both the Stellenbosch University CT Scanner Facility and the Collaborative Program for Additive Manufacturing (CPAM), further expertise and funding was utilized for the purposes of Non-Destructive Testing (NDT) and Design for Additive Manufacturing (DfAM) techniques. The Stellenbosch University collaborator (co-supervisor) has ongoing research efforts funded by the CPAM program, investigating the structural integrity of metal AM parts and evaluating their performance based on defects identified non-destructively, and ultimately enhancing the quality of such AM parts for critical applications. This student project forms an important aspect of this program, with great synergy between the expertise of the groups involved to make progress in this area of application. The primary aim of this study is to gain insight and experience in the AM production process to produce liquid rocket engine components according to specified requirements.

While the AM industry and its applications are growing extensively worldwide, the adoption of this technology is still in its infancy. Manufacturing aerospace components requires careful examination and inspection during the manufacturing process. This due diligence is lacking in the South African AM industry and requires a strict framework for manufacturing standards similar to those of the International Organization of Standardization (ISO). Aerospace components in particular have much stricter standards generally when compared to those of other industries, and the international standards related to AM components in the space industry are mostly new or still in development.

AM enables the creation of extremely complex geometries which are not possible using traditional manufacturing methods. As with all manufacturing methods, the final manufactured product may deviate from the design in several ways, or may feature unexpected defects or flaws. In AM, this occurs mostly in the form of geometric deviation (e.g. due to thermal warpage), surface roughness irregularity, internal porosity and cracks forming during the build process. These defects require advanced evaluation techniques such as Non-Destructive Testing (NDT) to assess the internal defects (porosity and cracks) and external defects (geometric deviation and surface roughness irregularity).

The quality of components manufactured using AM techniques also differs widely depending on the material used. Liquid rocket engine components are often made from high performance alloys which mostly can be manufactured using AM processes. However, due to the low margins for error and strict tolerances required on such components, extensive testing is required to ensure the quality of the AM process for each material under consideration.

AM presents some unique advantages in the design and manufacturing process in the form of rapid prototyping and cost-effective production at small scale production volumes. This synergizes with the space industry as most components are high-value and low production volume. All engineering components are somewhat limited by the production method used to fabricate them, and the associated capital investment required. Using AM to iteratively

design test components enables the creation of high-performance components with a very small initial investment when compared to traditional manufacturing techniques.

Considering the above, advantages of particular value to work undertaken in this study include the utilization of cost-effective production, rapid prototyping and improving performance through lightweighting and geometric complexity. The primary research objectives of this study are therefore as follows:

- To perform extensive research into the advantages and practical use-cases for AM in aerospace components
- To investigate material characteristics and manufacturing processes of various AM engineering alloys
- To design, manufacture and test iterations of pump components in order to assess the practicality of using AM for liquid rocket engine component fabrication
- To investigate and optimize the production of an injector flange component within the framework of AM process characteristics

1.5. Dissertation Outline

A detailed literature review on metal additive manufacturing in aerospace is presented in Chapter 2. It includes a historical perspective and recent developments in the field. The general AM manufacturing process is described in detail followed by the various specific AM processes that are widely used. The currently available materials for AM processes are then outlined in detail to give insight into which materials can be manufactured at high quality and why they are used. The specific advantages and challenges of utilizing AM processes are identified throughout Chapter 2. Chapter 2 also highlights various successful applications of AM in aerospace and more specifically, in the production of liquid rocket engine components. This chapter was written by a number of authors and published in the Elsevier Materials & Design Journal under the title *Metal additive manufacturing in aerospace: a review* [5].

Chapter 3 summarises the evaluation of various additively manufactured alloys which are under consideration for components associated with the SAFFIRE programme. These alloys were specifically selected due to their material performance properties; however, the full scope of materials available are limited by the South African AM material supply chain. These materials were manufactured using AM in specific samples to provide insight into the geometrical capabilities and quality of the AM process. Chapter 3 also provides insight into the possible defects that can arise due to various production issues, such as incorrect powder quality control methods.

Chapter 4 describes the iterative design of pump components intended for use in liquid rocket engines. The analytical evaluation techniques used to inform the design of further iterations are described. This chapter focusses on the design and testing of pump components and the validity of AM processes applicable to these components. NDE techniques are demonstrated in this chapter to exhibit the quality and repeatability of the AM process. Various advanced Design for Additive Manufacturing (DfAM) techniques are used throughout the iterative design cycle of the pump components to improve part quality and performance.

Chapter 5 addresses the design and manufacture of the AM-built injector flange component for the first iteration SAFFIRE liquid rocket engine. The effects of using advanced DfAM techniques such as build simulation and

geometric distortion are presented and evaluated using NDE. NDE methods are applied for this large scale AM-built component using high-power MicroCT techniques to fully characterise the quality of the AM process. This chapter focusses on the quality output of the AM process for larger scale components such as the injector flange.

The dissertation is concluded in Chapter 6 with a discussion of the study's findings and the process problems encountered, as well as recommendations for future work.

2. METAL ADDITIVE MANUFACTURING IN AEROSPACE: A REVIEW

2.1. Introduction

The aerospace sector encompasses commercial and military aircraft, space launch and in-space systems, missiles, satellites and general aviation. The recent COVID-19 pandemic has had a substantial negative impact on aerospace sector revenue, with a decline from \$342.2 billion in 2019 down to \$298 billion in 2020 [6]. This decline was primarily caused by air travel restrictions, social distancing protocols and other restrictions imposed on the commercial aviation industry during the ongoing COVID-19 pandemic. Despite this challenge, the aerospace sector is expected to grow to \$430.87 billion revenue by the year 2025 [6]. This growth is primarily due to the long-term demand for new commercial aircraft, increased global military expenditure, high market activity in the space sector and substantial research and development ongoing throughout the pandemic [6].

Manufacturing in the aerospace sector is subject to numerous interacting technical and economic objectives of: *functional performance*, *lead time reduction*, *lightweighting*, *complexity*, *cost management*, and *sustainment*. Each of these objectives have strong relationships to one another and considerations from each factor must be considered carefully when selecting an optimal design solution. The relative importance of these objectives depends on the specific aerospace application, but in general terms these objectives can be described as follows:

- The aerospace sector often requires the delivery of safety-critical components that must operate in their intended environment (*functional performance*) in small production volumes with relatively inflexible delivery schedule. *Lead time reduction* is therefore highly relevant to the aerospace sector as this allows rapid product certification and retains flexibility in design of high value components.
- *Lightweighting* is relevant to the technical and economic performance of aerospace structures. Specifically, the technical performance and allowable mission-defined payload of aerospace structures is physically limited, meaning that system mass reduction directly relates to enhanced economic and technical performance, including reduced fuel costs, lower emissions, larger payloads and increased range.
- The objective of lightweighting is tempered by the economic constraints of *cost management*, whereby a specific financial resource is available for a specific design objective. Cost management, lightweighting and lead time reduction objectives are interrelated, whereby the allowable system cost generally increases for solutions that are able to achieve lightweighting or lead time reduction objectives. Cost management applies to all aspects of component use, including certification and maintenance, and opportunities for cost management by reduced certification risk.
- A critical challenge to metal AM application in aerospace is the hurdle of *certification*; requiring that regulatory bodies be confident that the AM systems are fundamentally well understood, and can be repeatably designed and inspected such that reliability and safety expectations can be satisfied. These certification requirements vary according to the criticality of the proposed AM system as being safety or mission-critical or otherwise. Practical certification requires connection with existing standards for traditional manufacturing as well as to standards emerging for AM processes.

- Aerospace structures are generally *high-complexity*, low volume systems, where the challenges associated with *sustainment* can become substantial. Sustainment challenges include part availability for ageing aircraft, remanufacture and repair, and re-certification of existing aircraft for alternate uses.
- *Complexity* is inherent to the designs where often hundreds, or thousands of parts make up systems and subsystems to achieve the desired *functional performance* in the intended environment. These systems are integrated into the overall flight system that can often exceed millions of components. These components are high in complexity to allow for functional performance (structural, flow, thermal, reliability, durability, compatibility, etc.) as well as weight reduction [7,8].

In the drive to continuously improve efficiency through cost and lead time reduction and attempts to lower the mass of flight components, high-performance materials are used, with increasingly complex designs. This must be done within a reasonable cost and schedule to meet commercial orders or mission requirements. Traditional manufacturing systems and strategies have been developed over many decades to accommodate these aerospace design objectives for numerous application types; however additive manufacturing (AM) is and will continue to have a profound impact on the design and manufacturing. This AM digital transformation, often touted as Industry 4.0, will increase its market size in the aerospace sector to \$3.187 billion by the year 2025 with an average compound annual growth rate (CAGR) of 20.24% [9]. Research into AM in aerospace has also seen exponential growth over the last decade ¹. In addition to this scholarly literature, much relevant work exists in the form of technical reports, popular literature and publicity articles from commercial aerospace suppliers, with technical details sometimes restricted for commercial reasons.

AM, unlike conventional subtractive manufacturing techniques, utilizes layer by layer manufacturing based on a common feedstock, typically powder or wire that is melted or fused by a heat source and solidifies to produce the final geometry based on a digitally defined heat source trajectory [10]. The advantages of the use of additive manufacturing for aerospace components are the reduced lead time and associated cost, the ability to design and manufacture complex geometries that enable lightweighting, consolidation of multiple components and improvements in performance; within cost and timeline constraints, hence offering improved programmatic and technical risk management [11,12]. By utilizing the design freedom of metal AM, it is possible to optimize the material distribution, reducing mass while maintaining the mechanical and other performance requirements of the component. It is also possible to combine components, reducing risk and cost for multiple components and reducing potential failure modes across joints. Additionally, enhanced performance (above that of conventional manufacturing) is possible by utilizing mechanical, thermal and other optimization approaches for the design of complex parts that were previously impossible to manufacture, incorporating internal features such as conformal cooling channels on combustion chambers or turbine blades, for example [13,14]. While the reduced lead times are the present main driver for the use of AM in aerospace applications, specific manufacturing scenarios give AM advantages over traditional manufacturing as schematically illustrated in Figure 2-1 and discussed below.

¹ Scopus listed publications with the keyword *aerospace* combined with either *3D printing* or *additive manufacturing* have increased from 15 in 2010 to 490 in 2020.

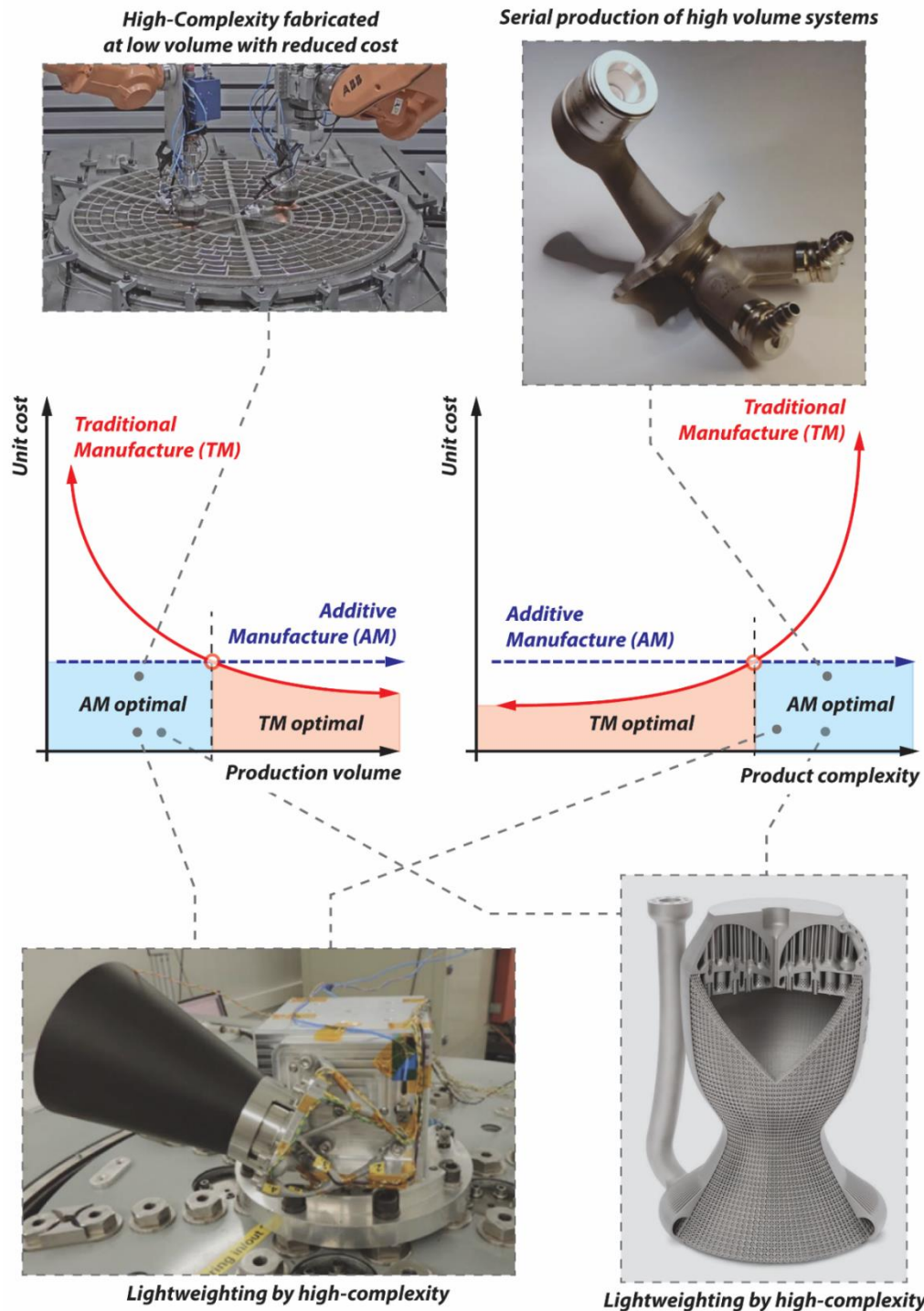


Figure 2-1: Unit cost versus production volume and production complexity identifying economically and technically optimal scenarios further developed in this monograph. Images courtesy of Fraunhofer, European Space Agency and GE Additive Copyright: ©SLM Solutions/CellCore. [15–18]

The *complexity* afforded by the AM process is a large advantage of this technology, as new designs can be realized for enhanced mechanical and thermal performance and reduced system mass, not possible by other manufacturing methods. The inherent capacity for complexity within AM design enables *lightweighting* by both the consolidation of multiple components into one and by the opportunity for increased technical efficiency. Despite persistent misconceptions regarding the constraint-free nature of AM technologies, they are highly compatible with high complexity outcomes; thereby enabling lightweighting by deploying material as required for the technical

response (structural, vibratory or thermal) rather than as constrained by the associated manufacturing process. A systematic review of AM design constraints is provided in [19]. It should be noted that the complexity should be appropriately traded within the design and throughout the lifecycle of AM components as it can add additional post-processing operations or unintended operational challenges if not fully understood. Algorithmic design methods, often referred to as generative design and opportunities for topology optimisation are reviewed in [20]. Part consolidation refers to the re-design of multiple interacting components as a single integrated system, thereby enabling substantial enhancement in technical performance. The integration of multiple components enables removal of mating surfaces and structures, thereby increasing structural efficiency and reducing the costs associated with inspection and certification of mating structures. Furthermore, the consolidation of multiple parts may substantially reduce overall fabrication costs. This cost reduction is enabled by the direct reduction of manufacturing lifecycle costs as well as the reduction in non-recurring costs associated with design, certification and risk-management of the failure modes that are associated with part interactions.

The aerospace sector relies heavily on machined forged and billet structures for high-value structural systems. This manufacturing methodology provides high certainty in final component quality, as billet materials are readily certified for porosity and microstructure, but adds substantial direct manufacturing costs and induced costs due to high production lead times. Forging requires the expensive design, manufacture and trialling of pre-forming dies and billet machining is inherently expensive with typical buy-to-fly ratios² estimated at 20:1. For example, a final product with a mass of 10 kg would require 200 kg of stock materials as discussed in [21]. Others claim this ratio is closer to 40:1 [22]. The unused material is waste and is recycled or reprocessed if possible, adding extensive costs to all projects.

By the nature of the layer-by-layer manufacturing process, AM produces little to no waste with buy-to-fly ratios of between 1:1 and 3:1 [23,24]. Using far less stock material by mass compared to traditional manufacturing techniques, AM has the potential to reduce the cost of manufacturing aerospace components substantially all while simplifying the recycling and reprocessing processes, subject to proper precautions. Many aerospace alloys can have long lead times to produce the required wrought starting stock material. Since many AM processes start with a powder or wire feedstock, which is readily available for common alloys, this lead time can be substantially reduced for commercially available materials. Additional costs and lead time in processing of aerospace components can include specialty tooling and fixturing [25]. Currently, the significant reduction in lead times is one of the major advantages of AM in this industry.

Historically, AM was used extensively for rapid prototyping in the form of part representation and initial functional testing before infusion into commercial end-use applications. While flight use of AM has roots back to the early 2000's, the decade from 2010 to 2020 expanded the use of AM to include final production, including mission-critical components [26]. This digital transition in manufacturing focus has been successful due to improvements in the underlying technology and in the improved understanding of the technology and its supply chain, as well as the rigor being applied to standardization and certification [27].

² Buy-to-fly ratio refers to the ratio of billet material purchased to material in the production component.

The basic principle of AM is to build up components layer-by-layer directly from 3D model data created using Computer-Aided Design (CAD) to near net or final shape. This is a complex procedure as a large multitude of process parameters have possible effects on the quality of produced components [19,28]. Continued improvements in the understanding of these parameters has steadily lessened the associated quality impacts, however. The process parameters including the power, scan speed, hatch spacing or overlap between tracks, scan strategy (e.g. use of contours, angle change of hatch tracks per layer, etc.), among many other parameters may influence the process and thus the quality and properties of the manufactured components. This includes surface roughness, porosity, residual stress and associated cracks or warping of components, unique microstructures, as well as anisotropy of the material. This traces back to the roots of materials characterization and allowing an understanding as part of the Process-Structure-Properties linkages ultimately leads to better performance [29]. Further disadvantages of AM include limited materials, uncertainty in material properties, specific design constraints inherent to AM, post-processing requirements, waste generation (used powder, build plates, failed builds), increased requirements of design skills to allow lower-mass components with feasible complex designs, often utilizing time-consuming topology optimization software workflows, as well as the need for strict quality control and certification of the process [30]. While accessibility to AM machines is increasing, the technology is not as readily available as traditional manufacturing techniques so the supply chain requires maturation. Post-processing is an area that challenges AM, requiring specialized and custom-developed or optimized thermal treatments, part cleanliness, and surface enhancements to improve rough surfaces.

All aerospace manufacturing is subject to strict quality controls such as Quality Systems - Aerospace SAE AS9100 and Standard for Additively Manufactured Spaceflight Hardware MSFC-STD-3716, NASA Standard 6030 [19,31–36], but this is further exacerbated by the complexity of the AM processes and many possible influences on component quality. This is especially true as the technology has matured only recently and many studies have reported widely disparate mechanical properties [37–39], especially fatigue performance [40–42]. This leads to uncertainties in the material performance and in the required controls which may be different from traditional manufacturing. Various international standards are currently under development to address this [19]. With appropriate quality controls and optimization procedures, the issues mentioned above can all be mitigated or minimized [19,43]. Significant post-processing is generally required such as powder removal, thermal processing, supports and baseplate removal, hot isostatic pressing (HIP), surface polishing, and often final machining.

Despite the disadvantages mentioned, specific scenarios exist where AM is well suited to aerospace applications, with clear technical advantages over traditional manufacturing methods. This paper provides a comprehensive review of applications of metal AM in the aerospace industry, thereby also providing a review of the current state of the art. Through demonstrating these successful examples, the benefits of AM and the challenges associated with AM in aerospace are systematically documented; thereby promoting the further application of AM in this sector, taking into full consideration the challenges and requirements.

2.2. Key Concepts Related to AM for Aerospace

2.2.1. Additive Manufacturing Technologies

AM, outlined by the international standard ISO/ASTM 52900 [44], comprises of seven process categories which include directed energy deposition (DED) and powder bed fusion (PBF), both allowing the fabrication of metal components with near full density. Other process categories are not discussed in detail as they are still evolving and still at lower technology readiness levels (TRL). Metal AM is reviewed in more detail in [45], including an overview of the processes, resulting structure/microstructure, and properties of manufactured materials. In particular, laser powder bed fusion (L-PBF) is gaining traction in industry due to its high resolution, high-quality part production capabilities, combining fine feature sizes with adequate build volumes for many part sizes required [19]. Once the process is optimized, the resulting components produced often boast mechanical properties exceeding those of conventional manufacturing techniques with proper procedures in place [39,46]. AM methods, key technologies and relevant materials for aerospace applications are briefly described below.

Laser Powder Bed Fusion Processes

The L-PBF technology utilizes high power laser(s) to melt and solidify individual layers of metal powder [19,47]. The L-PBF process involves powder feedstock delivered and coated onto a platform where the powder is selectively melted and solidified according to the CAD design. The laser scanning strategy and layer thickness, inert atmosphere and gas flow, and various other parameters are selected by the user and need to be optimized for the particular material and system used [19]. This is the most common form of metal AM with options for machines from a host of manufacturers and a large collection of materials such as aluminium alloys, steel alloys, copper alloys, nickel and iron superalloys, precious metals, refractory metals, titanium alloys and many more. Although L-PBF is known to produce fully dense components with high precision in a relatively short time [48], the manufacturing process is relatively expensive and is only applicable in industries with high-value components and where higher performance can result in cost reduction, such as the aerospace industry. Typical minimum features sizes of $\approx 0.2\text{--}0.4$ mm with maximum part size of about 300–400 mm cover a range suitable for most small-medium part size requirements [19]. Larger systems are now becoming available using multiple laser sources for enhanced productivity and with build volumes up to 500 x 280 x 850 mm [49]. Even larger systems (>1 m) are being developed but not commercially available or widely accessible yet.

Electron Beam Powder Bed Fusion Processes

Electron beam powder bed fusion (EB-PBF) is an AM process that uses a high-energy electron beam to melt metal powder layer by layer to fabricate fully dense solid components. The components are manufactured in a vacuum due to the electron beam requiring a vacuum. This reduces impurities in oxygen sensitive materials. These systems often use preheating by rapidly scanning the electron beam across the powder prior to melting of a layer, assisting to minimize residual stresses. Mechanical properties of components manufactured through EB-PBF can be similar to that of cast materials, demonstrating the suitability of components manufactured through this process as compared to traditional manufacturing methods [50]. The maximum build sizes are similar to that of L-PBF, with latest models having circular build diameter 350 mm and height 380 mm. These systems typically use larger

feedstock powder sizes, which results in larger roughness values. However, the systems are well developed and offer relatively short build times due to the high power and rapid scanning of the electron beam.

Directed Energy Deposition Processes

Dissimilar to powder bed processes, in directed energy deposition (DED) techniques, the stock material is deposited locally using powder or feeding wire feedstock directly into the meltpool which is created by an energy source, e.g. electrical arc, lasers or electron beams [46]. Since DED does not rely on a powder bed, it has some benefits in geometrical freedom compared to other AM techniques and are not limited to build size restrictions. Very large components can therefore be manufactured with this technique. DED machines are commonly used in the repair of high value existing components such as turbine blades. DED has a poorer resolution (minimum feature sizes of about 1 mm) [51,52] compared to that of powder-bed based technologies and is therefore used for larger components with reduced resolution [53], or where post processing machining operations can be performed.

There are various forms of DED techniques that each have different advantages and disadvantages. Laser Powder DED (LP-DED) is one of the most common using a laser as the energy source and a powder feedstock. The deposition head is attached to a robotic or gantry system and uses a local inert or fully purged chamber. In LP-DED the powder is blown into the melt pool and joined with the previous layer creating a freeform structure. An alternate DED method is Laser Wire DED (LW-DED), which uses the same concept, but the feedstock is wire. The wire feedstock may be off-axis or co-axial and has an advantage with high material efficiency, but at a reduced resolution. Other systems such as Arc Wire DED (AW-DED) and Electron Beam DED (EB-DED) use wire but with an electric arc or electron beam as energy sources, respectively. Each of these processes has differing feature resolution and deposition rates, which can result in varying heating and cooling rates and subsequent variance in metallurgical characteristics and in mechanical and thermophysical properties [54–57].

2.2.2. Materials for AM Aerospace

Distinctive materials for AM in aerospace are aluminium alloys, stainless steel, titanium alloys, nickel-and iron-based superalloys, copper alloys, refractory alloys, and steels, among other metallic materials. Most of these are used in pre-alloyed powder form, most often produced by gas atomization, or in wire form depending on the process [10,45].

Aluminium alloys have been an integral material since the inception of the aerospace industry. Their low cost, light weight, high strength to weight ratio and ease of manufacturing characteristics have resulted in aluminium being the most widespread material in aerospace until recent advances in composite technologies [58,59]. However, aluminium alloys suffer from poor elevated temperature capabilities, and in some high strength aluminium alloys, corrosion resistance is poor, thus limiting their applications.

Stainless steels are used in several aircraft and space components due to the high strength to weight ratio, excellent durability, hardness and good mechanical properties at elevated temperatures. Various classes of stainless steels are often used with AM including austenitic, precipitation hardened (PH), and maraging (martensitic and aged) [60,61]. These forms of stainless steel are used across components for engine and exhaust systems, hydraulic

components, heat exchangers, landing gear systems, and structural joints. Stainless steels also exhibit high corrosion, oxidation, and wear resistance depending on the environments [62].

Titanium alloys have been the subject of much interest in aerospace applications due to their high specific strength, excellent corrosion resistance and high-temperature stability [63,64]. Titanium alloys are also electrochemically compatible with polymer matrix-carbon fibre composites (PMCs) which are extensively used in modern aircraft [65]. Titanium alloys see substantial use in aircraft due to their high-temperature stability and specific strength [65]. Cryogenic applications, commonly found in rocket propellant tanks, are also potential use-cases for titanium alloys since they exhibit no ductile to brittle transition at low temperatures [65].

Nickel-and iron-based superalloys have become key materials for the manufacturing of disks and blades in high-pressure turbines for gas turbine engines. They are also used in many high temperature and cryogenic applications such as valves, turbomachinery, injectors, igniters, and manifolds [66,67]. Their excellent mechanical properties under extremely high temperatures, high pressures and in corrosive environments improve the efficiency of modern aircraft engines substantially [68]. Currently, over 50% of the mass of an advanced aircraft engine is comprised of nickel-based superalloys [69]. Iron-based superalloys are commonly used in high pressure hydrogen applications such as rocket engines to mitigate hydrogen environment embrittlement [70].

Copper-alloys are commonly used in heat exchangers, such as for combustion chambers for liquid rocket engines. This high heat flux environment requires a high strength and high conductivity alloy to properly cool the walls of the thrust chamber with a high pressure propellant or oxidizer. Common AM copper alloys used include GRCop-42, GRCop-84, C-18150 (Cu-Cr-Zr), C-18200 (Cu-Cr), and GlidCop [71–76].

Other metal alloys that can be used in aerospace applications include refractory materials such as niobium, tantalum, and tungsten and alloys such as C-103 [77]. These are used in extreme temperature applications such as in-space radiatively-cooled thrusters. Other materials such as Cobalt-based alloys including Co-Cr and Stellite are also used for a variety of elevated temperature applications [78,79].

Many of these materials exist as feedstock for metal AM machines and have seen varying levels of research into the material characteristics and properties produced using AM methods, with demonstrations of mechanical properties equal to, and sometimes exceeding those associated with typical manufacturing methods [80–83]. For this to be achieved, the process parameters and post-processing operations must be optimized to minimize porosity, residual stress, crack propensity and includes optimized post processing steps as required to enhance the material properties, including optimized heat treatments and hot isostatic pressing (HIP). For more information on AM materials and related issues, the reader is referred to [45] and Table 2-1 summarizes the more popular materials for AM in aerospace. Some of the alloys highlighted in this table include variations such as Metal Matrix Composites (MMC), including Al-MMC and Ni-MMC. Many of these alloys have also been used in test and flight applications as discussed in Appendix A: Prominent examples of AM applications in non-technical literature and Appendix B: Prominent examples of AM applications in technical literature.

Table 2-1: Metal AM materials and processes. [83,84]

Material	Metal AM Process									References
	L-PBF	EB-PBF	LP-DED	LW-DED	AW-DED	EBW-	CS	AFS-D	UAM	
Nickel-Based										
Inconel 625	X	X	X	X	X	X	X	X	X	[85–92]
Inconel 713	X		X							[93,94]
Inconel 718	X	X	X	X	X	X	X	X		[95–104]
Inconel 738	X	X	X		X		X			[105–110]
Inconel 939	X	X	X							[111,112]
Hastelloy-X	X	X	X		X				X	[113–118]
Haynes 214	X									[119]
Haynes 230	X		X	X	X		X			[120–125]
Haynes 282	X	X	X					X		[118,126–128]
Monel K-500	X		X	X	X	X	X	X	X	[129–131]
C-276	X		X		X		X		X	[113,130,132,133]
Rene 80 / 142	X	X	X				X			[130,134–136]
Waspaloy	X		X	X			X			[45,130,137–139]
Iron-Based										
SS15-5 GP1	X		X		X		X			[130,140–142]
SS17-4PH	X	X	X		X		X	X		[86,130,140,141,143–145]
SS304L	X		X	X	X	X	X	X	X	[45,129,140,146–151]
SS316L	X	X	X	X	X	X	X	X	X	[103,129,140,141,143,150–153]
SS321	X				X	X				[154–156]
SS347	X	X	X	X	X	X				[141,157,158]
SS410	X	X	X	X	X		X	X	X	[45,121,130,145,159–161]
SS420	X		X	X	X		X	X		[129,130,140,141,160,162]
SS440	X		X	X				X		[160,163]
4140 / 4340	X		X		X	X	X	X	X	[45,130,140,141,145,164–166]
Invar 36	X		X		X		X		X	[164,167–169]
JBK-75	X		X	X	X		X			[125,170–172]
NASA HR-1	X		X	X	X		X			[173–176]
Copper-based										
Cu110	X	X	X	X		X	X	X	X	[126,152,177–182]
GRCop-42	X		X				X	X		[176,183–186]
GRCop-84	X		X				X		X	[121,184,187–192]
C18150	X		X				X			[125,189,193–200]
C18200	X		X				X			[193,201–203]
Glidcop	X						X			[181,203,204]
Cobalt-based										
Haynes 188	X		X				X			[121,205]
CoCr / CoCrMo	X	X	X	X						[45,121,130,206,207]
Stellite 6, 21, 31		X	X		X		X			[144,208–211]
Titanium-based										
Ti-6Al-4V	X	X	X	X	X	X	X	X	X	[45,129,212–217]
Ti-6242	X	X	X	X			X	X		[118,130,218–221]
Gamma-TiAl	X	X	X	X	X	X	X	X	X	[147,222–229]
Aluminum-based										
AlSi10Mg	X		X		X		X			[45,230–233]
F357	X		X				X			[234,235]
1000	X		X		X		X	X	X	[236–240]

Material	Metal AM Process									References
	L-PBF	EB-PBF	LP-DED	LW-DED	AW-DED	EBW-	CS	AFS-D	UAM	
2024	X	X	X		X		X	X	X	[103,236,241–245]
2219	X				X	X		X	X	[103,235,246–248]
4043				X	X	X			X	[130,249,250]
6061	X		X		X	X	X	X	X	[126,130,152,214,250–253]
7075	X	X	X		X	X	X	X	X	[126,130,151,152,244,250,254,255]
7050	X		X		X		X	X	X	[232,235,250,256]
Scalmalloy	X		X				X	X		[230,232,257]
7A77	X									[235]
Refractory										
W-based	X	X	X		X	X	X		X	[45,258–270]
Mo-based	X	X	X		X	X	X	X	X	[263,267,271–278]
Ta-based	X	X			X	X	X	X	X	[45,91,214,263,267,279–284]
Nb-based	X	X	X			X	X	X	X	[214,263,267,285–287]

2.2.3. Development of New Alloys

Early adoption of AM techniques used materials that had traceability to common alloys manufactured by traditional processes. Many of these alloys are not the best-fit alloys for the applications in question, and have challenges in AM processing due to cracking or susceptibility to porosity formation, oxidation or other unwanted properties. AM offers a significant opportunity to introduce new and custom alloys that allow for more optimized processing to reduce cracking or to mitigate other challenges, while being further optimized for the end application. Examples of this include the introduction of custom aluminium alloys such as 7A77, 6061-RAM2, AlSi10Mg, Scalmalloy, and many others that offer high strength and successful AM builds with otherwise crack-prone alloys [288–292]. NASA also recognized the need for advanced alloys in AM and initiated development of the GRCop-84 and GRCop-42 copper-alloys for use in high heat flux applications, in addition to the iron-based superalloy NASA HR-1 alloy for use in high pressure hydrogen environments [293]. Many other examples of custom superalloys, aluminium, refractory metals and others being designed specifically for AM processing are being addressed as the industry need for more materials has been identified [294–297]. Beyond monolithic alloys, AM also provides the opportunity to create custom bimetallic and multimetallic material solutions, adding material locally within a design for optimization of thermal or structural loads. These can include a discrete material transition or a function gradient material [298–302].

2.2.4. Reduction in Processing Time

One of the key motivations for use of AM in aerospace is the significant cost reduction and improved lead times achievable compared to traditional manufacturing [303]. Aerospace components are complex to manufacture and often require unique alloys that have long procurement lead times in a wrought form and can also take significant processing time to form, machine, build, or assemble during the transition from the starting stock to a final part or subsystem. This lead time is compounded due to the various inspections and traceability requirements of material and parts throughout the process. The parts must follow a rigid process to ensure they are fabricated

safely and meet standards and certifications while also meeting the design intent. AM is no different and must still follow strict requirements to be flown safely in aerospace applications, but the lead time can be drastically reduced and the supply chain simplified. This is true for new part designs, but probably more important for spares or replacements that are out of production. This simplified supply chain and reduction in lead time, including logistical benefits, presents a substantial opportunity to reduce cost [304]. There are several examples that have demonstrated cost reductions of 50% and lead time reductions of 50% or more [14].

2.2.5. Process Certification

AM requires strict control of the entire process from powder through AM build and post-processing. This includes a detailed understanding of the process parameters that impact the microstructure and subsequent mechanical and thermophysical material properties. This certification process must be determined early in the design process to include both a methodology and traceability throughout all stages of AM. This includes implementation of adequate non-destructive evaluation (NDE) techniques, surface enhancements to improve fatigue performance, optimization of thermal processing, and other pre-, build and post-processing operations. Methodologies described by standardization and government organizations require strict control of machine parameters, powder lots, and include a detailed understanding of variations in powder or off-nominal build operations to ensure that parts meet the intended design and metallurgical characteristics.

Metal AM processes derive their flexibility and low-production costs to their fundamental unit material inputs (either powder, wire or sheet) which are then additively joined to generate the required as-manufactured geometry (including additional supporting structures) which are then post-processed as required to generate the final component. The production capabilities inherent to metal AM technologies allow previously unachievable system efficiency, however the fundamental certification challenge for these high-complexity AM components and systems is commensurately high. AM production phases are not compatible with aviation certification protocols derived for traditional manufacture [46,305], consequently the certification of these components is challenging; for applied engineering of AM products it is therefore necessary either to accommodate *existing* certification protocols or to actively engage with *emerging* protocols for AM structures.

Existing certification protocols are derived as a function of criticality and associated failure modes, consequently, the challenge for metal AM certification is lower for non-safety critical applications, and for applications that are subject to monotonic failure modes (such as static deflection) rather than for complex failure modes such as corrosion cracking or multiaxial fatigue. Therefore the certification challenge for a statically loaded assembly fixture may be less than for a non-safety critical structure, than for a dynamically loaded structural safety-critical component.

Emerging standards are under active development from government entities, industry groups, and standard organizations. These emerging protocols seek to find practical certification strategies that allow the benefits of emerging AM technologies to be applied without undue risk to safe aircraft or space vehicle operation. These emerging protocols include a combination of:

- Detailed understanding of material, process and design inputs on critical failure modes, potential defects and the classification of parts [306].

- Post processing standards, including for example, the application of hot isostatic pressing (HIP) to ameliorate the criticality of internal defects to component function and to improve the material microstructure and anisotropy [64,307].
- Safe-life design methodologies that presume the existence of internal crack-like defects and based on minimum critical flaw size calculations an allowable service life is defined [308].
- Non-destructive testing protocols, for example the application of computed tomography CT to quantify the existence and geometric attributes of internal defects [309].
- Material certification protocols and systematic quantification procedures to determine the consistency and fitness-for-purpose of metal AM inputs, especially for powdered metal.
- Developing fundamental requirements for process controls and part production including qualification of material processes, equipment controls, personnel training, and material property development.
- Allowable design and data management protocols to ensure geometric data is not corrupt and is appropriately encrypted and that appropriate file management procedures are followed.
- In-situ data acquisition provides an opportunity for increased confidence in internal AM structural integrity as well as potentially identifying geometric “high risk” regions that require enhanced resolution Non-Destructive Evaluation (NDE) or Non-Destructive Testing (NDT).
- Standards for witness coupon design and deployment to destructively validate integrity of a manufactured AM batch without damage to production AM components.

2.2.6. Topology Optimization & Lattice Structures

Topology optimization (TO) refers to a mathematical design method often used to optimize material layout in structural applications for given boundary conditions and a given set of loads and constraints [310]. The method is based on simulations and iterative shape optimization, realized in different forms in commercial software packages. Traditional TO utilizes the finite element method (FEM) to improve design performance [310] and was first popularized by Bendsoe and Kikuchi in 1988 [311]. The method has seen extensive development over the last three decades and has been a popular topic of continued research and development, for all manufacturing processes. It finds particular application in AM due to the manufacturing freedoms of the AM technologies, allowing optimization of shape and geometry for highly efficient designs with internal components and intricate features that are not feasible otherwise [312],[313].

Lattice structures are porous cellular architectures that typically refer to a network of solidly interconnected struts forming the faces or edges of cells [314]. Cellular porous material architectures are common in a wide variety of materials such as wood, bone, metals, and other microscopic structures. The conventional manufacturing of cellular porous structures includes direct foaming, spray foaming and vapour deposition techniques [315]. Both random stochastic structures and periodic designs exist as low-density cores in sandwich panel designs which are used in many industries. Additively manufactured cellular structures (or lattice structures) can mimic the geometry of these random stochastic and periodic structures, on both the macro and micro scale, creating new opportunities for complex designs with unique properties to be realized using the materials available for AM.

Metal lattice structures demonstrate high mechanical performance, offering high specific strength, good energy absorption capacity and robust thermal characteristics [316]. Periodic lattice structures typically exhibit better performance characteristics with respect to specific strength and energy absorption [317]. However, these periodic lattice structures typically have manufacturing constraints and therefore higher costs over stochastic structures [318,319]. L-PBF in particular can produce complex geometric structures at a high resolution with limited flaws and this shows potential for the manufacture of both periodic and stochastic cellular lattice structures for excellent performance components, with precisely controlled properties [320].

While these lattice structures allow for designs with interesting lightweight, thermal, acoustic and energy absorption properties, parameters for manufacturing still need to be improved to reduce common issues such as strut micro-porosity, surface roughness inconsistencies and poor mechanical properties that can occur [318]. These manufacturing errors and the overall choice of cellular geometry influence the fatigue performance of such structures, which is key for their use in aerospace and is discussed in more detail in [314].

2.2.7. Advantages of AM for Aerospace Applications

While the concept of lightweighting design exists in many industries, aerospace industry applications are particularly well explored due to the reduced fuel costs achieved through mass reduction on aircraft and spacecraft. Recently, a specific focus on the contributions of the aviation industry to the climate change phenomenon and environmental pollution has prompted on-going efforts to increase the energy efficiency of aircraft for the reduction of emissions.

With the target of 50% reduced aviation emissions by 2050, the International Civil Aviation Organization has invoked widespread technology development in more efficient aircraft systems. One of the most effective methods of increasing the energy efficiency of aircraft is through mass reduction techniques. For example, reducing the mass of a Boeing 787 by 20% would improve the fuel efficiency by between 10%-12% all while improving the acceleration and other performance characteristics [321].

AM techniques have been shown to help reduce cost and lead times and also for reducing the mass of components aboard spacecraft and aircraft. Many popular publicized examples of AM applications in aerospace boast mass reductions, among many other benefits. While this is an attribute that holds much promise, lightweighting is currently still not the primary driver for AM in aerospace. Lead time reduction is currently the main benefit, which can be significant for complex aerospace components often taking months or years of (traditional) fabrication time for complex systems. For example, the manufacturing freedoms offered by using AM techniques to manufacture the A320 nacelle hinge bracket, shown in Figure 2-2, enabled the TO design to reduce the CO₂ emission for the entire operational lifecycle of the nacelle hinges by 40% utilizing the mass reduction benefits of AM and TO [322]. The optimized design achieved a mass reduction of 64% from the original component, although the use of Ti-6Al-4V instead of the original HC101 steel accounts for about half of this mass advantage [322].



Figure 2-2: TO and AM of Airbus A320 nacelle hinge bracket. Left: TO design process. Right: Original bracket (top) and final TO optimized design (bottom) [322,323]. Copyright: Airbus.

Rotational components in aircraft and launch vehicle rocket engines also benefit highly from the application of lightweighting. While the reduction in mass of these components decreases the overall mass of aircraft and spacecraft, they have the added benefit of reducing rotational inertia which improves performance characteristics such as starting torque reduction, braking torque reduction and a lower interaction moment between the rotational components and the axle structures [324].

As demonstrated in [325], the potential for lightweighting of structural components, rapid prototyping in the design chain and complex designs of instrumentation hardware are all compelling applications for AM in the space industry. With more recent improvements in AM technology, new manufacturing opportunities have arisen for even the most technically demanding components. The technology synergises with space applications serving low quantities of components that exhibit mass and cost reductions crucial for increased development in space. With mission costs for space exploration exceeding €20,000 per kilogram, every gram saved translates to an increased payload capacity per launch and a consequent reduction in launch costs [326].

Besides lightweighting, part consolidation is another prominent benefit for the application of AM techniques in aerospace applications with nearly every example displaying massive reductions in part counts. Part consolidation, the act of consolidating multiple components into one piece without assembly operations, has widespread benefits across all industries where it is applied. For example, the Ariane 6 rocket injector head built through AM techniques achieved a part consolidation count of 248 individual components manufactured as a single component compared to traditional manufacturing methods all while reducing the mass of the injector head [327]. NASA has also demonstrated multiple applications of AM to rocket engine injectors since 2013 that significantly reduce part count and lead time [67]. One such example demonstrates a reduction from 115 parts to 2 parts and tested at full operational conditions with high performance equivalent to traditional manufacturing [328,329].

Part consolidation has the primary benefits of reducing the assembly operations and minimizing the usage of joining methods such as bolting, welding, brazing, soldering and chemical bonding methods [330]. Minimizing the use of these manufacturing methods in the production process drastically decreases the need for skilled labour throughout resulting in cost reductions [331]. The tooling necessary to manufacture components through traditional manufacturing methods is also reduced by using part consolidation through AM techniques. Another advantage is the reduced number of components requiring certification and associated documentation. Risk

mitigation is inherent in component consolidation as fewer joins and fewer processes are involved in the manufacturing process. The certification process is still a strict requirement though with AM and is being evolved across various organizations.

Tooling for traditional manufacturing techniques typically induce long lead times for manufacturing components and adds significant costs to the production process. AM techniques notably reduce the need for tooling to manufacture components. This has the benefit of improving lead times for manufactured components since tooling lines often take months, if not years, to set up. An example of this is the Pratt & Whitney stator blades which claim a lead time reduction of 15 months when compared to traditional manufacturing methods all while reducing the mass of the blades by 50% [332]. The design flexibility of AM machines also allows engineers to potentially manufacture multiple components on a single machine, sometimes in the same build process. This enables the tooling and assembly requirements to be reduced even further since multiple, already part consolidated components can be manufactured on a single machine. An example of a stator from a NASA liquid oxygen (LOX) turbopump can be seen in Figure 2-3, from [333].

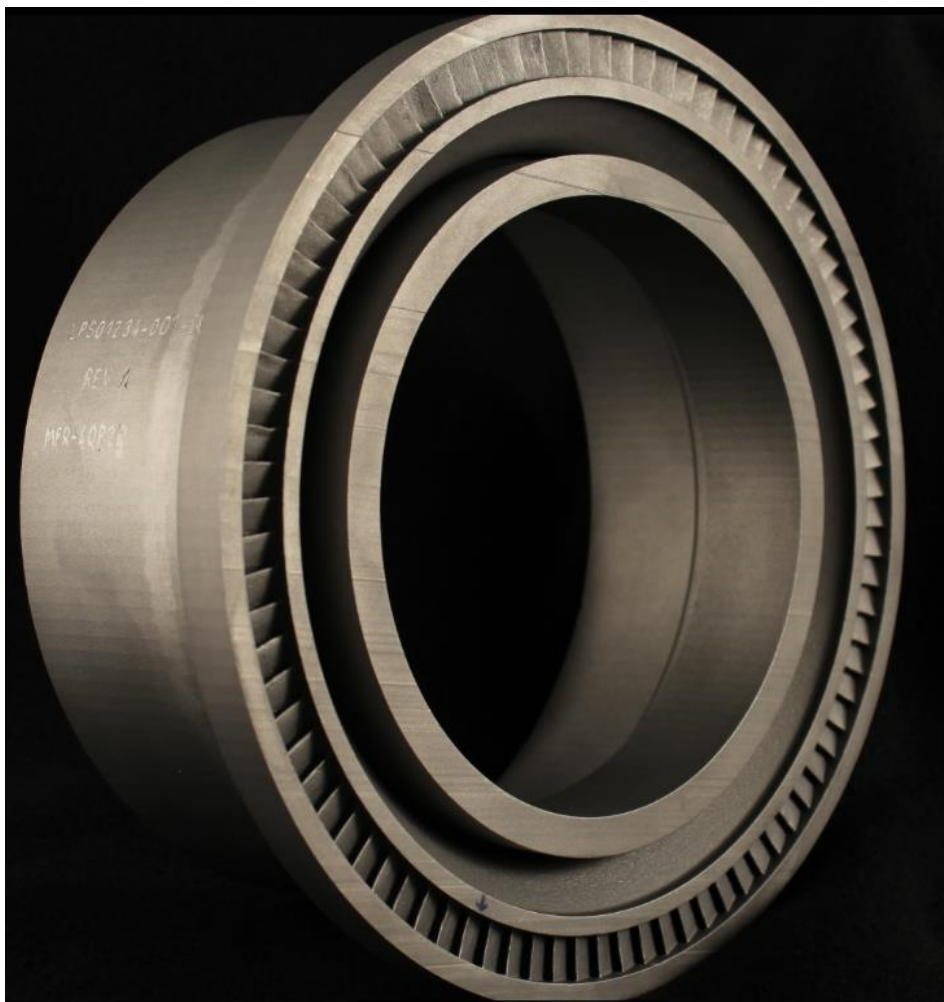


Figure 2-3. Additive Manufacturing Demonstrator Engine Liquid Oxygen (LOX) Turbopump Stator. Courtesy: NASA. [333]

Another benefit to part consolidation is the reduced need for warehousing for components and legacy components, including components no longer available. Reducing part counts through part consolidation directly reduces the need for large storage facilities and cataloguing of components. Furthermore, manufacturing and storing legacy

stock components for future use is all but eliminated since AM machines can produce components from a digital catalogue when they are needed. This is especially useful for unpredictable maintenance related components and legacy components that can be manufactured on demand, thereby eliminating the need for stagnant production lines [334].

Heat transfer device applications have a strong potential to disrupt traditional aerospace manufacturing processes by using AM techniques, due to complex designs and internal features required. Manufacturing these devices through AM allows for vast improvements in part count reduction, lead time reduction, lightweighting and cost reduction. For example, the GE9X heat exchanger boasts a 40% mass reduction, 25% cost reduction all while consolidating 163 components into a single part using AM techniques compared to traditional manufacturing methods [335]. Manufacturing heat transfer devices using AM also has performance benefits when compared to traditional manufacturing methods. As shown by Saltzman et al [336], replicating a traditionally manufactured aircraft oil cooler using AM techniques improved the heat transfer performance by up to 15% due to the improved design with larger surface area. Integrating heat transfer devices into structures and other components also benefits from the use of AM techniques. As shown in [337], the incorporation of a thermal anti-ice system into the structure of the leading edge of aircraft aerofoils has the potential to improve the performance of anti-ice systems. Advanced complex high-performance components and structures also see beneficial applications in aerospace using AM. The manufacturing demonstrator AM rocket engine design by Cellcore, shown in Figure 2-4, demonstrates the ability to produce a complex thrust chamber assembly in IN718 with integrated lattice internal cooling channels [18,338].



Figure 2-4. Cellcore prototype rocket nozzle featuring internal cooling channels. Copyright: ©SLM Solutions/ CellCore. [18]

NASA has established significant use of AM for heat exchangers, such as liquid rocket engine combustion chambers and channel-cooled nozzles. Various applications have been reported including combustion chambers using copper-alloys such as GRCop-42, GRCop-42, C-18150, and also Inconel 718 and Bimetallic Copper-Superalloy structures that have accumulated over 30,000 seconds and 400 starts of hot-fire time. Channel-cooled nozzles in various alloys including JBK-75, NASA HR-1, Inconel 625, Haynes 230, and bimetallic (Copper-Superalloy) structures have also been demonstrated by NASA and accumulated over 11,000 seconds and 250+ hot-fire tests [339]. While these combustion chambers and nozzles meet performance requirements, they also demonstrated significant cost and schedule savings for hardware delivery [340]. An example of combustion chambers and nozzles for liquid rocket engines can be seen in Figure 2-5 [341–343].



Figure 2-5. Examples of fully AM thrust chamber assemblies hot-fire tested at NASA MSFC including injectors, combustion chambers, and channel-cooled nozzles, including L-PBF GRCop-84, L-PBF GRCop-42, bimetallic L-PBF and DED, and LP-DED NASA HR-1. (Courtesy: NASA). [341–343]

Repairing high-value aerospace components using AM techniques also presents a major benefit to AM applications in aerospace. The high-performance components present in aerospace applications such as impellers, turbine blades and airfoils are typically expensive due to the use of high-value materials such as Inconel and Titanium alloys. These components are often subject to harsh environments such as corrosive environments, impact events, thermal cycles and high stresses which all contribute to reduced service life [344]. Traditionally these components were repaired through welding processes which impart residual stress formation and geometric distortion in the repaired components [345,346]. Using AM techniques to repair aerospace components has the benefits of inducing a lower heat input during the repair process compared to the traditional repair techniques and can meet all structural design margins and operational conditions when applied properly [347].

Hybrid manufacturing, i.e. combining AM with traditional manufacturing methods is widely used in different permutations. One good example is demonstrated in the AGENT-3D_IMProVe project, where the Fraunhofer IWS team and project partner MTU Aero Engines have developed a hybrid process chain using DED with wire filler material and conventional machining. The project focuses on saving semi-finished product costs by reducing both material quantities and machining times in order to cut tool costs. This was achieved by way of precise AM using DED on the conventionally produced substrate body. The project partners used a nickel-based alloy (Inconel 718) as a wire filler material. Several other examples have been provided across aerospace including bimetallic structures reported by Virgin Orbit and NASA to form combustion chambers [348].

2.3. Distinct Applications of AM in Aerospace

2.3.1. Structures and Brackets

Complex designs for AM of load bearing structures and brackets are often unconventional, seemingly organic geometries with exceptional strength to mass ratio characteristics. While these can be applied successfully, a thorough understanding of full load paths and environments must be well understood to take full advantage of the technology. Metal AM enables the manufacture of components of complex design within the build limits of the AM technology being used, in a variety of materials and with different design methodologies [349]. Brackets, structures, and frames are common examples of applications for AM and TO in aerospace. The production of organic lightweight structures was previously performed by conventional manufacturing methods with serious manufacturing constraints such as casting directions, symmetry, size, and tool path limits. Compared to typical manufacturing techniques, AM allows higher design complexities which can be utilized fully through TO to further reduce the mass of components in aerospace applications [321,350].

Aviation Structures and Brackets

The recent growth of metal AM and TO methods gave rise to a host of new projects to potentially reduce the mass of components by using TO together with AM without applying typical manufacturing constraints. The synergy of TO and AM techniques saw great success throughout many aerospace application areas in recent years [321,350]. One of the first examples of a structural, TO, AM component being used for the interior on a commercial aircraft is the A350 cabin bracket connector. Airbus utilized TO and AM to produce the bracket, as shown in Figure 2-6, from Ti-6Al-4V using L-PBF in 2014 [15,351,352].

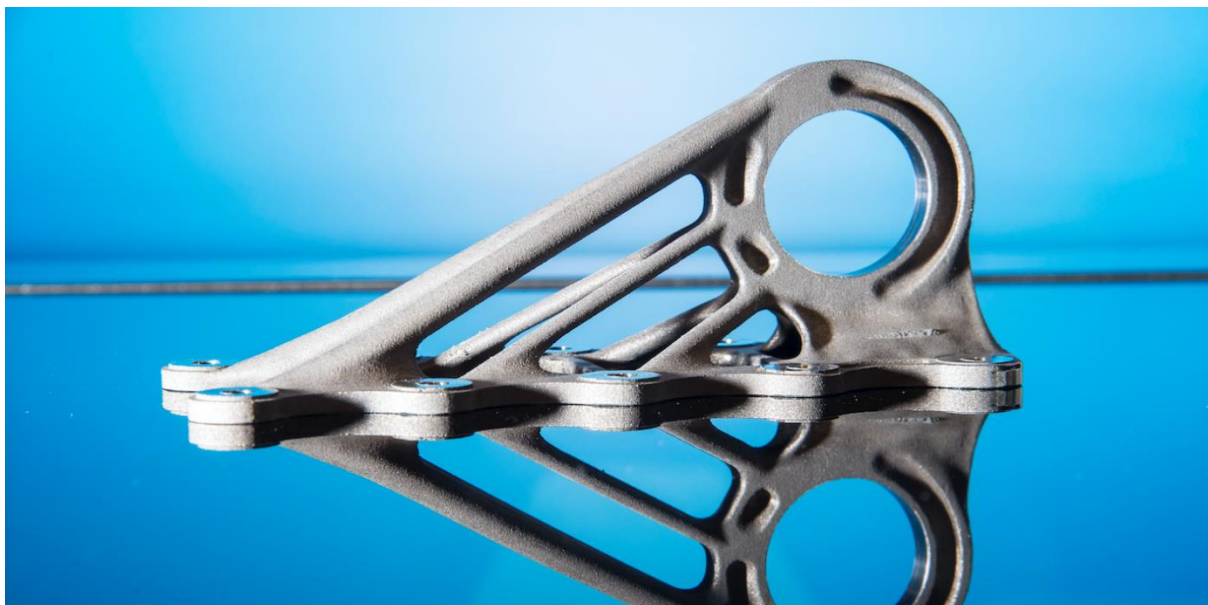


Figure 2-6. TO and AM of Airbus A350 XWB cabin bracket connector [15,351,352], by L-PBF of Ti6Al4V. Copyright Airbus.

In 2017, a Norwegian company Norsk Titanium was the first company to receive approval from the FAA to manufacture AM components, using DED techniques, that would experience in-flight stress loads [353]. These components, manufactured from Ti-6Al-4V, were the first structural AM components present on the Boeing 787

Dreamliner. Later in 2017, Airbus began production and installation of a titanium critical support bracket to be used on the A350 XWB. The component, manufactured via AM techniques, is part of the aircraft pylon, the junction between the wings and the engine [354].

A joint study between EOS and Airbus Group Innovations was carried out on the Airbus A320 nacelle hinge bracket shown in Figure 2-2 [322]. The component, to be manufactured from Ti-6Al-4V using L-PBF techniques, is intended to replace the current hinge brackets on the engine inspection nacelles with the goal of mass and cost reduction over current designs [322]. This study showed that compared to a conventional casting process, using L-PBF based AM techniques produced distinct advantages in mass reduction [323].

Mass reduction techniques have seen widespread adoption in the aerospace industry due to their inherent performance benefits. TO can be used as a powerful design tool for large scale structural components of aircraft for overall shape and mass optimization, when the appropriate environmental and combined loads are known. An early and popular application of TO is seen in the design of the A380 Droop Nose Ribs. Airbus UK, in partnership with Altair Engineering, used compliance-based TO intending to provide more stability to the existing Droop Nose Rib designs which resulted in a shear-web and truss design [323].

Recent applications of TO adaptation have allowed for shape preservation under given inputs which allow designers to optimize both for mass and, for example, warping deformation [321,350]. A good example of this is the cockpit windshield design. While the front fuselage of an aircraft has much potential for TO for mass reduction, damaging effects from bird strikes and extreme weather conditions need to be mitigated [323]. A study was performed which maintained warping deformation of the windshield to avoid cracking by constraining the windshield local strain energies [323,350]. The TO design with shape preservation dissipates the strain energies on critical loading from the elements surrounding the windshield [323,350]. The biggest challenges in incorporating TO into the design and manufacturing workflow for aerospace are the lack of well-defined inputs for loads and constraints, and the increased uncertainty in manufacturing quality for increasing complexity components – an area that will see much future development before AM techniques are used more routinely in aerospace applications.

Spacecraft Structures

The space industry commonly refers to economic activities related to orbital launch vehicles, manufactured components that go into Earth's orbit and beyond and other related services [355]. The space industry has often been reliant on new, advanced technologies to continue to push exploration and new missions. As discussed in [325], early applications using AM showed potential advantages in design and manufacturing for space applications. The first AM spacecraft structures were launched on the Juno mission to Jupiter in 2011. The components, a set of eight brackets, were used to attach the waveguide, which is used to transmit radio frequency signals between components [3]. Since then, AM technology continued to mature and confidence in metal AM throughout the aerospace industry increased significantly resulting in AM taking a larger role in the manufacturing of components for launch vehicles and satellites [3,356]. The Sentinel satellites, a series of earth observation missions from the European Space Agency (ESA), set out to develop AM solutions for the latest editions of the mission, the Sentinel-1C and Sentinel-1D [345,346]. The Sentinel-1A and Sentinel-1B, launched in 2014 and 2016 respectively, both utilized conventional manufacturing techniques which allowed a set of new brackets,

developed by RUAG and Altair, produced using AM techniques to be compared to the traditional design. Working together with EOS and Altair, RUAG designed and produced an optimized version of an existing Antenna bracket shown below in Figure 2-7 [345,346].

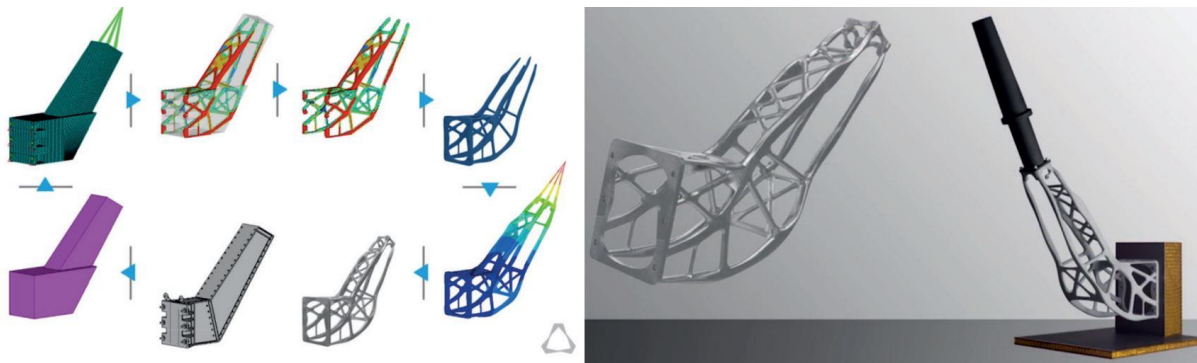


Figure 2-7. TO process of an antenna bracket for the Sentinel-1C and Sentinel-1D. Photo courtesy of Altair Engineering. [346]

RUAG in collaboration with Altair then went on to design and produce a star tracker camera bracket using TO and AM techniques. The precursor bracket, manufactured using typical subtractive methods, saw mass reduction advantages using TO and AM on the updated design. The bracket was manufactured in an EOS M290 L-PBF machine from AlSi10Mg. The bracket, in its testing configuration shown in Figure 2-8, underwent various testing procedures such as Computed Tomography (CT) scans, geometric analysis, tensile, microscopy and structural response testing [15].

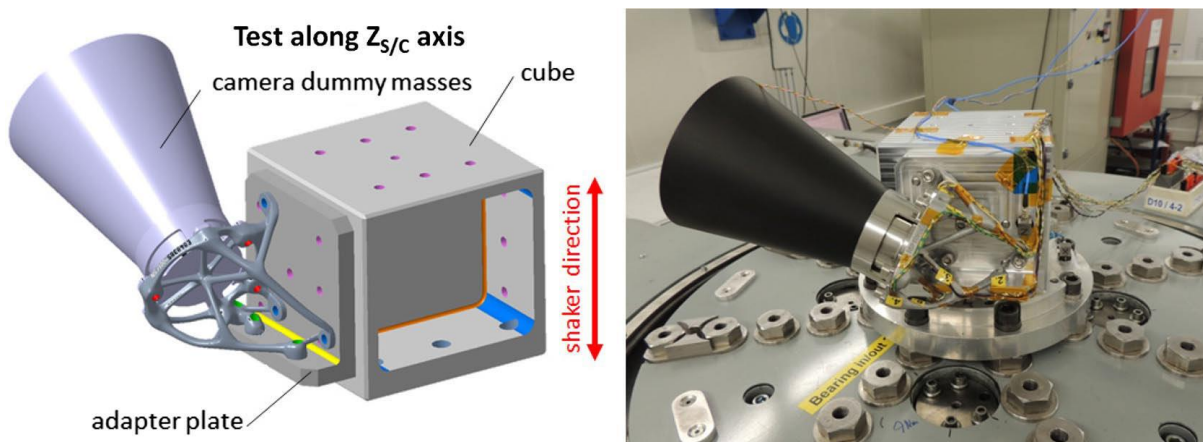


Figure 2-8. Test configuration of the bracket with adaptor plate and dummy masses. Copyright: The American Institute of Aeronautics and Astronautics (AIAA), Inc. [15]

The manufacturing of mirror components for satellite applications often push limits on several length scales. Besides the optical or sensor devices, the corresponding supporting structures are affected by high technical requirements as well. At the ESA L class ATHENA (Advanced Telescope for High ENergy Astrophysics) mission, Silicon Pore Optics (SPO) need to be aligned and carried by a supporting structure with a diameter of several meters. AM has been used at Fraunhofer IWS to print those supporting structures which require high process flexibility and scalability. Tailored processes and technologies are indispensable for overcoming the accompanied manufacturing restraints. A hybrid AM process has been developed in a cooperation between Fraunhofer IWS, ESA and Airbus merging the high flexibility of Laser Metal Deposition (LMD), another name for LP-DED, with the precision of cryogenic machining for the manufacturing of large volume titanium

components. So far a demonstrator of 1,5 m in diameter has been manufactured to demonstrate the feasibility of the whole process chain for this application, shown in Figure 2-9 [17].

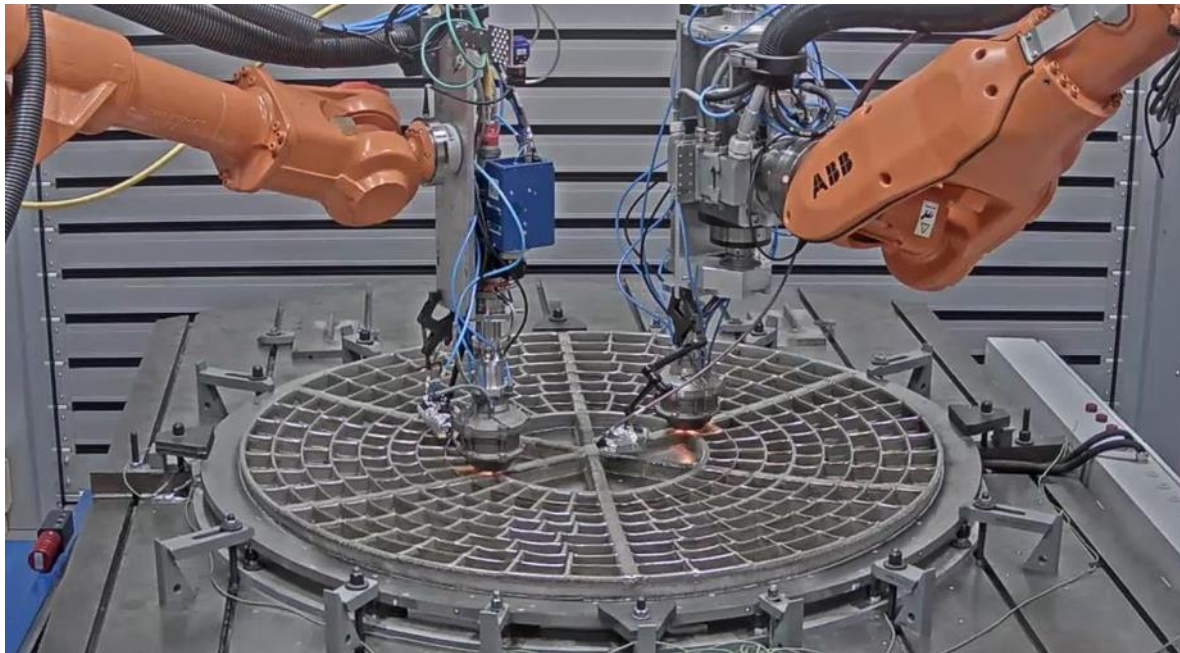


Figure 2-9. Example of 1.5 m mirror support being manufactured as technology demonstrator (image courtesy of Fraunhofer and European Space Agency). [17]

Lattice Structures

Recently, lattice structures have gained widespread interest from engineers and scientists for their multi-faceted design advantages [357]. Over the past decade, TO has seen commendation among designers to optimize patterns of both meso-structures and macro-structures, and sometimes including novel cellular (or lattice) structures (CSs) [358]. While lattice designs previously saw use in rectangular forms in heat exchangers and filter elements, it is now possible, using computational algorithms such as TO, to design far more complex structures and internal geometries using open-pored cellular / lattice structures [359]. While there is a lot of potential for these structures, they have numerous design and manufacturing constraints that need attention, limiting their application so far, this is discussed in depth in [314].

CubeSat's are types of miniaturized satellites designed to be manufactured and deployed in a modular system which allows for increased accessibility to space for small satellite manufacturers. These satellites are designed to be both lightweight and small enough to be carried as secondary payloads by satellite launch providers. Mass reduction of CubeSat structures presents the opportunity to increase the number of small satellites per payload and reduces the costs of launching these satellites to orbit. nTopology, in collaboration with the U.S. Air Force Institute of Technology (AFIT), completed a case study on the application of AM and lattice techniques for CubeSat bus structures. Built from Inconel 718 on a L-PBF machine, the structure has the goals of reducing mass, part count, lead times and increasing stiffness. The structure, shown in Figure 2-10 below, achieved a mass reduction of 50% and consolidated 150 components to less than 25 components which results in 6 times less possible failure locations [360].



Figure 2-10. nTopology and AFIT lattice CubeSat bus structure. Credit: nTopology. [360]

Thales Alenia Space, in collaboration with the European Space Agency (ESA), also used the combinations of AM, TO and lattice structure techniques to develop a solar panel deployment mechanism. The mechanism, intended for a satellite solar panel deployment application, uses TO to reduce the mass of the supporting structure and then further reduces the mass by lattice insertion techniques [361]. The structure, shown in Figure 2-11 (a) below, reduced the mass of the design by a factor of 5, the costs by a factor of 4, and reduced the total component count by 10 through part consolidation [361]. Another example of lattice use, in this case for thermal application, is shown in Figure 2-11 (b).

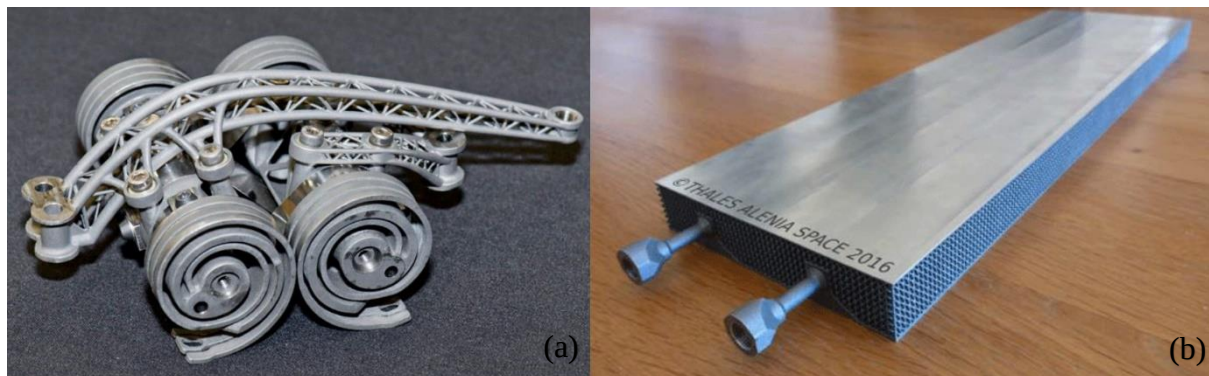


Figure 2-11. Thales Alenia Space examples including AM lattice structure usage: (a) satellite solar panel deployment mechanism – “Adel’light hinges for solar arrays of satellites” and (b) satellite sandwich panel “Diphasic heat spreader”. Courtesy of Thales Alenia Space. [361,362]

A collaborative project by Airbus, APWorks and Autodesk was created with the task of reducing the mass of a partition wall of the A320 series of aircraft. The Bionic Partition Project sought out to utilize modern manufacturing techniques such as AM, TO and lattice optimization to improve the partition design over existing designs. The new partition used generative design similar to TO to formulate an initial strut structure with the objectives of reducing mass while maintaining maximum displacement constraints [363]. Further mass optimization was then applied by converting the initial TO geometry to lattice beams, resulting in further mass reduction [364].

Combinations of conventional manufacturing and AM techniques present an interesting opportunity to reduce the impact of imperfect parameterization currently in metal AM solutions. Autodesk used both AM techniques and conventional metal casting processes to produce a lightweight frame design for aircraft seats. Using Autodesk's Netfabb software package, the geometry for an aircraft seat was produced using lattice and surface optimization, algorithms similar to TO techniques. These technologies were combined, producing positive molds for the seat frames containing complex lattice geometry, printed in plastic to reduce costs and time, and used to create ceramic casting molds in a similar method to the "lost wax" process. Finally, the frame was cast in magnesium allowing for the vast production quantities necessary for aircraft seat manufacture [365,366].

An obvious but less widely reported application of cellular lattice structures in metal AM is for internal fill geometries of components. TO techniques for AM applications can be further improved by shelling components to further reduce the mass, but this presents the challenges of internal voids which require internal support structures for any overhanging sections and can trap unprocessed powder increasing mass [314,318]. This challenge is often solved by using internal lattice geometry in printed components which replaces internal support structures with functional cellular lattice structures with load-bearing capabilities [367]. However, these internal lattices often do not eliminate all the unprocessed powder and holes need to be placed in a design to release this powder. Figure 2-12 demonstrates the applications of using an internal lattice fill to reduce internal mass. The branches of internal macro lattice designs are usually designed such that their angles did not exceed the minimum build angle of 45° common to metal AM machines [324]. Such mass reductions on impeller components reduces the rotational inertia and increases the performance.

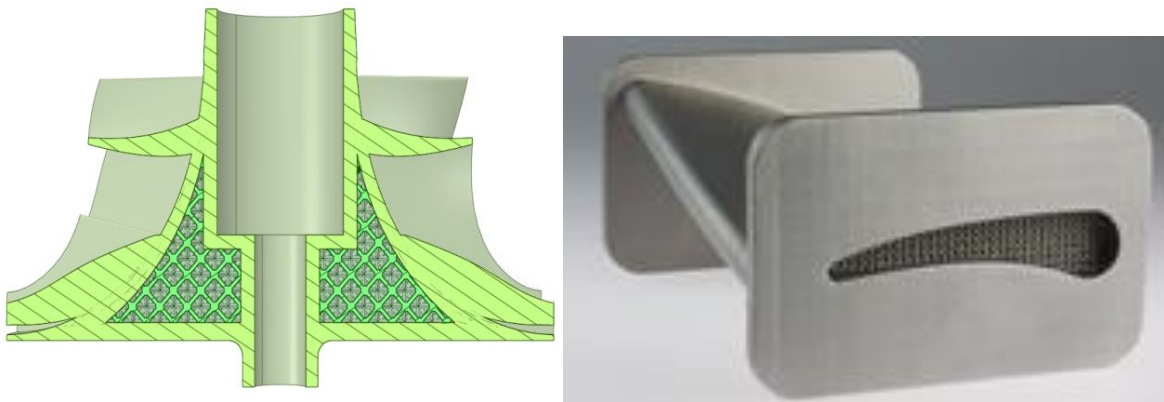


Figure 2-12. AM built impeller with internal lattice – design concept from [324] and a similar manufactured physical part with internal lattice from a different study. Courtesy: SLM Solutions.

2.3.2. Static and Dynamic Engine Components

The static and rotational components present in both aircraft and spacecraft rocket engines are subject to both extreme performance requirements and harsh environments such as elevated pressures, temperatures, and corrosive or embrittlement conditions. These performance requirements often lead to highly complex part geometries and special materials being used. Since components such as compressor blades, turbine blades, inducers and impellers have highly complex geometries, AM techniques can be used to increase their performance [21]. As complexity increases, traditional manufacturing techniques are limited by their constraints and part complexity is effectively restricted. However, AM technology enables more designer freedom from geometric

constraints commonly found using traditional manufacturing techniques allowing for more intricate final components [344], [303]. There are limitations of the AM process though in producing some of these geometries in build angles and also rougher surfaces than traditional machining. Typical build angle constraints limit overhangs to maximum 45 degrees relative to the vertical, with variation of surface roughness depending on the overhang angle. Down-facing surfaces often have rougher surfaces with excess material and irregular surface morphology. The build angle change affects the local thermal history as underlying powder (in powder bed fusion) does not conduct heat away, leading to higher temperatures in these regions. The inclusion of support structures may alleviate this and reduce the possibility for residual stress and warping, but this requires post process support removal and this may be difficult in complex components. These and other limitations are being solved incrementally by various researchers and manufacturers, to improve the quality of metal AM components to remove some design constraints.

An early case study in this domain was reported by Allison et al in 2014 [368] showing the potential for AM applications on turbomachinery components such as an impeller. The Impeller, manufactured from Inconel 718 using AM techniques, demonstrated promising results functioning as predicted during testing. Interestingly, during testing, the impeller test rig consisted of multiple other components manufactured through AM techniques. Both the inlet guide vane assembly and the aluminium volutes were manufactured using SLS with Duraform HST and additively manufactured sand casting molds respectively.

General Electric's next-generation GE9X engine, which made its maiden flight in January 2020 aboard the new Boeing 777X, is the largest and most powerful commercial jet engine ever produced [369]. The GE9X, a high-bypass turbofan engine, features a large number of additively manufactured components integrated directly into the core structure of the engine. Notably, 228 low-pressure turbine blades are manufactured from TiAl using EB-PBF manufacturing techniques for each engine with the express goal of mass reduction [335].

A case study by Siemens highlights the manufacturability of turbomachinery by the means of AM. Siemens, after initial investments in AM in 2009, installed their first AM components in gas turbines in 2013, began to work on hot gas path blades which were successfully printed and installed in 2017. They then went on to assess the viability of commercial applications of closed radial impeller wheels for turbo-compressors using AM techniques such as L-PBF and EB-PBF. Depending on successful testing and regulatory approval, the radial impeller wheels show promising results to improve the efficiency of turbomachinery [370]. While these are not directly aerospace examples, they are a good indication of the potential and capability for high performance components.

Engine maker Pratt & Whitney has produced more than 100,000 prototype components using AM over the last 25 years and currently produces compressor stator blades and ring synch brackets for the PW1500G engines which are part of the Bombardier C Series aircraft. These stator blades are used to guide airflow through the compressor of aircraft engines [332].

While still in their infancy, lattice structures are gaining interest into turbomachinery components to increase performance. A recent study done by [324] demonstrates the application of internal lattice structures on compressor impellers to reduce the mass and moment of inertia. Manufactured using AM with Ti-6Al-4V, the internal lattice impeller design attempts to both improve the performance of the impeller, and the AM manufacturing process by reducing residual stress build up in the printing process.

Large modern aircraft engines typically consist of tens of thousands of components and, while not all components are feasible to manufacture through AM techniques, many components besides the core turbine and compressor blades can be reasonable applications for AM. Honeywell Aerospace completed a study into various turbomachinery components for applications in their HTF7000 series of aircraft engines. The study details not only high-pressure turbine applications for AM but also designs for a tangential on-board injector, a second stage high-pressure turbine nozzle, an atomizer shroud, an inlet booster rake, and an engine mount. This demonstrates that a large quantity of aircraft engine components have the potential to be manufactured using AM [371].

Probably the most well-known AM component application in aerospace is the General Electric LEAP engine fuel nozzle. Beginning production in 2015, over 30 000 nozzles were produced by the year 2018 and it is still in production [372]. Utilizing L-PBF with a Cobalt-Chrome alloy, the fuel nozzle, shown in Figure 2-13, is installed on multiple engines currently in service on commercial aircraft.



Figure 2-13. The GE LEAP Fuel Nozzle. Courtesy of GE Additive. [372]

After the success of the LEAP engine fuel nozzle, GE then went on to produce a myriad of AM components for the new GE9X engine series [335]. The GE9X engine boasts 304 additively manufactured components across seven multi-part assemblies resulting in over one-third of the components on the GE9X series engines being produced using AM techniques. The engine will feature 28 of the LEAP engine fuel nozzles, 228 low-pressure turbine blades, a T25 sensor housing, a combustion mixer, 8 cyclonic inducers and a heat exchanger all manufactured using AM. All these components, except for the LPT blades and the heat exchanger, are manufactured from a Cobalt-Chrome (Co-Cr) alloy using L-PBF processes.

DED is being used to fabricate overhanging large scale metal structures. These structures can be built through multi-axis orientation systems that orientate the constructed component so that layers are always built with no overhangs. A five-axis DED system produced by The Welding Institute (TWI) is used in the production of an

Inconel 718 helicopter engine combustion chamber with effective overhang angle constraints of close to 90° [303,373].

AM techniques have been extensively applied to repair existing components [303]. Damaged components can be repaired using AM instead of replacing and scrapping them, this can result in significant cost savings for expensive components commonly found in aerospace applications, not only for saving on the cost of new components but also on the requirements for spare components inventories and new part manufacturing lead time savings [374]. This is also relevant to legacy and aging aircraft for which components might not be available at all. Further studies have shown that restoration through AM has, compared to traditional methods, a substantially lower environmental footprint [375]. Restoring the structural integrity of integrally bladed rotors, also known as Blisks, provides cost-effectiveness since these components can cost tens or even hundreds of thousands of dollars to manufacture. The T700 Blisk, made from AM355 steel suffers from erosion on its leading edge. This led to a case study by Optomec which utilized Stellite® 21, an erosion-resistant cobalt-based material to repair the leading edge. Passing spin and low-cycle fatigue tests, the repaired blisk showed positive metallurgical, tensile and erosion qualities enabling its certification [376]. Figure 2-14 below shows the repair process using LENS AM (LENS is a DED process) where (A) shows the LENS system repair process, (B) shows the T700 blisk after leading-edge repair and (C) shows the repaired blisk after finishing [376,377].

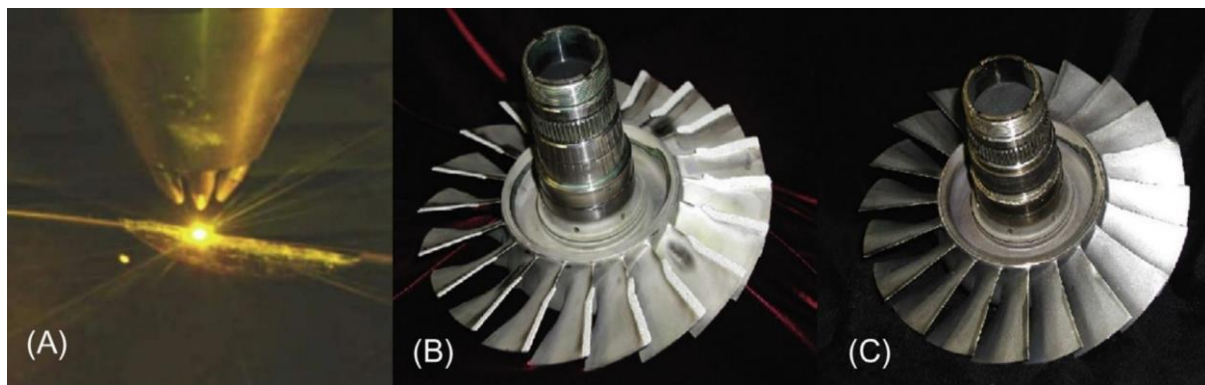


Figure 2-14. Blisk repair solution using LENS AM. Photo Courtesy of Optomec. [376]

The ability to produce impeller and airfoil components through the use of DED processes with high-quality mechanical properties compared to typical manufacturing techniques was demonstrated in [378]. They illustrate the ability of DED to repair turbine blades and a fuel nozzle swirler using Inconel-718, which is usually a difficult material to deposit using typical welding processes or laser deposition processes without causing cracks [378].

Complex turbomachinery has been demonstrated in various applications by NASA [379–382]. Impellers, pump volutes, turbine blisks, turbine stators, turbine exit guide vanes, and the turbine nozzle have all been demonstrated for an AM liquid rocket engine within a liquid hydrogen fuel (LH₂) and liquid oxidizer (LOX) turbopumps. The liquid hydrogen fuel turbopump was successfully manufactured with 45 percent fewer parts than traditional manufacturing and completed spin and engine testing operating at more than 90,000 revolutions per minute. A LOX pump demonstrated the use of AM within relevant test environments as well. It was noted that process improvements were required for these AM components include surface finishing, material properties, application of support structures during build, and dimensional tolerances [383].

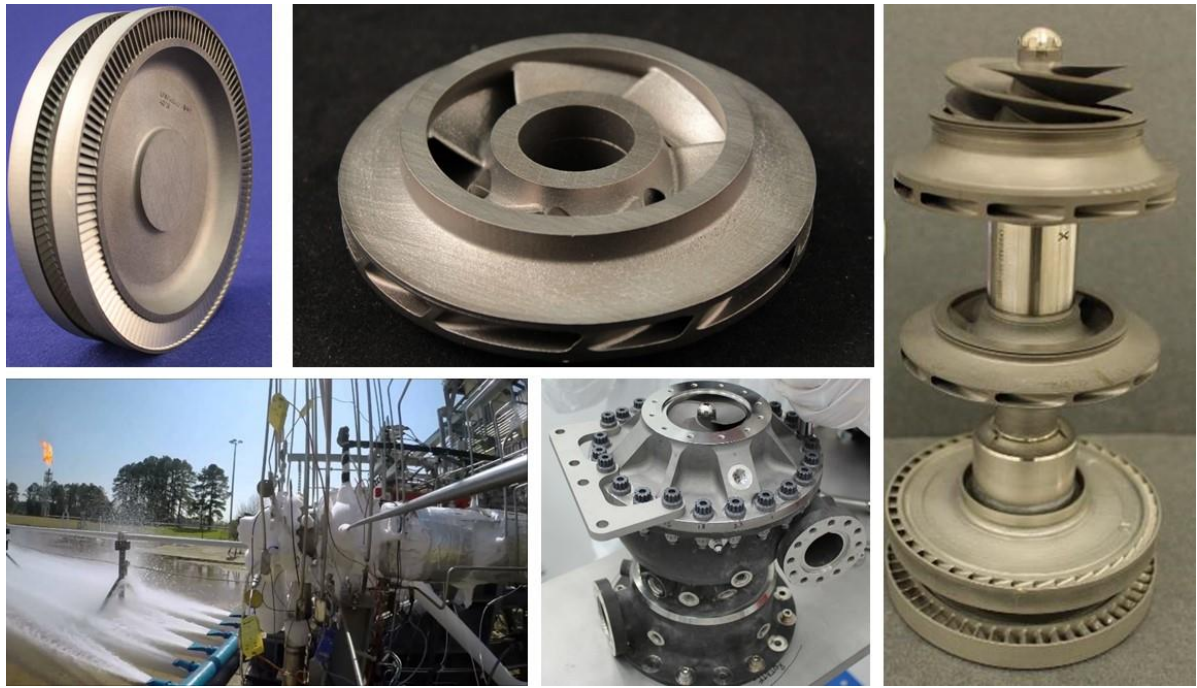


Figure 2-15. Examples of AM used within liquid hydrogen and liquid oxygen turbopumps at NASA MSFC. (Courtesy: NASA). [333]

2.3.3. Thermal Devices

Advanced heat transfer devices manufactured employing AM techniques have strong potential to reduce manufacturing time and improve the performance characteristics of heat exchangers (HXs), heat sinks (HSs) and heat pipes (HPs) mainly when used to manufacture complex free-form internal geometries [384]. In particular, HXs are crucial for components in aerospace applications such as the performance of aircraft and spacecraft engines, while HSs are typically used in crucial electronic applications present on both aircraft electronics and spacecraft electronics. Conventional manufacturing techniques for HXs and HSs typically include brazing techniques and CNC milling techniques respectively [385]. While brazing is an effective manufacturing method for HXs, it typically consists of a high amount of time-consuming skilled labour work, high tolerances for achieving reasonable gaps in joining, and often expensive materials which increases the cost of components. Furthermore, HSs are limited by the complexity achievable through the use of CNC milling techniques. For these reasons, AM techniques are being used to improve the performance of HXs and HSs by increasing their complexity and providing flexibility for designers to improve the surface area to volume ratios, all while eliminating the brazing process. Moreover, brazing techniques are typically limited to the materials compatible with the process while AM allows for a wider variety of materials to be used on these devices, although new material developments are a significant growth opportunity for AM. Gobetz et al, performed a study to demonstrate the feasibility of using AM techniques in a heat exchanger for aerospace applications with the goals of more direct manufacturing as opposed to the traditional manufacturing and assembly techniques present in current applications of HXs. Built from Al-10Si-0.5Mg on an EOS M280 L-PBF machine, the HX functioned effectively under an air-to-air testing setup after successful manufacturing [386].

A case study done by [336] highlights the ability for AM to be utilized on typical aircraft HX applications. The study compares the AM HX functionality relative to conventionally manufactured HX components with an

example of an aircraft oil cooler. The component in question was manufactured as a direct copy of an existing conventionally manufactured HX. Using L-PBF techniques, the component was realized with AlSi10Mg. Although the AM manufacturing process produced some defects in the component, under testing the AM component performed significantly better than the baseline, conventionally manufactured component demonstrating the functionality of AM HX in realistic applications.

More notably, General Electric has incorporated an AM HX into their latest GE9X engine series currently in operation [335]. Manufactured from Aluminium alloy (F357) on an L-PBF machine, the HX functions as a vital component in the engine with the goals of reducing size, reducing mass, reducing costs, and improving durability.

NASA's latest mission to Mars saw the delivery of yet another rover to Martian soil. The rover, called Perseverance, has the task of finding signs of ancient life and to collect samples of regolith and rock for future return missions to Earth. The rover, which landed on Mars in February 2021, carries 11 components fabricated using metal AM techniques. Of these 11 components, 6 are AM built heat exchangers used in the Mars Oxygen In-Situ Resource Utilization Experiment, or MOXIE. These heat exchangers are exposed to the Martian atmosphere and have to withstand temperatures of over 800° Celsius for extended periods of time. This resulted in nickel-based superalloys being used for the heat exchanger structures which simplified and improved the capabilities of the MOXIE system compared to if they were built using conventional techniques. Besides the 6 heat exchangers, the Perseverance rover features 5 other AM built components on its Planetary Instrument for X-ray Lithochemistry, or PIXL. To reduce the mass of the PIXL, AM was used to fabricate a two-piece titanium shell, a mounting frame and two supporting struts which all resulted between 3 to 4 times less mass compared to components manufactured through traditional methods [387].

Since AM techniques provide an unprecedented amount of complexity in components compared to traditional manufacturing methods, often there are limitations on design capability to fully optimize devices such as HXs and HSs. Optimization tools are gaining traction for their application in HXs and HSs through the application of AM techniques in the form of thermal TO methods. A paper by de Bock et al [388] exhibits the potential performance of using hybrid analytical thermal TO (HAATO) on an HS application. While still in their infancy, these optimization techniques have the potential to drastically increase the performance of HSs and HXs by increasing their geometric complexity. Figure 2-16 displays the incredible internal complexity of their AM HX designs intended for motorsport applications [389]. The highly complex structure is optimized for large surface area, enhancing the heat exchange. While this is not an aerospace example, the principles and advantages are the same for aerospace components.



Figure 2-16. Conflux sectioned F1 heat exchanger application. Picture Courtesy of Conflux Technology. [389].

Examples of internal cooling channels are present in cylinder blocks of internal combustion engines and internal cooling channels in high-pressure turbine blades in aircraft. These integrated systems are generally constrained by typical manufacturing techniques such as joining, brazing, casting, and CNC milling processes. AM techniques offer vastly more design complexity for these components for placement options for internal cooling channels and micro-lattice structures and related designs. Cobra Aero, a small UAV designer and manufacturer, sought out to improve their UAV engine using AM, optimization techniques and lattice structures with the goals of reducing mass and optimizing heat transfer characteristics [390]. Using AM, they produced a cylinder block leveraging multi-physics simulations which accounted for thermal, flow characteristics, stress and pressure factors resulting in an optimally designed cylinder block that was more thermally efficient than designs realised through conventional manufacturing methods (Figure 2-17).

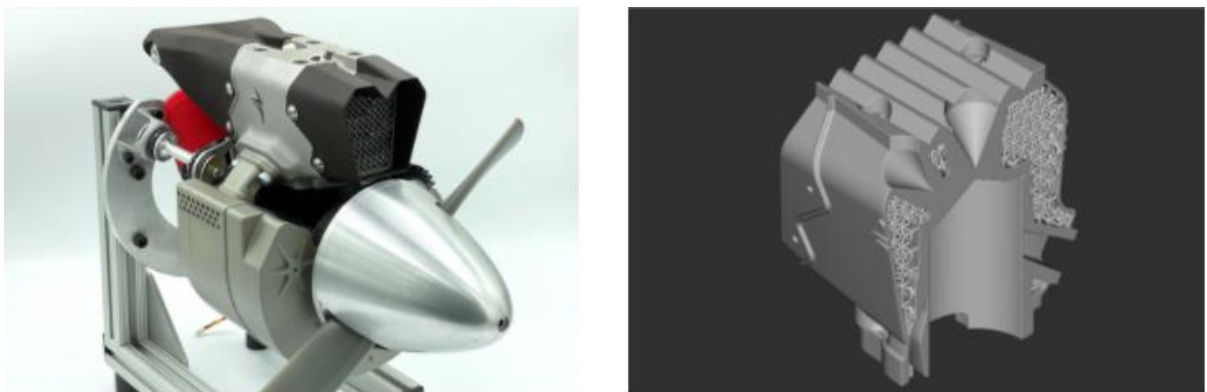


Figure 2-17. Cobra Aero AM and lattice cylinder block design. Courtesy of Cobra AERO. [390]

Thermal anti-ice is a widely used anti-ice system present on commercial aircraft that limits the ice build-up on the leading edges of aircraft airfoils. Bici et al [337] explored the application of lattice structures to a multi-functional panel intended for this application. The design focusses on a single piece airfoil, manufactured via AM techniques, with an internal lattice structure that distributes hot air to the leading edge of the airfoil to prevent ice build-up. Furthermore, the design mitigates structural stresses by using lattice structures to reduce the mass of the leading edge.

HiETA, an engineering company based in the United Kingdom, uses AM techniques such as L-PBF to produce a variety of products for the aerospace market with a particular focus on heat exchanging and cooling devices [391]. In particular, they produce an internally cooled radial turbine wheel and housing using AM techniques. The turbine wheel, manufactured from CM247 LC or Inconel alloys, features complex internal cooling channels that enable higher operating temperatures for turbine engine applications all while reducing the mass of components.

2.3.4. Liquid Fuel Rocket Components

Space-based activities have seen rapid increases over the last two decades in the form of space exploration, scientific endeavours, communication satellites, earth-observation satellites, permanent human presence on the International Space Station (ISS) and many other missions. All this increased activity has been assisted by programs such as NASA's Artemis program and many private companies entering the orbital launch capabilities market [392–394]. With so much competition among launch providers, innovation has soared and advanced manufacturing techniques such as AM are widespread among space launch systems and satellites. Satellites, space stations and space launch systems boast the most expensive man-made objects in existence with the ISS costing \$160 Billion alone and more sections are still being added [395]. The benefits of AM in cost reduction compared to conventional manufacturing methods has the potential to make space-based activities more accessible.

As launch systems provide the infrastructure to access space, the efficiency of these systems is vital to reducing costs for all space-based systems. At the core of these systems is the Liquid Fuel Rocket Engine which is widely used in most launch vehicle applications. This complexity is leveraged through the use of AM techniques in more modern rocket launch systems to increase the performance of the system. With unofficial quotes reporting the in-development SpaceX Raptor engine containing as much as 40% by-mass AM fabricated components [396], space launch systems are proving to be one of the primary applications for AM built components.

Injector systems on rocket engines typically consist of hundreds of individually manufactured components which are then brazed or welded into a single injector head [327]. These injector heads are generally time-consuming and expensive to manufacture through traditional manufacturing techniques. However, AM techniques can be employed to drastically reduce the cost and time needed to manufacture these injector components. Ariane 6, set to be launched in 2020, is the latest rocket in production by the Ariane Group and as part of their RAMS (Reliability, Availability, Maintainability and Safety) guidelines, they are looking for innovative manufacturing solutions. With an overall goal of 40-50% lower production costs over the Ariane 5, AM was employed throughout the project for increased cost efficiency and performance [327]. This injector core was produced with Inconel 718 on an EOS M 400-4 system utilizing 4-laser technology to rapidly decrease production time. The injector core is shown below in Figure 2-18.



Figure 2-18. Injector core for the Ariane 6 Rocket. Photo via Ariane Group/EOS. [327]

Copenhagen Suborbitals, an entirely crowdfunded and non-profit space company, has been assessing the feasibility of utilizing AM components into their workflow to reduce costs [397]. Their latest AM case study, swirlers for a coaxial swirl injector, have been manufactured using Digital Metal's proprietary binder jetting metal AM technique. A project was completed by DLR, a German Aerospace Centre, also involving coaxial injector applications of AM in aerospace [398]. The project was based on a small satellite launch vehicle (SSVL) program with the aims of reducing costs to orbit for small satellite producers. After being manufactured using L-PBF techniques, the injector was tested successfully showing good performance characteristics and is set to pave the way for further development at DLR. NASA has also demonstrated and reported hundreds of successful injector hot fire tests in a variety of fuels for liquid rocket engine applications [67].

At NASA's Marshall Space Flight Center (MSFC) in Alabama, much research and development is being performed into AM applications for rocket engine components. NASA, in collaboration with Stratasys Direct Manufacturing, developed and tested an injector system using L-PBF AM techniques [399]. The injector, manufactured out of Inconel 625, allowed for unique flow characteristics only achievable through AM technologies. The MSFC then went on to manufacture and test a demonstrator rocket engine that was comprised of 75 percent AM built components [399]. NASA has since developed and tested a myriad of AM components since 2010 such as copper alloy (GRCop-42 and GRCop-84) combustion chambers, liquid hydrogen and liquid oxygen fuel pumps, maintenance port cover on J-2X and the pogo accumulator assembly with integral pogo z-baffle [400]. Some of these components are set to be installed on the upcoming RS-25 liquid rocket engines respectively under the Space Launch System (SLS).

Aerojet Rocketdyne, an American rocket and missile propulsion manufacturer, has also been evaluating applications of AM in their new line of rocket engines, the AR-1 series. Intended to replace the currently used Russian-made RD-180 engine, the AR-1 utilizes AM methods for its pre-burner. The pre-burner is manufactured from Mondaloy™, a proprietary burn resistant nickel-based superalloy which eliminates the need for elaborate metal coating processes currently in use on oxygen-rich engines such as the RD-180 [401]. This is not the first component Aerojet Rocketdyne has manufactured and tested using AM techniques. In 2017, the space company

manufactured and tested a copper combustion chamber for their RL-10 rocket engine using AM and has developed significant qualification criteria for AM in space applications [402].

Arianne group recently manufactured and completed hot-fire testing of a full scale, demonstrator rocket engine with AM built components integrated. The engine featured a L-PBF built injector head and combustion chamber using the materials of a nickel-based alloy and stainless steel respectively [403]. As part of the Arianne 6 development program, these new AM built components are set to fly aboard the new variant of the Arianne rocket family in 2022.

AM applications are sometimes limited by the build area constraints of the L-PBF based machines which can only produce components as large as their maximum build volume (typically 300-600 mm cubed for single-laser commercial systems, although larger systems are in development). Built under conventional manufacturing conditions, rocket fuel tanks and aircraft fuselages are often built in several sections using thin-skinned supported structure designs called Isogrid [404]. The process of traditionally manufacturing these structures generates a lot of material waste in space structure applications since entire Isogrid designs are commonly machined, using milling techniques, from a thicker plate of stock material. This process results in extremely expensive components due to the high wastage of stock material and the long-lead plate stock material needed for larger rocket fuselages.

Since DED techniques are not limited by size, larger liquid fuel rocket components have recently seen development using DED. A specific application area is to use DED to manufacture entire fuel tanks, which typically also act as the structure for rockets. Using DED techniques, it may be possible to print entire rocket fuel tanks in a single piece or far fewer pieces with significantly less material wastage and much fewer tooling requirements which could significantly reduce the costs of these components. Lockheed Martin recently developed and tested a titanium dome for satellite fuel tanks built using EB-DED. The 1.16-meter diameter domes serve as caps for variable size fuel tanks aboard the LM 2100 satellite buses currently in production. Using these manufacturing techniques, Lockheed Martin is saving valuable production schedule time all while reducing material wastage significantly [405]. Relativity Space, an American aerospace company, plans to build their entire Terran 1 rocket using AM techniques from a proprietary alloy. Notably, the company plans to build the 2-meter diameter, dual-purpose fuel tank and fuselage using AW-DED, significantly reducing wasted materials and has demonstrated and ground tested pathfinder units [406]. With no theoretical limits to the height of components being produced using these methods, the component could potentially span the entire fuel tank section which is over 10 meters tall.

SpaceX, after successfully developing and launching an AM built main oxidiser valve aboard their Falcon 9 rocket, went on to develop the SuperDraco engine series that is used in their Dragon V2 spacecraft [407]. The SuperDraco engines are used for the mission-critical launch escape system designed to carry astronauts to safety if an emergency occurs during the launch sequence of the Dragon spacecraft. Featuring an AM built engine chamber made of Inconel, the SuperDraco is one of the earliest applications of crucial AM built components aboard human-occupied spacecraft which has had two successful missions launching astronauts to the ISS on the Demo-2 and Crew-1 missions [408]. Figure 2-19 shows the assembled Superdraco rocket engine and the engine on a testing stand in a hot-fire test.

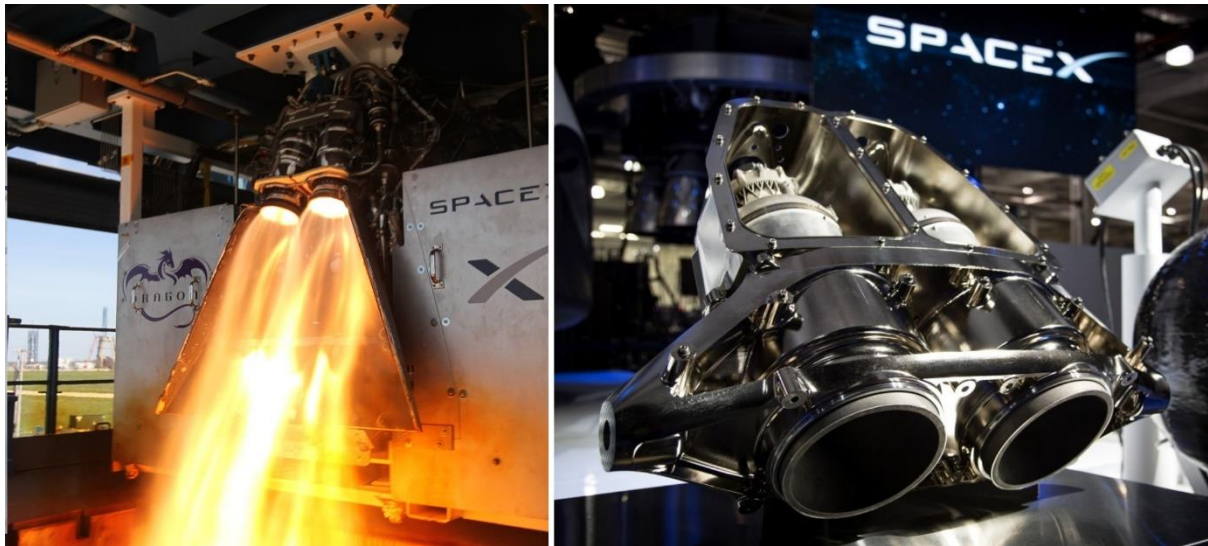


Figure 2-19. SpaceX AM built Superdraco rocket engine. Left: SuperDraco Test Fire by SpaceX [409]. Right: SuperDracos by SpaceX. [410]

Using AM techniques for manufacturing primary engine components has seen large-scale adoption at the California-based aerospace company, Rocket Lab. The company, a small satellite launch provider, has used AM technology to manufacture all of their primary rocket engine components aboard their Rutherford rocket engine since 2013. Aboard their Electron launch vehicle, the Rutherford rocket engine features a combustion chamber, injector, pumps, and main propellant valves among other components all manufactured using EB-PBF techniques and has been tested, manufactured, and launched on 18 missions to date [4].

Usually built using multiple materials and thousands of components, the combustion chamber, nozzle, and injector assemblies have garnered much interest recently for AM applications of single-piece manufacturing. An example of AM techniques being used for single-piece manufacturing is the Aeon 1 rocket engine built by Relativity Space. The Aeon 1 engine, to be launched on the upcoming Terran 1 launch vehicle, features its injector, ignitor, combustion chamber and nozzle all manufactured as a single component using AM techniques [411]. Another company, the New York-based Launcher Inc., plans to build the highest performance rocket engine in the small satellite launcher class using predominantly AM built components. Their rocket engine, the E-2, will be the largest single part combustion chamber, injector and nozzle assembly all built using L-PBF techniques from a C-18150 (Cu-Cr-Zr) alloy [412].

More recently, NASA has been developing new materials and application areas for AM for liquid fuel rocket engine components using a variety of AM methods. Prominent examples of this are development under NASA's Rapid Analysis and Manufacturing Propulsion (RAMPT) project and Space Launch System (SLS) projects evaluating new materials, GRCop-42 and NASA HR-1 developed at NASA's Glenn Research Center (GRC) and NASA MSFC, respectively. These materials, namely the GRCop-series of copper-chrome-niobium alloys and iron-nickel NASA HR-1 superalloy, were developed primarily for rocket combustion chamber and nozzle applications using AM. The 7k coupled-design features a GRCop-42 (Cu-4 at.% Cr-2 at.% Nb) L-PBF built combustion chamber with integral channels and a NASA HR-1 LP-DED integrated nozzle built directly onto the chamber to eliminate the manifold and joint. Combining these AM techniques and materials allows for tailored material characteristics in the combustion chamber, nozzle, and integrated cooling channels which could improve

the performance and reduce mass of rocket engines substantially while using AM techniques. The 40k, shown in Figure 2-20c below, is built using a L-PBF GRC-42 combustion chamber liner and a DED HR-1 jacket [413].



Figure 2-20. a: Bimetallic 7k Coupled Chamber using L-PBF GRC-42 with HR-1 DED Integrated Nozzle (b: complete with manifolds). c: Bimetallic 40k Chamber built using L-PBF GRC-42 liner and HR-1 DED Jacket. (Courtesy: NASA). [413]

NASA has also demonstrated many applications of large scale DED, specifically LP-DED. The focus at NASA has been rocket engine components, such as large scale channel wall nozzles and powerhead components. The nozzle has design similarities to combustion chambers with fine feature channels being incorporated to eliminate braze joints and extensive tooling often involved during assembly. NASA MSFC along with industry partners recently demonstrated a 65% scale RS-25 engine integral channel LP-DED nozzle at 1.52 m in diameter and 1.78 m in height [414]. Prior nozzles were also demonstrated with LP-DED integral channel features at smaller scales and several completed hot-fire testing with high duty cycles [52,415]. The nozzles demonstrate the potential to reduce part counts of the traditionally built nozzles from over 1,100 components to fewer than 10 components on full-scale designs [416]. Other components have been demonstrated using LP-DED including various large scale manifolds and complex components such as the RS-25 powerhead half shells. Many of these components were typically fabricated using forgings or castings with extensive machining operations and have shown significant cost and schedule savings with short deposition times using DED. Examples of these components can be seen in Figure 2-21.

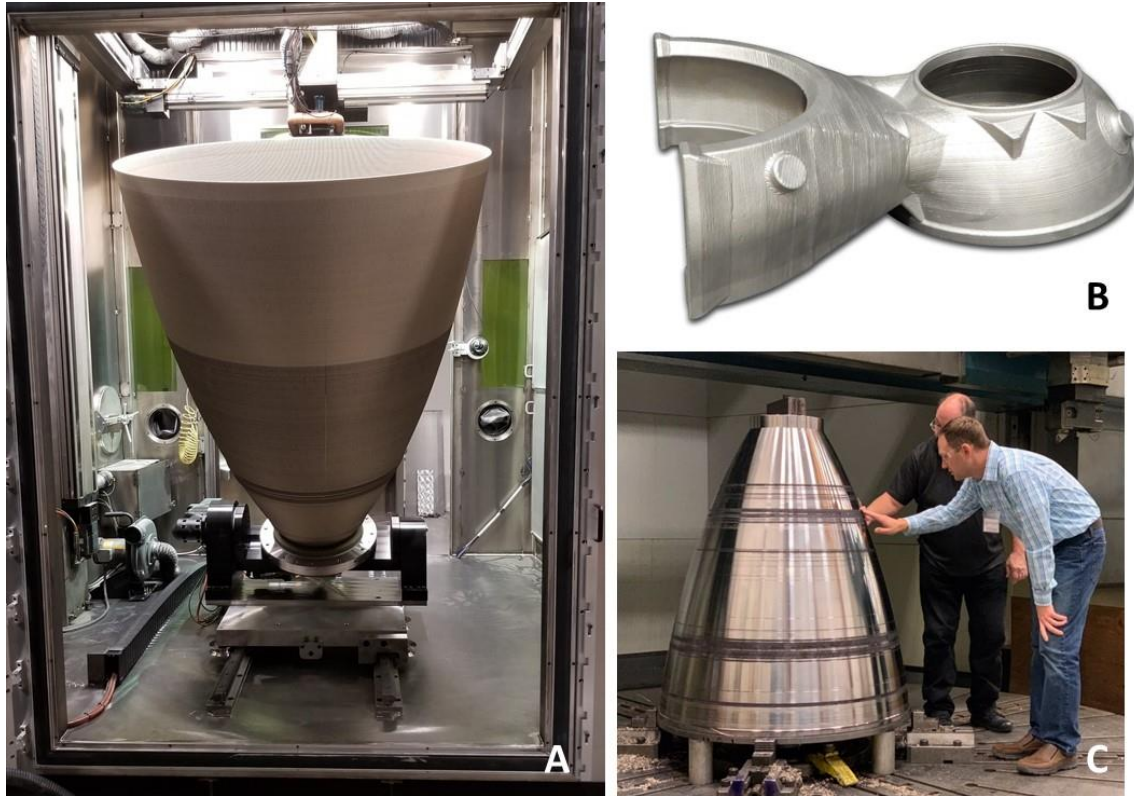


Figure 2-21. Examples of Large Scale DED at NASA. A) LP-DED integral channel nozzle with NASA HR-1 alloy, B) Powerhead half shell using Inconel 718, and C) LP-DED JBK-75 Nozzle. (Courtesy: NASA). [416]

Recently, there have been investigations in the use of AM and lattice structures in rocket nozzle components. Thrust chamber assemblies see various integrated mechanical, thermal, and integrated environments loads which presents difficulties in optimization. A collaboration between Cellcore and SLM Solutions produced an integrated thrust chamber design utilizing L-PBF to show potential applications in the aerospace industry [18]. The thrust chamber, as seen in Figure 2-4, features a structural lattice in its walls which has a second purpose as the cooling jacket for the engine. The one-piece engine is manufactured in just over three days on an SLM 280 printer. The design uses Inconel 718, a particularly difficult to machine material, and features very little post-processing avoiding costly tool wear, saving considerable costs while producing an extremely lightweight structure. The design also features a fully integrated injector further reducing the total part count and complexity of the final rocket design [338]. While this design shows an example of the potential integration of multiple parts, the overall loads must be well understood in the full environment to incorporate TO and also the entire AM process, including post-processing for successful application.

Due to the complexity present in the regenerative cooling channels of most rocket engines, thermal and structural optimization techniques have been gaining interest in applications for rocket engine designs. In one example, Hyperganic, a developer of AI-driven design software similar to TO, demonstrated optimization applications on a concept rocket nozzle using a top-down approach, integrating several components such as the combustion chamber, nozzle, and cooling channels into one single component [417]. The nozzle is designed to be printed using metal AM techniques allowing for a high complexity in design. The prototype, shown below in Figure 2-22, is designed to be manufactured from Inconel 718 on an EOS M400-4 printer. The nozzle features a gyroid minimal surface lattice design on its outer surface. These lattice structures are found in nature and are often self-supporting

and feature zero average curvature at any point on their surface, making for more even distribution of stresses within the structures [318]. These examples demonstrate some of the potential, but these designs are still not widely tested or adopted in flight components and currently are still in design and proof of concept phases.



Figure 2-22. Hyperganic prototype rocket nozzle featuring internal cooling channels and an external lattice. Photo courtesy of Hyperganic. [417]

2.3.5. Summary of Applications of AM in Aerospace

Many applications of metal AM exist throughout the aerospace industry. The attached appendices present a collection of prominent examples appearing in non-technical literature (Appendix A: Prominent examples of AM applications in non-technical literature) and a comprehensive list of articles, books, reviews, and conference proceedings appearing in technical literature (Appendix B: Prominent examples of AM applications in technical literature) in recent years. Appendix A: Prominent examples of AM applications in non-technical literature alphabetically lists the company involved, the application, the design approach, the material used, technology used and reference for further reading. Appendix B: Prominent examples of AM applications in technical literature chronologically lists the studies and includes information on the author/s, the title, the application area, the publication year, and reference for further reading.

2.4. Challenges and Opportunities

2.4.1. Overview

Metal AM technologies show great potential for technical and commercial advantages for aerospace applications, as is evident from the applications reviewed in the previous sections. Despite the technical and commercial opportunities enabled by metal AM, these technologies are subject to unique challenges for implementation in aerospace applications. This section highlights some of the current major challenges and some future opportunities.

2.4.2. Certification & Standards

A major current challenge is the lack of technological standards and certification approaches present for metal AM in the aerospace industry, which is caused by the rapid growth in AM technologies over the last decade. These standards need to be refined and agreed upon across industry to assure the repeatability, reliability and quality of

AM components for aerospace sector applications. Major governing bodies such as the European Union Aviation Safety Agency (EASA), NASA, European Space Agency (ESA), and Federal Aviation Administration (FAA) are imposing increasingly strict testing protocols and certifications that are required for the use of aerospace components in service for both mission-critical and non-critical applications [305]. These certification processes generally involve the repeatability of production processes and consistency in the quality of manufactured components which are both considerable challenges currently in the metal AM industry, especially when producing components in larger quantities. The primary challenge for the certification process of AM built components is the lack of prior knowledge, complete understanding and traceability of the AM process, detailed characterization and AM property databases, and data regarding the failure mechanisms. While materials with traditional processes are well understood and significant databases exist, AM materials lack a large database and agreed upon properties. Currently, ISO and ASTM develop a range of AM standards, described in more detail in [19]; these standards are evolving, and new standards are emerging, to accommodate the certification and design requirements for AM applications in the aerospace sector. A list of relevant current certification-related standards is provided in Table 2-2 below.

Table 2-2: Certification-related AM standards, including those under development (marked by *).

ISO/ASTM 52942-20	Qualifying Machine Operators of Laser Metal Powder Bed Fusion Machines and Equipment Used in Aerospace Applications
ASTM F3434-20	Installation/Operation and Performance Qualification (IQ/OQ/PQ) of Laser-Beam Powder Bed Fusion Equipment for Production Manufacturing
ISO/ASTM 52941-20	Acceptance Tests for Laser Metal Powder-Bed Fusion Machines
ISO/ASTM AWI 52937 *	Qualification of Designers
ISO/ASTM CD 52920 *	Quality Requirements for Industrial Additive Manufacturing Sites
ISO/ASTM AWI 52935 *	Qualification of Coordinators for Metallic Production
ISO/ASTM CD TS 52930 *	Installation, Operation and Performance (IQ/OQ/PQ) of PBF-LB Equipment
ISO/ASTM CD 52926-5 *	Qualification of Machine Operators for DED-ARC
ISO/ASTM CD 52926-4 *	Qualification of Machine Operators for DED-LB
ISO/ASTM CD 52926-3 *	Qualification of Machine Operators for PBF-EB
ISO/ASTM CD 52926-2 *	Qualification of Machine Operators for PBF-LB
ISO/ASTM CD 52926-1 *	General Qualification of Machine Operators
NASA-STD-6030	Additive Manufacturing Requirements for Crewed Spaceflight Systems
SAE AMS7032	Additive Manufacturing Machine Qualification
NASA-SPEC-6033	Additive Manufacturing Requirements for Equipment and Facility Control
NASA MSFC-SPEC-3716	Standard for Additivity Manufactured Spaceflight Hardware by Laser Powder Bed Fusion of Metals
NASA MSFC-SPEC-3717	Specification for Control and Qualification of Laser Powder Bed Fusion Metallurgical Processes

2.4.3. Structural Integrity

Structural integrity is of paramount importance for mission-critical aerospace applications and integrity includes dynamic loading in both high cycle and low cycle modes, thermal cycling and impact loading. Fatigue response to dynamic loading is of particular concern for AM aerospace applications. The static mechanical properties of

metallic AM built components are relatively well researched, with hardness and strength matching and exceeding that of traditionally manufactured material properties [80–83]. However, dynamic mechanical properties such as fatigue and creep have seen relatively less research and there remains a lack of test data reporting among aerospace companies [45]. The existing AM fatigue literature demonstrates test results that indicate porosity, residual stress and surface roughness are the largest concerns for high cycle fatigue (HCF) and low cycle fatigue (LCF) testing scenarios with overall fatigue properties generally substandard compared to conventional manufacturing processes [39,45,46]. Much progress has been made in mitigating these issues, allowing the improved performance of components with minimized defects, and achieving mechanical performance (including fatigue performance) on par with traditional manufacturing techniques [19].

Residual stress during AM builds result in possible part warping, cracking and reduced mechanical properties, and this build-up depends on the material used, the process parameters used, and the geometry and orientation in the build chamber, amongst other factors (specific thermal histories during the build). Similarly, porosity formation is strongly dependent on the process parameters as well as on local changes during the build. Various mechanisms for porosity formation in metal AM have been identified, which result in lack of fusion and keyhole porosity, with varying extent and 3D distributions depending on various factors [418–421]. The presence of porosity reduces the ductility of the material but is especially critical in fatigue loading as pores can act as stress concentrators and hence crack initiation locations, resulting in premature failures [422–424]. The surface roughness of the as-built components are generally irregular and vary with build orientation, process parameters and other factors – these also act as stress concentrators and crack initiation locations [306]. The obtained material microstructure is often anisotropic and the grain orientation depends on the building direction, affecting the mechanical properties. Appropriate post-process heat treatments and HIP are required to improve this.

All of these manufacturing-specific defects remain an issue to actively mitigate and be cognizant of, and therefore require extensive quality control across all stages of the AM process, from powder feedstock to ambient gas purity, process parameter control, process optimization, in-situ monitoring and post-processing for each individual process and feedstock combination required. Despite optimal processes and material quality, unexpected errors may still occur which require post-process inspection and often some kind of machining and further post processing is needed. All of these operations add cost to the produced components, but bypassing these steps leads to a loss in performance and reliability. HIP has been widely adopted in aerospace, and in AM in particular it is very useful for closing pores [307], homogenizing the microstructure, and improving the ductility of components. This also leads to greatly improved fatigue performance of metal AM components [425,426], leading to its wide adoption in aerospace manufacturing [39].

2.4.4. Design for Additive Manufacturing

Design for Additive Manufacturing (DfAM) can be used to optimize a design for the best manufacturing quality, minimizing support structures and post processing requirements [349]. This can also be done in combination with build simulations to determine optimal orientation of the part on the build platform, help determine build strategies to minimize residual stresses and resulting distortion, and to minimize support structures. Such simulations and DfAM methods are time consuming and dependant on the engineering skill of the AM engineer, and on the computing power of the computer used, but may be necessary. Simulations also require well defined inputs such

as alloy mechanical and thermophysical properties, often at the liquidus and solidus temperatures, which are not always available [427–429].

While TO and lattice structure applications are discussed throughout Chapter 2, there are several challenges that exists with these in practical applications. Since these techniques often optimize the design based on a known set of inputs and constraints, the load paths for aerospace components must be well understood. These load paths and combined structural, thermal, dynamic, and integrated environments are not always well defined or known for aerospace applications and designs can often include high design margins to account for these uncertainties.

2.4.5. Powder Removal & Post-Processing

Post-processing is a step in metal AM that is often underestimated. For complex parts with fine feature sizes, trapped powders can be a challenge to remove. This is compounded by the difficulty in handling such powders due to safety concerns. Typical approaches involve rotational movements, tapping and blowing powders out of surface cavities and channels. Trapped powders can be observed in CT scans, but may be permanently trapped if not removed prior to heat treatment.

Further post-processing steps might involve support removal, surface finishing and other heat treatments. For all post processing steps, care must be taken to prevent damage (e.g. when removing supports) to ensure the part conforms to the design and does not include surface damage. Surface polishing, machining or other processing is further challenged as AM part complexity increase and must be carefully designed for early in the process.

2.4.6. Non-Destructive Testing (NDT)

Due to the many challenges described before, NDT is prescribed for all critical flight components manufactured by metal AM. Various NDT methods used are described in detail in [430]. NDT is important for identifying flaws such as porosity or cracks in critical components and a variety of methods may be used including radiographic testing (RT), dye penetrant, eddy current, ultrasonic testing, amongst others. Some of the new challenges in this type of NDT for AM parts is that the complexity of the parts requires more challenging NDT approaches and some traditional tools are not relevant to such complex parts. In addition, the inherent surface roughness reduces the sensitivity of some traditional NDT tools. Despite these challenges, one method has proven to be able to overcome many of these challenges: X-ray computed tomography (CT). This method has been demonstrated to be successful for inspection of cracks, pores, trapped powders, deviation from design geometry, warping due to thermal processing, and more [309]. One particular advantage of the method is the ability to provide information on the changes in a part over time, with multiple scans at different times, providing information on crack formation or the extent of wear or other damage that can occur. The dimensional evaluation provided can be used to assess suitability for further use and even provide predictions of further life expectancy.

The major limitations of the technique are the relatively poor resolution for large parts, components with thick walls, challenges with alloys such as copper, and the time and cost involved. In addition, some metals are highly absorbing for X-rays, making their CT ineffective except for very small parts. In these cases the use of smaller coupon samples are recommended for checking process optimization conditions [420] and witness specimens built alongside larger parts give a good indication of the quality of the build [431]. For larger parts, it is technically

feasible to use higher-energy X-ray sources, but these are less widely available. For the above reasons, many in-process monitoring tools are being developed to improve the identification of flaws during the process, rather than post-process.

2.5. Conclusions

This review paper has demonstrated numerous successful examples of metal AM in aerospace applications. Various AM technologies are utilized for aerospace applications with L-PBF and DED as the most popular. DED is used for components with large build volumes and can be used to repair existing components in addition to building components. L-PBF is the most widely used for applications in aerospace with its capabilities to produce fully dense components at high resolution with complex geometries, with small to medium build volumes [405]. Most applications of metal AM in aerospace, demonstrated in Appendix A: Prominent examples of AM applications in non-technical literature and Appendix B: Prominent examples of AM applications in technical literature, have shown advantages for cost and schedule reduction, some applications see mass reduction improvements and many part consolidations from previous designs. However, metal AM has the challenges of part certification, unique quality control requirements, poor high-volume production rates, limited materials in use, post-processing challenges, potential reduced fatigue properties, supply chain maturity issues, high cost of machines and expertise required to produce functional components.

Many open research questions remain, with opportunities for enhanced understanding and performance and further developments in the near future. The most important areas of current development are listed and discussed below:

- New alloys developed for AM and for aerospace applications
- In situ monitoring for digital twin and accurate flaw identification
- Build simulation for identification of risks
- In-space and non-terrestrial AM
- Wider usage of architected cellular structures (lattices)
- Usage of optimization techniques such as topology optimization (TO) and hybrid analytical thermal optimization (HAATO)
- Multi-functional components such as integrated electronics and sensors in AM processes

Significant future growth areas include the application of novel custom alloys, bimetallic and multimetallic processing and characterization, detailed process-structure-property understanding, databases for AM materials, process certification, design optimization, and process simulations. The application of TO and the use of lattice structures has potential for mass reduction, which is a requirement for aerospace. In both cases, the growing understanding of the design and successful fabrication of such structures leads to improvements in components with increasingly complex designs. TO has already seen some applications throughout aerospace with emphasis on lightweighting of components. Lattice structures have shown much potential for improvements in design for aerospace applications. With unique and precisely designed properties such as energy absorption characteristics, lightweighting and heat transfer characteristics, lattice structures are expected to be a popular topic of interest for research in aerospace applications in the near future. It is expected that these optimized geometries are also

simultaneously optimized for build quality optimization, minimizing the need for support structures or overhanging regions, thereby enhancing the surface quality and minimizing the defect formation. Build simulation is expected to play an increasingly important role in optimizing the build quality by finding the optimal orientation and highlighting regions of potential thermal build-up, which can inform the design process. While the potential for TO and lattice structures could be used for mass reduction, there needs to be a significant understanding of mechanical and thermal load paths in addition to integrated load, which are not always defined with high certainty in aerospace. This is another area of research to ensure the techniques are applied appropriately and allow for fully safe and successful missions.

As highlighted here, the main advantages of metal AM in aerospace is cost and lead time reduction. Mass reduction is also a significant opportunity area with optimized design or use of multiple alloys, however these techniques need to be well understood and properties well defined. In addition, part consolidation is a major advantage in this industry. Part complexity capabilities inherent in metal AM allows for high complexity within components, including internal features such as channels and high surface area for heat transfer applications. AM has also been demonstrated in many large-scale applications, so scale is becoming far less of a limitation. These and other advantages of metal AM in aerospace hold great potential for wider adoption of this technology, which will further drive the remaining challenges to be overcome.

2.6. Acknowledgements

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3. AM PROCESS AND MATERIALS CHARACTERIZATION FOR AEROSPACE APPLICATIONS

3.1. Introduction

Processing of Additive Manufacturing (AM) materials by Laser Powder Bed Fusion (L-PBF) and other AM techniques has advanced significantly in recent years, particularly in terms of process optimization and material property characterization. A wide range of aerospace, biomedical, and industrial fields are interested in the as-manufactured material properties characteristic of AM due to the performance and cost-effectiveness of the process. Despite this, AM components often contain defects, which affect mechanical properties [432]. These defects occur due to a variety of causes such as imperfect processing parameters, unoptimized geometry, incorrect post-processing and contamination.

For the aerospace and space industries, AM has been shown to provide benefits such as cost savings and high performance when compared to traditional manufacturing techniques (Chapter 2). In addition to typically being high-cost, lightweight and high-stress, spacecraft components are often exposed to multi-physics loading conditions that create local stress states that are hard to predict. As a result of the uncertainties associated with AM materials, design processes for components are often highly test dependent. Before production of flight-weight, full-scale components, small-scale components are often designed, manufactured, and tested.

As part of the Aerospace Systems Research Group (ASReG), the South African First Integrated Rocket Engine (SAFFIRE) programme aims to produce functional rocket engines for the purpose of powering an orbital class rocket for delivering small satellites to orbit. Due to the high-stress nature of spacecraft components, the fidelity of mechanical properties of their materials of construction is very important. Incorporating new technologies such as AM into a rocket engine poses many challenges with respect to confidence in material properties. A particularly effective way to increase the confidence in manufactured components through AM is to use non-destructive testing (NDT) techniques such as micro computed tomography (Micro-CT). It is common to use MicroCT in the AM industry, but the process is often time-consuming and costly, thus decreasing its potential impact and benefits. Standardized test procedures can reduce analysis time and costs associated with Micro-CT and improves confidence regarding results [433].

As opposed to traditional manufacturing processes, AM parts and material processes have not yet had many years of incremental refinement by third parties. In addition, a range of factors affect the end quality. In order to ensure consistency and manage risks associated with the process, a disciplined methodology must be established to control these variables. While standards related to the qualification of AM materials and processes do exist, most were developed recently or are still in development. Table 2-2 lists several examples of these.

This chapter describes the process by which the AM materials evaluated in this study were tested and classified through NDT techniques, in order to improve the confidence in the mechanical properties of the materials considered for use in the SAFFIRE programme.

3.2. Additive Manufacturing Materials

AM materials, and particularly, metal AM materials, present distinct advantages for aerospace applications. While AM is still a relatively recent technology, many processing technologies and materials have been developed with high technology readiness levels (TRL) and extensive research is being performed to increase the range of materials and processes available to the AM industry. Due to the particular advantages that AM introduces to aerospace applications, the majority of the materials with extensive development in the AM industry are useful aerospace engineering alloys. Section 2.2.2 provides an in-depth review of AM materials available and their use-cases for aerospace applications.

A critical aspect of spacecraft component success is the selection of materials, which is often dependent on the design methodology and manufacturing process used. The materials used for spacecraft are often exposed to severe conditions such as extreme temperatures and heavy loads. Typically, temperatures vary from cryogenic (-253 °C) to extremely high (over 1,000 °C). The specific environments that spaceflight components are subjected to vary widely and depend on the nature of the component in question, which may fall into the following categories: rotating turbomachinery, injectors, valves, pressurized tanks, combustion chambers, feedlines, and various structural components. In spacecraft component environments, propellants or combustion products are often present and under high pressure which poses the risk of corrosion, embrittlement, or oxidation. Additionally, a reliable determination of material properties is dependent on both the maturity of the material and the maturity of the processing associated with the material. With respect to the SAFFIRE programme, there exist potential AM applications for all of the above-mentioned component categories.

The South African AM supply chain, while still in its infancy, enables the manufacture a variety of useful aerospace materials with high levels of maturity. This is, however, limited to common Laser Powder Bed Fusion (L-PBF) platforms such as the EOS M280 (see details in Appendix C: EOS M280 Additive Manufacturing Machine Specifications). Other AM processes such as Directed Energy Deposition (DED) have some potential applications in the SAFFIRE programme, however, the supply chain in South Africa for this method is underdeveloped. In the context of the SAFFIRE programme, L-PBF introduces the most relevant advantages such as feature resolution, surface finish and mechanical properties, and is therefore the primary AM process under investigation in this chapter. The testing methods proposed here can however be applied to AM processes such as DED.

The materials investigated as part of the work presented in this chapter were also selected on the basis of their potential value to the SAFFIRE programme. These four materials are Ti-6Al-4V, 316L Stainless Steel, MS1 Maraging Steel and Aluminium 6061-T6 RAM2. Each of these materials is readily available via the South African AM supply chain for the L-PBF process. The specific application areas, advantages and maturities are listed below.

Titanium Ti-6Al-4V is a widely adopted aerospace alloy and one of the most researched AM materials. It has high specific strength, excellent corrosion resistance and high-temperature stability [434]. These characteristics provide significant lightweighting opportunities for aerospace applications, however, Ti-6Al-4V is highly reactive in oxygen-enriched environments and therefore its application areas are limited to those not exposed to oxygen-rich

fluids. This material has the greatest maturity both in the South African AM supply chain and globally [38,435,436].

316L Stainless Steel is a common alloy used in most industries due to its relatively low cost, durability, corrosion resistance, high-temperature stability, cryogenic temperature stability and oxidation resistance [62]. 316L is compatible with liquid oxygen (LOX) and is therefore of interest to application environments where LOX is present. This material exhibits low relative specific strength at room temperature when compared to other AM materials and should therefore be used sparingly where lightweighting is of importance. 316L is a commonly manufactured alloy on L-PBF systems and has high maturity.

MS1 Maraging Steel is an AM-specific alloy similar to tool steels such as European 1.2709 and German X3NiCoMoTi 18-9-5, which are used in many industries. It has high specific strength, hardness, corrosion resistance, stiffness and durability [60,61]. In its heat-treated state, this material exhibits excellent strength characteristics, however, it is highly brittle in this state. The maturity of this material is high but limited to the EOS M280 platform.

Aluminium 6061-T6 RAM2 (Appendix G: Elementum3D 6061-T6 RAM2 Powder/Material Specifications) is a general-purpose AM alloy that features a combination of high specific strength, good thermal conductivity, high ductility, low-cost, and low-temperature stability. The alloy was recently developed by Elementum3D and is one of the first 6061 alloys successfully developed for AM processes. However, due to the novelty of the material, its mechanical properties and AM process quality are not well known and are thus a particular focus in this chapter.

In this study, all parts made from the above alloys were manufactured on EOS M280 systems operated by various AM suppliers. The following heat-treatment processes were employed for each material evaluated:

- Ti-6Al-4V – Stress relieving at 650 °C for 3 hours. Annealing at 941 °C for 2 hours
- MS1 - Solution treatment at 940 °C (1724 °F) for 2 hours. Age-hardening at 490 °C (914 °F) for 6 hours
- 316L - Stress relieving at 900 °C for 2 hours, followed by water quenching. Solution annealing at 1150 °C hold for 1.5 hours, followed by water quenching
- 6061-T6 – Solution treatment at 500 °C for 9-10 hours, followed by water quenching. Precipitation hardening at 160 °C (320 °F) for 18 hours, followed by air cooling. Further precipitation hardening at 177 °C (350 °F) for 8 hours, followed by air cooling.

3.3. Common AM Defects

The formation of defects in AM-built components used in critical structural applications is of major concern. Therefore, knowledge of the mechanisms of defect formation in L-PBF and other AM processes is crucial to determining the appropriate use cases specific to the selected alloy system and processing technique. It has been demonstrated that powder-based AM can be applied in many fields such as the aerospace and medical industries [437]. A suitable AM part should have at least the same, or preferably better, quality as a conventionally manufactured part. Detailed knowledge of the defect types, their mechanisms of occurrence, and associated detection methodologies is essential to the evaluation and control of mechanical properties. In fusion-based processes, defects can include keyhole collapse, gas porosity, solidification cracking, solid-state cracking, and

surface-connected porosity [437,438]. It is possible to eliminate such defects using defect-mitigation techniques such as post-process machining, surface treatment, and post-processing HIP [437,439]. The two primary defect types that are considered in this chapter are porosity and surface roughness.

3.3.1. Porosity

Voids are typically defined as regions containing “nothing” either inside, or outside of an object under inspection. In manufacturing, pores, similarly to voids, can be defined as regions where the intended material/s is not present. For AM processes, pores are often spherical in shape, and contain process gasses such as Argon. These pores can be formed within a single layer, between adjacent layers and/or on the external surface of the build part. In addition to pores, other microstructural inhomogeneities can occur in AM-built parts, such as impurities (inclusions, contaminations, and metal oxides), un-melted or partially melted particles, and abnormal grain growth. In this chapter, the effects of pores and how pores can be identified through MicroCT imaging are specifically examined. Pores can be formed for several reasons which are discussed in greater detail in [440–443].

The presence of manufacturing faults, such as porosity, can adversely affect the mechanical properties of final products, regardless of whether they are manufactured by additive manufacturing or not. In terms of mechanical properties, a range of specific properties are of concern, including modulus of elasticity, yield strength, ultimate tensile strength, ductility, toughness, and hardness, under static/quasi-static conditions. It is typically desirable to have both a high number of crack initiation cycles and high crack growth resistance under dynamic loading, both of which are affected by porosity characteristics. There are typically standards for wrought or cast materials for each material type that can be used as a guide to ascertain how L-PBF materials should be manufactured. The earliest L-PBF optimization studies reported, at best, a relative density of 87% (or 13% porosity) [444], with future improvements resulting in significantly higher relative densities, typically reaching values greater than 99% (or less than 1% porosity) [445]. It is, however, important to know how the desired mechanical properties are affected by porosity, pore size, and distribution, and the extent to which these characteristics threaten critical applications, such as those in the medical or aerospace fields.

The defects present on AM-built materials often resembles that of cast metals, and consideration of cast metals provides useful insight into the effects of defects on mechanical properties. Casting porosity is well known to result in poor mechanical properties such as reduced tensile strength and ductility, according to studies such as those highlighted in [446]. According to static tensile tests monitored by X-ray tomography, failure almost always occurs at the largest pore, and pore size is inversely related to effective yield strength and ductility [447]. For AM process, and with properly selected process parameters and scanning strategies, the significance of porosity associated with L-PBF can be greatly reduced, to the extent that the mechanical properties of the material aren't significantly affected. According to [448], porosity values of up to 1% tended to have a small effect on tensile strength and elongation to failure. Furthermore, even with pores up to 4 mm in diameter, yield strength was still comparable to the wrought standards [447].

Porosity is of particular interest in this study for the purposes of determining the specific defects present in the AM materials evaluated here, and how they influence mechanical properties. These defects have mitigation

strategies such as optimal heat treatment or HIP techniques, and characterising porosity for specific component and material combinations informs critical design choices to enable safe component optimization.

3.3.2. Surface Roughness

Surface roughness, residual stress, and microstructures also influence the mechanical properties. The relative importance of these factors in determining the mechanical properties varies from material to material. Fatigue properties of L-PBF materials are of particular importance in high-end applications such as aerospace and medical devices. Researchers have demonstrated that surface roughness, inclusions and microstructure have a significant effect on fatigue life [449]. As-built surfaces with high surface roughness exhibit poor fatigue properties regardless of the defect populations, indicating that surface roughness is especially important in the context of fatigue. Although machined samples with smooth surfaces did show poor fatigue properties, this was so only for those that had many fusion pores [425,450]. Combining machined and low-porosity samples yields the best results. Surface enhancement processes are highly important due to the subsurface pores and surface roughness of as-built specimens. A variety of methods, including shot peening and laser shock peening, have been shown to enhance mechanical properties [451]. It has been widely assumed that this improvement is due to the compressive stresses at the surface and the improved surface finish, but recent research has demonstrated that pores near the surface can also be closed through laser shock peening [452].

Surface roughness is characterised to a limited extent by most AM service providers (powder manufacturers and machine manufacturers), but for quality assurance purposes in relation to this study, surface roughness characteristics specific to the South African AM supply chain are quantified here. Evaluations were performed on both cube samples (Sections 3.6.1 and 3.7.1) and overhang samples (Sections 3.6.2 and 3.7.2).

3.4. Unsupported Surfaces

Since the L-PBF process is layer-by-layer, the accuracy, surface detail and smoothness of parts produced by it are limited. Heating and cooling at high rates can cause the materials to become stressed and distorted. Support structures can be used as anchors and heat dissipators to reduce the risk of warpage in overhanging structures. In an AM build platform, the design of the support structures affects the amount of deformation, build time, and support removability of the parts. Support structures, when excessively utilized, can directly impact powder consumption, as well as build and post-processing times. Additionally, surface irregularities resulting from support removal can lead to more post-processing being required. As a result, it is highly valuable to find ways of building L-PBF specimens without supports. Observations, however, have revealed that supported specimens show significantly higher fatigue resistance than unsupported specimens [453]. When complex parts without supports are manufactured, surfaces may be excessively rough on the downward side. When deformation occurs during processing, it results in porosity and increased roughness of the surfaces, or even failure.

In AM processes, down-facing surfaces are often referred to as “downskins” and up-facing surfaces are referred to as “upskins” both with reference to the build plate. Downskin surfaces are typically rougher than upskin or vertical surfaces depending on the overhang angle. Because of this, and because it is not always possible to machine or smooth complex or hidden surfaces, it is important to evaluate and optimize the surface finish for as-

built components for different surface overhang angles. Generally, in the design of AM components with overhangs, surface angles below the critical angle toward the build plate necessitate the use of support structures. The critical angle provided by most AM service providers is 45 degrees. While it is generally advised to use support structures for surfaces under the provided critical angle, components can be built at lower overhang angles. However, parts produced with angles under 45 degrees often have a poor surface quality due to heat accumulation that causes powder to agglomerate on the underside.

It is certainly of importance and demand to develop AM processes that can produce complex geometric shapes with overhang surface angles below 45 degrees without supporting structures, thus having better surface qualities. In order to achieve the highest quality L-PBF parts with overhanging surfaces at low inclination angles, various processes are being developed by AM machine manufacturers and AM service providers with the intent of mitigating warpage, high surface roughness and defects on low-angle overhanging surfaces. This can be done using special software, closed-loop control, and a special noncontact powder recoating systems [454]. This latter method provides significant advantages over traditional L-PBF processes, but to fully explore its advantages, more studies are needed.

In the context of this study, determining the characteristics of low-angle, unsupported surfaces on as-built AM components would provide clarity on the surface characteristics of potential SAFFIRE engine components. This is particularly relevant in the case of components having internal surfaces which are difficult or impossible to machine (for example, internal fluid channels).

3.5. NDT Through X-Ray Tomography

In the recent past, X-ray microcomputed tomography (MicroCT) has become an established method for testing and analysing AM-built components. Researchers and engineers are becoming increasingly interested in this non-destructive analysis method, but its capabilities are still not fully utilized and are still being developed. The use of MicroCT is crucial for quantifying porosity values, pore distributions, and measuring dimensions non-destructively. Ideally, this process can be carried out before mechanical tests in order to pinpoint the impact of defects - relating unexpected failures to certain types, locations, or distributions of defects. In addition to surface roughness, residual stress and microstructure, other factors can influence as-built mechanical properties. The relative importance of each of these factors varies with the material and cannot always be predicted [447,455].

The principle of MicroCT is to irradiate a sample with X-rays, measure the subsequent absorption X-ray image, and acquire such images repeatedly as the sample rotates. Due to the penetration of X-rays, X-ray absorption images provide a view of the sample from many angles. The acquired images can then be used in a mathematical reconstruction to create a volumetric data set [456]. The volume is made up of voxels (volumetric pixels), with the brightness of each pixel related to the density of the material it represents (X-ray density is determined by the material's density and effective atomic mass). Figure 3-1 illustrates the X-ray tomography process for the case of an additively manufactured sample with an internal lattice. Powder is observed in the sample between the lattice struts based on X-ray imaging results.

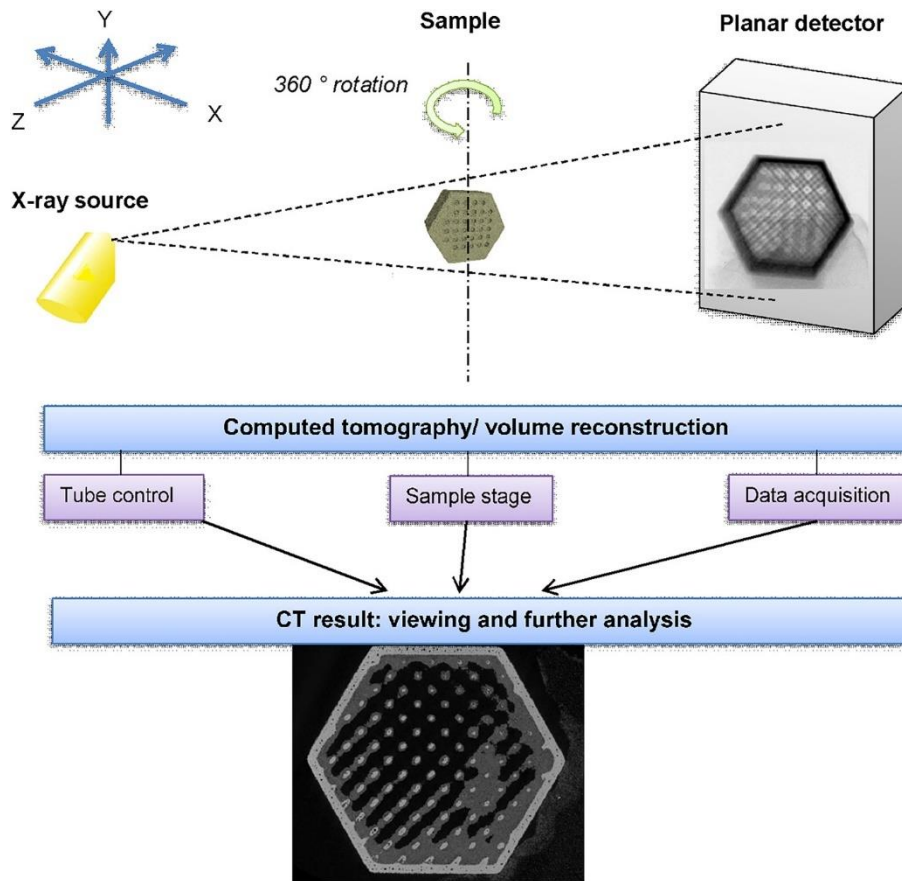


Figure 3-1: Illustration of the process of X-ray micro computed tomography taken from [457].

Once the data is scanned and reconstructed, it is visualized and analysed, typically using specialized software. A more in-depth review of MicroCT technologies and the applications of such can be found in [455–457].

3.6. Methodology

A specific methodology was used to characterise the four AM materials evaluated in this study (Ti-6Al-4V, 316L, MS1, 6061-T6), their respective manufacturing processes and the associated supply chains. This methodology comprised the manufacture of standardized cube samples (for porosity and surface roughness analyses), overhang samples (for surface roughness and defect analyses) and wall-thickness specimens to be individually analysed using MicroCT methods. These analyses were performed with the intent of characterising defects or lack thereof in AM materials for potential use in SAFFIRE engine components.

The MicroCT facility used was the Stellenbosch University CT Scanner Facility. As part of the Central Analytical Facilities (CAF) of the university, it offers non-destructive X-ray computed tomography (CT) and high-performance image analysis services. A MicroCT scanner system (Figure 3-2) for large samples and a NanoCT scanner for small samples are both available in the facility, both manufactured by General Electric Sensing and Inspection Technologies/Phoenix X-ray. In this study, all samples smaller than 10 mm in the longest-axis dimension were scanned using the NanoCT instrument. Larger samples, having longest-axis dimensions of 10 mm to 300 mm, were scanned using the MicroCT instrument. Further details on the facility and its capabilities can be found at [458].

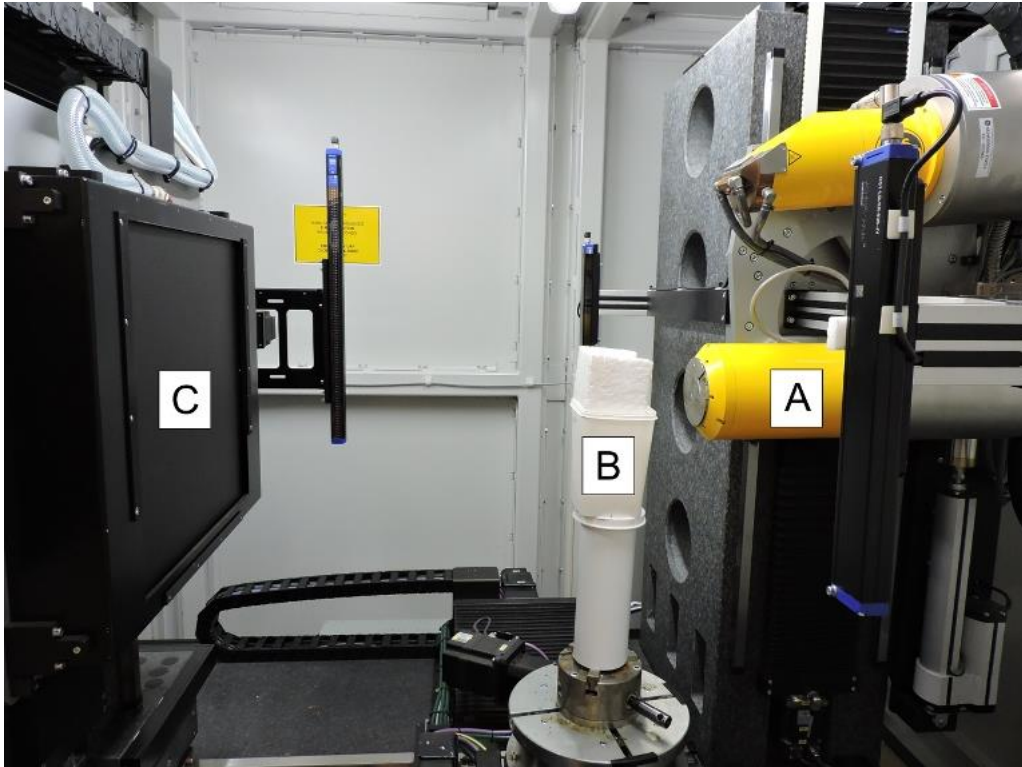


Figure 3-2: This photo illustrates the fundamental components of the setup used during the study. Taken from [459].

In a MicroCT scanner as shown in Figure 3-2, the x-rays are generated by an x-ray tube (A) that emitted x-rays and transmitted through a sample (B), which is recorded by a detector (C). The resolution at which X-ray MicroCTs can inspect AM-built parts non-destructively is limited by the size of the part. As a general rule, MicroCT resolution scales linearly with part size. For example, a 30 mm part yields a 30 μm resolution, and a 5 mm part yields a 5 μm resolution. Because some types of defects in additive manufacturing may be small and therefore missed in a scan of a large complex part, it is ideal to optimize processes before building critical parts. Small test samples can be used in small-scale optimization by means of MicroCT.

For this study, various cube and overhang samples were manufactured using L-PBF on EOS M280 platforms employed by a variety of service providers. Since EOS provides highly developed processing parameters for their material catalogue, parts employing EOS-specific sample materials (i.e. Ti-6Al-4V, 316L, MS1) were manufactured using the company's standard processing parameters. While Elementum3D's 6061-T6 powder is designed to be manufactured on EOS platforms, it's specific processing parameters for the EOS M280 are still under refinement. The 6061-T6 samples were therefore manufactured using the processing parameters provided by Elementum3D.

3.6.1. Cube Samples

All cube 6061-T6 sample tests utilized a standard coupon sample of 10x10x10 mm as shown in Figure 3-3 (left). Aerospace and medical industries employ this procedure for Ti-6Al-4V, but it can be applied to other lower-density metals and plastics equally well. For steels and other higher-density metals, a higher voltage must be applied to provide sufficient X-ray penetration. A stronger beam hardening correction must be also applied during reconstruction. Due to the limitations of the Stellenbosch University CT Scanner Facility, for the Ti-6Al-4V, 316L

and MS1 samples, a standard coupon size of 5x5x5 mm was used as shown in Figure 3-3 (right) to allow for sufficient X-Ray penetration and high-quality scans. While providing the ability to image all types of additive manufacturing defects (10-15 μm), these sample sizes also allow for a large enough sample size for practical purposes and provide sufficient field of view.

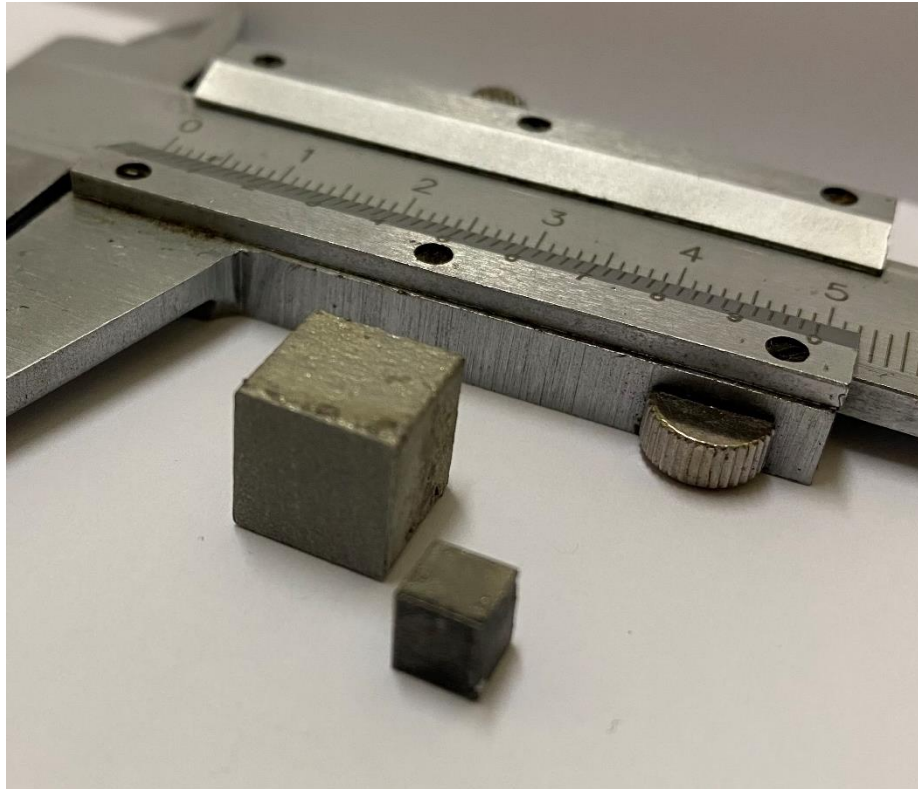


Figure 3-3: As-built Cube samples with scale. 10 mm 6061-T6 cube (left) and 5 mm MS1 cube (right).

The MicroCT study was conducted using the CT Scanner Facility at Stellenbosch University, with parameters optimized according to the guidelines provided in [459]. For each material and sample size, different MicroCT parameters are required due to the differences in densities. Table 3-1 shows the different MicroCT parameters used for each material. A total of 2400 step positions were used in a full 360° rotation. Two subsequent images are averaged at each step position after discarding the first image and the overall scan time is approximately one hour. After scanning, the data were then analysed in Volume Graphics' VGStudioMax 3.5 (VGStudio Max 3.5 manual) software with the aims of determining both porosity and surface roughness.

Table 3-1: Cube sample MicroCT parameters.

Material	Sample Dimensions	Voltage (kV)	Current (μA)	Voxel Resolution (μm)	Beam Filter Size (mm)	Image acquisition time per image (ms)
Ti-6Al-4V	5x5x5 mm	200	70	10	0.5	500
6061-T6	10x10x10 mm	140	100	10	0.5	500
MS1	5x5x5 mm	230	40	10	1	500
316L	5x5x5 mm	230	40	10	1	500

Several scanning and analysis workflows and algorithms are possible for MicroCT-based porosity measurement. This is due to the widespread use of MicroCT. The results of image processing can vary from one scan to another based on the quality of the scan (whether there are artefacts, noise, image blurring, etc.), the type of de-noising applied, and the analysis workflow engaged. The coupon sample used here is small enough to allow high resolution and quality scanning, and in addition to assessing porosity and surface roughness, average density can also be determined for the same sample. This is demonstrated in [460].

To accurately determine porosity through MicroCT techniques, specific image processing techniques need to be used. The image processing steps are described here briefly; for more detail of the procedure, the reader is referred to [461]. With these image processing steps, the aim is to remove the effects of exterior air, but include all materials and air to separate the results of the surface roughness from the pores within the data set. Different methods and software may be used to achieve this, since the goal is to create an accurate segmentation of the edge of the cube, regardless of the size of internal pores or possible external noise particles. Segmentation is accomplished by via the following proces:

1. Apply a basic automatic surface determination
2. Create an expanded (dilated) region of interest to eliminate surface roughness from the selection
3. Create a region of the air and noise particles surrounding the expanded region
4. Invert the region of air to select the entire part containing only internal pores
5. Advanced surface determination using the inverted region to achieve an accurate closed surface containing only internal pores
6. Create a contracted (eroded) region of interest to again remove the surface roughness from the selection

The analysis of quantitative defects (porosity/voids) can be done using a number of methods, but the one presented here works without a special algorithm, so it is less likely to yield errors from noisy data that is often present in MicroCT scans. Using the region of interest determined from the above process, a porosity/void analysis was performed using the “from defect ROI” setting. This analysis provides a vast amount of data with respect to pore sizes, distributions and other criteria. The primary aim of these porosity analyses is to determine the total percentage porosity and maximum pore size in the samples. Another important insight offered by porosity analyses is the indication of contour porosity. Contour porosity is an accumulation of pores close to the surface of a component and is a common defect in AM, often caused by incorrect processing parameters. Figure

3-4 shows a typical porosity analysis result with pores visualised. In this analysis, a significant amount of contour porosity is indicated.

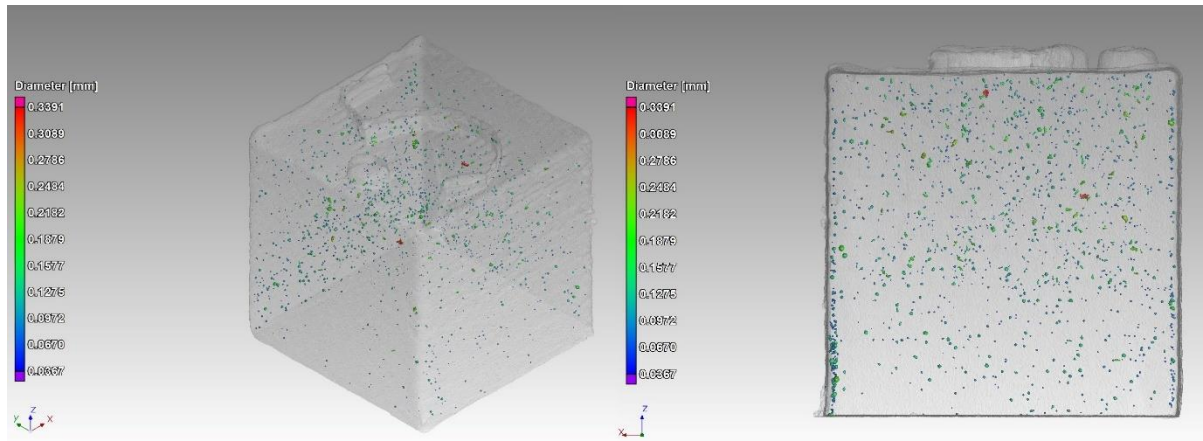


Figure 3-4: Porosity analysis indicating pore size. Contour porosity is clearly visible here.

In a number of recent studies of complex parts [462–464], surface topography was measured with MicroCT. As-built surface roughness analysis can be performed using MicroCT on cube coupon samples. Surface roughness is historically measured with tactile probes or with non-contact interferometric techniques. However, AM surfaces may be difficult to access and may be very rough, making it hard to accurately measure surface roughness. With traditional techniques which are designed to measure smoother surfaces, it can be difficult to accurately measure undercuts and deep, narrow troughs. In this study, the procedure used for determining surface roughness is standardized and is only described briefly. Reference [465] contains the complete workflow details and results-accuracy information.

To obtain an accurate surface determination for surface roughness calculations, a similar process to the porosity segmentation processes, mentioned previously, is used:

1. Apply a basic automatic surface determination
2. Create an expanded (dilated) region of interest to eliminate surface roughness from the selection
3. Create a region of the air and noise particles surrounding the expanded region
4. Invert the region of air to select the entire part containing only the external surface
5. Conduct advanced surface determination using the inverted region to achieve an accurate closed surface and close all internal pores

For surface roughness measurements, specific areas must be analysed with respect to the mean flat surface in the vicinity. For these analyses, a standard size of 3x3 mm for the analysis area is used on both the vertical and top (opposite the AM build plate) surfaces. These surface areas are separated from the entire surface using regions of interest placed roughly in the centre of each face. A flat-plane geometry element is then created to represent the mean flat surface in the vicinity. This is done by using a best-fit surface measurement probe on a minimum of fifteen points surrounding the region of interest. This flat surface can then be referenced in a “nominal/actual comparison” analysis. This nominal/actual comparison calculates the deviation of the determined surface when compared to the flat-plane geometry element in the region of interest specified. The analysis creates both a visual result (Figure 3-5) and a significant amount of data on the deviation of each voxel from the mean surface.

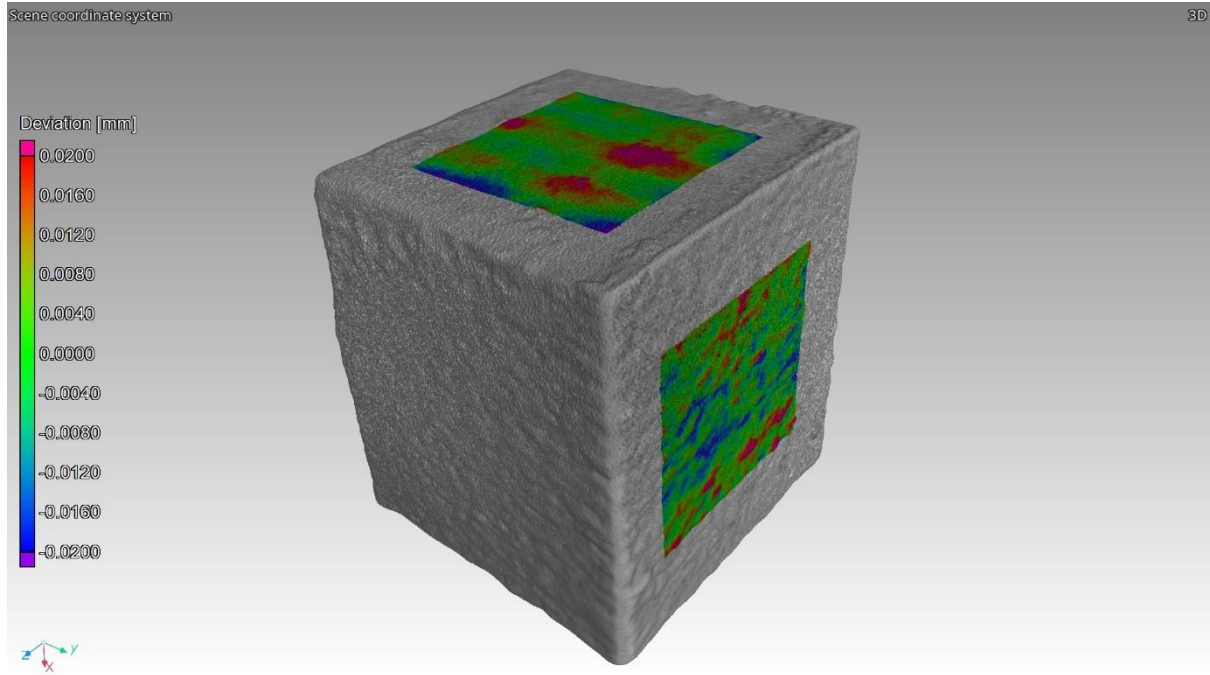


Figure 3-5: MicroCT surface topography for selected surfaces on both the side and top of a cube.

The MicroCT data were then extracted to a CSV file for further analysis. The data extracted captures surface deviation (both positive and negative) and the surface area of the elements at each deviation. These data are further processed to determine arithmetical mean height (Sa) values which are defined by the following equation:

$$Sa = \frac{1}{A} \iint_A |z(x, y)| dx dy \quad (3-1)$$

To calculate Sa values, for each deviation, the absolute value of the deviation is multiplied by the surface area, and the sum is divided by the total surface area. In the same way that roughness is often calculated as the arithmetical mean height of a line (Ra) values, linear measurements can be found by downscaling the data or by using a line fit instead of a surface area calculation. With 3D data, roughness measurements can be performed for both Ra and Sa values, but the Sa measurement is usually preferred, because it is less subject to variation.

3.6.2. Overhang Samples

For the purposes of testing the defects present in low angle, unsupported surfaces on AM as-built components, “overhang” samples were manufactured for all candidate materials (Ti-6Al-4V, 316L, MS1, 6061-T6) under investigation for the SAFFIRE programme. The characteristics of unsupported surfaces built using AM for build angles above 45 degrees are well known and therefore, the primary low angle range of interest was chosen as 25 – 55 degrees. This was achieved using samples (Figure 3-6) that contain various flat surfaces at angles stepped in 5-degree increments from 25 – 55 degrees from the build plane. Importantly, these samples contain a constant wall thickness at each angle increment to ensure no wall thickness defects are present when comparing angles. The scale of these samples was determined to ensure that the MicroCT scanning quality was high, and to find relevant data for similar-scale geometric features present in SAFFIRE programme designs.

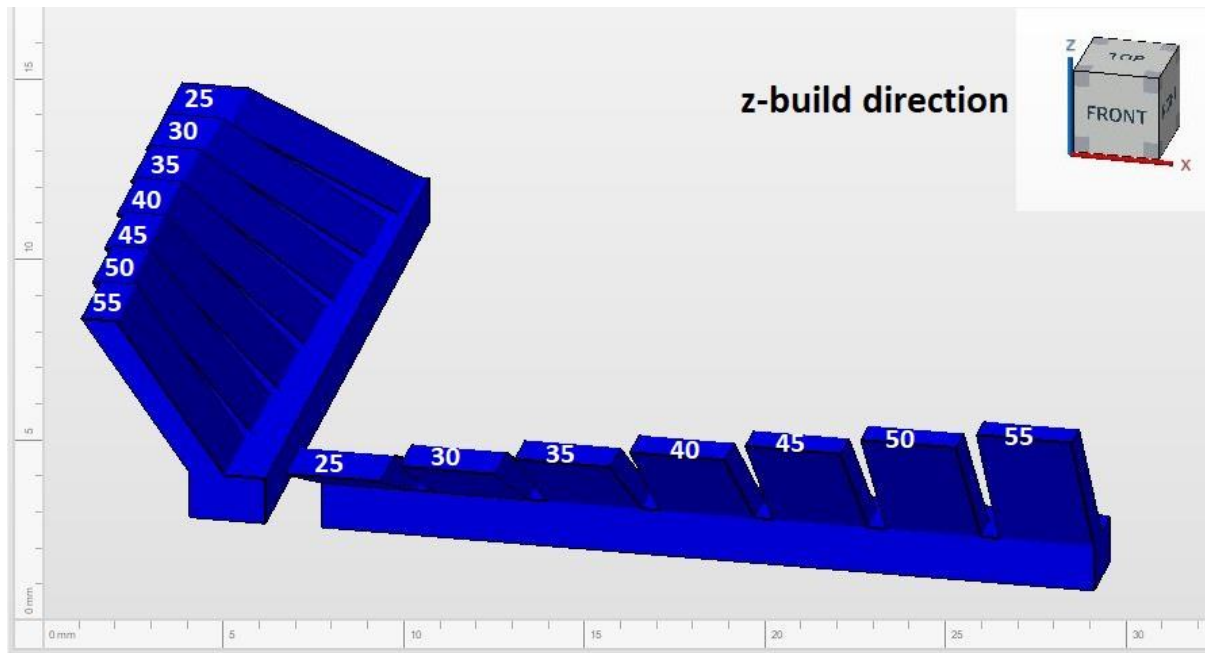


Figure 3-6: A set of overhang samples showing unsupported surface angles (degrees from build plane), build direction (z-axis) and scale (mm).

All samples were manufactured on an EOS M280 AM platform with standard processing parameters for each material. All samples were then brought to their final, heat-treated state as applicable for applications in the SAFFIRE programme, described in section 3.2. No other post-processing was performed on the samples prior to MicroCT analyses.

In this study, the MicroCT parameters were optimized in accordance with guidelines provided by [458,459]. Depending on the density of the material, MicroCT processes require different parameters. The MicroCT parameters used here are shown in Table 3-2. In total, 2000 step positions were used on a full 360° rotation. At each step position, two consecutive images are averaged after the first image is discarded, with scans taking approximately 40 minutes. Data were then processed by Volume Graphics' VGStudioMax 3.5 software to determine surface roughness, geometry deviation (warpage), and near-surface porosity.

Table 3-2: Overhang Sample MicroCT Parameters.

Material	Sample Dimensions	Voltage (kV)	Current (μ A)	Voxel Resolution (μ m)
Ti-6Al-4V	$\approx 20 \times 10 \times 5$ mm	120	120	15
316L	$\approx 20 \times 10 \times 5$ mm	120	120	15
MS1	$\approx 20 \times 10 \times 5$ mm	120	120	15
6061-T6	$\approx 20 \times 10 \times 5$ mm	100	100	15

As discussed previously in section 3.6.1, the surface topography of AM components can be measured through MicroCT techniques. Here, the procedure used for determining surface roughness is standardized and is only described briefly. For the purposes of surface roughness measurements, the methodology used follows that outlined in [465]. To obtain an accurate surface determination for surface roughness calculations, a similar process to the porosity segmentation process, mentioned in Section 3.6.1, is used:

1. Apply a basic automatic surface determination
2. Create an expanded (dilated) region of interest to eliminate surface roughness from the selection
3. Create a region of the surrounding air and noise particles surrounding the expanded region
4. Invert the region of air to select the entire part containing only the external surface
5. Conduct an advanced surface determination using the inverted region to achieve an accurate closed surface and close all internal pores

Specific areas of each angled surface were analysed with respect to the fitted CAD geometry in the respective vicinities. On each of the angled, unsupported surfaces, a standard size of 3x3 mm is used as the analysis area. In order to separate these areas from the overall surface, regions of interest are placed roughly in the centre of each face, as shown in Figure 3-7 (a). The CAD geometry is imported into the analysis software to be used as a nominal analysis reference. The CAD geometry is then registered (orientated) to the MicroCT scanned data which is done using a function called “best-fit registration” shown in Figure 3-7 (b). This aligns the CAD geometry iteratively to best represent the deviation of surface voxels from their associated CAD locations. After referencing this CAD object, a "nominal/actual comparison" analysis can be performed. With this nominal/actual comparison, the deviations of the determined surfaces are calculated by comparing them to the CAD geometry element in the given regions of interest. A visual result is created by the analyses and a significant amount of quantitative data is produced with respect to the deviation of each voxel from the CAD surfaces.

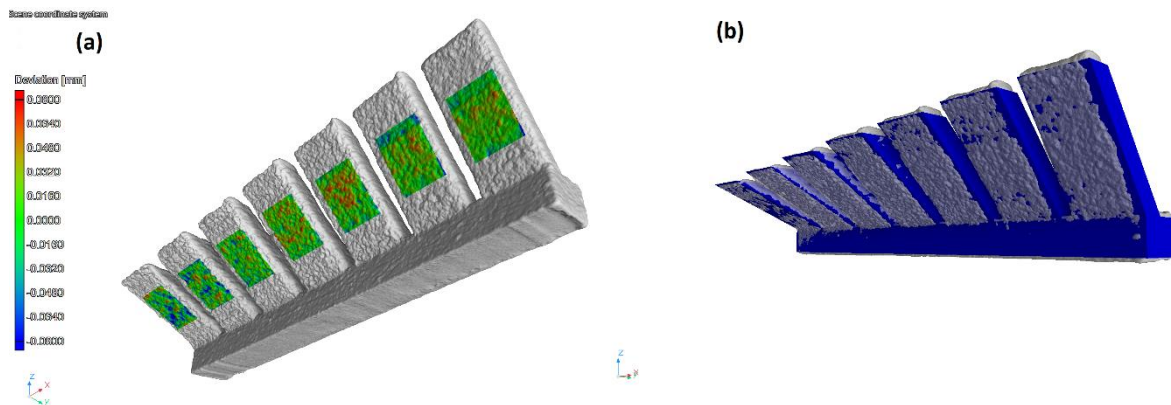


Figure 3-7: MS1 Overhang sample displayed showing both: (a) Regions of interest on downfacing surfaces, and (b) CAD overlay on MicroCT scanned part (blue CAD body, grey scanned body).

For further analysis, the MicroCT dataset was then extracted to a CSV file. Extracted data are represented as surface deviations (both positive and negative) and element areas associated with each deviation. Using Equation (3-1), these data are further processed to give arithmetical mean height (S_a) values for each region of interest (angle). The procedure for calculating S_a values from the extracted datasets is described previously in Section 3.6.1.

The above procedure can similarly be used for the purposes of calculating the geometry deviation (warpage) in MicroCT scanned components. Deviation is described here as the absolute maximum difference between a measured surface voxel's cartesian location when compared to the intended location of the surface (CAD). The deviation present in as-built AM materials arises from a variety of sources such as AM processing parameters, heat treatment processes, material thermal expansion coefficients, residual stresses etc. The causes and mitigation strategies associated with geometry deviations are detailed in [466].

The procedure used to calculate total deviation follows the same procedure used for surface roughness calculation, with the following differences in analysis methods. Instead of separating the downfacing surfaces for each unsupported angle, the entire surface of the measured samples was compared to the aligned CAD geometry. This was done by creating a region of interest which encompassed all the extended surfaces, apart from the base structure (to eliminate errors in the base). This region was then processed using the same “nominal/actual comparison” analysis used previously. The maximum distance for measurements was set to 0.5 mm for the Ti-6Al-4V, 316L and MS1 samples. The 6061-T6 sample contained significant deviation, and the maximum distance for measurements had to be increased to 3 mm. Both qualitative (Figure 3-8) and quantitative data were produced by these analyses.

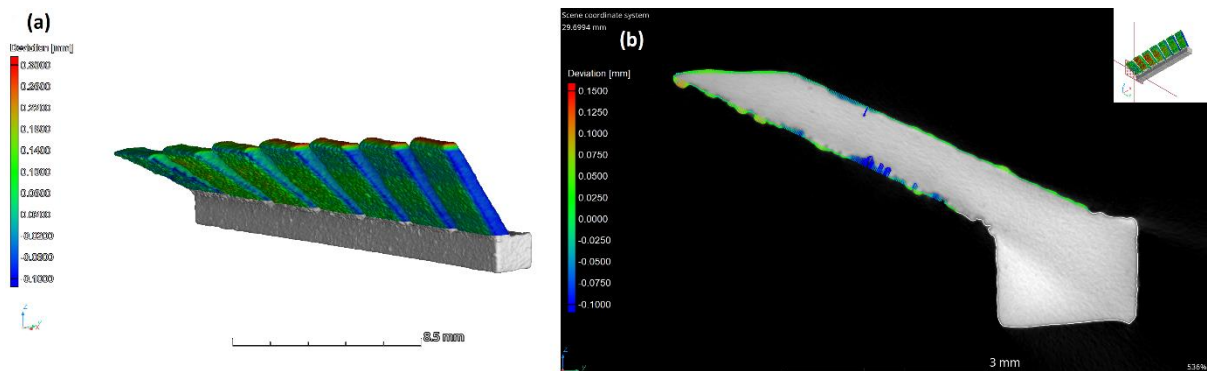


Figure 3-8: Overhang sample displaying both: (a) Total deviation on all extended surfaces, and (b) Deviation on the 25-degree overhanging section.

The results of the “nominal/actual comparison” analyses, for the purposes of deviation measurement, were quantified in the following ways. The minimum and maximum deviations were reported as well as a cumulative distribution function in the form of 90/95/98% percentile occurrences of deviations.

Near-surface porosity and total porosity are also important factors in unsupported surfaces and the former has been previously studied in [467]. It was found that there is not necessarily a strong correlation between either near-surface porosity and unsupported surface angle, or near-surface porosity and surface roughness. For the purposes of this study, the near-surface porosity and total porosity were not quantified due to both scan quality and lack of porosity in most samples. The porosity was, however, visually inspected to identify any outlying porosity defects in samples. This is discussed in Section 3.7.2, and in further detail in Section 3.8.

3.7. Results

3.7.1. Cube Samples

Based on the 10 mm and 5 mm cube analyses, a percentage value is provided for the detected porosity. The minimum detectable pore width in this case was chosen to be two voxels, which equates to 20 μm and 10 μm for the 10 mm cubes and the 5 mm cubes respectively. Therefore, a quantification of all pores larger than 20/10 μm was performed, which provided, among other parameters, information concerning the distribution of pore sizes, the number of pores, the largest pore size, as well as specific parameters such as pore shape (sphericity, roundness) and the distance from the surface or the nearest next pore. Figure 3-9 displays the a graphical representation of

the results of the porosity analyses for each sample, while Table 3-3 presents the quantitative results from each sample with added comments.

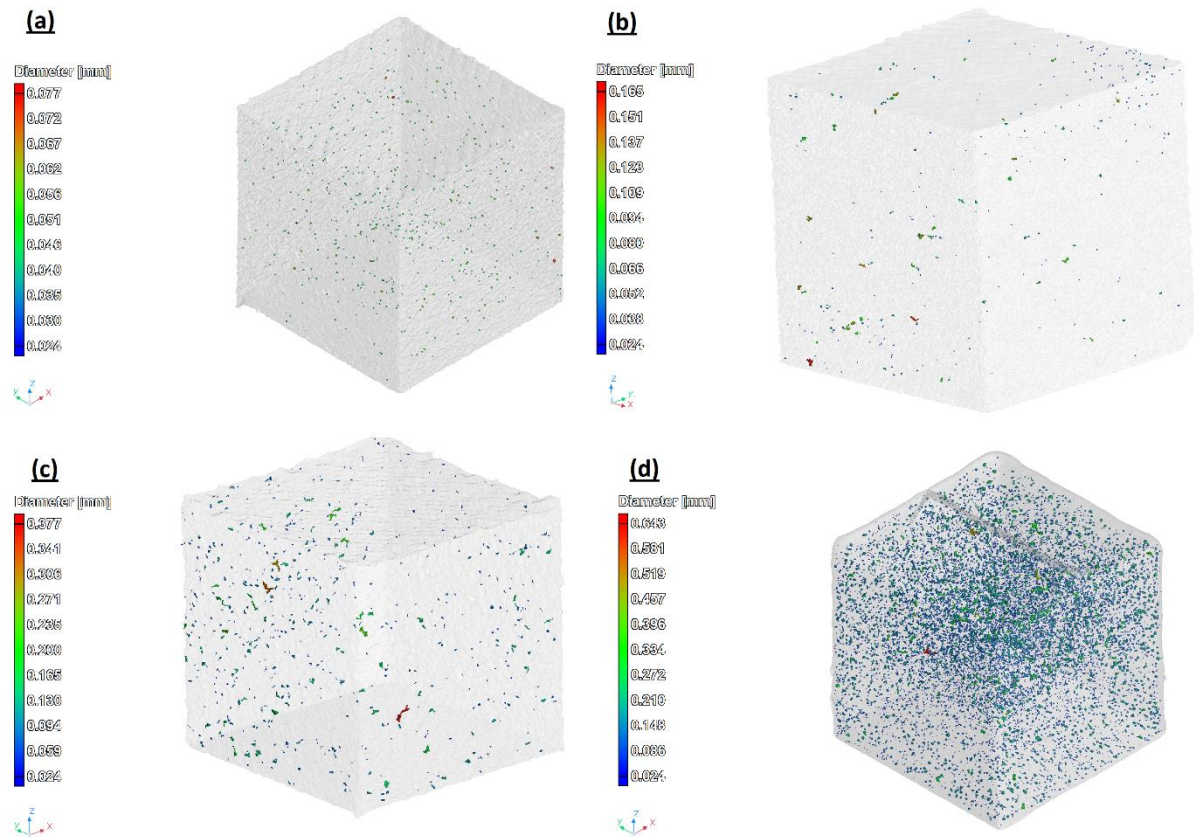


Figure 3-9: Cube porosity images for: (a) Ti-6Al-4V, (b) MS1, (c) 316L, (d) 6061-T6. Each sample is indicated with a different colour scale depending on the data set. The 3D porosity distribution is seen in each sample.

Table 3-3: Cube Samples: Summary of Porosity Results.

Sample Material	Cube Porosity (%)	Number of Pores	Largest Pore (mm)	Equivalent Spherical Diameter of largest pore (mm)	Comments
Ti-6Al-4V	0.01	178	0.0775	0.06	Evenly distributed, small pores
316L	0.02	74	0.165	0.115	Evenly distributed pores, with some larger pores
MS1	0.02	149	0.376	0.195	Evenly distributed pores with large, oblong pores present
6061-T6	0.25	5378	0.643	0.458	High porosity mostly distributed close to cube surfaces (contour porosity) and dense inclusions present

The 3D porosity distributions and quantitative results shown display the differences in material characteristics for each sample. The Ti-6Al-4V, 316L and MS1 show relatively low levels of porosity with good characteristics such

as even distributions, no excessive contour porosity and no clusters of pores. 6061-T6 displays higher levels of porosity with a large portion of the pores present close to the surfaces (contour porosity). These high levels of contour porosity, shown in Figure 3-10, are of critical concern in respect of part acceptability. As such, this material is further analysed in depth in Section 3.8.

The distributions, sizes and clustering of the pores for each material can be attributed to a variety of factors such as material chemical composition, AM processing parameters, relevant heat treatment processes, powder quality and scanning quality. The Ti-6Al-4V sample presents the lowest levels of porosity with no major defects present. This correlates accurately with the powder/material/AM process specifications, which can be found in Appendix D: EOS Ti-6Al-4V Powder/Material Specifications ($\approx 99.99\%$ compared to $\approx 100\%$ relative density for the observed results and provided data respectively).

The 316L and MS1 samples display slightly higher percentages of porosity, also showing accurate correlation with the material specifications (Appendix E: EOS 316L Powder/Material Specifications, Appendix F: EOS MS1 Powder/Material Specifications). The 316L comparative results indicate relative densities ranging between $\approx 99.98\%$ and $\approx 99.985\%$ in reference to the provided data, while the MS1 results indicate relative densities of between $\approx 99.98\%$ to $\approx 100\%$ in reference to the provided data. The 6061-T6 samples, even with significant defects present, also show an accurate correlation in measured and reference densities ($\approx 99.75\%$ to $\approx 99.7\%$). Even though the 6061-T6 results are within specification according to the manufacturer, the nature of the porosity was concerning, which prompted further analyses as discussed in Section 3.8.

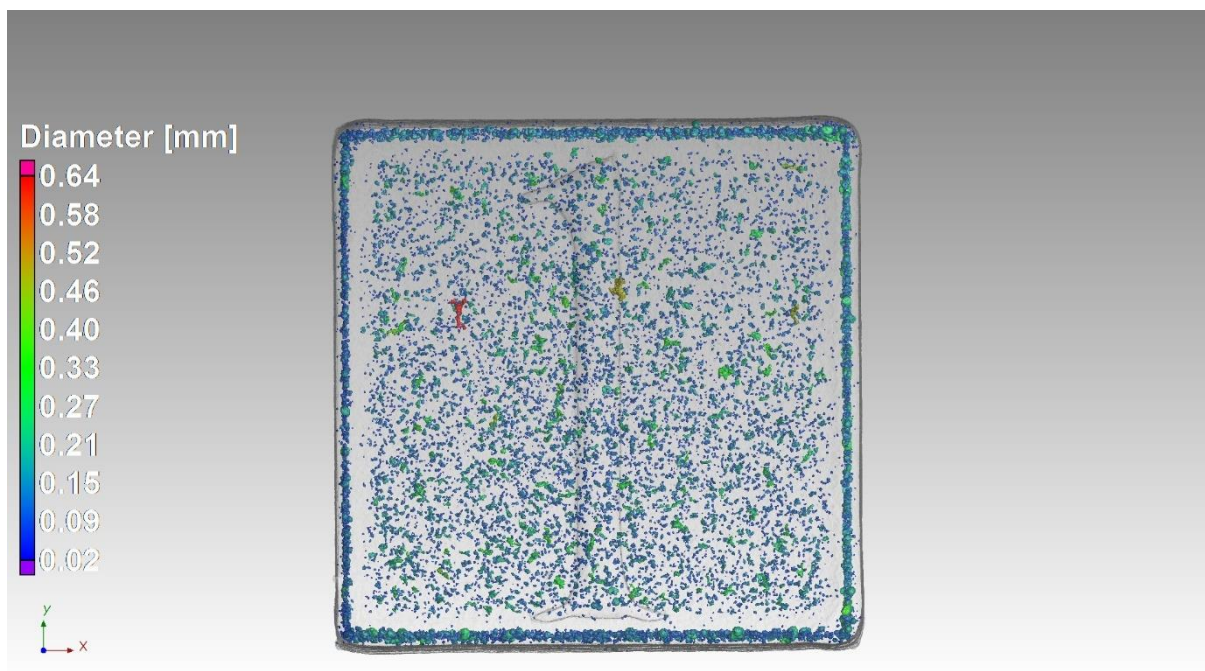


Figure 3-10: High levels of contour porosity present in the 6061-T6 sample. Top View.

The distance between pores relative to their size is another specification in aerospace applications besides total porosity, the largest pores, and the pore distribution. It is necessary that the distance between pores exceeds the diameter of the largest pore of the two considered [461]. An analysis of this may be automated by considering the “gap distance” analysis result that provides, for each pore, and the distance between the next nearest pore space.

Under this failure criterion, the Ti-6Al-4V, 316L, MS1 and 6061-T6 samples have 6, 4, 17 and 3556 criterion violations, respectively.

For surface roughness results, both graphical (Figure 3-11) and quantitative (Table 3-4) results were obtained. Visual inspection reveals notable similarities between the 316L, MS1 and 6061-T6 samples, in that their surface roughness features similar topographies, both in terms of magnitude and distribution. The Ti-6Al-4V sample is particularly rough when compared to the other samples in question.

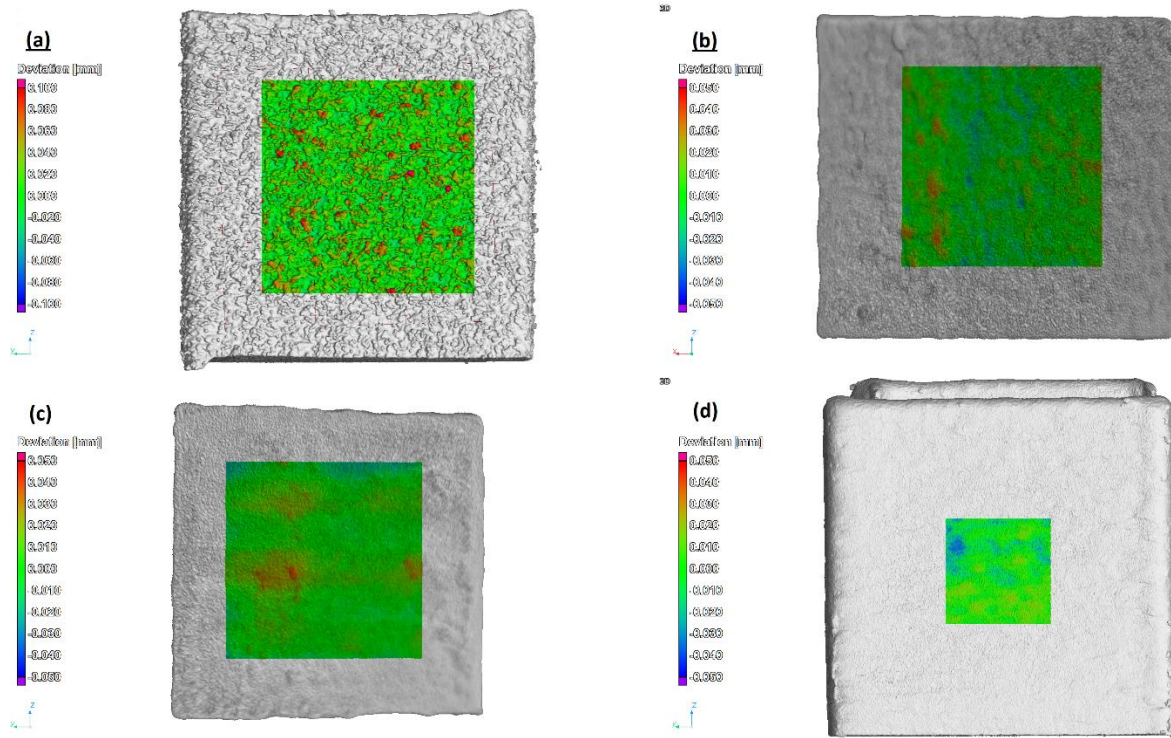


Figure 3-11: MicroCT surface tomography results on vertical walls for: (a) Ti-6Al-4V, (b) 316L, (c) MS1, (d) 6061-T6. Each sample is depicted with a colour scale depending on datasets.

The quantitative results confirm the qualitative inspection results, with the Sa values being similar for the 316L, MS1 and 6061-T6 samples, while the Ti-6Al-4V sample has significantly higher roughness. When compared to the manufacturer specifications, all samples display significantly higher surface roughness based on the measured results. Some deviations in these results could arise from various factors such as measured area, method of measurement (MicroCT or traditional techniques like tactile probe), slight differences in heat treatment processes, post-processing handling errors, and other factors. For the 6061-T6 sample, even though the porosity results are of concern, the surface roughness values obtained are the closest to the manufacturer specifications, suggesting that the processing parameters and build process are in fact as specified by the powder supplier.

Table 3-4: Surface roughness results for each sample on a 3x3 mm area.

Sample Material	Ti-6Al-4V	316L	MS1	6061-T6
Measured Sa (μm)	31.76	16.95	14.64	12.32
Manufacturer specified Ra (μm)	9 – 12	9.8	9	7.8 $\pm$ 0.31

3.7.2. Overhang Samples

For the overhang samples, the surface roughness at different inclinations was the primary area of interest from the analyses. For each material, these results were quantified on a per-angle basis to determine the impacts of low angles from the build plane on surface roughness. These analyses were performed on areas of 3x3 mm at the centre of each unsupported surface. An example of this segmented region is displayed in Figure 3-12, showing the deviation of the MicroCT measured surface from the relative flat-plane geometric element.

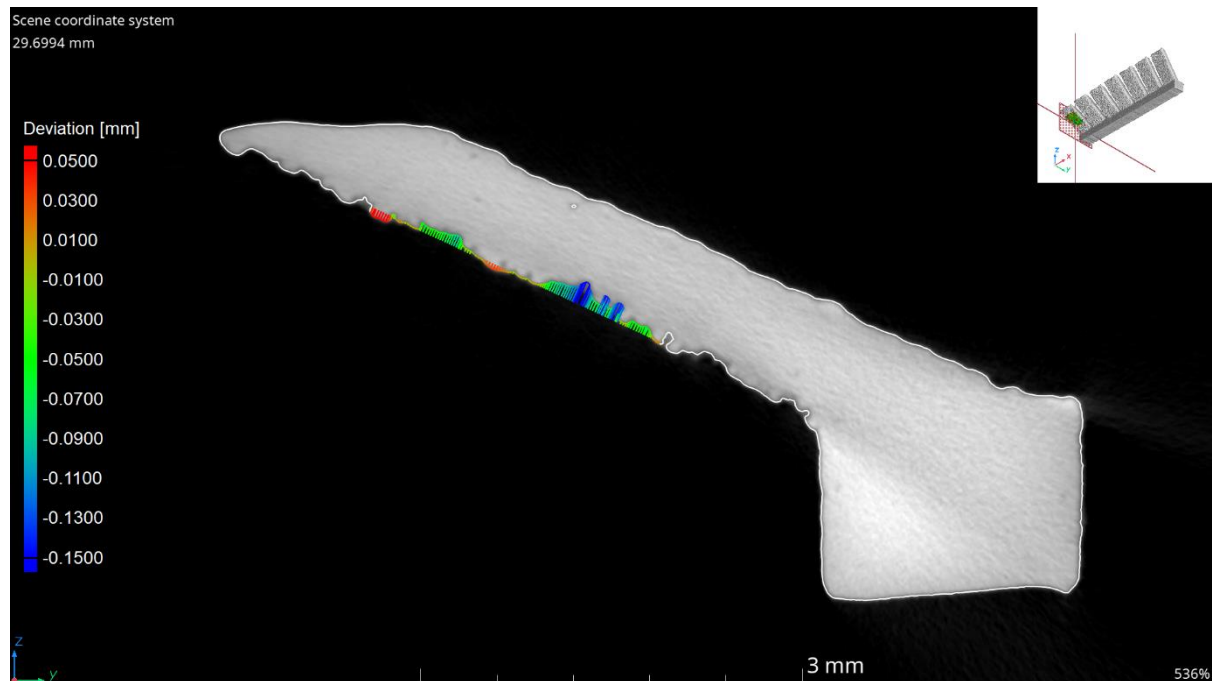


Figure 3-12: MS1 overhang sample segmented region surface roughness analysis (25° inclination angle).

Using the methods for data extraction presented in Section 3.6.2, S_a values were obtained for each angle and material, and are shown quantitatively in Table 3-5. These values are for downskin (downward facing) unsupported surfaces. The specific surface roughness values vary between samples, with the maximum and minimum surface roughness values occurring on the 6061-T6 and MS1 samples, respectively. These data show a clear correlation between surface roughness and inclination angle, with all samples showing rougher surfaces at low inclination angles.

The manufacturer-provided datasets for inclination angle surface roughness are either limited or non-existent, and of these data that are provided (only for 316L and 6061-T6), the lowest inclination angles tested are at 45° and 40° for the 316L and 6061-T6 materials, respectively. Comparing the measured surface roughness data to the manufacturer-provided data for the 316L and 6061-T6 samples indicates significant deviations. The corresponding surface roughness values for 316L at 45° are 62.68 μm (measured) versus 18.5 μm (provided), while those for 6061-T6 at 40° are 110 μm (measured) versus 18.91 μm (provided). These differences could be attributed to a variety of factors, such as differences in sample geometry, heat treatment processes, post-processing techniques, AM processing parameters, and measurement techniques. For the 6061-T6 sample in particular, the surface roughness is exceptionally poor and is discussed further in Section 3.8.

Table 3-5: Surface roughness (Sa) at different inclination angles.

Angle (°)	Sa (μm)			
	Ti-6Al-4V	316L	MS1	6061-T6
25	104.6	98.8	34.2	179.4
30	84.4	90.14	30.8	161.6
35	72.2	71.06	21.4	144.4
40	58.8	77.2	23.4	110
45	53.8	62.68	25.6	92.2
50	40.4	28.2	21.8	76.4
55	31	23.8	17.8	59

On a per-sample basis, the percentage change in surface roughness between the angles of 55° and 25° provides a clear indication of the significant sensitivity of surface roughness to inclination angle within the low inclination angle range (Table 3-6). With a 315.1% increase in surface roughness between the angles of 55° and 25°, 316L shows particular sensitivity in this regard, while for a 92.1% increase, MS1 is shown to be fairly insensitive. All samples, except the 316L sample, demonstrate a relatively smooth, linear increase in surface roughness as inclination angle decreases, as displayed in Figure 3-13. 316L, in particular, shows a sharp increase in surface roughness between the inclination angles of 50° and 45°.

Table 3-6: Percentage change in measured surface roughness for each sample in the range of 55° to 25° inclination angles.

	Ti-6Al-4V	316L	MS1	6061-T6
Percentage change (%)	237.4	315.1	92.1	204.1

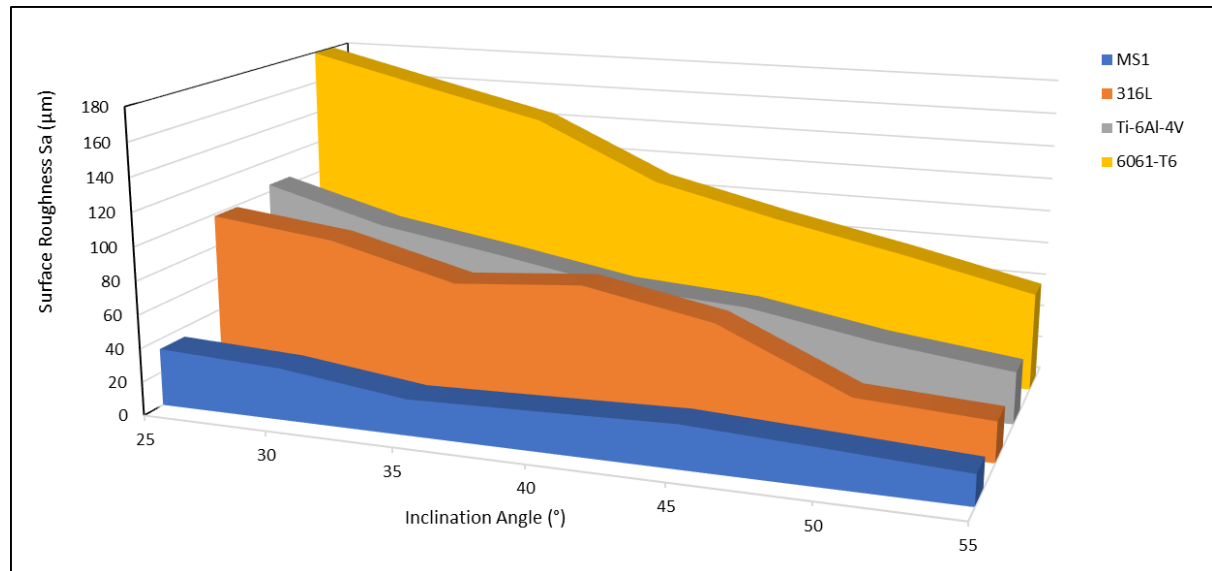


Figure 3-13: Surface roughness per inclination angle 3D graph for all materials.

Geometry deviation data were obtained through similar analysis techniques, with the primary difference being the use of surface deviations of the entire component in reference to the nominal CAD geometry (Figure 3-8). By processing these data using averaged distributions of deviations, meaningful data were acquired for the purposes

of comparing material warpage characteristics. Cumulative distributions of deviations are statistical functions which represent the probability of deviation measurements occurring under measurement thresholds. For example, the result of a 90% cumulative distribution of deviations would represent the maximum deviation measurement of the lowest 90% of measurements. This provides an effective characterisation of average deviation measurements excluding possible large maximum results.

Table 3-7 shows the minimum and maximum deviations and the 90/95/98% cumulative distributions of deviations for each sample. While these maximum and minimum data for each sample can be useful, it was found that these values all occurred on the tips of the overhang samples, where either a drooping effect occurred or, in the case of the 6061-T6 sample, a large bulbous effect was observed (Figure 3-15). The cumulative distributions of deviations for each sample provide a more effective means of determining the specific warpage characteristics of each material. It is, however, important to note that the cumulative distributions do contain the errors due to the tip effects described above.

Table 3-7: Overhang sample deviation data showing minimum, maximum and cumulative distributions of deviations.

Sample Material	Minimum Deviation (mm)	Maximum Deviation (mm)	90% Cumulative Distribution (mm)	95% Cumulative Distribution (mm)	98% Cumulative Distribution (mm)
Ti-6Al-4V	-0.1524	0.4951	0.1002	0.1194	0.1453
316L	-0.1997	0.321	0.0872	0.1052	0.1279
MS1	-0.1793	0.3128	0.1179	0.1567	0.2332
6061-T6	-0.4392	>3	0.3183	0.3627	0.3884

Comparing these data on a per-material basis, it was found that the MS1 sample demonstrated the smallest minimum and maximum deviations, while conversely, the 6061-T6 sample displayed the highest minimum and maximum deviations. However, due to the extent of the 6061-T6 bulbous defect and the sample's dissimilarities to the other samples, the data sets for 6061-T6 are not easily comparable to the other material data sets.

The cumulative deviation distribution is expressed in the form of the 90/95/98 cumulative percentiles of deviation for each sample. For example, for Ti-6Al-4V, these measured voxel data shows that 90% of the surface lies within a ≤ 0.1002 mm deviation, 95% within a ≤ 0.1194 mm deviation, and 98% within a ≤ 0.1453 mm deviation. As such, discarding the 6061-T6 sample data, the 316L and MS1 materials are shown to have the best and worst warpage characteristics, respectively. This is in stark contrast to the results from the inclination angle surface roughness measurements where 316L exhibited poorer surface roughness characteristics and higher inclination angle sensitivity compared to the MS1 sample. This suggests that while AM-built samples may have specific surface roughness and warpage characteristics, these characteristics may not necessarily be interdependent.

Determining near-surface porosity and total porosity of the overhang samples quantitatively was deemed unnecessary, due to the scan quality achieved and the frequency of porosity defects in the samples. However, manual qualitative analyses were performed on each sample to determine whether any significantly large, or high quantities of, porosity-based defects were present. Figure 3-14 shows the 25° inclination angle surface of each overhang sample as a slice view to annotate worst-case porosity defects. As shown, for the Ti-6Al-4V, 316L and MS1 samples, the quantity and size of porosity defects is minimal and not of particular concern for engineering

applications. Qualitative analysis shows that these porosity defects do not appear to change relative to the inclination angle measured, which is similar to the findings in [467]. These scans, however, are of low quality and further analyses at higher resolutions should be carried out to definitively analyse the porosity characteristics of these materials.

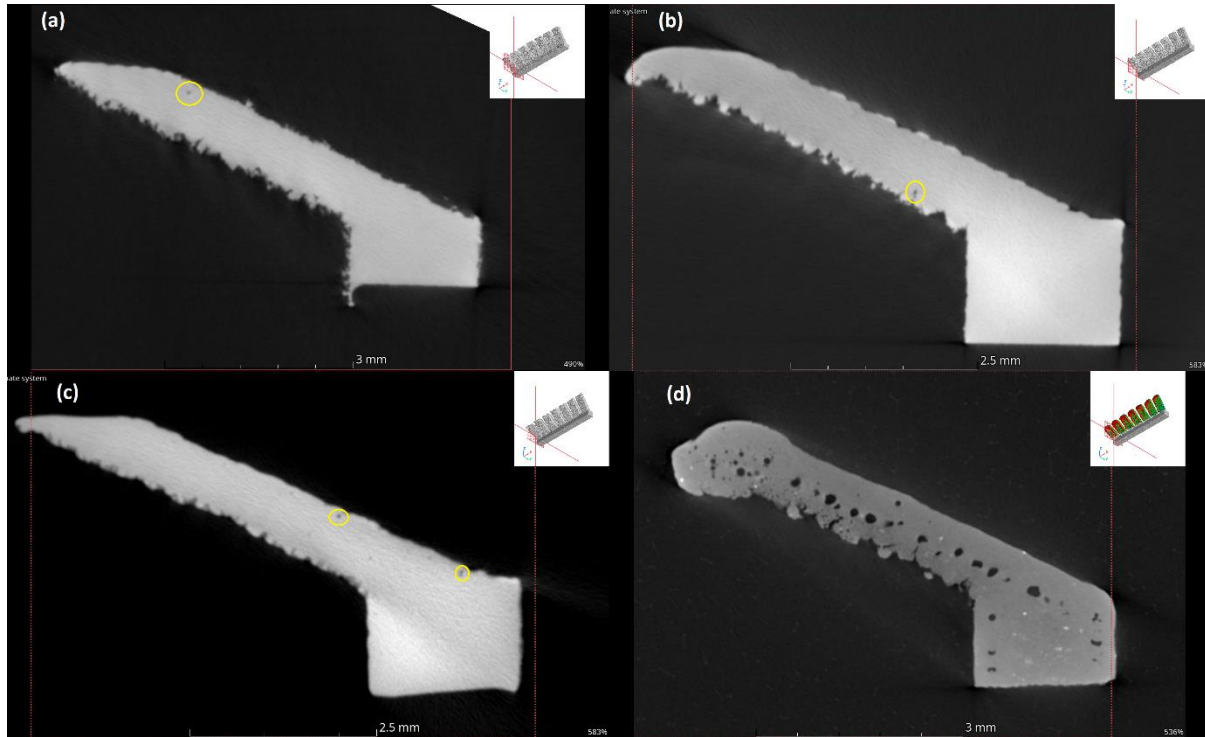


Figure 3-14: Overhang sample slice views showing scan quality and porosity defects outlined by yellow circles: (a) Ti-6Al-4V, (b) 316L, (c) MS1, (d) 6061-T6.

The 6061-T6 sample displays excessive porosity along all unsupported surfaces at equidistant locations from all surfaces. This is reflected for all inclination angles (Figure 3-15) and is a clear indication of contour porosity defects. Interestingly, these contour porosity defects occur at a consistently larger magnitude on the unsupported surfaces when compared to the base of the sample. This could be due to the consistent nature of the contour porosity found in the 6061-T6 samples (all occur at roughly 0.25 mm from surfaces). Since the overhang samples have effective wall thicknesses of roughly 0.6 mm, these contour tracks appear to be combining, with the defects being exacerbated further as a result.

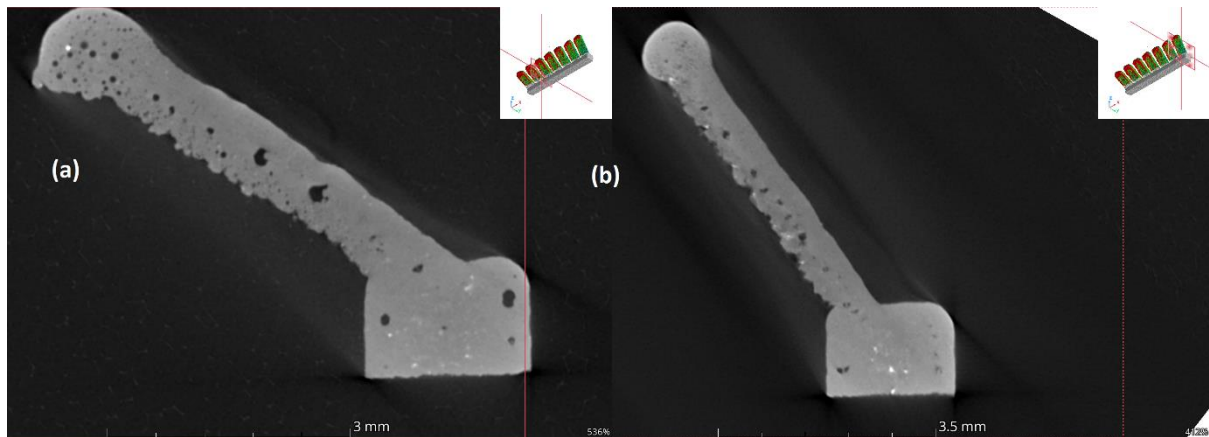


Figure 3-15: 6061-T6 overhang sample slice view displaying contour porosity and inclusions: (a) 35° inclination, (b) 55° inclination.

3.8. Elementum Aluminium 6061-T6

Due to the contour porosity defects, high surface roughness and bulbous features present in the 6061-T6 overhang and cube samples, an investigation was conducted focussing on the quality control associated with the material AM supply chain. Since both the overhang samples and cube samples are relatively small-scale components, it was important to first check whether the scale and proximity to the build plate were important factors contributing to the defects found in Section 3.7.

This evaluation was performed by using MicroCT techniques on some larger components. These components, an example of which is shown in Figure 3-16, all displayed the same form of contour porosity found in the cube and overhang samples. All sample components were observed to contain high levels of contour porosity, and, while some general porosity is present, pores within the bulk regions of the components (away from the contour tracks) were found to be widely distributed and small in size. This led to the specific investigation of strategies for contour porosity mitigation which is presented in Section 3.8.1. MicroCT techniques were used as the basis of this investigation and the specific scanning parameters are not detailed. All associated samples processed through MicroCT followed the scanning quality strategy outlined in [458,459].

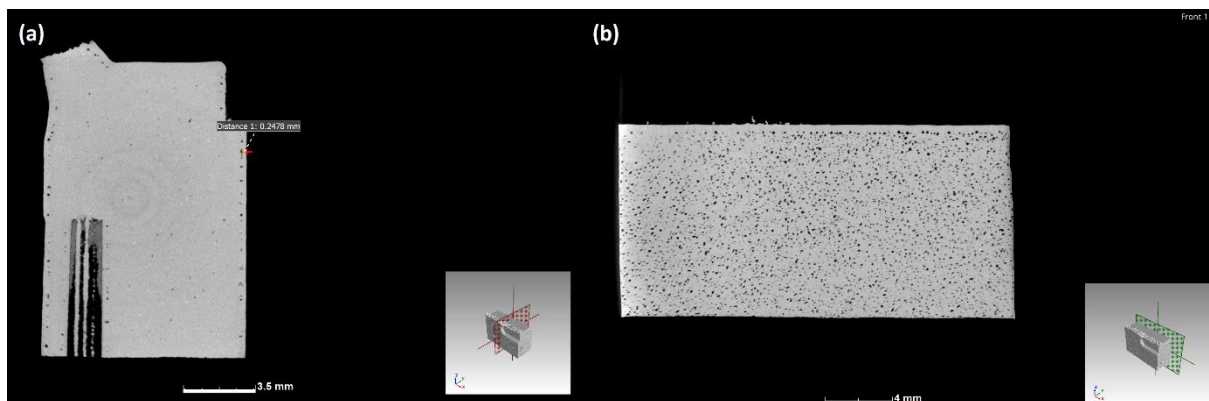


Figure 3-16: 6061-T6 off-cut sample slice view showing contour porosity and locations of pores: (a) Slice view showing contour porosity equidistance from surfaces, (b) displaying slice view through contour porosity indicating regularity of defects.

3.8.1. Cube Sample AM Processing Parameter Optimization

Due to the rapid-prototyping nature of AM processes, samples can easily and rapidly be designed, built and tested under different configurations. This, when combined with MicroCT techniques, allows for a rapid and strict quality control programme to be developed for relatively low cost. For the purposes of eliminating the contour porosity characteristics in the 6061-T6 material, a methodology to test various AM processing parameters was used. These processing parameters were applied to 10x10x10 mm cube samples, all built using the same AM build process, in order to accurately compare the effects of processing parameters on cube samples. The powder manufacturer (Elementum3D) was informed of the defects present in the samples found in Section 3.7 and consequently provided a new set of processing parameters.

The old processing parameters (used in Section 3.7) were tested alongside the new processing parameters referred to above to assess the differences in contour porosity mitigation techniques. A total of fourteen cubes, in two sets of seven cubes, were built using AM techniques where Set 1 was subjected to changes made to the old processing parameters, and Set 2 was subjected to those made to the new processing parameters. The AM processing parameters used for Set 1 and Set 2 cube samples are listed in Table 3-8 and Table 3-9 respectively. Set 1 sample 1 and Set 2 sample 1 are baseline samples for both the old processing parameters, and the new processing parameters respectively.

Table 3-8: AM processing parameters Set 1 samples of the 6061-T6 process parameter optimization study.

Set 1					
	Hatch Track		Contour Track		Additional Changes
Sample Number	Laser Power (W)	Laser Scanning Speed (mm/s)	Laser Power (W)	Laser Scanning Speed (mm/s)	
1	370	1300	370	300	
2	370	1170	370	270	
3	410	1170	410	270	
4	370	1300	370	300	Three contour tracks (default two)
5	370	1300	370	300	Four contour tracks (default two)
6	370	1300	370	300	Total remelt of each layer
7	370	1300	370	300	Reduced corridor value (reduced to 0.001 mm)

Table 3-9: AM processing parameters Set 2 samples of the 6061-T6 process parameter optimization study.

Set 2					
	Hatch Track		Contour Track		Additional Changes
Sample Number	Laser Power (W)	Laser Scanning Speed (mm/s)	Laser Power (W)	Laser Scanning Speed (mm/s)	
1	370	1300	260	167	
2	370	1170	260	150.3	
3	410	1170	286	150.3	
4	370	1300	260	167	Three contour tracks (default two)
5	370	1300	260	167	Four contour tracks (default two)
6	370	1300	260	167	Total remelt of each layer
7	370	1300	260	167	Reduced corridor value (reduced to 0.001 mm)

These samples were manufactured using AM techniques to the same heat-treatment state as used previously in Section 3.6.1. The same MicroCT process was used for all subsequent evaluations to ensure consistent sample comparison. All samples were scanned and then analysed both quantitatively and qualitatively using the porosity techniques outlined in Section 3.6.1. Graphical representations of the best and worst cases of contour porosity for each set are shown in Figure 3-17, and the quantitative results for percentage porosity and largest pore, as well as associated comments are presented in Table 3-10 and Table 3-11.

It is clear from these data that Set 2 outperforms Set 1 with respect to porosity characteristics. This is likely due to the increased energy input rates on the contour tracks (reduced laser scanning speed increases energy input per unit area). For Set 1, the best improvement in porosity characteristics is observed when remelting each layer of the sample. For Set 2, increasing the number of contour tracks was found to be optimal. Both of these alterations (remelting of layers and increased contour tracks) increase the time taken to manufacture each layer, resulting in longer build times.

Various other conclusions could be reached concerning the specific correlations between porosity characteristics and processing parameters, but for the purposes of this study, finding an acceptable set of parameters for the manufacture of mechanically-sound components was the objective. In this regard, Sample 5 of Set 2 was found to achieve acceptable material characteristics.

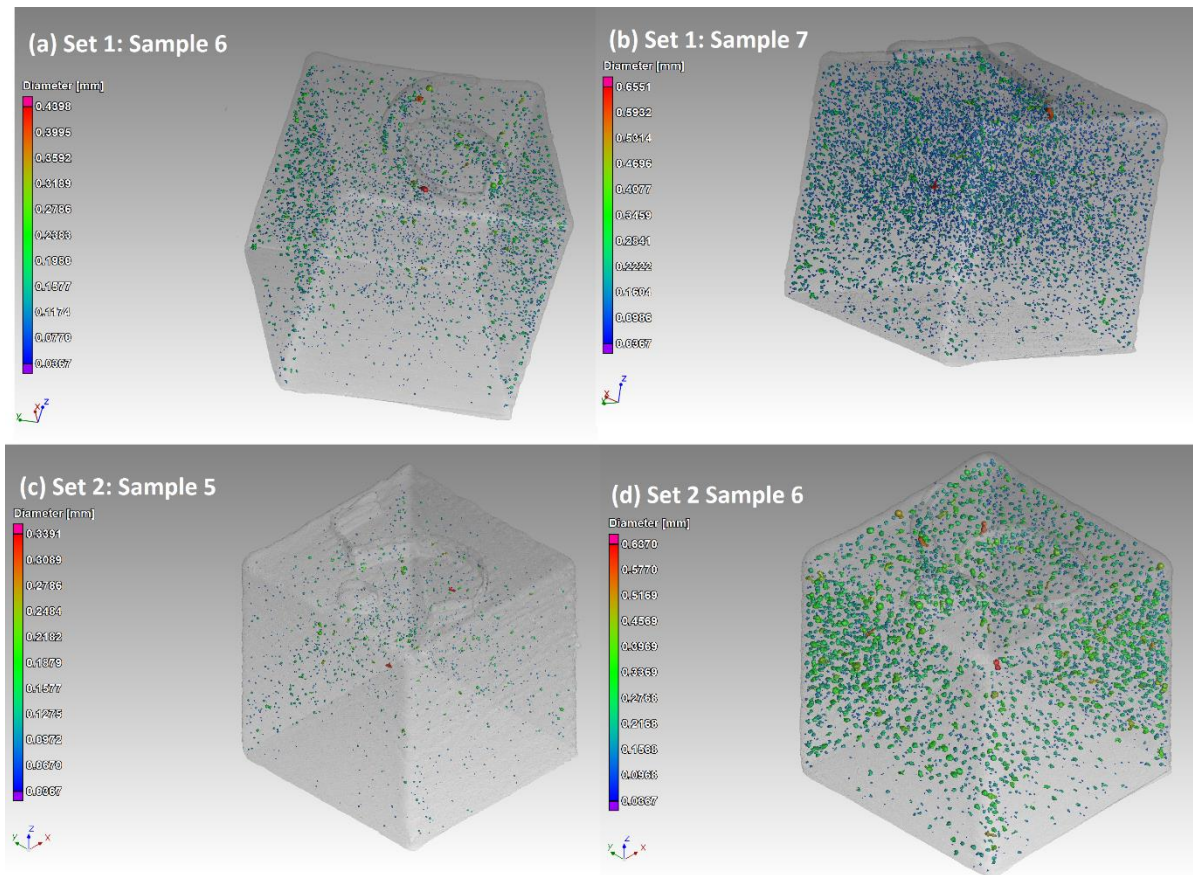


Figure 3-17: 6061-T6 cube sample optimization: best-cases (a, c) and worst-cases (b, d) for each set.

Table 3-10: Set 1 6061-T6 cube sample optimization: largest pore, percentage porosity and comments.

Set 1			
Sample Number	Largest pore (mm)	Percentage Porosity (%)	Comments
1	0.64	0.25	High contour and bulk porosity with large, oblong pores present
2	0.39	0.22	High contour porosity and low bulk porosity
3	0.52	0.31	High contour porosity and very low bulk porosity
4	0.53	0.33	High contour and bulk porosity
5	0.46	0.29	High contour and bulk porosity
6	0.44	0.12	Moderate contour and bulk porosity
7	0.65	0.31	High contour and bulk porosity with some large pores

Table 3-11: Set 2 6061-T6 cube sample optimization: largest pore, percentage porosity and comments.

Set 2			
Sample Number	Largest Pore (mm)	Percentage Porosity (%)	Comments
1	0.58	0.08	Low contour porosity, moderate bulk porosity and some large pores
2	0.53	0.08	Moderate contour and bulk porosity with some large pores
3	0.5	0.26	High contour porosity and very low bulk porosity
4	0.46	0.04	Moderate contour and bulk porosity with some large pores
5	0.34	0.03	Moderate contour and low bulk porosity with few large pores
6	0.64	0.41	High contour porosity and very low bulk porosity
7	0.47	0.07	Moderate contour and bulk porosity with some large pores

Another interesting observation was made through qualitative analyses. The cube samples were generally found to exhibit good surface roughness characteristics in correlation with poor contour porosity characteristics, with the converse also being true. 6061-T6 typically has a good quality surface finish when compared to other as-built AM materials, and this presents an interesting trade-off between surface finish and mechanical properties. It could be useful, where good quality surface finish is required, to optimise AM process parameters for surface finish, while managing the associated degradation in mechanical properties due to increased porosity via design modification.

3.8.2. Powder Quality Control

Traditional powder analysis is performed using a laser diffraction method described in ASTM B822-17 [468]. While laser diffraction provides particle sizes, it assumes that particles are spherical regardless of the fact that particles may often be not as spherical as expected. Using microscopy and image analysis, studies have examined the morphology of metal powder particles, but the method can only obtain pseudo-3D images, which limits qualitative analysis and does not reveal internal porosity.

Using synchrotron tomography, researchers have demonstrated that porosity within powder particles can be transferred to the melt pool and consequently to the final product [469]. Additionally, particle size distribution and sphericity influence powder flowability, in turn influencing the characteristic of the powder bed, such as spreading and packing density.

Performing MicroCT scans on AM powders is an effective way to characterise various specific properties of AM materials. The methods and purposes of scanning powders via MicroCT are described in detail in [470,471]. MicroCT methods typically require little to no sample preparation, reducing the complexity and therefore cost of scanning powders. For the purposes of this study, the methods used in [470] were applied to two sets of 6061-T6 powders.

In AM processes, the un-melted powder present in the build chamber is often recycled to various degrees, and different quality control methods are applied dependant on powder and AM machine characteristics. Metal powder analysis has been used to detect changes in powder quality after repeated re-use. It was shown in [472], for instance, that powder particles become less spherical as the number of re-use cycles increases, and that their surface becomes increasingly rougher. Additionally, other types of powder may partially fuse, which decreases flowability in the powder bed process. Re-used powder should be characterized in any case to ensure that its properties are maintained.

After the findings of porosity defects in Section 3.7, the AM service provider employed for manufacture of the 6061-T6 samples conveyed that the powder batch from which the samples were made had undergone a large number of re-use cycles. This finding led to a quick investigation into the powder characteristics associated with all samples manufactured, including those discussed in Section 3.8.1. In support of this investigation, a new batch of 6061-T6 was ordered from the powder supplier and samples of both this new powder and the old powder were analysed using MicroCT techniques. This effectively enabled a comparison of the 6061-T6 powder in two states: new powder with no re-use cycles, and old powder with high re-use cycles. Both powders were scanned using the method described in [470].

After MicroCT scanning, the powders were analysed with respect to each other. However, due to many inclusions (high-density particles) present in both powders' samples, the MicroCT scan quality was poor, as can be seen in Figure 3-18, and very little quantitative data could be extracted on this basis. In MicroCT, these higher-density inclusions distort the X-rays and cause scattering in the vicinity. Since powder scans using these techniques require a particularly good scan quality to determine powder sphericity, no quantitative data of this nature could be obtained. It was however found that the inclusions present in the sample of old powder (Figure 3-18, left) were far more numerous, and thus the scan quality was significantly poorer. This suggests that a higher quantity of high-density particles was present in the old powder sample, which could be a possible cause of the defects described throughout Sections 3.7 and 3.8.1.

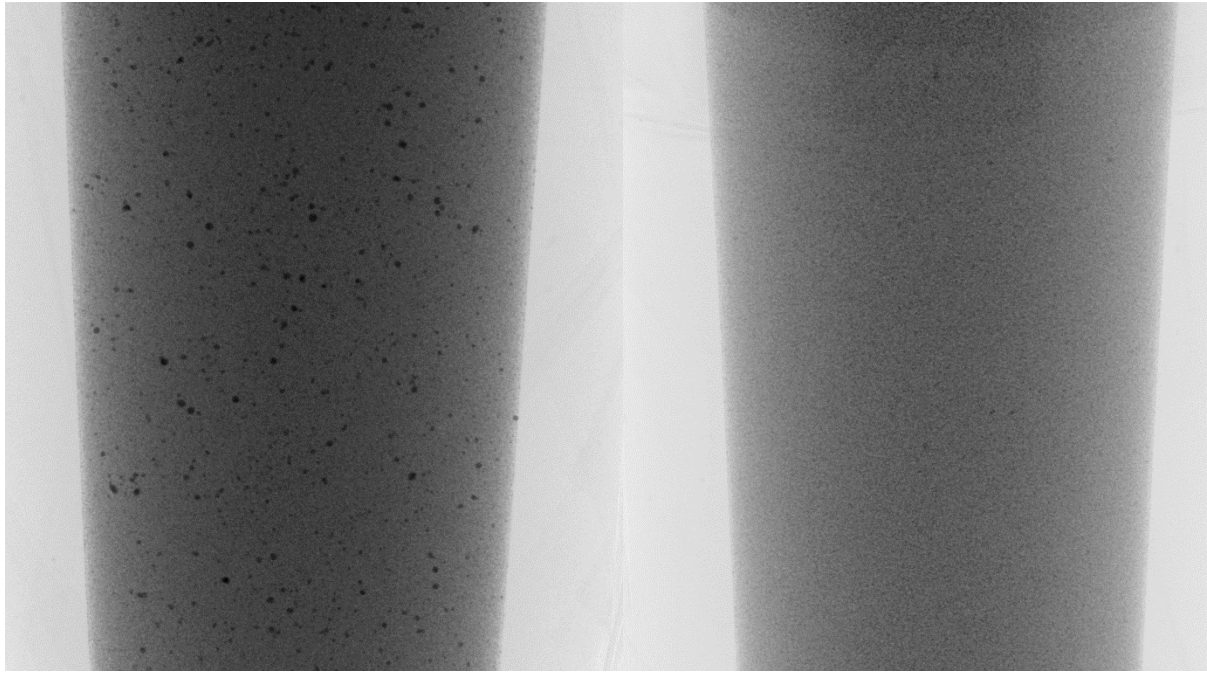


Figure 3-18: 6061-T6 powder MicroCT scan images of both old (left) and new (right) powder samples.

Due to the proprietary chemical composition of the 6061-T6 powder, it is difficult to arrive at definitive conclusions concerning the nature of the differences between powder samples. Of significance, however, the powder supplier did indicate that an unknown ceramic forms $\approx 2\%$ of the powder's composition (see Appendix G: Elementum3D 6061-T6 RAM2 Powder/Material Specifications). It is well known that Aluminium 6061, in its traditional chemical composition, manufactures very poorly on AM systems. The incorporation of ceramics in difficult-to-manufacture AM materials is an emerging technique used to enhance associated manufacturing characteristics. Since the density and sphericity of these ceramics often differ from those of the a powder's metallic alloying elements, it is possible that through repeated re-use, the ratio of ceramic content to metallic content within such powders may change. Some assumptions can be made with respect to this information:

1. Ceramic particles may be larger or smaller than metallic particles, and may therefore get separated from the powder during the sieving phase of the re-use process
2. During mixing phase, the higher density metallic particles and the lower density ceramic particles may not necessarily become evenly distributed

There are also some other potential causes for these high-density inclusions in the powder samples. One of these could be the partial fusion of metallic particles that occurs during repeated use. Over time, this could result in an accumulation of particle clusters. Partially fused or fully fused particles have a higher effective density than virgin powder, and as such, they would be detected as inclusions.

The results of this investigation highlight key potential causes of defects in AM processes which stem from inadequate powder quality control. Specific quality control measures should be established to ensure that powder quality does not reduce over repeated re-use cycles. This could, for instance, take the form of periodic MicroCT analyses of both powder and cube samples to detect defects which may arise from repeated powder re-use.

3.8.3. Further Sample Testing

Following on from the various studies and investigations described in the previous sections, new samples were manufactured and analysed using MicroCT techniques. These samples were all manufactured using the optimal parameters found in previous sections; namely, new powder (Section 3.8.2) and new processing parameters (Section 3.8.1). The same cube samples as used previously (10x10x10 mm) were manufactured for the purposes of both contour and bulk porosity analyses. An updated version of the overhang samples was introduced with the intention of eliminating the tip defects (bulbous and drooping effects) identified in Section 3.7.2. These new overhang samples, shown in Figure 3-19, were modified by mirroring the geometry present in the original overhang samples, and joining these mirrored objects at the tips.

Apart from measuring the downskin surface roughness per angle of inclination, as well as porosity defects, these samples were also intended to measure geometry deviation and detect possible defects in the converging geometry regions, i.e. where the “tips” meet. Furthermore, new wall thickness samples were designed and manufactured with the intention of measuring both the minimum acceptable wall thickness, and possibly the converging contour porosity defects described in Section 3.7.2. These wall thickness models, shown in Figure 3-19, comprise a vertical wall varying in thickness from 2 mm at one end, to an infinitely small thickness at the other.

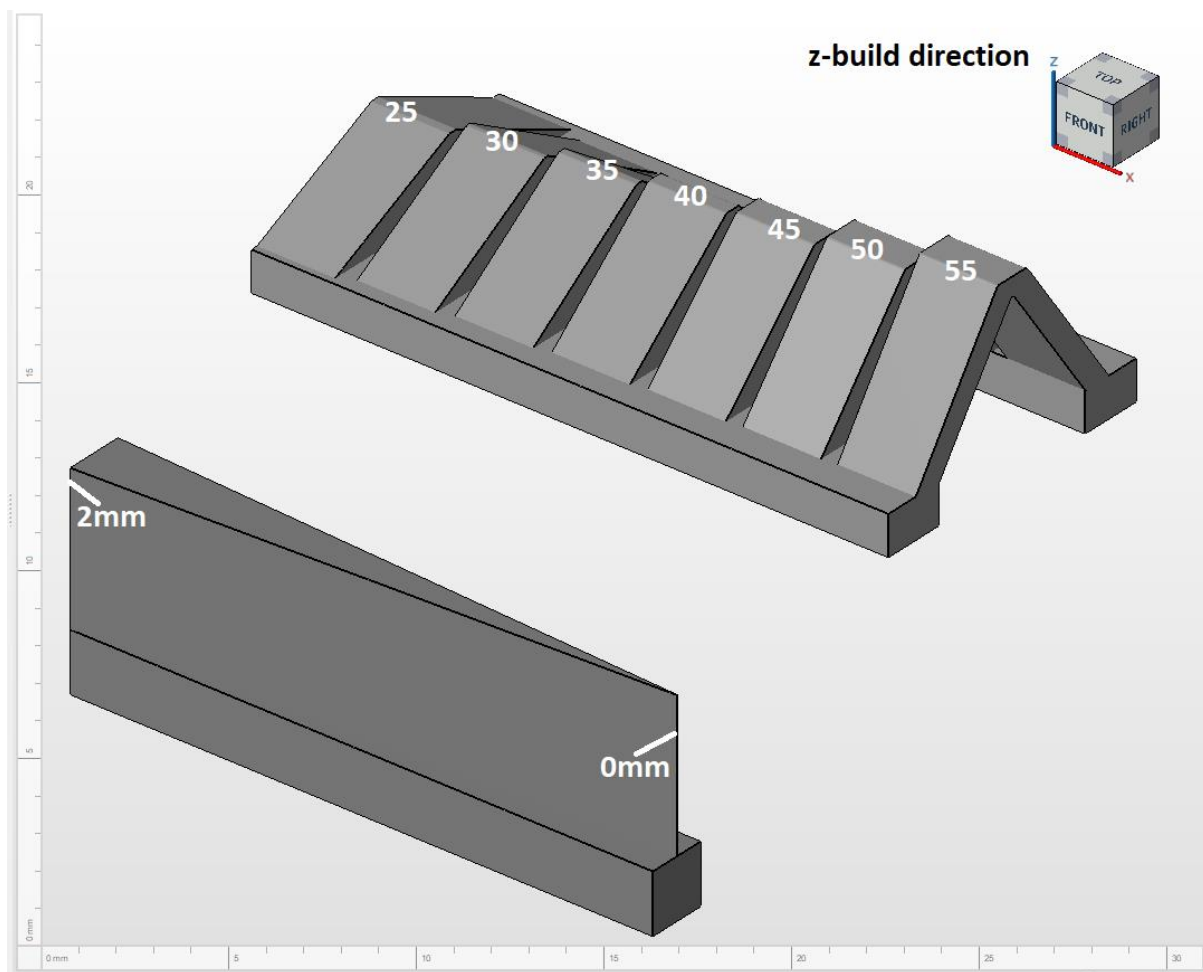


Figure 3-19: Joined Overhang sample showing various inclination angles and wall thickness sample showing minimum and maximum wall thicknesses. Scale in (mm).

All samples were manufactured on the same AM platform used previously (EOS M280) and brought to the same heat-treated state (Section 3.2). The samples were then scanned via MicroCT using the same methods described in Section 3.6. The cube samples and overhang samples were subjected to the same image analysis methods used in Section 3.6 for the purposes of porosity and surface roughness evaluation.

Figure 3-20 shows both the porosity and surface roughness for the revised 6061-T6 sample. The porosity results were of high-quality, with the quantitative results showing a maximum pore size of 0.151 mm, a maximum pore effective diameter of 0.138 mm, 0.01% total porosity, and no contour porosity. It is however important to note that the quantitative analysis for porosity was limited to a minimum pore size, and that smaller pores are likely present but undetectable under this resolution. Further analysis, using higher resolution scans on smaller samples (5x5x5 mm cube), could yield more accurate results for porosity measurements. The surface roughness results indicated the achievement of high surface quality, with an Sa measurement of 11.244 μm .

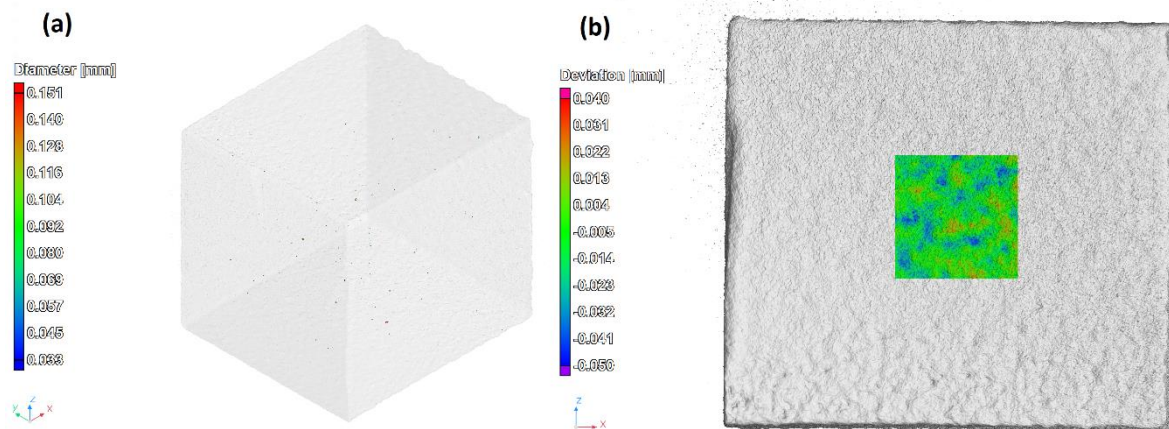


Figure 3-20: 6061-T6 cube sample qualitative analyses indicating porosity (a) and surface roughness (b).

The inclusions present in all 6061-T6 samples were also observed. Figure 3-21 shows MicroCT slice views with inclusions and porosity being visible within both the cube sample (a) and the joined overhang sample (b). Interestingly, the cube sample featured little to no porosity compared to the overhang sample, where porosity is clearly visible in both the unsupported overhang regions and the base region. Much like previous samples, the overhang sample contains no relevant increase in porosity defects in unsupported overhang regions when compared to the base regions.

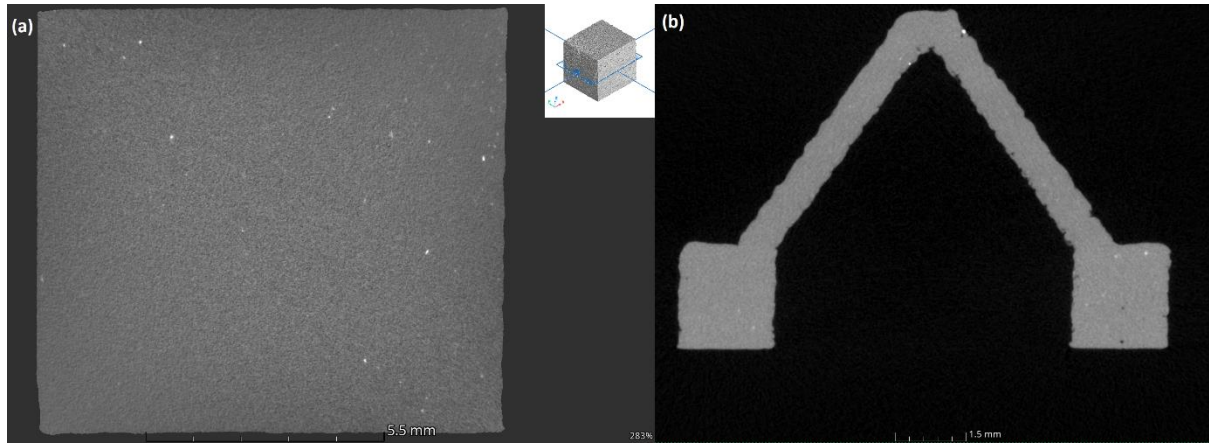


Figure 3-21: 6061-T6 cube and overhang sample slice images displaying inclusions and minimal porosity.

Graphical representations of surface roughness characteristics (a) and geometric deviation (b) of the joined overhang samples are displayed in Figure 3-22. As in the case of the previous overhang samples, surface roughness is shown to increase as the inclination angle of the downskin surface decreases, as quantified in Table 3-12. For all inclination angles, these surface roughness characteristics are considerably better than those exhibited by the previous 6061-T6 overhang samples. It must be noted, however, that the alteration in sample geometry makes direct results comparison difficult.

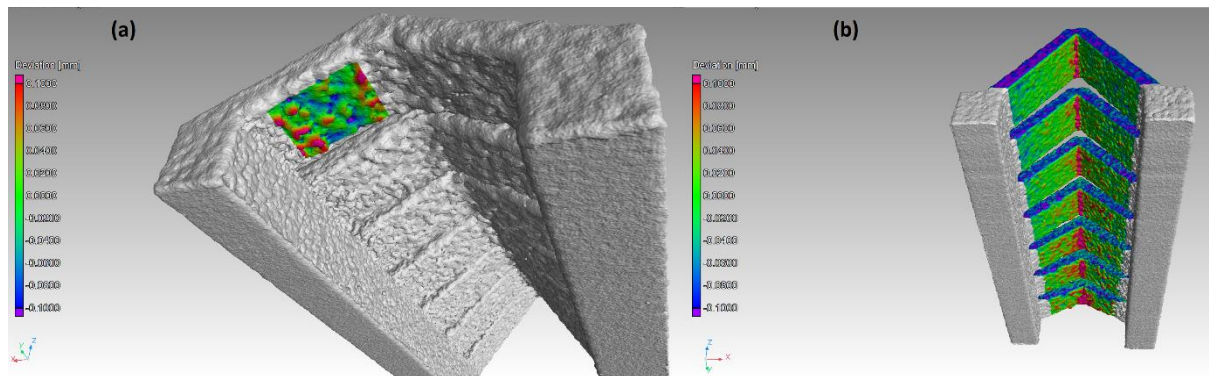


Figure 3-22: 6061-T6 overhang sample surface roughness and deviation qualitative results.

Table 3-12: 6061-T6 joined overhang sample surface roughness (S_a) values per inclination angle.

Inclination Angle (°)	25	30	35	40	45	50	55
S_a (μm)	42.1	38.1	35.6	31.9	29	26.4	24.5

The geometry deviation characteristics of these overhang samples are also improved when compared to the previous samples described in Section 3.7.2. The specific quantitative deviation results showed a minimum and maximum deviation of -0.2958 mm and 0.345 mm respectively. Furthermore, the cumulative distributions of deviations showed 90% of the surface measurements lie within a ≤ 0.0841 mm deviation, 95% within a ≤ 0.0943 mm deviation, and 98% within a ≤ 0.1154 mm deviation. These cumulative distributions show less deviation in the revised 6061-T6 samples when compared to the samples in Section 3.7.2. However, these values are ultimately incomparable due to the drooping and bulbous defects found in the first samples. The graphical results presented

in Figure 3-22 (b) also suggest that surface roughness and deviation significantly increases as the inclination angle is lowered.

The wall thickness samples were analysed using a method similar to that used for the surface roughness measurements in Section 3.6.1. Once an accurate surface determination was made, a “wall thickness” analysis was performed. This analysis used the sphere method, which uses a rolling sphere to calculate the wall thickness of a given body. Using the same method for geometry deviation analysis shown in Section 3.6.2, the nominal CAD geometry was imported, registered, and then a “nominal/actual comparison” analysis was performed to calculate deviation.

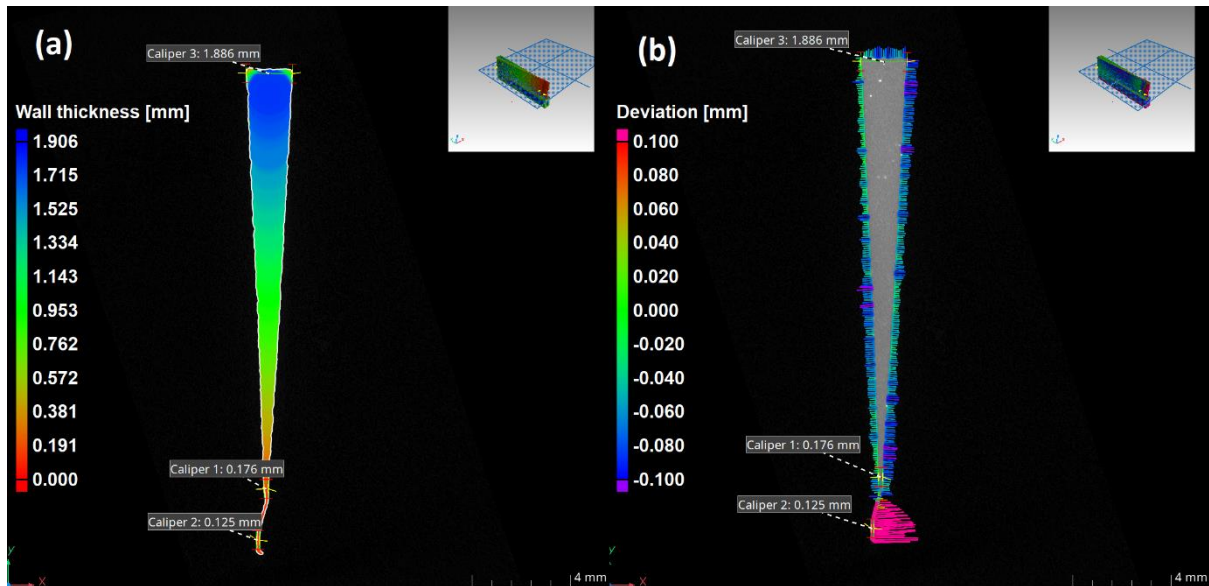


Figure 3-23: 6061-T6 wall thickness sample showing wall thickness results, wall thickness measurements and deviation results.

As shown in Figure 3-23 (a), the measured wall thickness was found to range between 1.886 mm and 0.125 mm. When comparing this to the CAD geometry, both the maximum and minimum results deviate roughly 0.125 mm from the nominal thicknesses (1.886 mm vs 2 mm and 0.125 mm vs 0 mm). On the maximum end, this is likely due to the sharp edges present on the CAD geometry which are not translated perfectly into the AM build preparation software. On the minimum end, the same sharp geometry issue is present, but also laser scanning track widths become relevant. In Figure 3-23 (b) the deviations at these minimum wall thicknesses (Caliper 2) tend to increase with distance from the solid structure in the vicinity. This suggests that although these thin walls can be manufactured, the warpage and defects present are significant.

The measurement indicator, Caliper 1, shows the minimum wall thickness at which no significant deviations or defects occur. This observation displays the minimum possible, defect-less wall thickness manufacturable under these machine and material configurations. This is, however, limited to vertical surfaces and may also be influenced by various factors in altered designs. Typically, minimum wall thicknesses for AM materials range from 0.3-0.4 mm, as specified by AM service providers. Nonetheless, these results show that thinner walls are manufacturable if careful design practices are applied and stringent testing procedures are introduced.

3.9. Conclusions and Recommendations

Standardized testing procedures were applied to quantitatively analyse AM-built samples for the Ti-6Al-4V, MS1, 316L and 6061-T6 materials candidates considered, allowing for direct comparison of material properties and characterisation of defects. Detailed analyses were performed to gather the porosity, surface roughness and unsupported surface characteristics for each material. 10 mm and 5 mm cubes provided insight into the nature of porosity defects in each material and further process optimization was performed (Section 3.8.1) to clarify the possible mitigation and quality control procedures which can be applied when poor porosity characteristics are identified. These porosity distributions extend into larger more complex parts (overhang samples) as well. Surface roughness analyses showed both the comparison of surface characteristics between materials, and the correlation of surface roughness variations with varying unsupported surface inclination angles. The geometry deviation characteristics of materials was quantified for overhang samples, which allowed for insights into cross-material deviation. AM powder quality control methods were outlined with suggestions for future quality control methodologies. A wall thickness sample was analysed to gain an understanding of the design limits for AM materials related to the AM supply chain in question.

Recommendations for further work:

The effects of heat-treatment processes on porosity and deviation are not analysed here and could be studied further for each material. Post-processing such as Hot-Isostatic Pressing (HIP) are known to improve porosity characteristics and could be applied to contour, bulk and subsurface porosity present on all materials if the design requires. Further improvement of porosity characteristics may be possible for all materials through the iterative process parameter optimization outlined in Section 3.8.1. Some AM materials were found to feature inclusions, and further analyses of these inclusions may lead to correlations for porosity measurements.

Overhang samples showed defects such as drooping effects or bulbous features and further testing with optimized samples may enable reduced surface roughness and geometric deviation. Furthermore, overhang samples could be modified by varying wall thicknesses at each inclination angle, to give insight into the effects of wall thickness variations on both surface roughness and near-surface porosity for different inclination angles. Although literature suggests that near-surface porosity is not correlated to inclination angle, further quantitative analyses could be performed to compare the near-surface porosity characteristics of different AM materials. Deviation analyses on unsupported surfaces could be performed on a per-inclination angle basis to provide data and correlations relating to the specific warpage characteristics of AM materials. Due to the relatively high-quality material characteristics found at low inclination angles, further testing could be undertaken at lower inclination angles. MicroCT surface roughness results may also be further verified through traditional measurement techniques, such as the use of tactile probes or non-contact interferometric techniques.

In optimising the AM process parameters for 6061-T6 aluminium, higher-quality porosity characteristics were demonstrated than those achieved using process parameters provided by the AM service provider. The corresponding changes may affect mechanical properties and should therefore be tested.

4. PROCESS CHARACTERIZATION OF ADDITIVELY MANUFACTURED PUMP COMPONENTS

4.1. Introduction

Aerospace industry segments, including commercial and military aircraft, space applications, and missile systems are transforming with the use of additive manufacturing (AM) technologies. The AM process has enabled such a transformation because it is able to fabricate parts with complex designs and reduce production costs (material waste, assembly costs due to part consolidation, and the need for tools and fixtures). In addition, AM processes are able to produce parts in small quantities with short turnaround times and low cost. Through AM, advanced part designs can be realized, such as multifunctional parts, multi-material parts, consolidated parts, and difficult-to-machine parts. AM is ideal for the space industry due to its ability to fabricate freeform designs in short lead times.

The AM of metal components is promising in industrial applications due to the high complexity of part geometries that can be achieved, allowing improved performance to be realized for functional components, including lightweighting, part consolidation, the minimization of joints and connections between different parts, and allowing shorter lead times and easing design iterations. Despite the high cost of once-off part manufacturing, the above advantages find niche applications in specific industries such as the space industry, as discussed in [473] and Section 2.3.

The growth of the space industry in recent years is especially driven by the increased demand for small and medium sized satellites. This demand has led to the increased use of metal AM to manufacture various rocket components including combustion chambers with complex geometries. Due to the vast complexity of rocket engine components and the high performance required for optimal function, such components are traditionally tested extensively via many iterations. Using traditional manufacturing techniques, this has historically been a remarkably expensive endeavour. With the advantages of design complexity come new challenges in quality control and inspection. Metal AM processes are highly complex, with numerous interrelated parameters and significant potential for manufacturing errors or flaws, including internal defects (various forms of porosity, inclusions, cracks, residual stresses, unwanted microstructures) as well as high surface roughness and dimensional errors. Typically, process optimization is combined with quality control efforts to ensure high quality parts are produced as discussed in Section 2.2.5. High quality here refers to mechanical properties reaching at least the minimum levels as set out by aerospace standards. The presence of manufacturing flaws will compromise mechanical properties and may have catastrophic consequences, if undetected.

Traditional non-destructive evaluation (NDE) methods are not well suited to metal AM, as the typical flaw sizes are much smaller than in traditionally manufactured components. Furthermore, the typical complexity of the parts, the rough surfaces and the unique microstructures all play roles in limiting the efficiency of traditional NDE methods [474]. X-ray micro computed tomography (MicroCT) is well established and has proven to be best suited to the inspection of such parts, as concluded in [457]. While the method is used widely, it is mainly used for small objects of low-density metals such as aluminium and titanium alloys. This is due to one of the well-known challenges faced by this method: it requires sufficient X-ray penetration of the evaluation object, which limits its

use for dense materials such as steel. When an object is too dense, the resulting MicroCT images are noisy and contain image artifacts, limiting further data processing and the ability to make a reliable evaluation. Due to these reasons, it is widely considered impossible to evaluate large, high-density components using typical laboratory MicroCT systems (225 kV, typically). High-power CT systems with voltages exceeding 300 kV as well as synchrotron tomography beamlines (with high flux) are better-suited but are not widely available.

Rapid prototyping, rapid tooling, and repairs are some of the most important applications of AM in the space industry, together with final part manufacturing of metal, plastic, ceramic, and composite parts. Final part manufacturing is the application with the fastest growth currently. AM-fabricated components can be designed and manufactured on demand with significant mass and cost-reductions, without sacrificing their mechanical properties. In the space industry, rapid prototyping plays an important role. Despite the growing prevalence of virtual prototyping, rapid prototyping remains essential for activities such as evaluating and testing design concepts iteratively and at low cost compared to traditional techniques. A primary advantage of using AM for rapid prototyping is the reduction of time-to-market by accelerating prototyping, the reduction of development costs, and the possibility of increasing innovation through design complexity enabled by AM design.

For the purposes of the SAFFIRE programme, utilizing AM for both rapid-prototyping and final part manufacturing can enable significant cost reductions and performance enhancements via mass reduction and part consolidation. This is especially advantageous for components where specific performance and operational characteristics are unknown, and iterative design and testing is required to both verify and optimize the components. The flaws present in AM processes are well-studied and common practices are available for design for additive manufacturing (DfAM) activities. However, these practices are limited and the specific geometry of the components in question heavily affects the occurrence of defects, especially in cases where highly complex geometry is present.

The primary focus of this chapter is to evaluate the AM manufacturing process for functional rocket engine propellant pump components under development in the SAFFIRE programme. These pump components are iteratively designed and evaluated through NDE techniques to assess the quality of the manufacturing process and to inform designers of the possible design complexities achievable through AM, and the benefits of doing so.

4.2. Liquid Rocket Engine Propellant Pumps

At present, South Africa, and Africa as a whole, lacks the capacity to independently launch payloads into Low Earth Orbit (LEO). In response to this capacity shortfall, the Aerospace Systems Research Group (ASReG) of the University of KwaZulu-Natal is incrementally establishing the technology building blocks that are required to develop an indigenous orbital launch vehicle. In this regard, the first major challenge to overcome is the development of the propulsion systems required by such a vehicle. The SAFFIRE liquid rocket engine development programme was initiated by ASReG as a means of doing so [475,476].

In the majority of traditional liquid rocket engines, the propellants are pressurized by turbopump systems prior to being admitted to the combustion chamber. Standard engine system cycles include the gas generator, staged combustion, and expander cycles. Turbines are used in all these systems to drive the pumps mechanically through a shaft. What differentiates these systems is how these turbines are powered.

With the recent advancements in battery technology, the SAFFIRE programme intends to use an alternative approach. Rather than relying on a turbine, high-speed electric motors will be used to drive the propellant pumps. Figure 4-1 shows the SAFFIRE engine configuration with electric motor driven propellant pumps (left) and the full stack of 9 engines (right) on a concept launch vehicle [475]. The electrically-driven pumps have been referred to as ‘electropumps’ which, under certain circumstances, have many advantages over turbopumps. In this case, there is no need for the turbine, gas generator/pre-combustor, drive shaft, and any gearboxes which all add significant mass to the rocket engine. However, these electropumps require significantly powerful battery systems which are heavy, especially when larger scale engines are considered. Simplified system control, start-up sequences and engine throttling are key advantages of electropump-based engines. The Rutherford engine is the most prominent example of a electropump-driven liquid rocket engine that is actively used to launch payloads into orbit. This engine is used to propel Rocket Lab’s Electron launch vehicle [4].

AM, while still a relatively new manufacturing technology, is being used extensively in spacecraft applications and almost all in-development liquid rocket engine systems use this technology to some degree. The most prominent examples of this are the SpaceX Raptor, the Rocket Lab Rutherford and the Launcher E-2 rocket engines. Each of these engines feature substantial use of AM; for example, SpaceX’s Raptor engine contains as much as 40% by-mass AM fabricated components (Section 2.3.4). Rocket Lab’s Rutherford engine (Figure 4-2) in particular uses AM for the majority of its components too, including the combustion chamber, nozzle, valves, pipes and propellant pumps.

Propellant pumps, which typically consist of a centrifugal-type impeller and a corresponding volute, are high complexity components which typically require advanced manufacturing methods to achieve the geometrical quality needed for performance. The first-iteration SAFFIRE engine, currently under development, features two propellant pumps; one liquid oxygen (LOX) pump and one kerosene pump. Both pump systems have different performance characteristics in terms of hydrodynamic power, rotational speed of the impeller and mass flow rates, and therefore, each pump system requires an individual design. Both the LOX and kerosene pump systems comprise a pair of impellers and volutes which have vastly different designs due to the different working fluids. Initial hydrodynamic analyses and evaluation of conceptual LOX and kerosene pump systems were previously undertaken as discussed in [477,478]. Following the initial conceptualizations, small-scale electropump systems were designed, manufactured and tested in two projects concurrently as described in this chapter, and in [479]. The focus of this chapter is to describe the AM processes used, design for additive manufacturing (DfAM) techniques and MicroCT-based inspection techniques primarily for quality control of final components. The focus of the work described in [479] was hydrodynamic design, manufacturing and testing of these small-scale electropump components to determine performance characteristics.



Figure 4-1: SAFFIRE liquid rocket engine concept (left) and 9-engine configuration for a concept launch vehicle (right). [476]

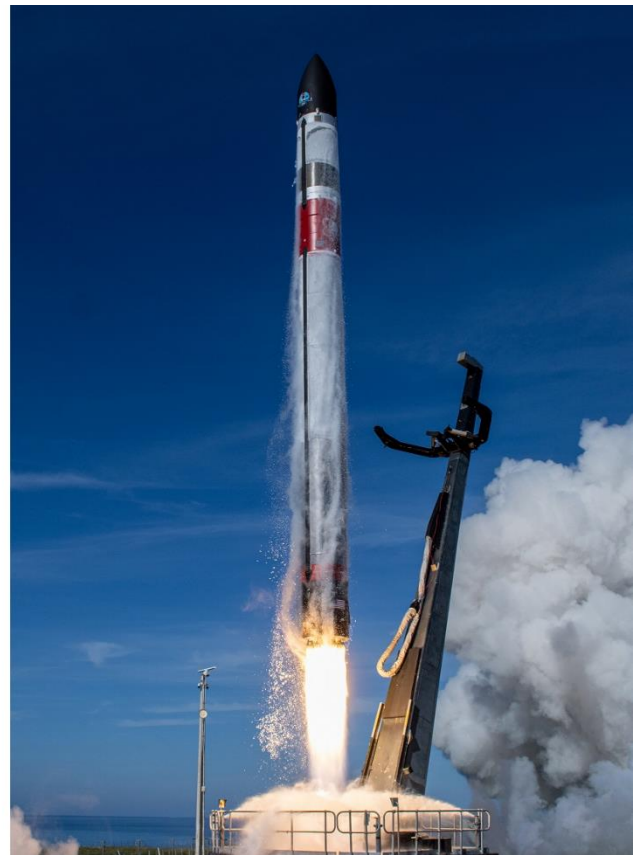


Figure 4-2: Rocket Lab Rutherford liquid rocket engine (left) and the Electron launch vehicle (right). [480,481]

Volute and impeller components, which are the primary components of a pump system, are traditionally manufactured using methods such as investment casting and 5-axis CNC machining. Centrifugal pump impellers

are relatively small and highly geometrically complex components. The design of pump impellers poses a variety of manufacturing challenges in the pumps industry [482]. Additionally, there is a size limit below which impellers are no longer manufacturable with any conventional method, mostly since investment casting is not suitable for 3D geometries at very small scales.

Volutes are the fixed elements of centrifugal pumps and are used to direct the fluid output, while concurrently decreasing the speed and increasing the static pressure of the fluid. While volutes can be manufactured by 5-axis machining methods, casting methods are generally used due to the relative costs associated with mass manufacturing. However, these costs move in favour of 5-axis machining methods at lower production volumes and under prototyping due to the large costs associated with casting tooling.

AM techniques provide benefits for manufacturing impeller and volute geometries in small-scale production runs, and for rapid prototyping. This is primarily due to the low costs associated with tooling requirements when using AM methods. The manufacturing lead times for AM enables fast iterations of testing components, and batch manufacturing of functional parts. Furthermore, the geometric complexity permitted by AM techniques allows for more complex, higher-performance designs to be realised.

The minimum size of pump components manufactured via traditional processes (casting and 5-axis CNC machining) is constrained by the difficulties associated with producing complex shapes at small scales, including the requirement of tooling. Conversely, AM can produce very complex geometries at small-scales and AM-built small-scale components often have better quality compared to casting methods and non-machined surfaces. This is limited, however, by the minimum achievable resolution of the AM process and material combination, which is usually around 0.3 mm for L-PBF systems.

Material selection for pump components plays an important role in the performance that can be achieved. Components of LOX pumps in particular have many associated risks, such as cryogenic temperature failures and LOX ignition events with some materials. LOX systems are subject to extremely low temperatures and proper material selection is essential as the correct material for the application can reduce the probability of system failure.

The mechanical properties of metals at cryogenic temperatures varies greatly. Certain materials weaken upon cryogenic cooling, while others can strengthen significantly. Marked changes in toughness can also occur, with appreciable embrittlement being more common than not. In addition, pump components exposed to LOX must be manufactured from materials possessing a degree of oxygen compatibility sufficient for exposure to high-pressure environments where part rubbing is a possibility.

Some materials are much better-suited than others in respect of the above requirements. From the findings of [477,479], Inconel 718 was selected as the optimal LOX pump system material and is considered for the flight SAFIRE engine configuration. Kerosene pump components are not subject to the same level of material compatibility constraints compared to those of LOX pumps, and a vast range of materials are available for use. Ti-6Al-4V was selected as the optimal material choice for the kerosene pump system due to its excellent specific strength characteristics.

For the purposes of this chapter, small-scale pump components were manufactured using the local AM supply chain in South Africa. Stainless Steel 316L was chosen for the purposes of the first two iterations of these small-scale pumps, and Ti-6Al-4V was assessed in the third iteration. These small-scale pumps are all designed to be tested on a hydrodynamically-similar, water-based pump test rig to compare achievable performance with theoretical performance.

4.3. Methodology

For the purposes of testing the quality of the AM process and materials employed, as well as the hydrodynamics of the pump components themselves, an iterative design, manufacturing and testing programme was created.

The first step of the programme entailed the manufacture of a small-scale impeller and two volutes using AM techniques and the evaluation of these parts and production processes using MicroCT techniques. The designs of these first iteration parts were not derived on the basis of specific DfAM techniques. MicroCT techniques were employed to detect defects specific to the complex geometries in question, and to evaluate associated design choices.

The second iteration component set was then designed specifically using DfAM techniques to maximise geometric accuracy, and to mitigate the first iteration flaws identified through MicroCT. The second iteration components were manufactured with some slight geometric changes, but the scale and purpose of the components, compared to the first iteration, were close to identical. MicroCT techniques were again used to evaluate the quality of the components and to verify the defect mitigation strategies applied. AM build simulation strategies were also applied to predict potential flaws in the AM process with respect to the specified geometries. The results of these simulations were compared to the outcomes of qualitative inspections and the MicroCT analyses, in order to inform designers of the potential advantages of using AM build simulation to mitigate defects in AM processes.

A third iteration component set was then designed and manufactured using the lessons learnt from previous iterations. In this iteration, a similarly designed, but larger scale propellant pump component set was considered, which employed further DfAM techniques for defect mitigation. The third-iteration components were designed for use in a full-scale water-analogue pump having vastly higher-performance than the first and second iteration pump designs. Build simulation techniques are again applied to detect possible defects in the DfAM process, and MicroCT techniques were used to evaluate the defects present in the components.

While standard methodologies do exist for MicroCT analyses of AM built components (Section 3.6), these methods typically require standardized geometry samples. For more complex components, such as impellers and volutes, dedicated methodological workflows needed to be adapted for each specific component. This is especially the case for components of different sizes and material compositions.

4.4. First Iteration

The first iteration component set incorporated two different volute designs, and a single impeller design. All of these components were designed as described in [479], and were manufactured by means of AM techniques on

an EOS M280 platform, and using the Stainless Steel 316L material and standard (recommended) processing parameters.

4.4.1. Volute Design

Typical volute structures consist of evolving cross-sections in the fluid path. Many volute cross-section types exist, each having particular advantages and disadvantages. For the purposes of the SAFFIRE propellant pumps, two primary volute cross-section types were investigated: namely, the circular and double circular cross-sections. Both feature an equal cross-sectional area at each angle of volute fluid path progression. Figure 4-3 shows a sectioned view of both cross-section types. Besides the differences in cross-sectional profile, these two volute designs are identical.

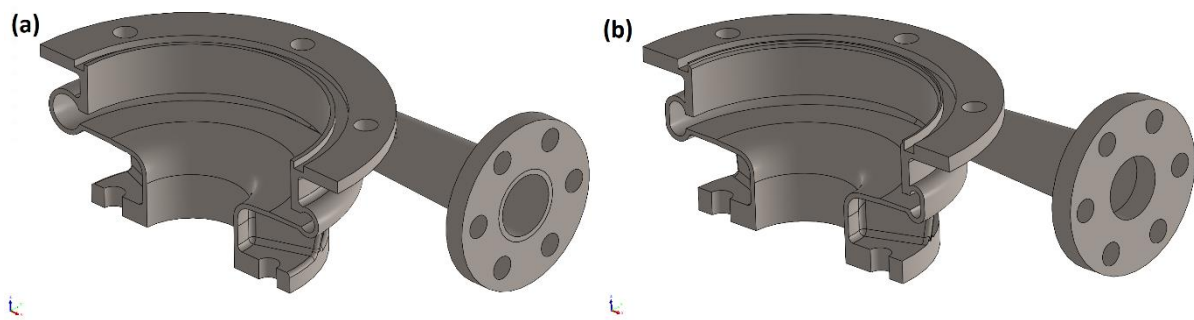


Figure 4-3: SAFFIRE first iteration volute CAD cropped cross-section. (a) circular volute design and (b) double circular volute design.

To effectively manufacture components using AM techniques, it is important to use AM build preparation tools to mitigate potential errors in the build process. The software package used for all the build preparation analyses in this section is Autodesk Netfabb 2023.0 [483]. Before sending AM designs to the AM manufacturer, it is important to follow some build preparation steps. The first step is to repair CAD geometry defects for the purposes of layer-wise model preparation. Automated repair algorithms are used to ensure CAD models are closed, contain no significantly small geometric sliver elements, and repair many other features. After repairing the CAD models, the components must be oriented relative to the build plate. In this case, as shown in Figure 4-4, the volutes were oriented parallel to the build plate with reference to the flat, circular faces of the volutes.

Following orientation, the support structure generation process can begin. The first step to support generation is to perform an “upskin/downskin analysis”. This analysis displays both qualitative data (Figure 4-4 red areas) and quantitative data on the characteristics of the areas with inclination angles below the specified setting. For the purposes of these volute designs, a minimum inclination angle of 45° was specified, which is the standard recommended angle for most metal AM processes and material service providers (Section 3.4).

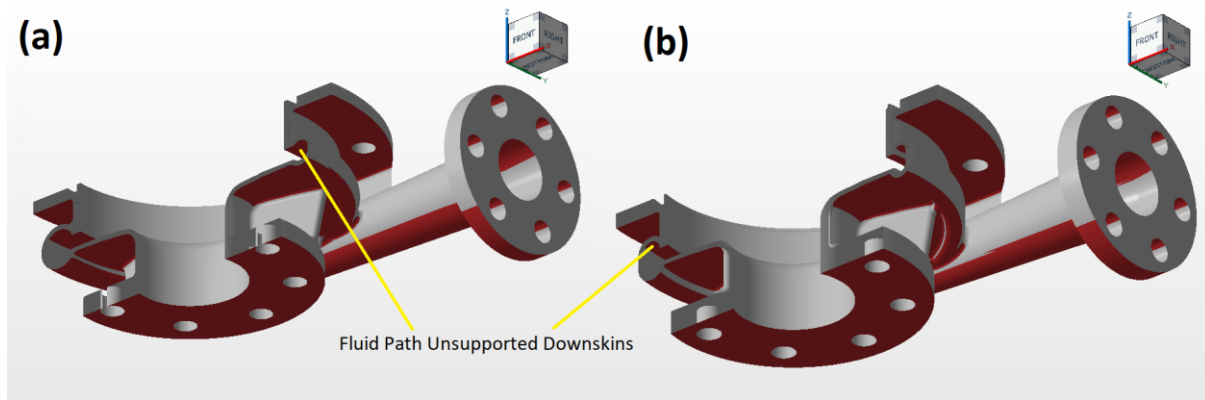


Figure 4-4: Upskin/Downskin analysis for AM build preparation. (a) - Circular volute cross-section downskin analysis and (b) - Double circular volute downskin analysis. Red areas depicted are areas with inclination angles below 45° to the build plate. Z-axis build direction.

For these downskin areas, an AM standard support structure was generated using a simple algorithm which generates support structures on all surfaces which have inclination angles of lower than 45° to the build plate. It is important to note that while automated support structure algorithms are effective for simpler components, some components may require custom support generation scripts or manual editing for desired results. This is typically done to accommodate complex features such as internal fluid paths and lattice structures.

Since the cross-sections of the volutes' internal fluid paths are difficult or impossible to reach with machine tooling during post-processing, both designs featured no support structures on the internal fluid path regions. Figure 4-5 shows the intended build direction, support structures and lack of fluid path support structures for the circular volute design. The support structure method applied to the double circular volute design was identical. Following the AM build preparation steps outlined above, the volute designs were both processed through layer-wise decomposition to prepare the geometry for the AM machine.

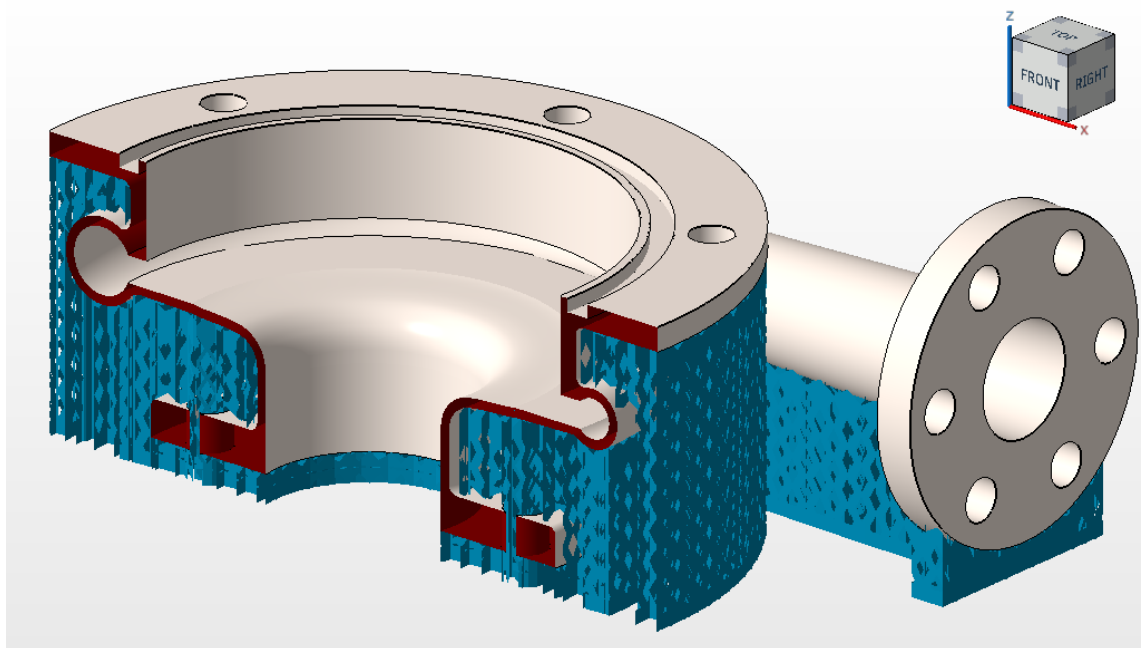


Figure 4-5: Circular volute design cross-section displaying support structures (blue) and lack of support structures on internal fluid paths. Z-axis build direction.

4.4.2. Impeller Design

AM can produce the complex geometry present on impeller components, which typically contain complex features, particularly on the blade structures. These blade structures are defined by meridional flow path shapes which are highly complex and difficult to machine. The impeller considered in this study was designed to be additively manufactured with minimal post-processing requirements, and with blades that require no machining. This was achieved by avoiding direct support of the blade surfaces.

The AM build preparation technique used for the impeller followed a similar process to that described in Section 4.4.1. The part geometry was imported into Autodesk Netfabb 2023.0 and repaired, following which “upskin/downskin analysis” techniques were applied. For the impeller design, the upskin/downskin analyses indicated that no regions of the blades required support structures, based on the recommended build inclination angle threshold of 45°. Figure 4-6 shows the CAD model of the impeller in the as-built state (a) and the post machined, final part state (b).

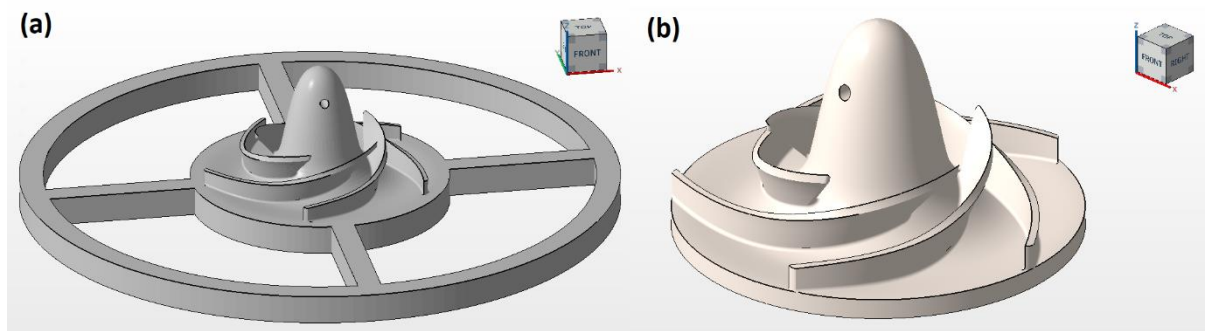


Figure 4-6: SAFFIRE first iteration impeller test design showing: (a) AM as-built CAD and (b) final part geometry. Z-axis build direction.

The impeller design has both extended surfaces on the faces closest to the build plate, and an exterior ring to be used as a grip for machining. The part was manufactured directly off of the build plate. Following the repair, analysis and orientation of the impeller, the design was exported as a layer-wise decomposition and sent to the AM machine for manufacturing. Following the AM build process, the impeller design was machined to the final tolerances as described in [479].

4.4.3. MicroCT Image Quality Evaluation

Following post-process machining of the impeller and volute components (Figure 4-7), evaluation of the components through MicroCT methods was performed. Before scanning and investigating the components, the MicroCT scanning parameters needed to be optimized to minimise the noise and number of artifacts visible in the scans. This was done by both qualitative inspection and quantitatively using the methods set out in [484]. MicroCT was performed with a typical laboratory system as described in [458] and Section 3.5.

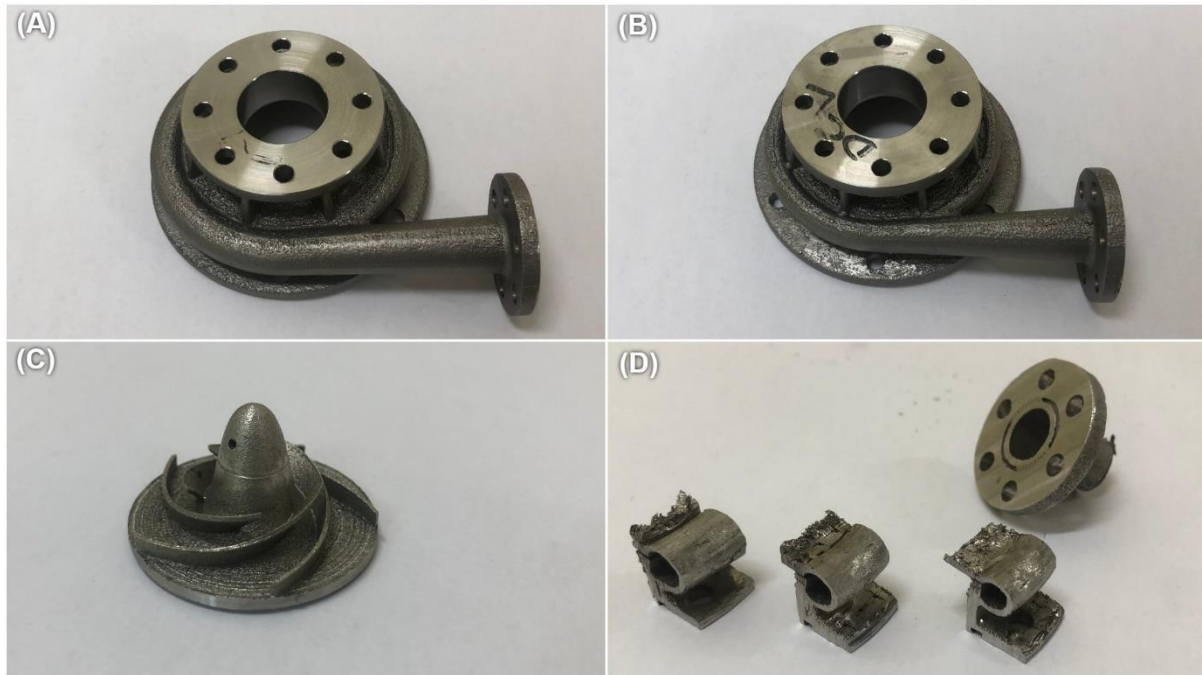


Figure 4-7: First iteration AM-built components after post-process machining: (a) Circular volute, (b) Double circular volute, (c) impeller, (d) cut-off sections of circular volute.

Since the resolution achievable through MicroCT methods varies depending on the bounding box size of the component being evaluated, a second circular volute was manufactured and cut into smaller sections (Figure 4-7 (D)). This was done to further investigate the internal defects and build adherence to design, using higher resolution scans (small cut-offs produce higher resolution as well as less noise, making quantitative evaluations possible). A total of four cut-offs were scanned in addition to each complete part, for the purposes of this image quality process. The bounding box dimensions of each volute were 95x85x30 mm, while those of the impeller were 40x40x25 mm and those of each cut-offs were 10x10x10 mm.

The volutes were the largest components in question and due to their similarities, image quality analysis methods were applied to only the circular volute design. This was done by scanning the part with various MicroCT parameters, measuring the image quality obtained from each and optimizing the parameters based on the results. The image quality is measured by evaluating the “grey values” present on MicroCT slice images. These grey values represent the lack of X-ray absorption on the detector of a MicroCT system, where higher grey values represents higher density material in relation to lower density materials in the vicinity. Image quality is an effective measurement of the grey value averages present in the solid material in comparison to the surrounding background (air), where higher quality represents more accurately defined surfaces of the solid material.

The circular volute was scanned four times using the CT parameters shown in Table 4-1. For each scan, 2400 images were used and various averaging and skipping parameters were varied. Skipping and averaging refer to the way in which subsequent images are combined (averaging) and the first image is discarded (skipping). For example, for Scan 4, two subsequent images were averaged at each step position and the first image was discarded. The MicroCT optimization process scans were then imported into Volume Graphics VGSTUDIO MAX 3.5 where these data was analysed further. Careful alignment of these four scans allowed for their direct comparison, as shown in Figure 4-8, with identical cross-sections selected.

Table 4-1: MicroCT image quality calibration parameters.

Scan Number	Voltage (kV)	Current (μ A)	Voxel Resolution (μ m)	Beam Filter Size (mm)	Averaging	Skipping
Scan 1	200	140	70	0.5	1	0
Scan 2	200	200	70	1	1	0
Scan 3	200	200	70	1	1	1
Scan 4	200	200	70	1	2	1

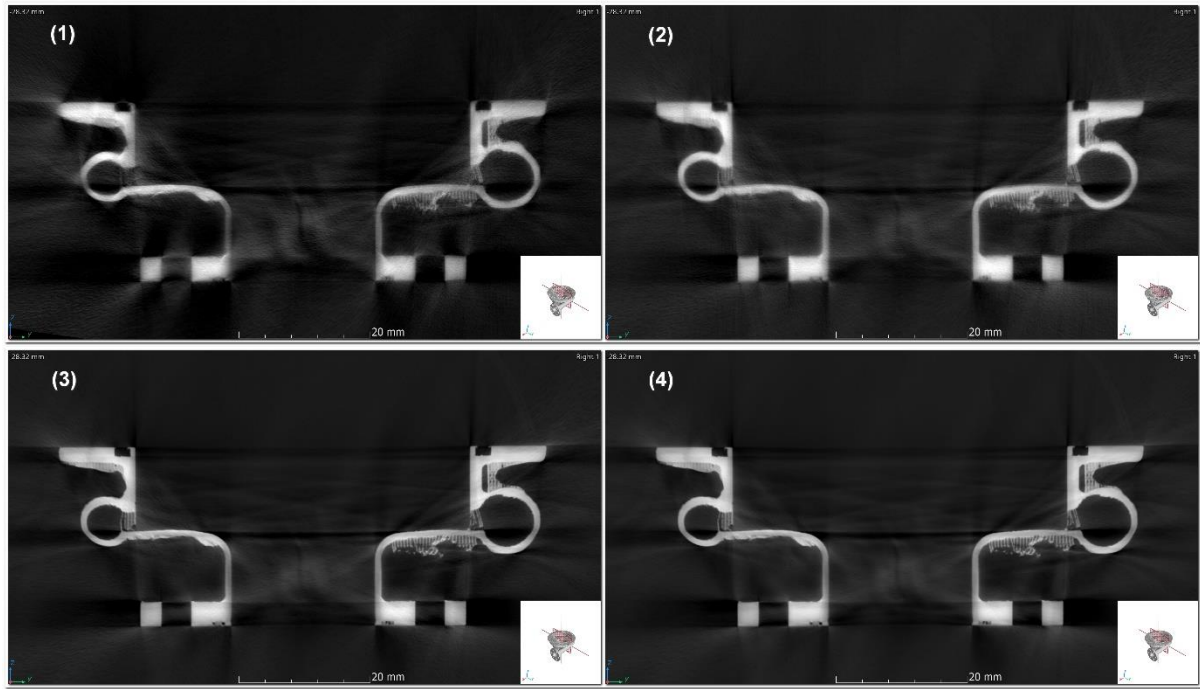


Figure 4-8: Identical MicroCT cross-section slice views of parameter optimization scans with scan numbers indicated for each view.

Upon qualitative inspection, scan 4 showed the highest image quality, with the least deviation of grey values present in the background and within the component. This was confirmed quantitatively using the methods described in [484]. To accurately determine background and internal grey values, background and material regions were separated using a surface determination, and regions were eroded and dilated by two voxels on either side of the surface in question, similar to the methods described in [484] and Section 3.6. Table 4-2 shows the mean grey values, standard deviation and image quality factor for the background and internal grey values of each scan. As indicated, all scans showed reasonable image quality without major differences.

Table 4-2: Quantitative quality assessment of calibration scans as per the methods described in [484].

	Scan 1		Scan 2		Scan 3		Scan 4	
	Background	Component	Background	Component	Background	Component	Background	Component
Mean Gray Value (μ)	15760.98	43428.59	14031.68	46546.14	14511.37	45046.31	14143.15	46026.15
Standard Deviation (σ)	1549.98	6527.69	1118.42	6173.66	911.36	6013.99	954.49	6046.28
Quality (Q)	4.12		5.18		5.02		5.21	

These quality metrics then informed the correct CT parameter selections needed for each component further, however, due to variations in component scale, slight changes were made for each component to obtain higher resolution MicroCT scan data.

Scanning smaller scale versions of components also provides insight into the AM process quality since higher resolutions can be obtained through MicroCT. This approach can be used to compare the relevant sections of components to verify the quality of the large-scale scans. On this basis, a circular volute was scanned using the parameters above and then cut into small sections for higher quality scans (Figure 4-7 (D)). These sections were scanned at higher resolutions, and overlaid onto the original component scans to check for differences in quality, and for possible defects which are not detectable at lower resolutions and on lower image quality scans.

When comparing the quality of the cut-offs to the whole circular volute component, no major differences in the outer surface determination quality were found, assuming that the manual surface determination techniques used for the whole component were sufficient. However, internal defects such as porosity and cracks appeared much more clearly on the small scale cut-offs – this is especially the case for very small porosity defects.

4.4.4. MicroCT Process and Defects

The MicroCT parameters used for each component here were based on the findings in Section 4.4.3. Each scan data set was then further processed to remove any remaining noise and artifacts from these CT data. This was done using various techniques such as filtering, erosion and dilation and manual infill using grey value ranges. The specific methods used for each component surface determination are not described here in detail due to the complexity of the process. However, a basic outline of the process is as follows:

- Import scan data into Volume Graphics VGSTUDIO MAX 3.5
- Register dataset depending on desired orientation
- Apply filtering with manual inputs based on best quality outcome
- Manually determine surfaces based on scan data set background grey values and internal grey values
- Create regions of interest from surface
- Manually erode and dilate regions based on best quality outcome
- Use manual region growing and infill techniques to optimize surface selection

Another method for acquiring an accurate surface determination is to import the component CAD geometry into the image analysis software and apply the software's CAD/Mesh function to identify associated surfaces. This method was used where better surface determination results were required. Its application presented some challenges, however, due to partially incorrect registration alignment.

Due to the scan quality associated with the components examined, quantitative defect analyses for porosity or surface roughness were mostly inconclusive and were therefore omitted from this section. The primary method for inspection of the quality of these components was through analysis of the surface deviation from the actual component represented by the imported CAD geometry. This comparative approach enabled both deviation analyses and wall thickness analyses, and is described in the section to follow.

4.4.5. Volute Quality and Defect Evaluation

After sufficient MicroCT quality was achieved and accurate surface determinations were obtained, qualitative inspection was performed on the volute components. These components had a wide variety of defects observed mostly in the form of high deviations from the nominal surface of the CAD body, and thinner-than-nominal wall thicknesses in particular locations. The defects present in both volutes reflect the similarity of their respective geometries.

Figure 4-9 shows the thin-wall defects present in both volutes in similar fluid path locations, where support structures were connected to the downskin (downward facing) surfaces. For both components, the wall thickness of the fluid path regions is constant according to the nominal CAD geometry, but varied significantly in these MicroCT scanned data. Insufficient support structures can often be the cause of deviation and defects on surfaces of AM built components and is the suspected cause of these defects. After manufacturing through AM techniques, incorrect support removal techniques can also result in defects such as this, due to excessive material removal by abrasive grinding tools.

While the detected defects have a significant bearing on geometric accuracy, both volute components were heavily over-designed from a structural perspective, and consequently, they would not be expected to cause operational failures. However, for components under higher stress loads in similar regions, these defects would need to be mitigated by either increasing the wall thickness in defect-affected regions, or by using DfAM techniques to prevent defects occurring at all.

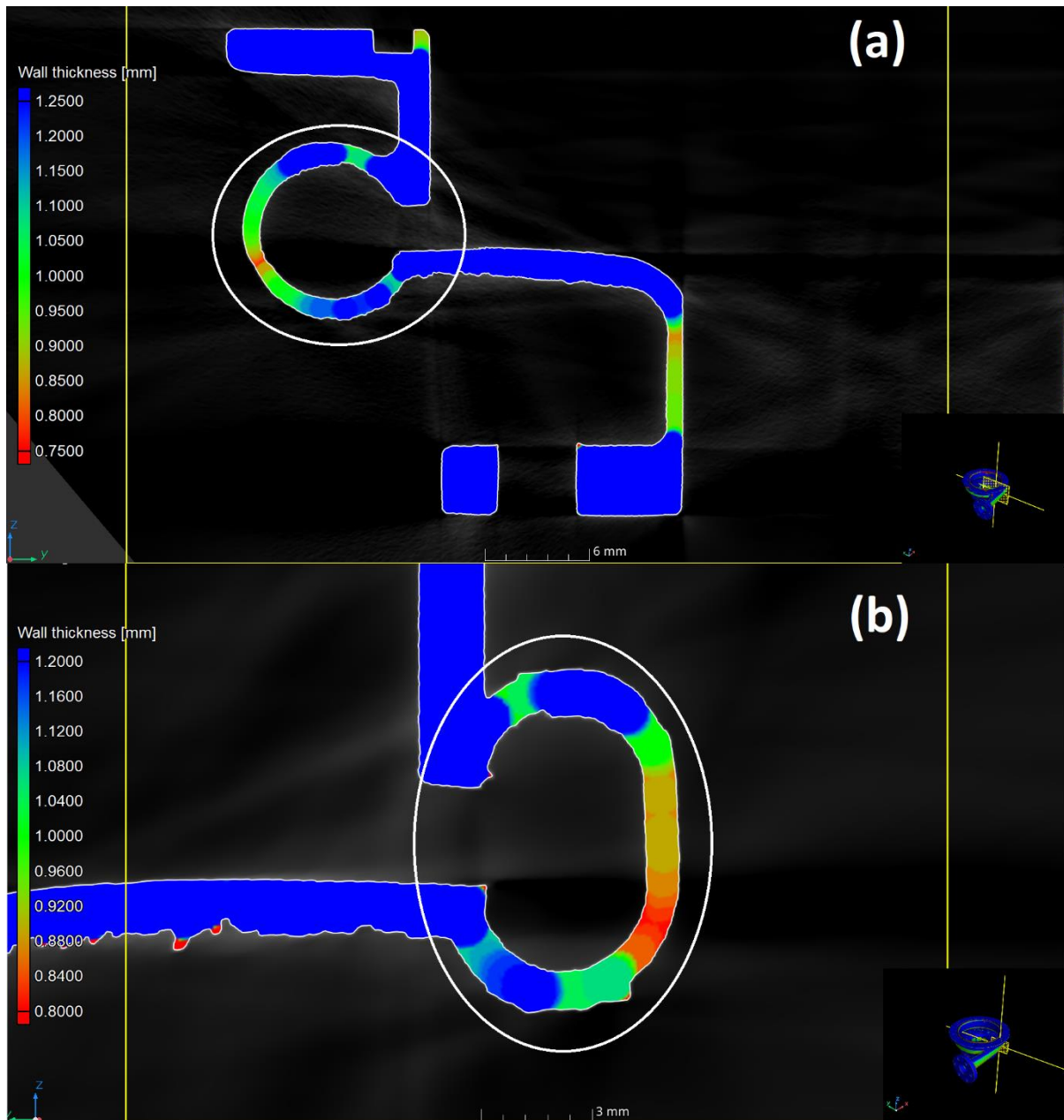


Figure 4-9: Volute thickness thinning defects for (a) circular volute and (b) double circular volute. z-axis build direction.

During post-process machining of AM-built components, careful consideration needs to be applied by both the design engineer and the machinists to ensure defects are not induced during this phase of manufacture. Common defects of this nature include swarf generation, inaccurate machined surfaces and unintended material removal in complex internal features. Figure 4-10 shows the cross-section of (a) the circular volute, and (b) the double circular volute, with a “nominal/actual deviation analysis” qualitative result for all surfaces. Swarf generation defects are observed in both components as highlighted. This swarf generation is suspected to have been caused by the machining procedure used to remove the support structures.

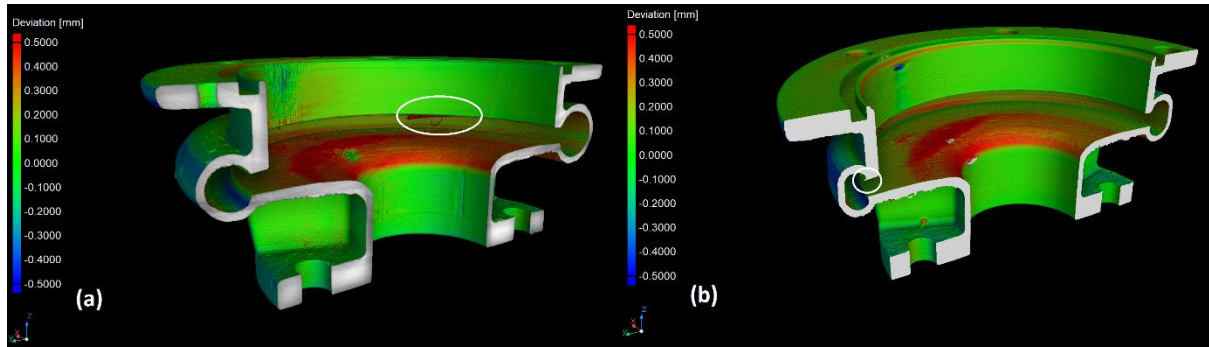


Figure 4-10: Volute cross-sectioned deviation qualitative result for (a) the circular volute, and (b) the double circular volute. z-axis build direction. Swarf defects highlighted.

For both volutes, defects were also found on the cutwater sections, which are critical geometric features of centrifugal pumps. The cutwater has the purpose of guiding the working fluid leaving the impeller into the discharge diffuser, as well as reducing recirculation back into the volute. The placement of the cutwater, and its length, thickness, leading edge profile and flow angle all significantly influence volute casing efficiency, and hence overall pump performance. The cutwater surfaces for both volute casings, as shown in Figure 4-11, deviate highly from the intended design, and swarf elements are also observed. The suspected cause of this defect is again improper post-process machining of the gap, resulting in machine tools contacting the cutwater. This defect alone would drastically reduce the performance of the pump as a whole, and its mitigation in further design iterations was pursued.

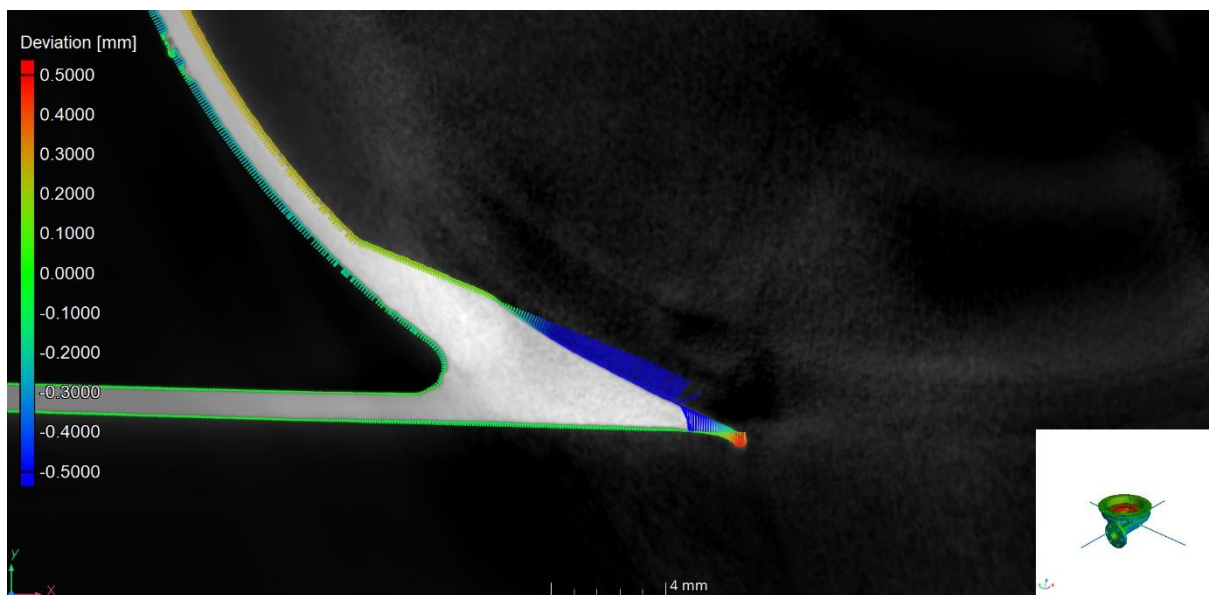


Figure 4-11: Circular volute MicroCT slice view and nominal/actual deviation analysis showing cutwater defect.

The relative roundness of the volute fluid path cross-sections was also investigated through MicroCT techniques. Relative roundness is an important characteristic for these volute designs. Deviations in roundness when compared to the nominal CAD design has various effects on pump performance. The primary factors which influence pump performance are the surface roughness of the fluid path region and deviations in total cross-section area. A poor surface roughness in the fluid path results in increased flow drag, while deviations in total cross-section areas can result in inefficient volute performance. Both of these characteristics are important to mitigate in volute design.

Relative roundness measures the specific tolerance of a surface when compared to a circular geometric element in the relative vicinity. In this study, relative roundness measurements were performed via a surface best fit analysis, where a circular geometric element is fitted to the circular surface profile in question. The relative roundness is then calculated by comparing the maximum and minimum diameter circles which touch the actual surface, resulting in a circular deviation tolerance. This analysis acts as a local surface roughness measurement and provides quantitative results for cross-sections under investigation. Figure 4-12 shows the relative roundness measurements for both volutes at specific cross-sections.

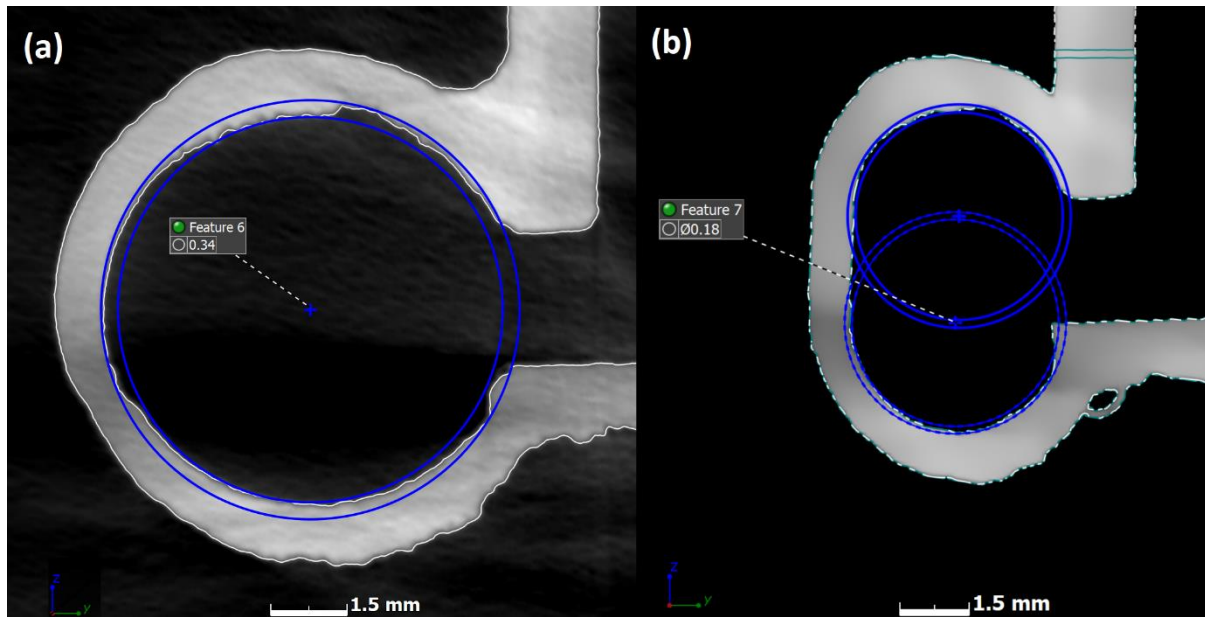


Figure 4-12: Relative roundness analysis results on selected cross-sections for (a) the circular volute and (b) the double circular volute. Z-axis build direction.

For each volute design, the roundness tolerances were investigated via both qualitative analysis and quantitative analysis. The specific roundness characteristics observed show that, for both samples, the uppermost (highest positive z-direction) surfaces at all cross-sections deviate highly and have a particularly rough surface. This is expected due to the inclination angles present on these surfaces where the surfaces transition from a vertical inclination to a horizontal inclination angle over the circular profiles. The surface roughness was observed to be dependant on the size of the cross-section examined, where larger cross-sections show higher surface roughness.

In addition, significant deviation was found in the areas computed at each cross-section when compared to those derived from the nominal CAD models. As-built areas were found to be consistently smaller than the associated nominal areas. The suspected cause of this deviation is the drooping effect which manifested in low inclination angle regions. Comparing the two volute designs, the double circular volute was found to present a higher level of design conformance in respect of surface roughness and cross-section areas throughout the fluid paths. This is to be expected, since the regions in which low inclination angles are present in the double circular volute are less extensive than those in the circular volute.

Furthermore, the defects described above were found to be distributed within the components consistently: Wall thickness thinning was consistently exhibited throughout the volute paths, numerous swarf generation defects appeared throughout both volute casings, both cutwaters were shown to deviate highly throughout their structures,

and relative roundness evaluations indicated both poor surface finish and significant cross-section area deviations. Other, less critical defects were also identified, but due to their relative lack of importance, they are not discussed here. On this basis, it is clear that in order to ensure the achievement of nominal pump performance and to prevent component failure during operation, DfAM techniques and defect mitigation strategies must be implemented by designers.

4.4.6. Impeller Quality and Defect Evaluation

Once the MicroCT scans of the impeller were performed, these data was processed through visual analysis methods. The impeller component, while presenting complex geometry elements, was found to have no major defects throughout its structure. The overall accuracy of the manufacturing process was found to be of high quality. Figure 4-13 shows the surface deviation for all the critical surfaces of the component. These surface deviation results were obtained through a “nominal/actual comparison” analysis, using the nominal CAD geometry as a reference. Certain observations were made during this process, including the presence of general surface roughness on complex surfaces, in addition to the presence of warpage. Determination of surface roughness relies on the mean surface quality in the vicinity of a region under analysis. While the general surface roughness was clearly visible, quantitative data analysis could not be performed due to the slight warpage characteristics present.

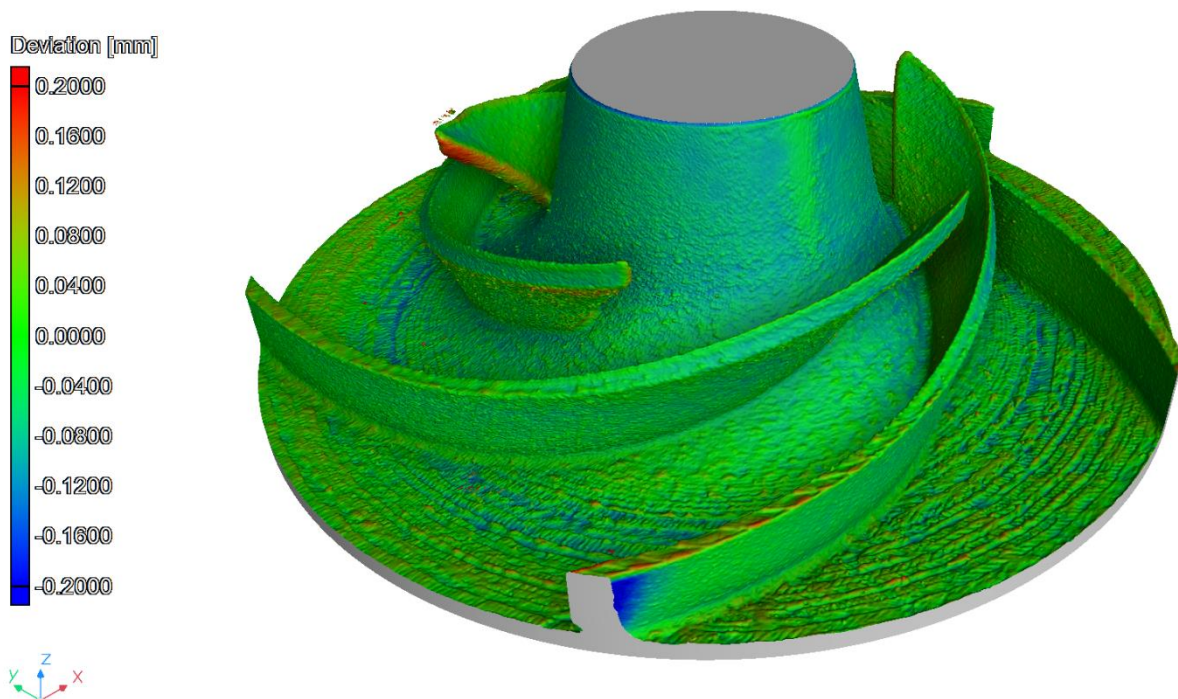


Figure 4-13: First iteration impeller MicroCT surface deviation analysis. z-axis build direction.

Qualitative inspection also revealed the warpage characteristics of the impeller, with slight warpage being observed throughout the structure. This warpage occurred at higher magnitude on the impeller blades, with the blade tips displaying the greatest warpage. Figure 4-14 shows the deviation of the blades at the blade base (lowermost z measurement) as well as at the blade tips (uppermost z measurement). It is clear from these inspections that the blades deviate at an increasing magnitude based on both the distance to the base structure, and the inclination angle relative to the build plate.

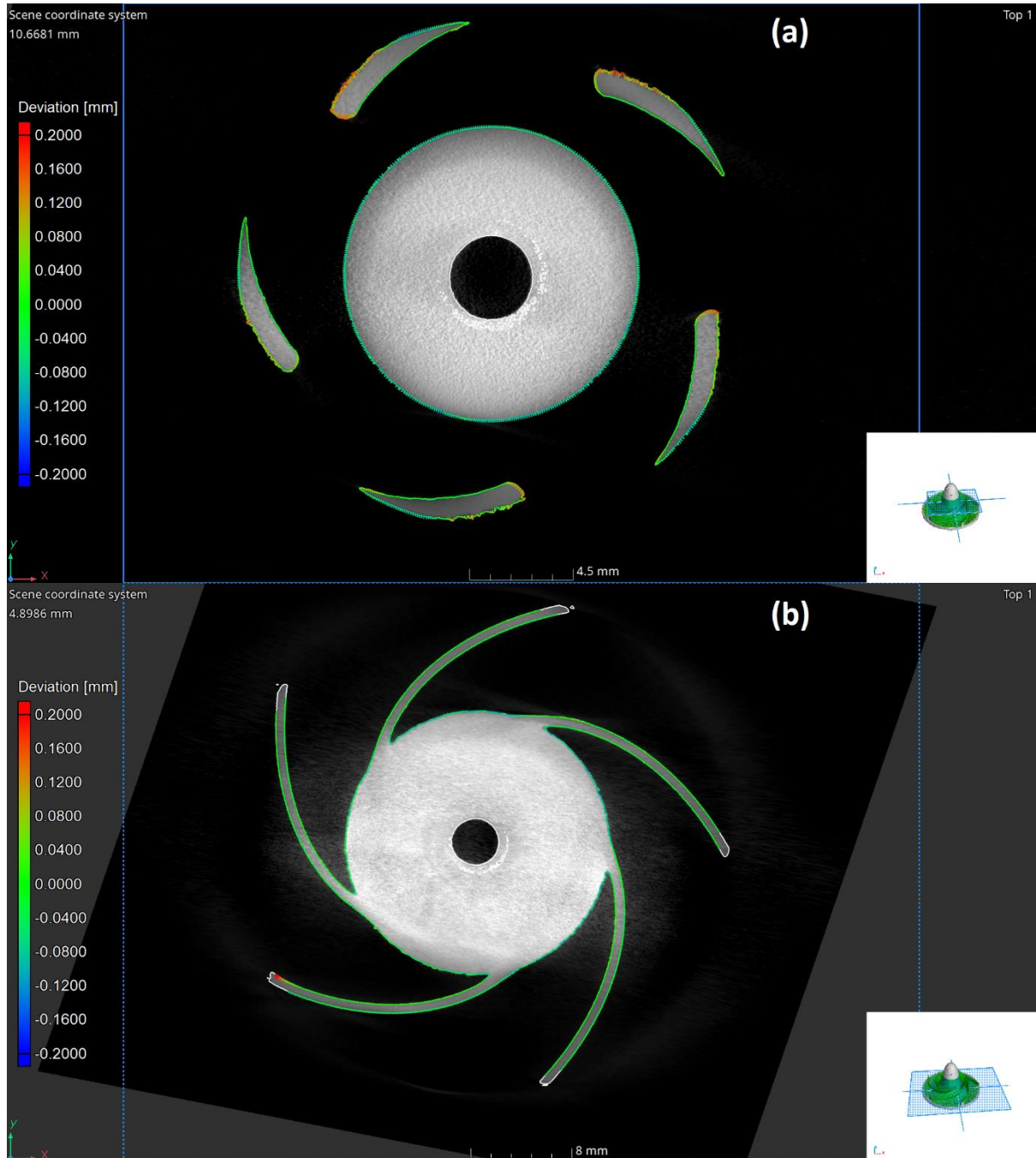


Figure 4-14: MicroCT slice views with deviation analyses on first iteration impeller at (a) the blade tip cross-section and (b) the blade base cross-section. Z-axis build direction.

The overall quality of the impeller component was found to be high with no critical defects observed. Warpage and surface roughness characteristics were identified, the nature of which was noted for further design purposes. The specific influence of blade surface roughness and warpage may be detrimental to the performance of the pump, and designers should account for these using DfAM or post-processing techniques, if necessary.

4.5. Second Iteration

Following the results and insights obtained using MicroCT techniques on the first iteration pump components, various design methodologies were adapted to mitigate the defects found. This was achieved through a variety of methods for the volute components. The volute types under investigation (circular and double circular) revealed differences in the cross-sections observed in the form of surface roughness and area deviation. While the double circular volute presented higher quality results in these respects, all other defects found were similar when comparing the volute designs. After reviewing hydrodynamic analyses, the circular volute design was determined to be the best volute type choice. From this iteration forward, only the circular volute was consequently investigated.

The overall quality of the first iteration impeller was good, and the design was not altered whatsoever in this iteration in the specific regions for analysis (blades and fluid-contact surfaces), although external machining interfaces were altered to improve the performance of the pump. An impeller was manufactured and analysed in this iteration. However, since the specific regions of interest were not changed, the results are not discussed in this section (although they are discussed further in Section 4.7.2).

4.5.1. Volute Design

Based on the defects identified in Section 4.4.5, various DfAM techniques were applied to mitigate the defects present in the volute and consequently improve the overall performance of the pump. The primary defects identified for mitigation were the poor surface quality of the volute fluid path, the area deviation of the volute fluid path, wall thickness thinning, cutwater defects and swarf generation defects. Other minor defects were found and mitigated in this design, but the associated methods are not detailed here due to the relative ease and importance of changes.

One of the primary defect mitigation techniques applied was to change the build orientation of the volute structure to improve the wall thickness consistency and the surface quality in the volute path, to reduce area deviation in the volute path, and to improve the cutwater geometry quality. This was achieved by orienting the volute at 45° to the build plate (relative to the circular flat faces). Figure 4-15 shows the relative sizes of downskin surfaces for both the standard 0° inclination setup used in the first iteration, and the 45° inclination used in this iteration.

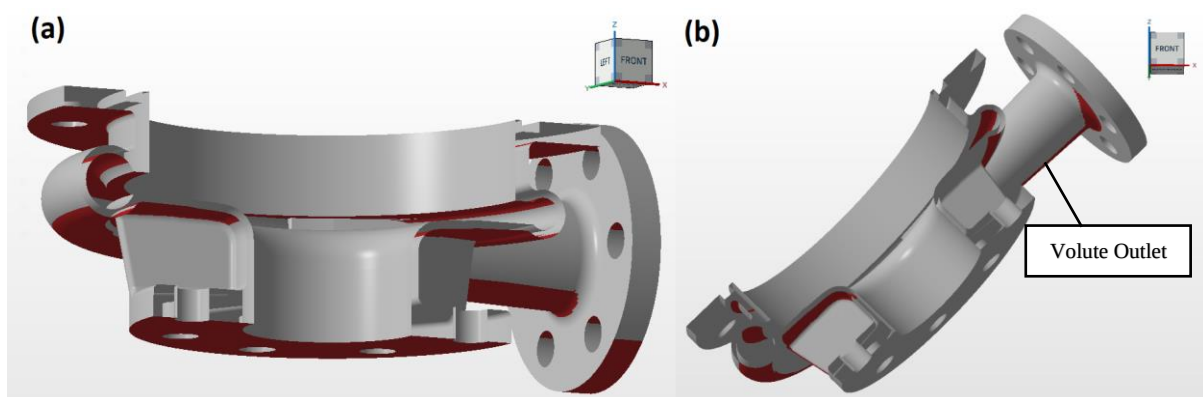


Figure 4-15: Circular volute downskin analysis sectioned views showing areas (red) with inclination angles under 45° for both the orientation angles of 0° (a) and 45° (b). z-axis build direction.

Under this orientation angle, the unsupported surface regions (shown in red) were largely reduced, resulting in a surface area of 19.263 cm² at 45° orientation compared to 82.611 cm² at 0° orientation. While this reduction is mostly associated with the external surfaces of the structure, the 45° orientation is the best-case orientation in terms of unsupported areas on the internal volute fluid path surfaces. Furthermore, under this orientation, the volute outlet contains no surfaces with inclination angles below 45° drastically improving the potential surface quality, with no need for post-process machining.

Although the unsupported surfaces were considerably reduced under this orientation angle, there still existed large areas with low inclination angles in the volute fluid path region. A mitigation strategy commonly used in fluid channels for AM built components is to apply a “teardrop” shape on the cross-sections, with the tip of the teardrop pointed upwards in the vertical build direction, which is the z-axis in the case shown in Figure 4-16. These teardrop shapes commonly limit the angle of surfaces on the cross-section to at least 45° with respect to the build direction.

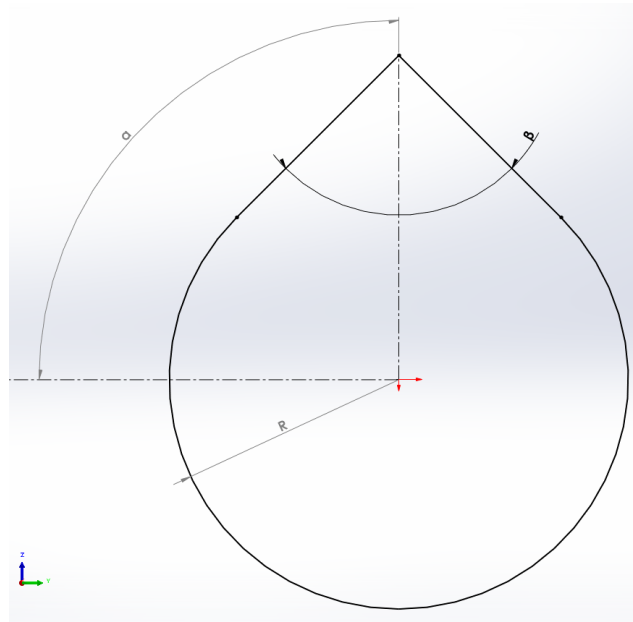


Figure 4-16: Teardrop fluid channel shape used in DfAM techniques for downskin inclination angle limiting. α , β and R geometric parameters shown.

For the 45° orientation angle, the specific plane angle for each volute cross-section, with respect to the build direction, varies and therefore the parameters α and β change over the volute rotation to maintain the 45° inclination angle for all surfaces. α represents the relative inclination angle of the teardrop profile with respect to the volute rotation axis frame, and β represents the teardrop surface angle required for a 45° inclination angle relative to the build plate. The functions which describe the parameters α and β , for each volute rotation angle (θ), are shown in Equation (4-1) and Equation (4-2) respectively. These functions are only applicable to a revolved circular profile, at a 45° orientation angle and with surfaces at 45° inclinations, both relative to the build plate.

$$\alpha = 45 \sin(\theta) + 90 \quad (4-1)$$

$$\beta = \begin{cases} 90 \cos(\theta + 90) + 180, & 0 < \theta < 180 \\ 90 \sin(\theta) + 180, & 180 < \theta < 360 \end{cases} \quad (4-2)$$

For these teardrop profiles, and for efficient fluid flow through the volute fluid path, the area of each cross-section must be equal to the equivalent circular profile area. Since the area of each cross-section per volute rotation angle is already known from previous design parameters, Equation (4-3) is used to calculate the radius (R) for any teardrop profile where A and β are known.

$$R = \sqrt{\frac{A}{\frac{3\pi}{4} + \frac{\pi(\beta - 90)}{360} + \frac{1}{\tan\left(\frac{\beta}{2}\right)}}} \quad (4-3)$$

Using these equations, a volute fluid path profile was generated which maintained a minimum 45° inclination angle for all surfaces relative to the build plate. Figure 4-17 shows the generated profile CAD model of the final volute structure. Also shown is the downskin surface analysis which displays the reduction in these surface areas in the volute fluid channel. Under this orientation angle, and with the teardrop DfAM methods applied, the resulting downskin analysis shows a further reduction in unsupported surface area, with the total area reduced from 19.263 cm^2 to 13.523 cm^2 .

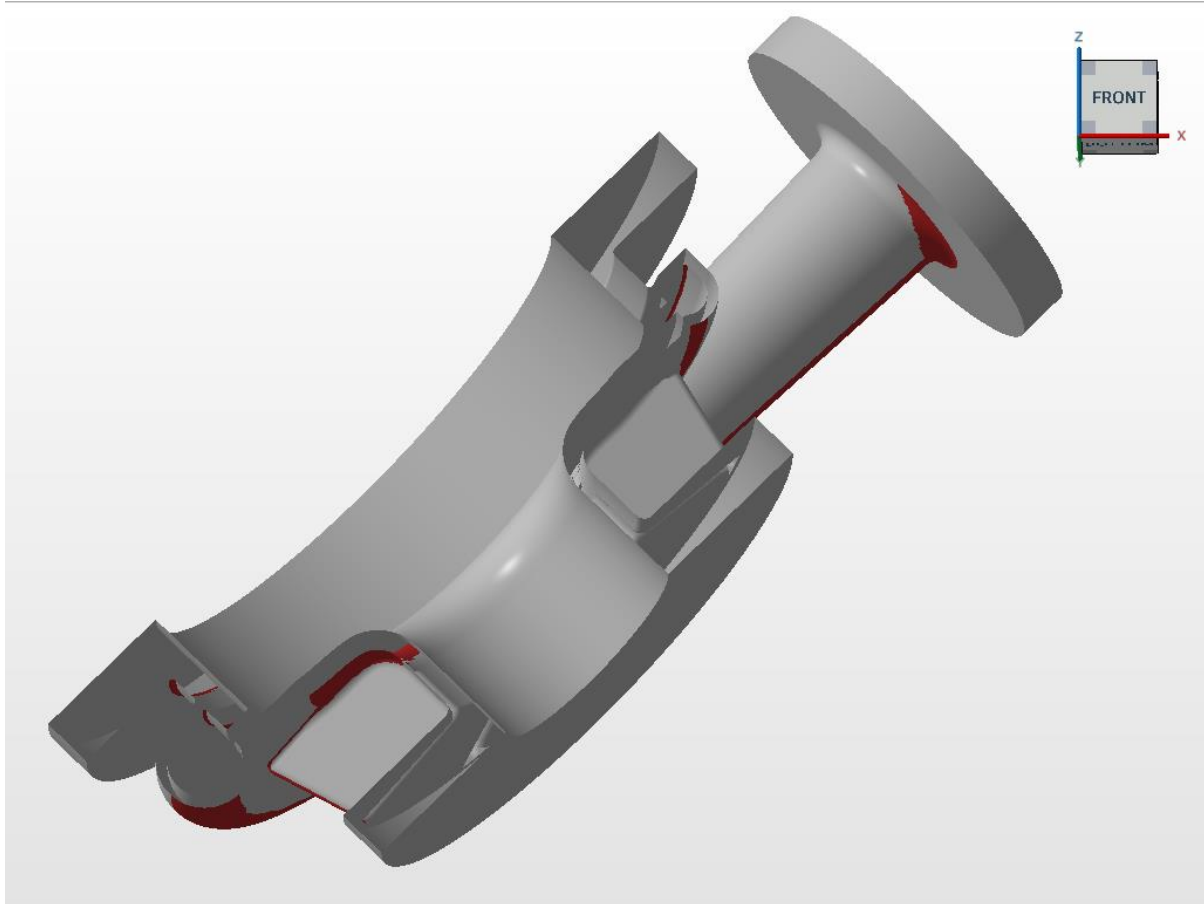


Figure 4-17: Teardrop DfAM edited volute structure cross-section showing downskin analysis. Z-axis build direction.

Generating support structures for a component at this orientation, and with the complex geometry and post-processing required, necessitates the generation of custom support structures. A lenient support structure generation algorithm was used here with a minimum inclination angle setting of 35° to ensure that all external surfaces were well supported. This algorithm, however, generated excessive support structures on the internal

surfaces and these support structure groups were deleted and replaced by custom structures on specific areas where post-process machining could take place. Figure 4-18 shows the final support structure layout on the final volute geometry with no support structures included in the fluid path, custom structures in the fluid gap, as well as custom cone supports (yellow) visible.

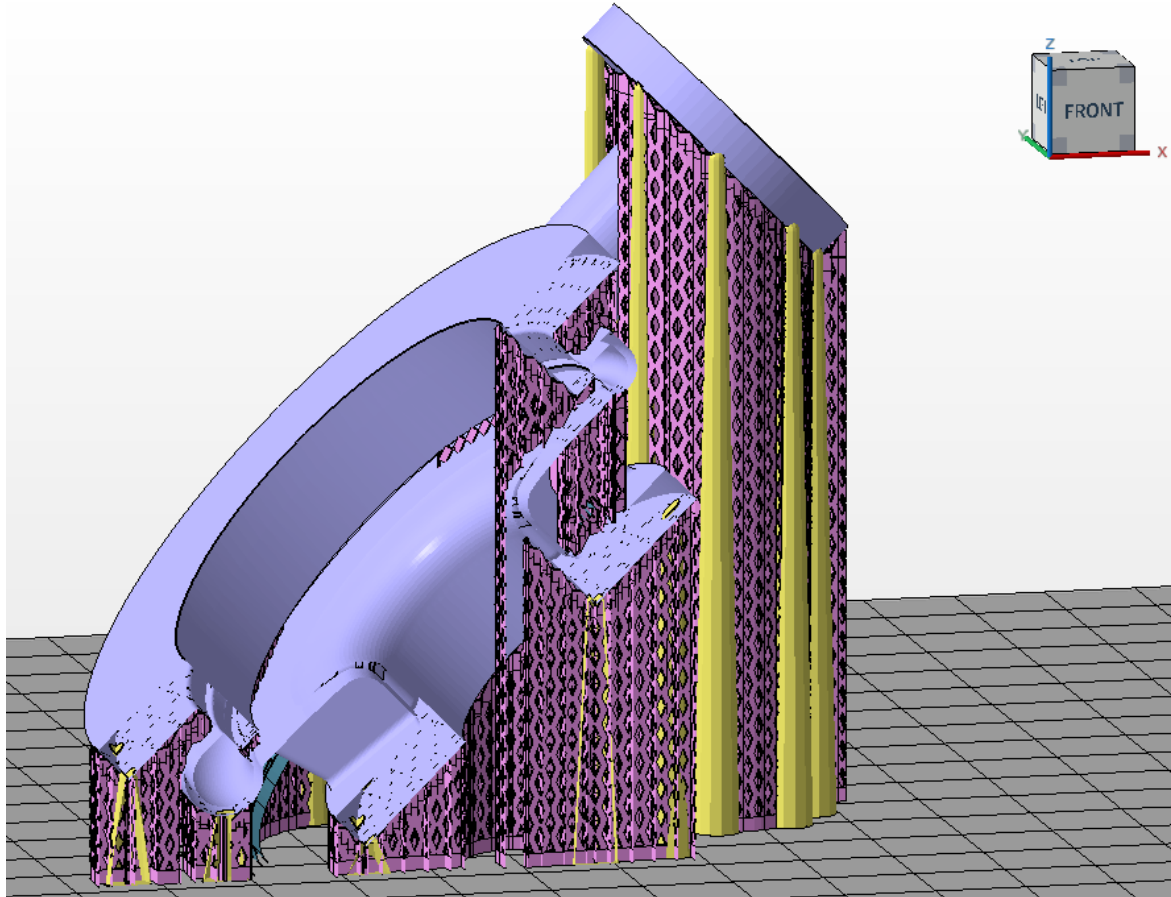


Figure 4-18: Second iteration volute cross-section in final orientation with various support structures visible. Z-axis build direction.

To adequately prepare for AM processes, build simulations provide a powerful tool to gain insight into the potential defects and the causes of these defects for AM-built components. ANSYS Additive 2021 R1 [485] was used to simulate to AM build process of the volute structure and support structures, to check for possible failures. The build simulation parameters were calibrated using the recommended guidelines in ANSYS Additive 2021 R1 and are shown in Table 4-3. This analysis accounted for the L-PBF process type, material used and geometry under investigation to provide a variety of results such as distortion, blade crash severity, high strain locations, amongst others. These results provide a variety of insights into potential defect locations, where distortion shows surface/feature inaccuracy, “blade crash severity” highlights potential locations where the component distorts vertically and damages the re-coater blade, and “high strain locations” represents excessively high expected thermal strains locations which often result in cracking.

From the results of the build simulation, some potential defect locations were noted with respect to blade crash severity. These possible failure locations were conveyed to the AM service provider before manufacturing, and the service provider suggested that such defects would not cause significant damage to either the component or the AM machine and therefore, manufacturing was initiated.

Table 4-3: Second iteration volute build simulation parameters.

Analysis Type	Voxel Size (mm)	Voxel Sample Rate	Material	Strain Scaling Factor	Strain Warning Factor
Assumed Uniform Strain	0.25	5	EOS 316L	1	0.8

The volute structure was then manufactured on an EOS M280 platform using 316L and standard processing parameters. Due to the variety of geometric and support structure changes present compared to the first iteration, numerous post-processing changes were made to ensure the geometric tolerances of critical features. This is described further in [479].

4.5.2. Volute Quality and Defect Evaluation

Following manufacturing through AM techniques, the volute component was found to contain defects in the as-built condition in the form of streak patterns, as seen in Figure 4-19, indicating the occurrence of blade crash events in the AM process. These blade crash events occur when excessive deformation of the component arises due to high thermal gradients, and the re-coater blade contacts the component leaving notches in the blade. These notches are then persistent throughout the build process, and subsequent powder layers are unevenly distributed along the notch effective path. The primary suspected cause of these blade crash defects is the lack of adequate support structures in the volute gap region. The origin locations of these notched re-coater blade defects coincide with the build simulation results for blade crash severity and high deformation in the vertical direction shown in Figure 4-20.

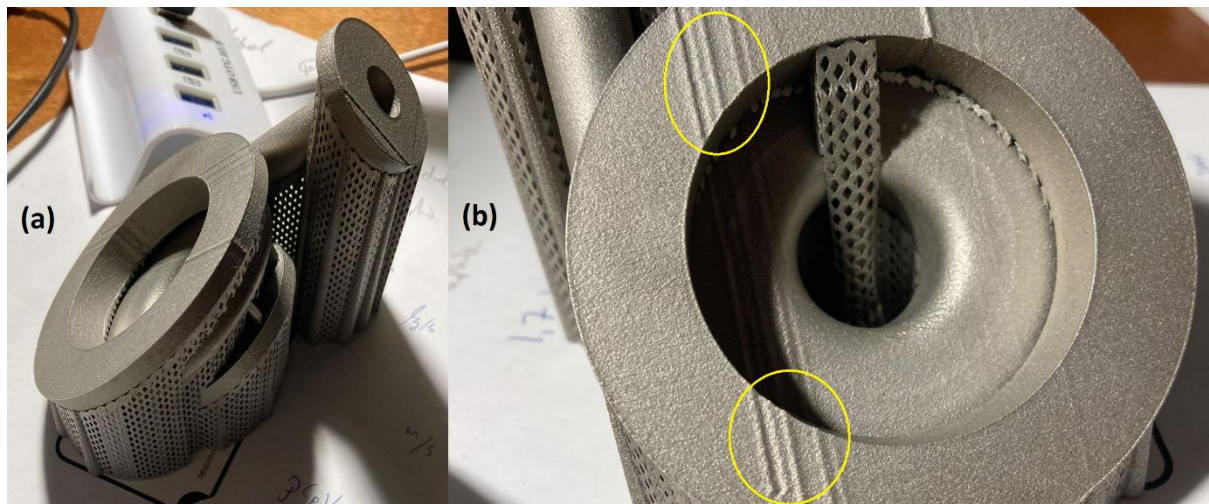


Figure 4-19: As-built second iteration volute with re-coater blade notch effect defects.

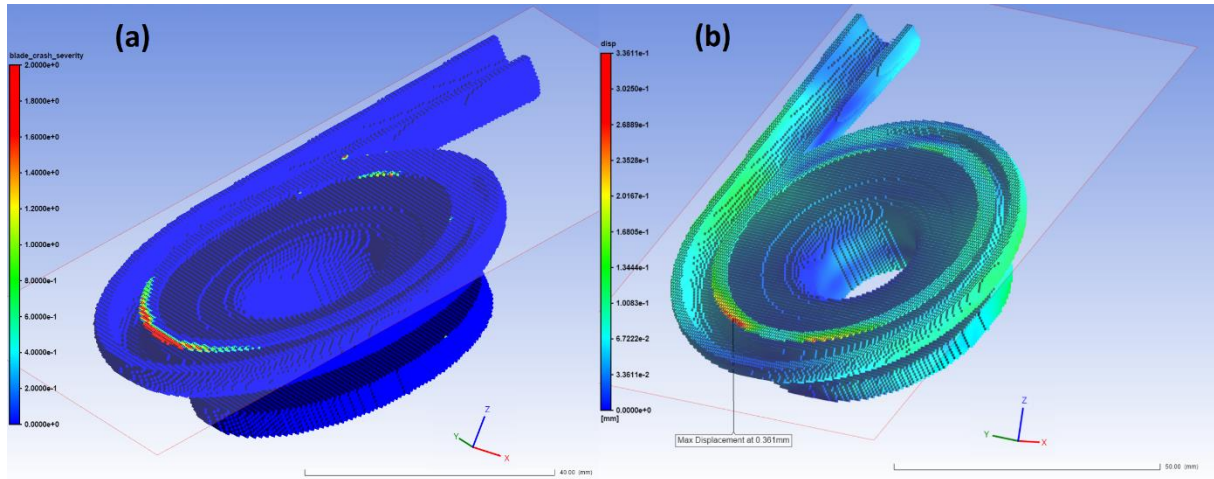


Figure 4-20: Second iteration volute AM build simulation showing; (a) re-coater blade crash severity and (b) displacement (deformation) during build process. z-axis build direction.

Various post-processing techniques were then applied to the volute to remove the support structures, and to machine the relevant faces to the final geometry. The streak patterns described above were isolated to faces that were machined during post-processing and therefore these defects were mostly eliminated. Following post-processing, the volute and impeller components were investigated through MicroCT techniques. The CT parameters used, shown in Table 4-4, were calibrated using the same methodology found in Section 4.4.3, and the specific analysis methods used were similar to those used in Section 4.4.4.

Table 4-4: Second iteration volute and impeller MicroCT parameters.

Scan Number	Voltage (kV)	Current (μ A)	Voxel Resolution (μ m)	Beam Filter Size (mm)	Averaging	Skipping
Volute	200	200	70	1	2	1
Impeller	200	140	30	1	1	0

MicroCT scan data were then imported into Volume Graphics VGSTUDIO MAX 3.5 and visually inspected for defects. The blade crash defects mentioned above can be observed again in regions which are not machined during post-processing, as shown in Figure 4-21 (a). These defects are also shown to appear in multiple locations along the volute fluid path regions, which would therefore be classified as critical defects with respect to volute and pump efficiency. Figure 4-21 (b) shows a particular location where a re-coater blade crash event is suspected to have taken place, resulting in high deviations from the intended CAD design. These defects are shown to occur consistently on the re-coater blade notch paths, but only in the volute gap region.

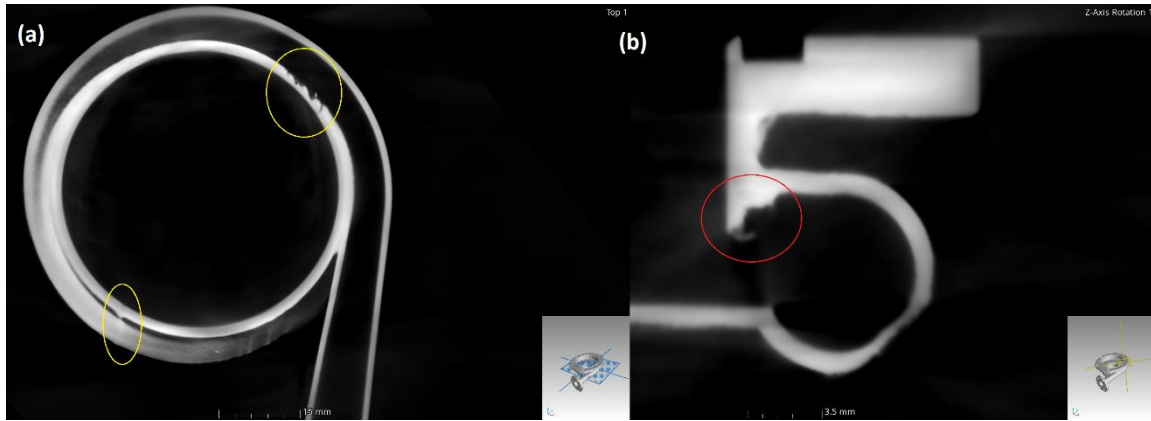


Figure 4-21: MicroCT observed re-coater blade notch defects in the volute fluid path. (a) top cross-section view showing notch streaks and (b) fluid path cross-section showing defect magnitude and wall thinning effects.

The internal surfaces of the volute fluid path were found to be of high quality, with no excessive drooping effects, good quality surface finish and no significant deviations being observed, aside from regions affected by blade crash events. Wall thickness analyses also yielded good results with no major variations in wall thicknesses throughout, except in the lowest region of the external surface. This lowest region, annotated in Figure 4-22 (b), is the effective start of geometric elements in the neighbouring vicinity and therefore relies on good quality supporting structures to start from. Figure 4-22 (a) shows a cross-section of the wall thickness in this region and the thinning effects. Both the wall thickness thinning observed here, and the re-coater blade defects mentioned previously, are suspected to be caused by a lack of adequate support structures for geometry element start locations. These defects should be possible to mitigate via additional support structures or other mitigation techniques such as thickened surfaces.

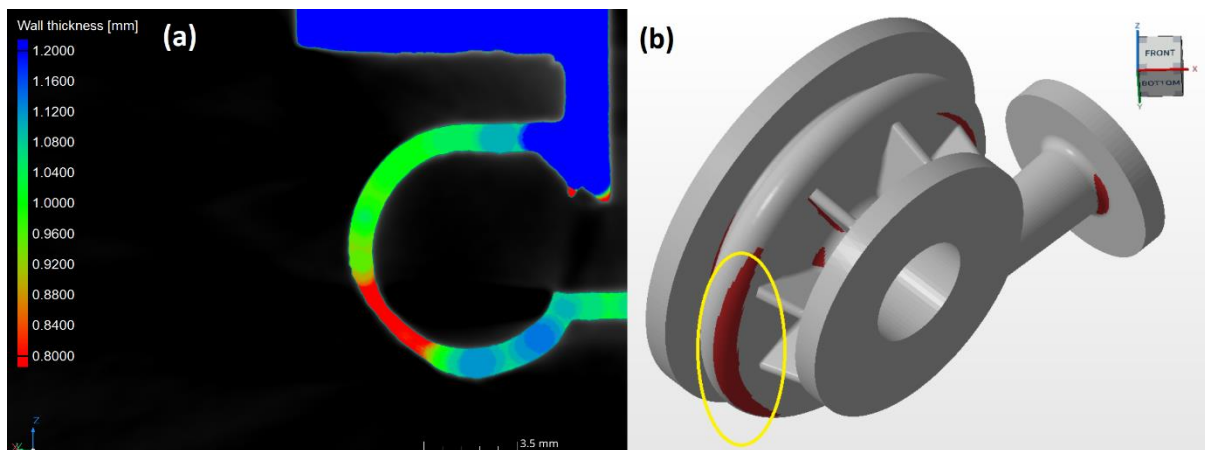


Figure 4-22: Second iteration volute wall thickness thinning effects on lowest supported surface. (a) displaying wall thickness thinning effects and magnitude. (b) region of lowest supported surface with z-axis build direction.

Selecting datum reference surfaces for post-process machining of AM-build components is challenging due to the geometric deviation caused by thermal warpage during the build process. Typically, the build plate is referenced by machinists and further datums can be created based on the build plate datum. However, for the volute component built at a 45° orientation angle, a specific as-built surface with minimal deviation needed to be selected as a reference. The build simulations done previously were used to select a surface with minimal deviation as an initial datum reference. The deviation of this datum reference surface is suspected to be the cause of the machining

misalignment shown in Figure 4-23 (b). The sharp surfaces caused by this misalignment would disturb the fluid flow passing through the volute gap. This defect would thus cause a reduction in pump efficiency and should therefore be mitigated through improved datum reference techniques.

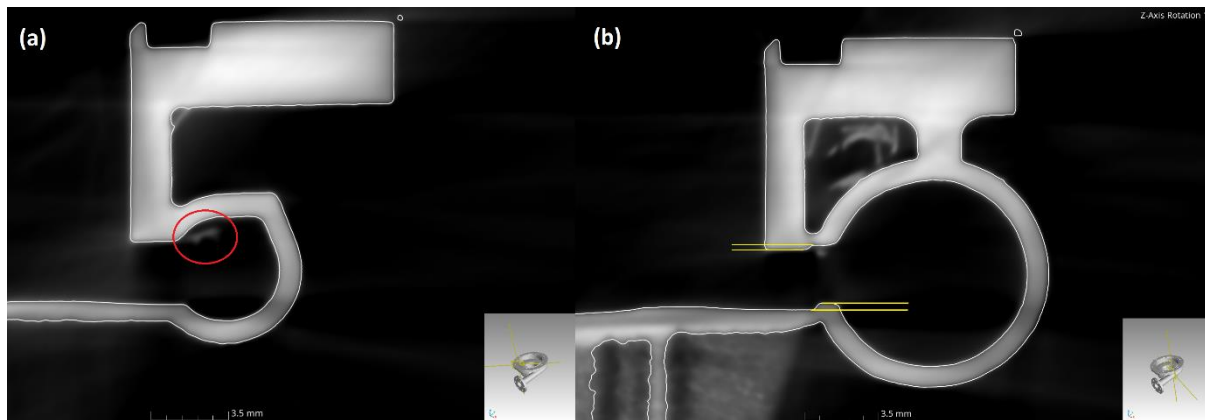


Figure 4-23: Second iteration volute fluid path cross-section views showing (a) swarf generation example and (b) datum machining misalignment.

Swarf generation defects (Figure 4-23 (a)) were also found throughout the volute structure, however, at a lower quantity and magnitude when compared to the first iteration. This suggests that while the post-processing techniques applied reduce the occurrence of these defects, further steps should be taken to mitigate them entirely.

While isolated instances of defects are present in this volute iteration, the overall quality was found to be vastly superior to the first iteration volute in terms of wall thickness defects, swarf generation defects, as well as fluid path surface quality and surface deviations. Furthermore, other defects not mentioned here were completely mitigated, particularly the cutwater geometry deviation. While the first iteration defects were mitigated almost entirely, unexpected new defects were found which were all primarily due to the 45° orientation angle, in addition to the associated support structures and post-processing used to subsequently remove these structures.

4.6. Third Iteration

Building on from the mitigation strategies applied to the second iteration volute, and the specific defects which arose during the production of this part, additional DfAM optimization techniques were considered for further iterations of the SAFFIRE pump components. The second iteration volute and impeller set were tested on a test bench setup, as described in [479], and performed relatively well considering the defects that were present. These data acquired from this testing adequately confirmed the hydrodynamic performance of the pump components, as well as the AM process used. Following these findings, the third-iteration production of “full-scale” the SAFFIRE pump components was the logical next step.

The term “full-scale” is qualified, since the impeller and volute components in question were designed for a water-analogue version of the full-scale kerosene pump, to enable safe, low-complexity preliminary pump testing [479]. The AM process is also assessed here for larger-scale configurations, with slightly more complex geometric elements. Compared to previous iterations, these components would experience considerably higher stresses and thus sound structural design would be required. Furthermore, the defects which may arise during part production could result pump component failure, therefore high manufacturing quality would be vital. The material used for

the pump components evaluated in this section is Ti-6Al-4V. This material is the intended choice for the full-scale kerosene pump configuration and AM process data for this material, on similar full-scale components, would thus be insightful for the final design parameters.

4.6.1. Volute Design

The volute structure in this iteration was similar to that of the previous iterations with respect to volute type (circular volute), typical fluid region methodology [479] and cutwater geometry generation methods. The general process used to create the volute revolved fluid path region is the same as that employed for the second iteration, utilizing the teardrop geometry process. However, the CAD geometry of this volute is significantly larger since this design has a much higher fluid mass flow rate, resulting in a greater volute fluid path volume.

The third iteration pump would be subjected to much higher magnitude loading conditions when compared to the first and second iteration pump configurations. The primary applied force on the volute structure is the internal pressure, which is raised to 4.62 MPa, as compared to 1.32 MPa on the first and second iteration pumps [479]. Due to these much higher force, and therefore stresses, careful structural analyses needed to be performed to ensure that the volute structure would not fail in operation.

4.6.2. Topology Optimization Methods

Since AM processes are being utilized for the volute structure component, this enabled the use of advanced design techniques such as Topology Optimization (TO). TO techniques, as described in Section 2.2.6, use advanced optimization algorithms to create free-form geometries, often with the objectives of reducing mass and reducing part count through part consolidation. The free-form structures generated through TO techniques synergize further with AM processes since these structures often contain smooth curvatures throughout, eliminating the defects which commonly arise on sharp edges.

TO techniques were used in this pump component iteration with the objectives of reducing the mass of the final components, gaining insight into the AM process of these complex structures, and to assess the quality of the geometry through inspection. The TO techniques used here were applied using a standard workflow methodology in the software package nTopology 3.20.3 [486]. TO methods are typically applied via the following procedure:

1. Preparation of design space and non-design space regions
2. Definition of forces and constraints
3. Definition of TO input parameters such as material properties, optimization objectives, optimization constraints and a variety of other possible parameters
4. Specification of TO output parameters and geometry parameterization
5. Structural analyses to verify structural feasibility

To prepare the volute structure for TO methods, the design space and the non-design space regions were first defined. The design space regions refer to the optimization region where no forces or constraints are applied, while the non-design space region refers to faces or structures where boundary conditions are not applied. Figure 4-24 shows a cross-section of the separated design/non-design spaces for the volute structure pre-topology

optimization. This process constrained the optimization to only create and iterate geometry elements within the design space (annotated in grey), and to maintain all geometry elements in the non-design spaces (blue and orange regions), all with the purpose of finding a local, optimal geometry solution for the given optimization objectives. These design and non-design space regions were set with the “bonded” contact condition applied to all mating surfaces.

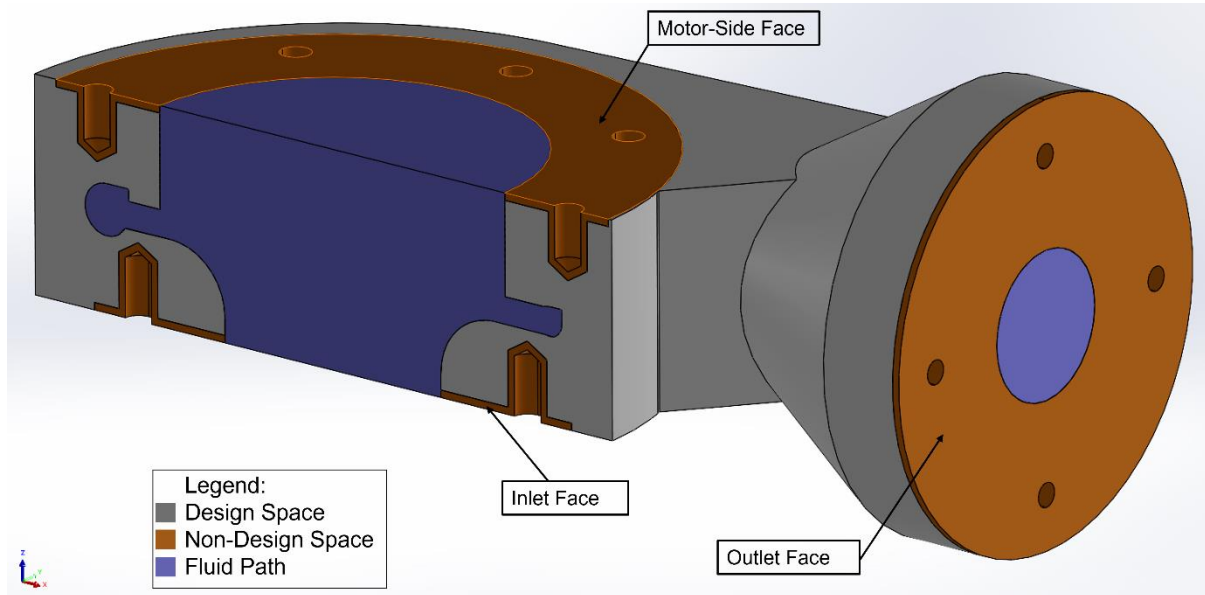


Figure 4-24: Third iteration volute topology optimization showing design and non-design space regions as well as constraint locations.

The primary force acting on the volute structure would be imposed by the pressure of the contained fluid in the fluid path region. Since this pressure is non-uniform and increases over the progression of the volute fluid path, the surface pressure distribution can be approximated by raising the surface pressure from the volute inlet location through to the volute outlet starting at 0.2 MPa and increasing to 4.62 MPa. The most accurate definition of this pressure would have been to map the volute part model and the surface pressures predicted by an associated computational fluid dynamics (CFD) model. However, for the purposes of this analysis, a conservative approach was used and the maximum pressure of 4.62 MPa was applied on all internal surfaces.

There are two primary effects of this pressure and fluid flow, namely the pressure on all internal walls, and the outlet thrust generated at the outlet structure location. Figure 4-25 shows the forces acting on all surfaces due to internal pressure, the outlet thrust location and the bolted constraints, all with respect to the design space geometry. For the purposes of this analysis, the volute structure was constrained by bolted joints on the inlet face, outlet face and the motor-side face (Figure 4-24). The specific constraints for these joints are: motor-side face bolts fixed in all 6 degrees of freedom (DOF), inlet face bolts fixed in 5 DOF (z-translation free) and outlet face bolts fixed in 5 DOF (x-translation free).

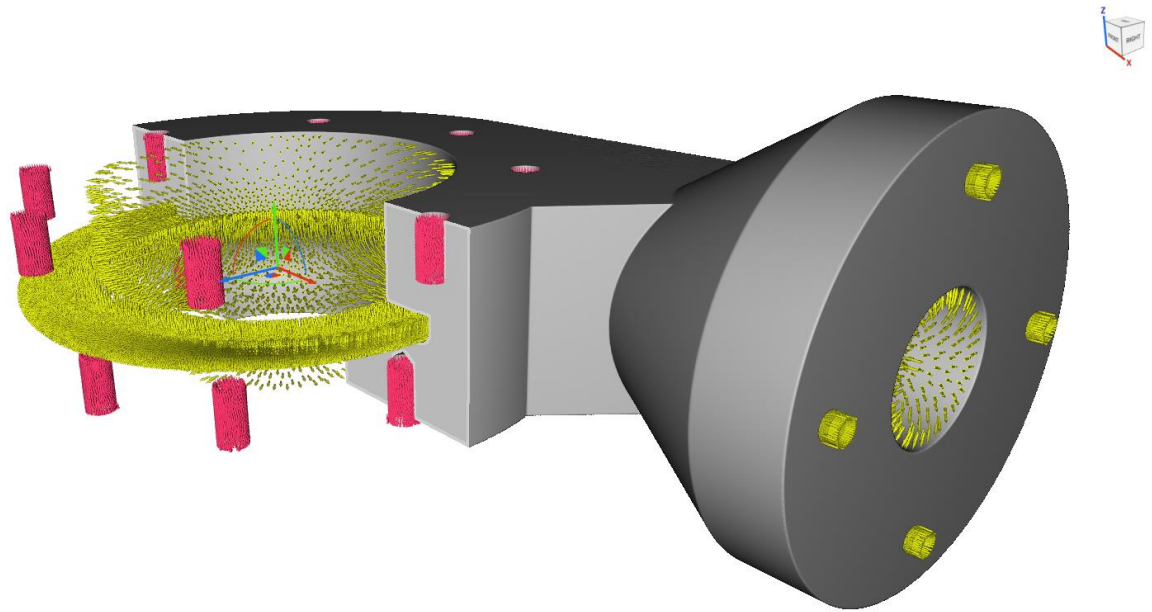


Figure 4-25: Cross-sectional view of forces and constraints applied to third iteration volute for topology optimization and finite element analysis setups.

Following the determination of forces and constraints, the geometry was then prepared for TO processing by creating discrete finite element meshes for each component (design and non-design spaces). Various input parameters exist for TO processes on different software platforms. Generally, the main TO process objectives used are either mass minimisation or stiffness maximisation. The constraints associated with TO processes refer to the limits applied to the optimization objectives, geometric constraints and manufacturing-based constraints. Table 4-5 shows the TO parameters for the volute structure. Once convergence is achieved on a local optimal geometry solution, a TO-generated structure is available for further processing. These structures generally require geometry smoothing and integration with non-design space features. Figure 4-26 shows the resultant topology optimized volute design after smoothing and joining with non-design spaces.

Table 4-5: Third iteration volute topology optimization parameters.

Material	Optimization Objective	Volute Fraction Constraint	Boundary Penalty	Maximum Iterations	Minimum Objective Change
EOS Ti-6Al-4V	Maximize Stiffness	0.3	0.25	200	0.003

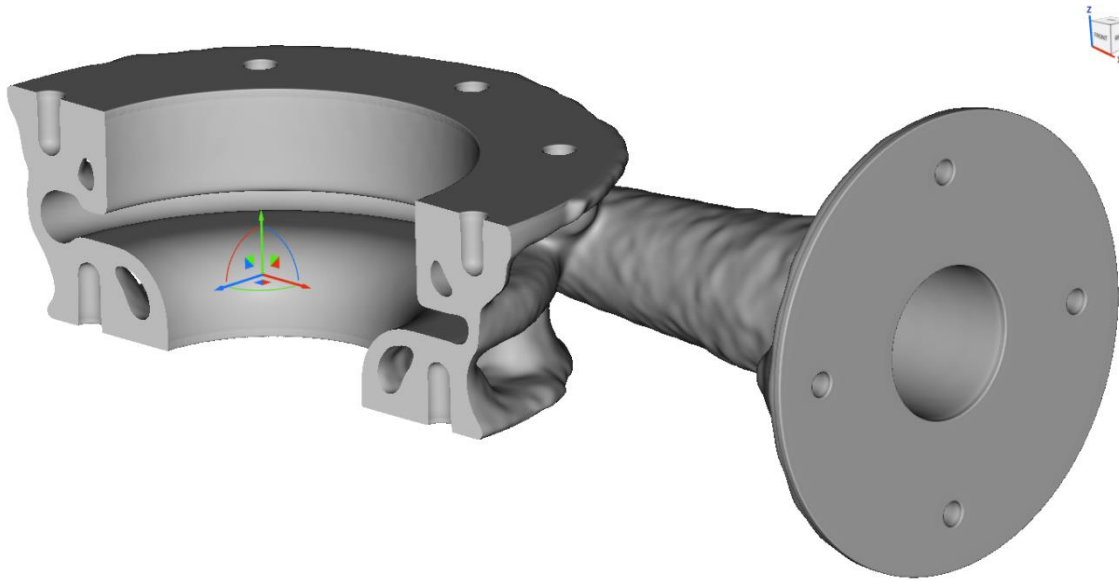


Figure 4-26: Cross-section of topology optimized third iteration volute structure after smoothing and joining techniques.

The next step taken was to verify the structural response of the topology optimized structure through finite element analysis (FEA). Associated analyses were performed on the smoothed TO volute structure, using a quasi-meshless static structural analysis method. All forces and constraints applied previously during the TO process were replicated in the FEA models, and resultant stresses and displacements were obtained as shown in Figure 4-27. The maximum equivalent von Mises stress (MPa) obtained from this analysis was 684 MPa, and the maximum displacement was 0.516 mm. The maximum stress locations for the structure were all shown to exist in the vicinity of the interface conditions (bolt constraint edges) which is a typical FEA inconsistency due to mesh effects. The stresses observed in primary geometry regions of concern (the volute path structure and supporting ribs) were predicted to be much lower than this. The maximum allowable stress for the volute design was specified as 700 MPa (material yield strength of 1120 MPa and safety factor of 1.6) and therefore the design was verified through FEA techniques in nTopology 3.20.3.

It is important to note that TO techniques are commonly iterated when FEA results show either very low- or high-magnitude stress distributions. This iteration is performed by varying the input parameters for the TO process, and therefore steps 3-5 mentioned previously were repeated until a mass optimized structure was found. This was performed for the volute structure, where the volume fraction constraint was varied between the limits of 0.2 and 0.5 until FEA results determined an optimal result. This volume fraction constraint represents the target volume of design space after TO in comparison to the initial design space volume.

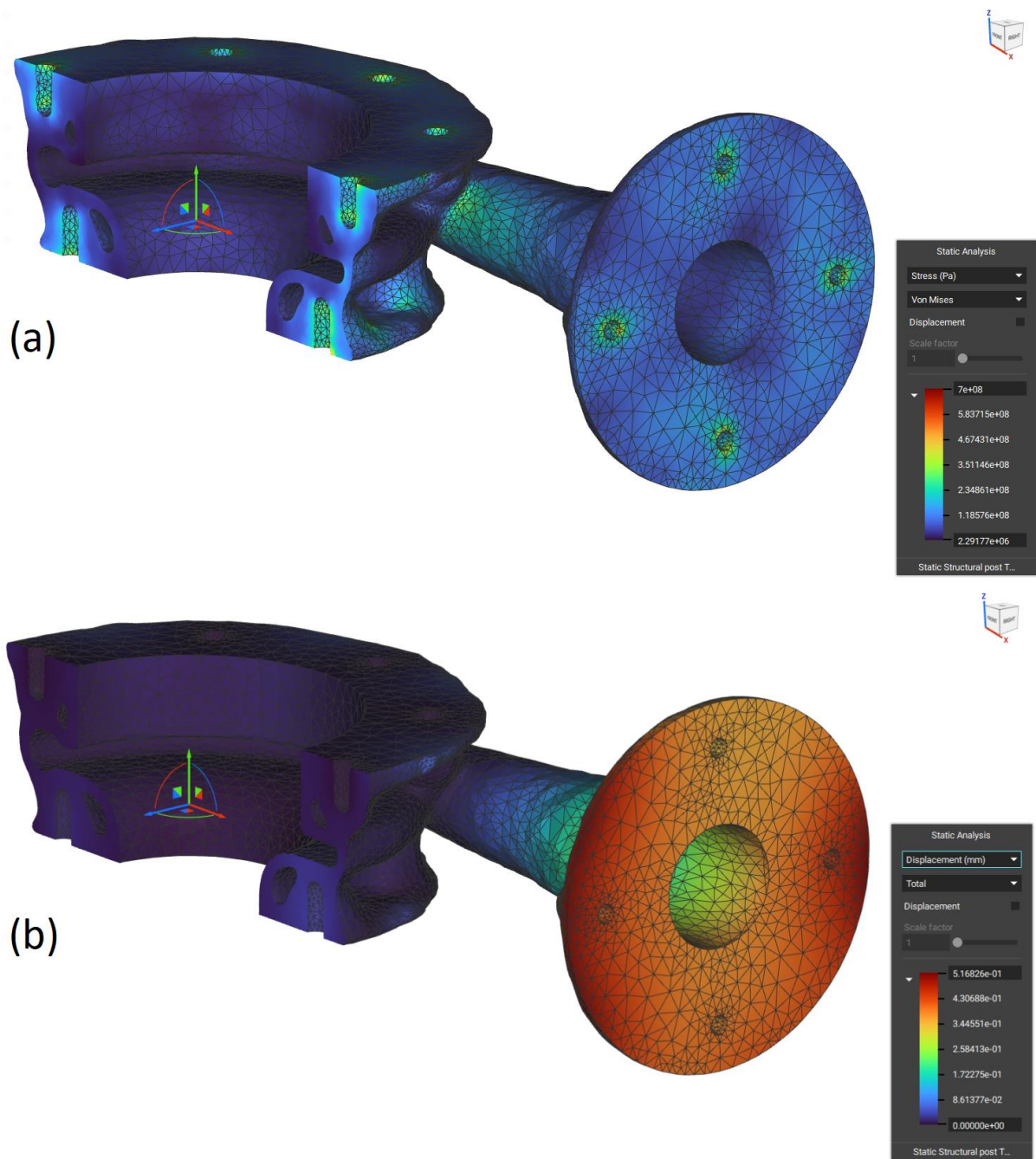


Figure 4-27: Cross-section of finite element analysis results for topology optimized volute structure. (a) - von Mises equivalent stress (Pa) result plot and (b) - total displacement (mm) result plot.

4.6.3. Additive Manufacturing Build Preparation

To adequately prepare for the AM process, various techniques were applied to finalise the volute geometry before the design was submitted for manufacture. The TO structure was exported into a parametric design platform where the geometry was edited to further refine the surface smoothness, eliminate unnecessary geometric elements and ensure tolerances on critical surfaces.

Further build preparation techniques were applied to mitigate the defects found on the second iteration volute in Section 4.5.2. The defects in question are the volute gap defects, and the volute wall thickness defects on the

lowest external surfaces, which were both caused by insufficient supporting structures. The volute gap defects were mitigated using a completely solid support structure, and the external surface defects were mitigated by thickening the surfaces and applying more robust support structures. Figure 4-28 (a) displays the final geometry with the extended external unsupported region and the volute gap solid support structure. Also shown in Figure 4-28 (b) is the final orientation and support structures applied before the final geometry was sent for manufacturing.

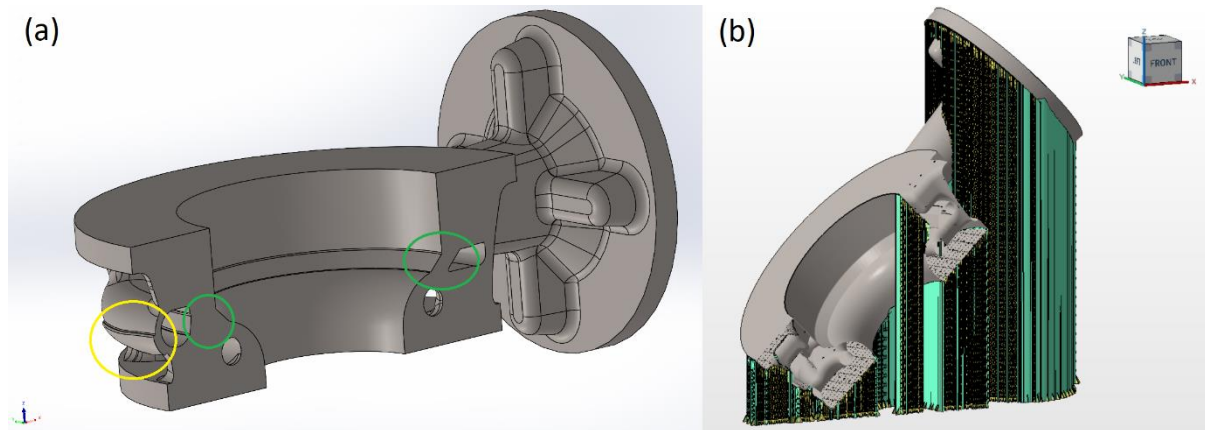


Figure 4-28: Topology optimized volute structure in AM build prepared state. (a) final geometry annotating external surface extension (green) and volute gap solid support structure (green). (b) final geometry orientated at 45° with support structures.

4.6.4. Impeller Design

The impeller design for this iteration was arrived at via various changes with respect to the size of the impeller, and the geometry of the blades. Following the hydrodynamic testing of the second-iteration pump components, the tolerance of the gap between the impeller and volute was deemed to need improvement. The tolerance of this gap is constrained by the as-built surface of the impeller blade edges annotated in Figure 4-29. To improve the performance of the pump, the impeller blade edges were extended, and machined after AM processes.

The only DfAM changes applied to this impeller iteration were the blade extensions. This impeller iteration design featured blades with lower inclination angles relative to the build direction. A typical downskin analysis, shown in Figure 4-29, was run on the impeller structure to characterise the inclination angles on the blades. While the blade tips exhibit small areas (3.01 cm^2) which are below the minimum recommended inclination angle, it was expected that these blades will not contain significant defects after the AM process since the areas are small and are present on surfaces with smooth curvature. However, some warpage is expected in these small overhanging regions.

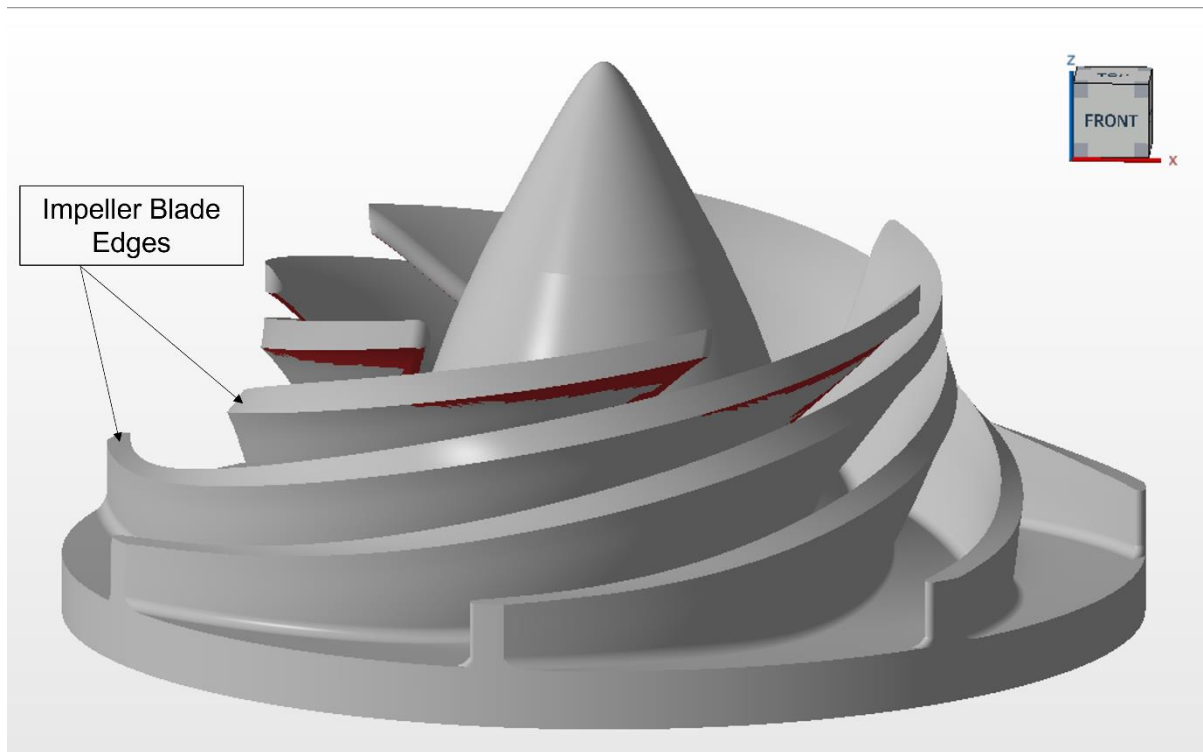


Figure 4-29: Third iteration impeller downskin analysis showing unsupported regions (red) with inclination angles below 45°.

Since the impeller blades, and other surfaces in contact with fluid, are not machined post-process in this design, surface finishing techniques were applied to the impeller after AM processes. The surface finish technique used was Dry Electropolishing on a DLYte 100H+ Ti system. This process entails mechanical polishing using a ceramic medium.

4.6.5. MicroCT Process and Defects

Both the impeller and volute components in this iteration were inspected using MicroCT techniques before significant post-process machining took place. This was done primarily due to project design changes where the final machined geometry of components was subject to change and therefore, further machining was put on hold. This, however, provided the opportunity to gain insight into the geometry of the impeller and volute before machining operations are performed, potentially enabling subsequently insights to be gained concerning the direct geometry effects of the machining if later verified by MicroCT techniques. Figure 4-30 shows the AM-built components on the build plate with the supports still attached. Once removed from the build plate, both the impeller and the volute were scanned as shown in Figure 4-30 with supports still attached.

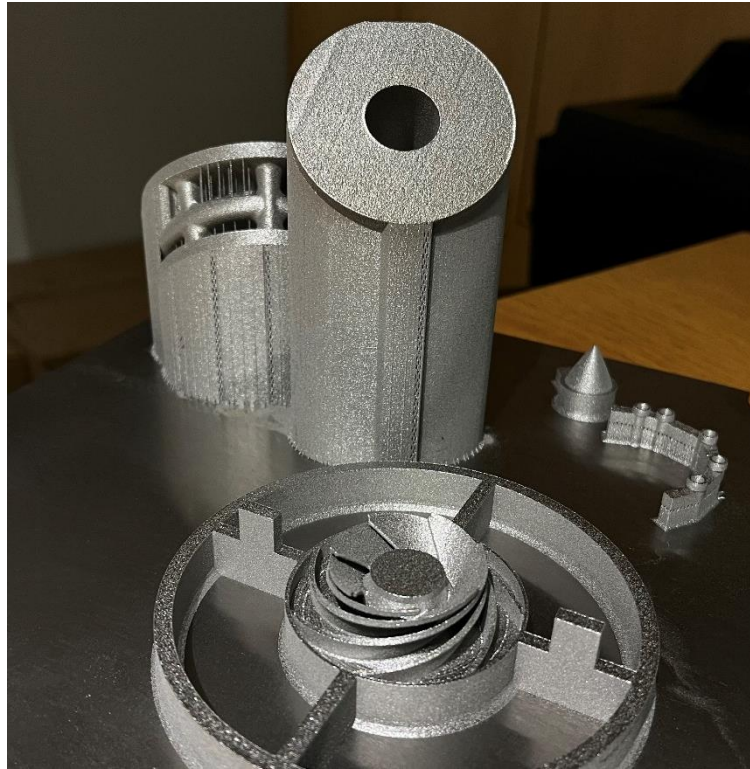


Figure 4-30: Third iteration impeller and volute as-built on the build plate.

The MicroCT system used for these components was the Nikon XT H 225. The specific CT parameters used for both components were derived using the methods described in Section 4.4.3 and [484]. The CT parameters for each component are shown in Table 4-6. While the impeller component was scanned in a single process, due to its size, the volute was scanned in two sections and stitched together during reconstruction. The scans of both components were then inspected qualitatively and quantitatively in Volume Graphics VGSTUDIO MAX 3.5.

Table 4-6: MicroCT parameters for third iteration volute and impeller components.

Component	Voltage (kV)	Current (μ A)	Projection Images	Voxel Resolution (μ m)	Beam Filter Size (mm)
Volute	215	419	2000	74	1.5
Impeller	200	150	2984	30	1.5

4.6.6. Volute Quality and Defect Evaluation

After typical image processing and surface evaluation techniques as described in Section 4.4.4, the volute component was visually inspected for defects, including those found in previous iterations, namely wall thickness thinning, volute fluid path section deviation and surface roughness. None were detected and the component displayed high conformance to design. Figure 4-31 shows the deviation (a) and the wall thickness (b) at particular sections where good quality was observed. These qualities are consistent throughout the volute fluid path regions, and overall quality for all surfaces is similar. The highest deviations shown in Figure 4-31 were measurements of the still-connected support structures and are not relevant. The volute gap defects found in the first and second iterations were also entirely mitigated in this iteration, with no defects found.

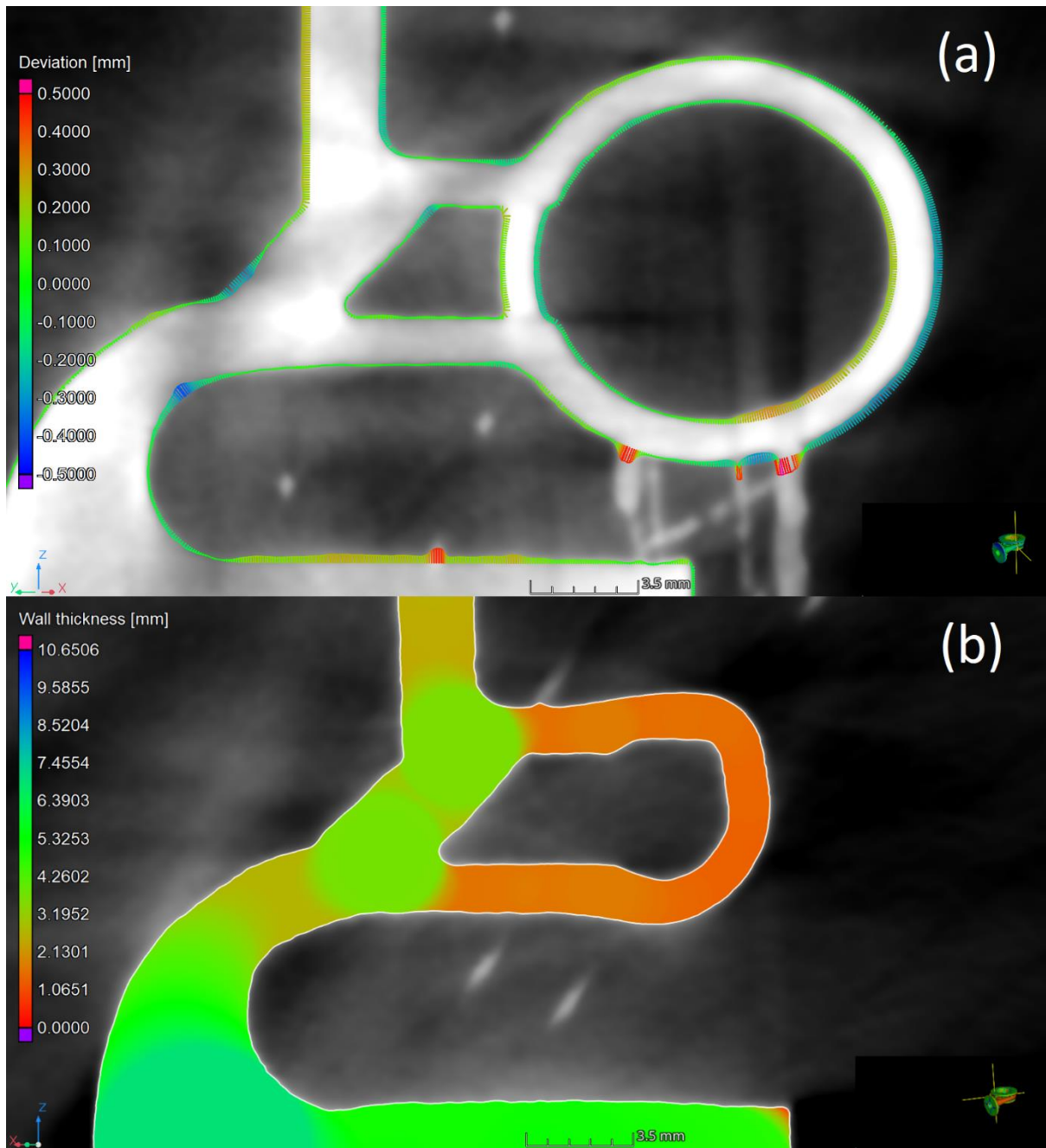


Figure 4-31: Third iteration volute MicroCT-derived deviation analysis (a) and wall thickness analysis (b).

Since the volute still required post-process machining, the presence of machining defects found in both the first and second iteration (swarf generation and datum misalignment) could not yet be assessed. However, these MicroCT data acquired here could be used as a reference for more accurate procedures during later machining. The overall quality of the volute component was found to be excellent with no detected defects and, pending evaluation of post-process machining, the volute AM process was evaluated to be highly qualified for pump operation.

4.6.7. Impeller Quality and Defect Evaluation

Similarly to previous impeller design iterations, the MicroCT derived qualitative and quantitative results here showed a high conformance to design with no significant defects present. Figure 4-32 shows the MicroCT-derived deviation analysis of the impeller.

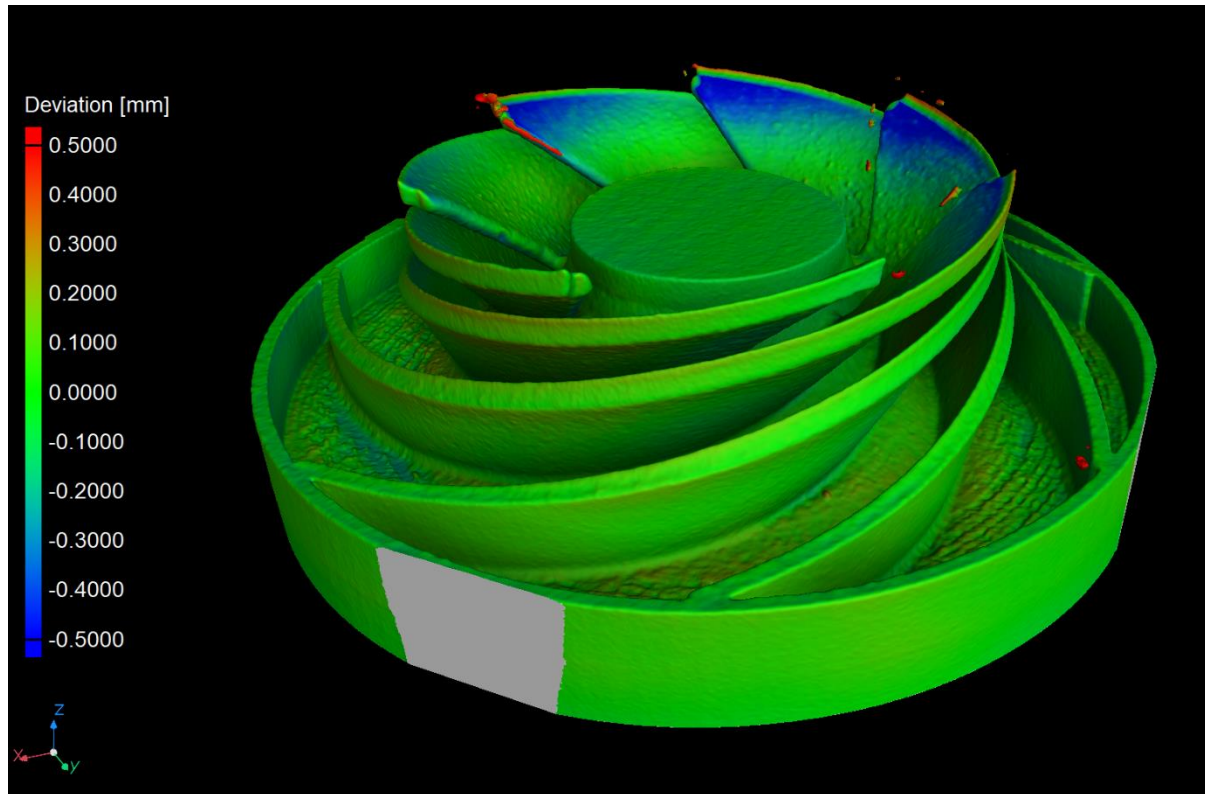


Figure 4-32: Third iteration impeller 3D MicroCT deviation qualitative plot. Z-axis build direction.

While no specific defects were detected for this component, blade tip warpage was observed. With unsupported structures extending far from a base structure and low inclination angle surfaces, the blade tip geometry elements make for particularly challenging shapes for AM processes. These geometry characteristics result in a drooping effect on the geometry where the blade tips are lower than the nominal CAD geometry position (Figure 4-32). The magnitude of this effect is shown in Figure 4-33 on the cross-section where the highest deviation occurs. As shown in Figure 4-33, these high deviations only occur on the furthest most tips of the impeller blades, and the rest of the blade structures have high conformance to design. The expected impact of these blade tip deviations on pump performance is unknown, and mitigation techniques should be applied if necessary.

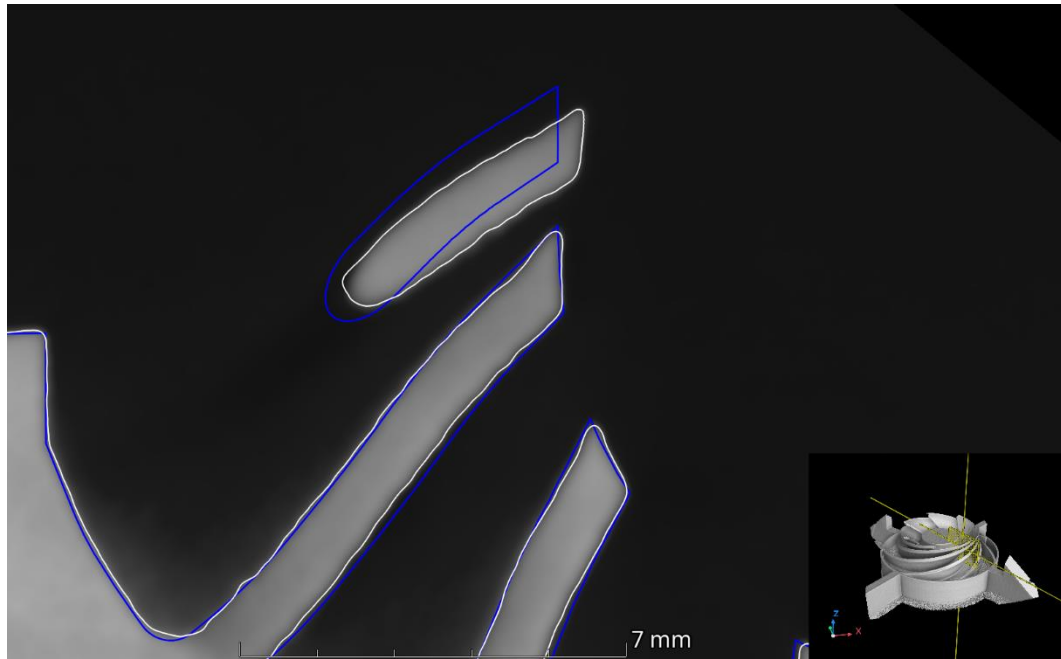


Figure 4-33: MicroCT cross-section of impeller showing deviation analysis result worst case. White lines represents MicroCT measured surface and blue lines represent CAD geometry.

4.7. Challenges and Summary of Findings

This section details the methods used and observations found for complex geometry components built through AM techniques. Components were iteratively manufactured and analysed using MicroCT techniques to characterise defects, and sequentially these defects were mitigated through various DfAM techniques for each iteration. These techniques were applied to varying component geometries, at different scales and with different materials. The DfAM techniques used clearly improved the overall quality of each component through the iterative process, and the knowledge gained from this process informs engineers as to the causes and potential mitigation techniques for similar components under consideration.

4.7.1. Volutes

Comparing each iteration volute component progressively, through MicroCT based qualitative analyses, clearly showed reductions in defects such as wall thickness thinning, critical surface deviations and post-process machining defects. Quantitative results were also obtained through surface deviation analyses to compare the volute components per iteration. The minimum surface deviation, maximum surface deviation and cumulative distributions of surface deviations are shown in Table 4-7 for each volute iteration. These deviations were analysed in two separate regions of the components, namely in the whole component (Figure 4-34 (a)), and in the volute fluid path structure (Figure 4-34 (b)). These comparative analyses showed that the overall averaged accuracy of surfaces (cumulative distributions) was greatly improved between the first and second iterations in both the whole component and the volute path regions. These distributions were further improved in the third iteration for the volute path region, but not for the whole component.

While these comparative results are relevant between the 1st and 2nd circular volute designs, due to the increased scale and different geometry, the 3rd volute does not provide a suitable basis for comparison with respect to the

previous volutes. For the 3rd iteration volute, many quantitative analysis errors are introduced, mostly in the form of support structures still attached to the volute, which produced errors in the surface deviation analyses, especially in the minimum and maximum deviation results.

Table 4-7: Volute maximum, minimum and cumulative percentile surface deviations for each iteration and configuration. CV – Circular volute, DCV – Double circular volute.

Whole Component					
Iteration and volute type	Minimum Deviation (mm)	Maximum Deviation (mm)	90% Cumulative Distribution (mm)	95% Cumulative Distribution (mm)	98% Cumulative Distribution (mm)
1st CV	-0.979	>2	0.382	0.4467	0.5705
1st DCV	-0.9127	>2	0.378	0.4494	0.6216
2nd CV	-0.8579	0.7068	0.2662	0.3206	0.4216
3rd CV	-1.54	1.68	0.377	0.4503	0.4958

Volute Path Structure					
Iteration and volute type	Minimum Deviation (mm)	Maximum Deviation (mm)	90% Cumulative Distribution (mm)	95% Cumulative Distribution (mm)	98% Cumulative Distribution (mm)
1st CV	-0.9967	>2	0.4135	0.4783	0.8251
1st DCV	-0.6257	1.96	0.3899	0.4226	0.4679
2nd CV	-0.4922	0.5227	0.3188	0.3723	0.4396
3rd CV	-1.54	1.68	0.3098	0.3612	0.4231

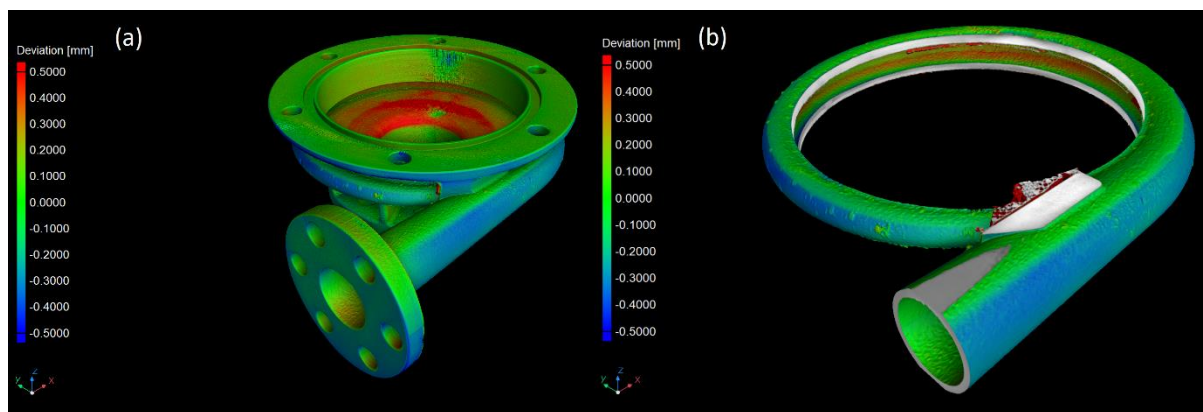


Figure 4-34: Volute example of whole region and separated volute path region surface deviation analysis.

Another observation made for all volute iterations is that the surface deviations are directed towards the centre of the geometric model/centre of mass. This is most likely caused by thermal contraction after the AM process once all heat sources are removed, and the component cools to room temperature. This phenomenon, shown in Figure 4-35, is observed for each iteration even though the AM-built components were manufactured using different materials, and by different service providers. Furthermore, the MicroCT techniques and machine used for the volute iterations varied. This suggests that AM-built components have the tendency to contract towards the centre of geometric elements or volumes due to thermal warpage, which is briefly discussed in [487,488]. For the purposes of these volute components, this effect does not cause any significant defects, however, for larger

components, this phenomenon may cause significant deviation which may affect final tolerances and wall thicknesses of surfaces.

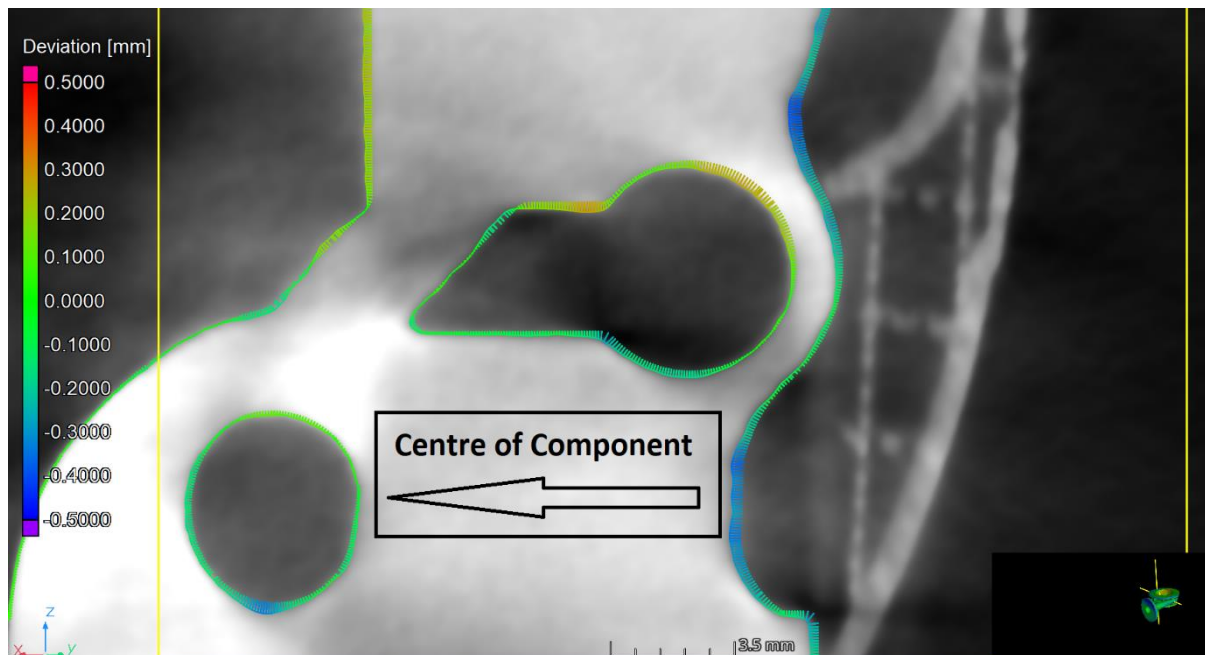


Figure 4-35: MicroCT slice view with deviation analysis showing geometric contraction towards centre of geometry/volume.

4.7.2. Impellers

For all iterations of impeller components, no significant defects were found. The only observed AM process defect which may potentially impede the performance of the pump is the thermally-based warpage effects on the impeller blade tips. To quantify the deviation in these blades, and for the overall structure, surface deviation analyses were performed and the maximum, minimum and average cumulative distribution data were extracted. To do this, the blades were separated from the base structure of the impeller for each iteration, using image processing tools. The regions for this analysis are exemplified in Figure 4-36 where (a) shows the entire component and (b) shows the separated blades.

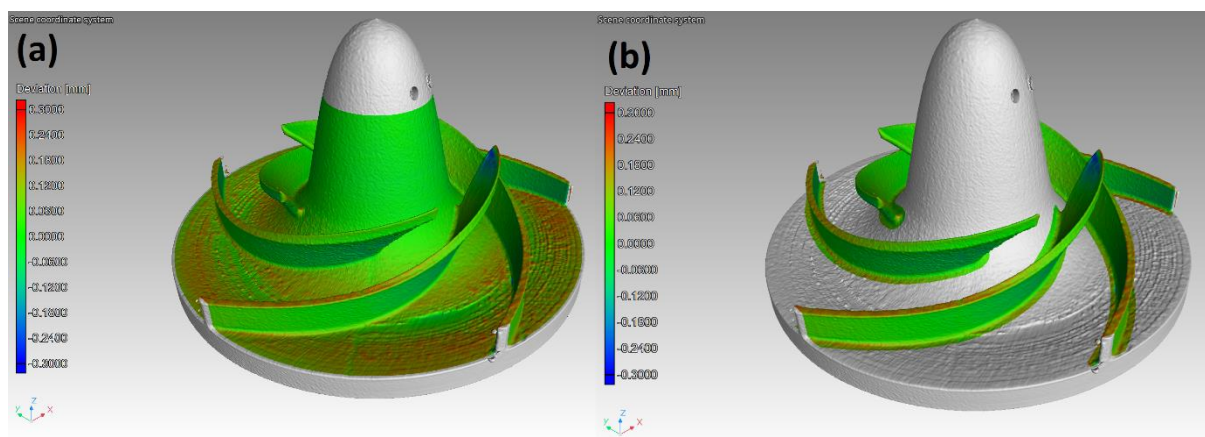


Figure 4-36: Second iteration volute surface deviation analysis showing (a) whole component region and (b) separated blades region.

The quantitative results from these surface deviation analyses are shown in Table 4-8. Some interesting conclusions can be drawn from these data, primarily when comparing the first and second iteration impellers which are identical in nominal (CAD) geometry. Comparing the deviation characteristics for these two iterations shows a percentage difference of 4.45% and 5.38% when comparing the 98% cumulative distribution for the whole impeller and the separated impeller blade surfaces respectively. This demonstrates the repeatability of the AM and MicroCT processes combined, when assessing the 98th cumulative percentile of all surface deviations. Many potential errors in the MicroCT and image analysis could account for this percentage difference such as mounting orientation in the MicroCT machine, surface determination grey value inputs, and other minor causes.

Table 4-8: Impeller MicroCT surface deviation results for each impeller iteration on both the whole component, and the separated impeller blades region.

Whole Impeller					
Iteration number	Minimum Deviation (mm)	Maximum Deviation (mm)	90% Cumulative Distribution (mm)	95% Cumulative Distribution (mm)	98% Cumulative Distribution (mm)
1st	-0.2499	0.2665	0.0826	0.0928	0.1076
2nd	-0.2652	0.2702	0.0935	0.1024	0.1125
3rd	-0.6254	0.6012	0.1438	0.1959	0.322
Separated Impeller Blades					
Iteration and volute type	Minimum Deviation (mm)	Maximum Deviation (mm)	90% Cumulative Distribution (mm)	95% Cumulative Distribution (mm)	98% Cumulative Distribution (mm)
1st	-0.2499	0.2665	0.0784	0.1102	0.1519
2nd	-0.2652	0.2702	0.0835	0.1251	0.1603
3rd	-0.6254	0.6012	0.1531	0.2151	0.3088

4.7.3. Recommendations for Future Work

While the impeller and volute components were successfully manufactured to an adequate quality, many unanswered research questions remain, and further sample testing cycles could be introduced for further defect mitigation and to inform engineers on applicable DfAM techniques for future component scenarios.

Since these pump components are all precursors to flight weight, propellant-based pump systems, further testing under the specific operating conditions of these flight weight configuration pumps is the next step. The primary differences in these pump designs are the scales and temperatures at which they operate. The Kerosene pump configuration consists of a highly similar geometry, and the same material to that of the third iteration components discussed in Section 4.6, thus the design is largely confirmed for both impeller and volute components.

The flight weight LOX pump has a geometry slightly larger in scale, and is to operate at cryogenic temperatures, which may cause significant thermal contraction. This effect, combined with the observed contraction effect from AM processes, could cause significant tolerance deviations on critical surfaces of the pump components, and DfAM techniques should be applied to mitigate these effects if necessary. The LOX components also require a

material composition that is LOX compatible, and AM process testing should be performed on the materials under consideration (Inconel 718 or Stainless Steel 316L). Furthermore, to test a wider variety of AM supply chain materials, LOX compatibility tests could be performed to broaden the material selection choices for LOX based pump components.

While good MicroCT observations were made relating to larger defects and geometric inconsistencies for all pump components discussed, the specific MicroCT scan quality is relatively poor and at low resolution. This inhibits the ability for MicroCT techniques to detect smaller, potentially critical defects such as cracks, near-surface porosity and bulk porosity. A potential method to detect possible AM process defects of this nature is to build witness specimen samples concurrently in the same AM process as the components in question. These specimens, as discussed [489], provide the opportunity to perform MicroCT evaluations of possible layered porosity distributions and general AM process defects.

Volutes

The volute structures, while exhibiting good conformance to design, presented poor surface finish characteristics which may degrade pump performance; the extent to which remains to be determined. While the majority of the volute structure's surfaces are machined after the AM process, the revolved volute fluid channel surfaces cannot be machined due to inaccessibility. A potential post-process technique to mitigate this effect is using Abrasive Flow Finishing (AFF) [490]. This technique involves sealing the volute structure, and forcing an abrasive, highly viscous media through the fluid channel to reduce surface roughness.

The exact operating conditions associated with the flight weight LOX and Kerosene pump designs are still being determined through iterative testing. This presents challenges to performing accurate and insightful structural simulations of component operation. For the volute structure TO processes, a deeper understanding of the operational conditions would improve the degree to which TO could be used to minimise volute mass. In addition, TO analyses employing CFD simulation input data may enable significant volute fluid path cross-section optimization.

The AM process contraction effects observed in the volute structures evaluated here could also decrease the performance of pump components. Tailored CFD analyses incorporating as-built flow domains could be used as a means of quantifying the extent of any degradation incurred. If performance is shown to be significantly compromised, DfAM methods such as build simulation and geometry pre-distortion could be used to mitigate these effects.

The DfAM techniques used for changing the build orientation of the volute structures in the 2nd and 3rd iterations led to a clear increases in design conformance. However, further CFD analyses and AM process evaluation could be performed on different volute build orientation angles to assess both the pump performance, and the associated cost of manufacturing.

Impellers

The impact of impeller blade geometry warpage and surface roughness on pump performance is unknown, and requires further consideration. Both of these effects could be characterised and potentially mitigated by performing AM process sample tests on impeller blades. In this regard, one suitable method might be to

manufacture single blade geometries through the applicable AM process, and to characterise them through high-resolution MicroCT techniques for both surface roughness and warpage.

Blade warpage reduction could be mitigated by utilizing AM process build simulations to generate pre-distorted geometries for subsequent AM and geometry conformance evaluation via MicroCT analysis. To mitigate surface roughness, samples could be subjected to different surface finishing processes employing varying abrasive media, and characterised through MicroCT-based surface roughness analysis. The results of these analyses would shed light on the degree of quality enhancement that could be achieved using such mitigation techniques.

Additional structural, dynamic, and CFD analyses could also be performed on the impeller components which could further inform engineers on potential design changes, and optimization techniques applicable to the impellers. Similarly to the volute components, TO techniques could be applied to reduce the mass of the impeller components. However, further-refined operational conditions for both the LOX and Kerosene pump configurations would be needed to adequately perform optimizations.

5. INJECTOR FLANGE DESIGN FOR ADDITIVE MANUFACTURING AND TESTING

5.1. Introduction

During the past few years, additive manufacturing has gained traction as a mainstream production method. The large-scale deposition of polymers has made particularly strong progress. Currently, metal parts can be manufactured using commercial AM devices with volumes on the order of several cubic feet, compared to hundreds of cubic feet for polymer parts [491]. To achieve large-scale metal AM effectively, many factors must be considered such as feedstock availability, robustness of the process, quality amongst others.

Commercially available metal AM systems fall into two main categories, powder bed fusion (PBF) devices and directed energy deposition (DED) devices. DED systems have great potential for large scale AM applications due to the large build volumes, and fast deposition rates of up to 80 cm²/h [491]. However, DED systems typically have poor resolution when compared to PBF systems. While systems such as laser powder bed fusion (L-PBF) realize higher resolution components compared to DED, these systems are limited by slower deposition rates (roughly 16 cm²/h) and smaller build volumes (typical systems are roughly 250x250x300 mm) [492].

L-PBF machines have recently seen much development with newer, larger and faster machines being developed. Technologies such as multi-laser systems for increased deposition rates, and vastly increased build volumes have enabled much larger components to be feasible on L-PBF systems. The most prominent example of a commercial large-scale L-PBF machines is the AMCM M 4K which has a build volume of 450x450x1000 mm with an optional quad-laser configuration [493]. While these larger PBF based AM systems are available, the associated initial costs are substantial both for the machine cost, and the amount of powder required to fill the build volume.

Large-scale components built using AM techniques can contain defects having a variety of sources. While the common AM based defects such as porosity, cracking, delamination, oxidation and surface finish defects can all occur on larger scale AM-built components, thermal warpage is a particular defect which can be exacerbated on larger, further overhanging structures.

The use of non-destructive testing (NDT) to observe the quality of large-scale metallic AM components presents many challenges. Common NDT techniques used are MicroCT based systems, which are widely available. These systems typically are limited by the X-ray source power, and larger component designs made from denser materials could produce poor scan quality due to insufficient X-ray penetration [494]. Higher power MicroCT systems have seen much development recently, and commercial systems are available for larger, denser components.

For the purposes of the SAFFIRE programme, a variety of potential large-scale AM applications are under consideration, such as the combustion chamber and nozzle casing, vacuum engine extension nozzle, regeneratively cooled combustion chamber and nozzle, thrust vector control attachment band, as well the injector flange. These components all have specific manufacturing requirements and while some of them can be manufactured using L-PBF techniques, DED techniques are needed for the larger components which, at present, would not fit in any commercially available L-PBF system. Due to the lack of knowledge on specific defects

which manifest in large-scale AM processes, the injector flange, which is the smallest component of those listed above, was chosen as an initial AM project to gain insight on this topic.

This chapter outlines the AM development cycle used to manufacture an injector flange component to the required tolerances needed for a functional part. Design for additive manufacturing (DfAM) techniques were applied to mitigate potential defects and improve component quality. The component was then characterised using NDT techniques on a high-power MicroCT system, and defects were investigated further.

5.2. Design and Methodology

5.2.1. Injector Flange Design

The injector flange component's primary purpose is to create a metallic interface between the SAFFIRE engine's ablative composite combustion chamber and its metallic propellant injector. Figure 5-1 shows the assembly of the ablative combustion chamber and nozzle joined to the injector flange. The flange essentially transfers gas expansion thrust to the injector, and acts as a sealing interface between the combustion chamber and the injector.

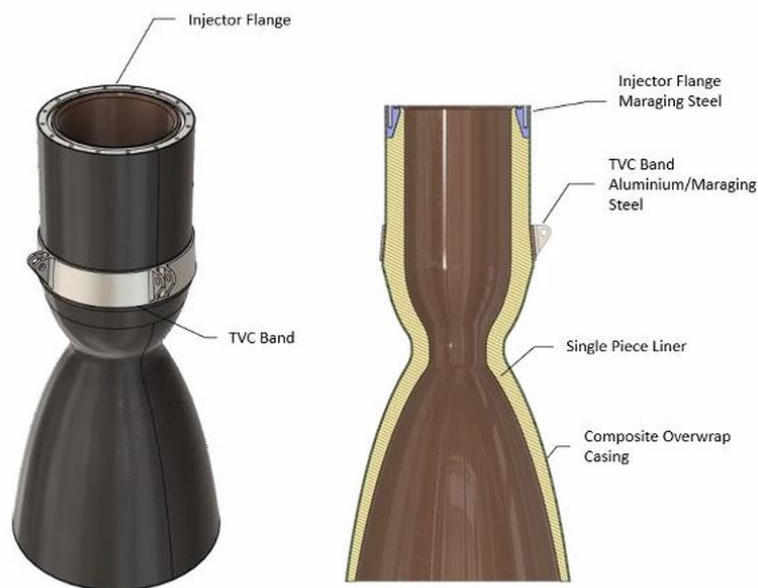


Figure 5-1: SAFFIRE composite overwrap combustion chamber and nozzle design with annotated injector flange.

As shown in Figure 5-1, and more specifically in Figure 5-2, the injector flange interfaces with three specific components, namely the injector, the ablative liner and the composite overwrap. These interfaces all have different bonding conditions based on the materials associated with each pair of mating components. The injector seal interface is a metal-metal interface, which requires precision machining and grinding for adequate sealing. The composite overwrap is interfaced with the injector flange via an adhesive bond as well as the mechanical interference arising from the circumferential ribs protruding from the injector flange, while the ablative liner is lightly bonded to the injector flange with an elastomeric adhesive. Both the composite overwrap and ablative liner are fibre-reinforced polymers (FRPs), with the ablative liner being a silica-phenolic composite, and the composite overwrap being a glass-epoxy composite.

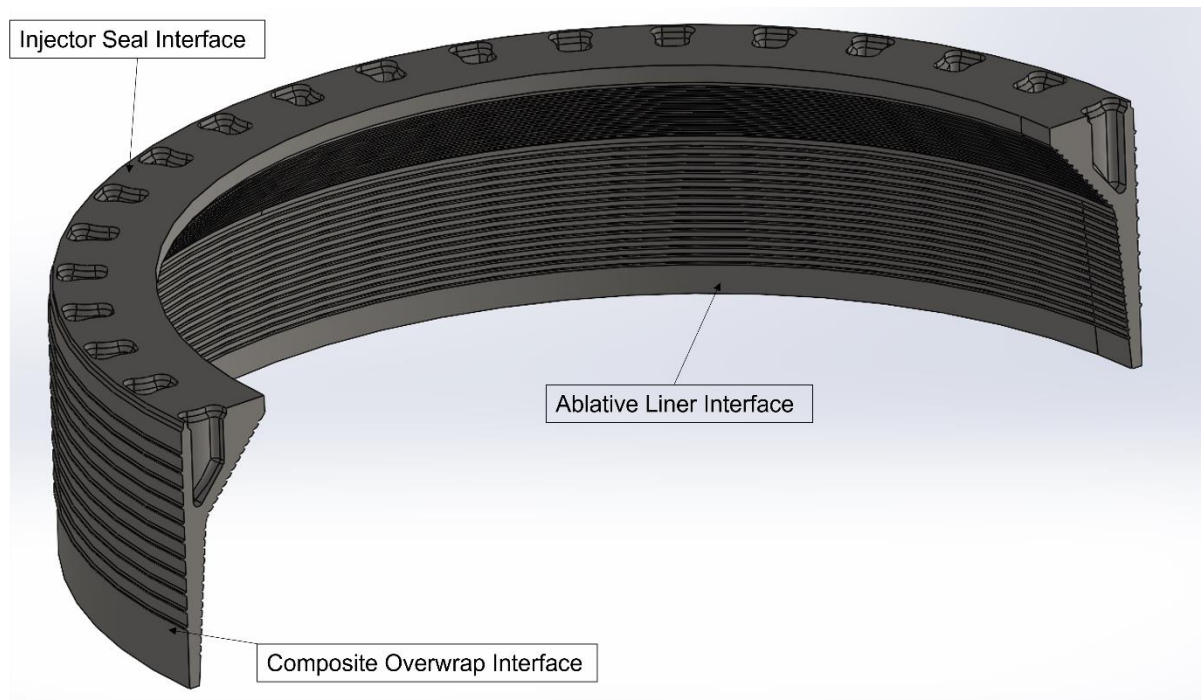


Figure 5-2: Slice view of CAD design for injector flange with interface conditions annotated.

For FRP and metal parts being combined in composite applications, various challenges arise. Due to the different properties and manufacturing methods of the two materials, it can be challenging to establish effective bonding between them, and careful considerations need to be made concerning corrosion compatibility, strain rate matching and surface adhesion. Various joining methods exist for metal to FRP parts such as bolts, rivets, press-fit joints and adhesive bonding. Adhesive bonding presents an effective lightweight method for joining, however, various factors strongly influence joint strength, such as overlap-length, joint-end configurations, fibre-orientation and surface preparation [495]. Surface preparation relates to creating surface conditions on the metal joint surface to allow for improved adhesion, and is an effective method to improve joint strength. Prominent metallic surface preparation approaches include the intentional roughening of surfaces, and the inclusion of protruding surface features such as pins or ribs which penetrate the fibre layers without fibre destruction.

Using AM techniques to manufacture components designed for direct surface adhesion to FRP provides various benefits for joint strength [496]. Metal AM techniques typically produce components with inherently rough surfaces, which enables good surface adhesion to FRP. Furthermore, intricate protruding surface features such as pins or ribs can easily be manufactured using AM for little to no extra cost compared to traditional manufacturing techniques. Both of these surface preparation techniques were leveraged in the injector flange design, with FRP contact surfaces being left in the as-built surface condition, and featuring protruding rings as seen in Figure 5-2.

The injector flange was designed to be built on an EOS M280 L-PBF system using MS1 Maraging Steel, with full material strength subsequently being achieved according to the following heat treatment cycle: stress-relieving at 940 °C (1724 °F) for 2 hours followed by age-hardening at 490 °C (914 °F) for 6 hours. In the age-hardened state, MS1 exhibits very high hardness which makes for difficult machining. Post-processing operations such as machining therefore take place before the age-hardening process. Before the AM process, standard DfAM methods were applied using Autodesk Netfabb 2023.0 in order to characterise the potential defects, mitigate defect

generation, repair geometry and determine the optimal build orientation. After performing a standard “downskin/upskin analysis”, no surfaces contained substantially overhanging regions in the standard build orientation, and no further alterations were therefore needed.

5.2.2. Build Simulation

To characterise potential defects which may arise during the AM build process, build process simulations were performed on the injector flange component. All build simulations reported on in this section were conducted using the Ansys Additive 2021 R1 software package. Employing the nominal CAD geometry for the injector flange, build simulations were run using the parameters shown in Table 5-1.

Table 5-1: Injector flange initial build simulation input parameters.

Analysis Type	Voxel Size (mm)	Voxel Sample Rate	Material	Strain Scaling Factor	Strain Warning Factor
Assumed Uniform Strain	0.28	5	EOS MS1	1	0.8

These parameters were setup using the highest resolution possible on a workstation PC with 128 GB RAM (higher resolution is possible with higher RAM). The analysis type that was employed, Assumed Uniform Strain, is one of many simulation types available on the Ansys Additive 2021 R1 platform. The other types available are Scan Pattern and Thermal Strain simulations. Thermal strain simulations in particular are more accurate than assumed strain, but require full process parameter settings from the AM service provider. In this case, EOS processing parameters are proprietary and can only be used for build simulations on a PC with an EOS Print license enabled. Due to unavailability of this license, the assumed uniform strain analysis was used for all simulations reported on here.

The results of the initial build simulation showed no potential defects from high strain regions, potential blade crash severity or any other defects. The thermal warpage, as shown in Figure 5-3, is substantial, with the entire upper ring section contracting cylindrically towards the centre of the component by a maximum magnitude of 0.72 mm. The plot presented in Figure 5-3 (a) shows displacement in the x-direction, while (b) shows the y-direction displacement. The resultant displacement of these two directions shows relatively uniform cylindrical contraction towards the centre of the component, and this displacement only varies along the height of the component (z-direction).

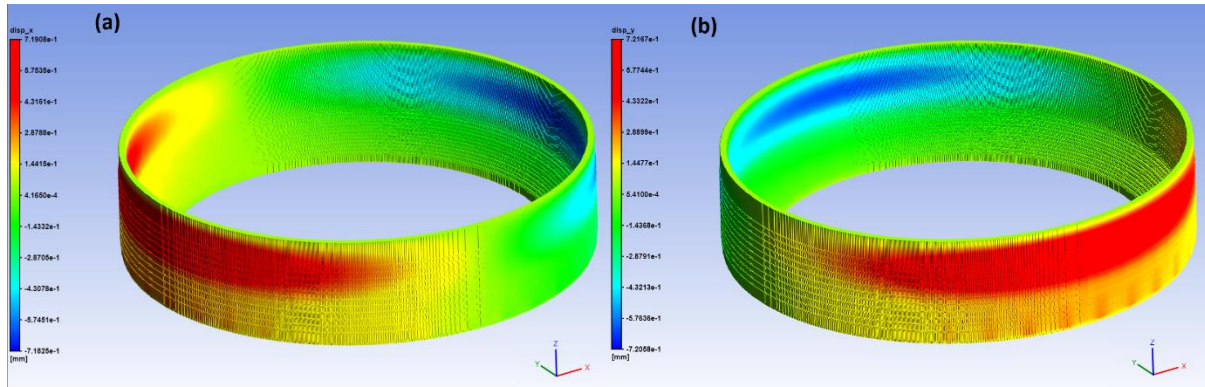


Figure 5-3: Injector flange initial model assumed strain build simulation displacement results. Displacement results in x-direction (a) and displacement results in y-direction (b), z-axis build direction.

The final configuration of the injector flange was intended to have as-built surfaces on all interfaces except for the injector interface. As shown in the analysis above (Figure 5-3) these as-built surfaces were predicted to deviate significantly and were of particular concern considering the tolerances required for the selected surface preparation techniques. This deviation of roughly 0.72 mm is particularly high with respect to the filament wound composite combustion chamber casing. This deviation reduces the concentricity of the wound composite layup, which would have to be considered in the composite design process.

There are two potential mitigation strategies for these deviations. Both the wound composite casing thickness and the ablative liner thickness could be adjusted or machined to match the deviation of the injector ring. Alternatively, another mitigation strategy, which was used here, relies on build simulations to produce a pre-distorted geometry model, which will distort toward the nominal CAD geometry during the AM process. In this approach, the accuracy of the build simulation is calibrated using samples and measurement techniques, allowing for more accurate build simulations to be performed, and subsequently, an appropriate pre-distorted geometry to be generated.

Small-Scale Sample and Build Simulation Calibration

Distortion calibration processes are highly recommended for ANSYS Additive simulations so that the simulation is best suited to the specific manufacturing scenario. Due to differences between manufacturers and suppliers of L-PBF machines and powder materials, actual distortion values can vary across different machine and material combinations. The guidelines provided by the Ansys Additive Calibration Guide [497] were generally followed in this study. In broad terms, the associated process comprises several steps:

1. An easy-to-build calibration component with comparable distortion to the final component is chosen or designed
2. The calibration component with the same AM processing parameters, and post-processing as the final component is manufactured
3. The distortion of the component at locations of interest is physically measured
4. Build simulations are run on this calibration component using different selected input parameters
5. Simulated distortion values in the same locations of interest as the physical measurements are obtained
6. The physically obtained distortion is compared to the simulated distortions using the Ansys Calibration Sheet

7. Simulation input parameters are iterated if the error is large
8. Custom material groups are created in Ansys Additive containing the iteratively optimized input parameters

The specific calibration component used for this study was a 1:0.15 scale version of the injector flange. The Ansys Additive Calibration guide suggests that using full-scale sections of components is the optimal calibration sample type. However, due to the size of the full-scale injector flange, the costs of building a sample would have been prohibitive and therefore a small-scale version was chosen. Since the initial distortion observed was to be correlated to the cylindrical features present on the component, the use of a similar cylindrical component was therefore ideal.

For optimal calibration, the specific AM process used for the small-scale sample must match the full AM process intended for use on the final, full-scale component. The AM process applied in this study employed the EOS M280 system using the standard MS1 material processing parameters. Heat treatment was undertaken according to the above-mentioned cycle.

Another potential difficulty for the calibration process is the physical measurement of the distortions of the small-scale calibration component. In this study, the "physical" measurements were made using MicroCT techniques, which enables data from all distortions to be compared to the same distortion results from build simulations. The MicroCT system used here was the system described in [458], and optimal CT parameters were determined through the methods outlined in [459]. The MicroCT parameters used for the small-scale sample are shown in Table 5-2. The MicroCT analysis methods applied were similar to those described in Section 3.6 and 4.3, where these scan data were imported into Volume Graphics VGStudioMax 3.5, processed using filtering methods, the surface was determined, the CAD geometry was imported and a "nominal/actual analysis" was performed to acquire surface deviation data. This data were then exported for further analysis.

Table 5-2: Small-scale injector flange sample MicroCT parameters.

Sample Dimensions	Voltage (kV)	Current (μ A)	Voxel Resolution (μ m)	Beam Filter Size (mm)	Image acquisition time per image (ms)
29x29x7 mm	160	90	16	0.5	500

Following the manufacturing and MicroCT analysis of the small-scale calibration sample, the sample build was simulated in Ansys Additive 2021 R1 using the Assumed Strain simulation type. For the calibration process for Assumed Strain simulations, the only input parameter that varies is the strain scaling factor (SSF). For the purposes of this calibration process, three build simulations were performed with iteratively obtained SSF values of 1, 1.058 and 1.066. The build simulation parameters used were the same as those used in the initial full-scale build simulation (Table 5-1), except for a different voxel size of 0.042 mm. This voxel size was the minimum size achievable within the computing power constraints.

Both these MicroCT data and the build simulation data sets were extracted from their respective platforms and analysed further in Paraview 5.9.1. The distortion results associated with both of these data sets were found to mimic the locations of distortions observed in the initial full-scale build simulation, shown in Figure 5-3. To enable equivalent positional measurements of the distortion portrayed in both the MicroCT and build simulation

data, both data sets were reduced to only include data points in the location range of 6.05 – 6.55 mm z-direction distance from the base of the sample (indicated by the white outline in Figure 5-4). This specific range was selected to gather data on the region where the maximum distortions occurred, and to eliminate possible errors from data near the base of the component. The average distortion is then calculated in this range for both data sets to further reduce possible errors in the measurement technique. These average distortion values are then used to iteratively calibrate the SSF build simulation input parameter as shown in Table 5-3.

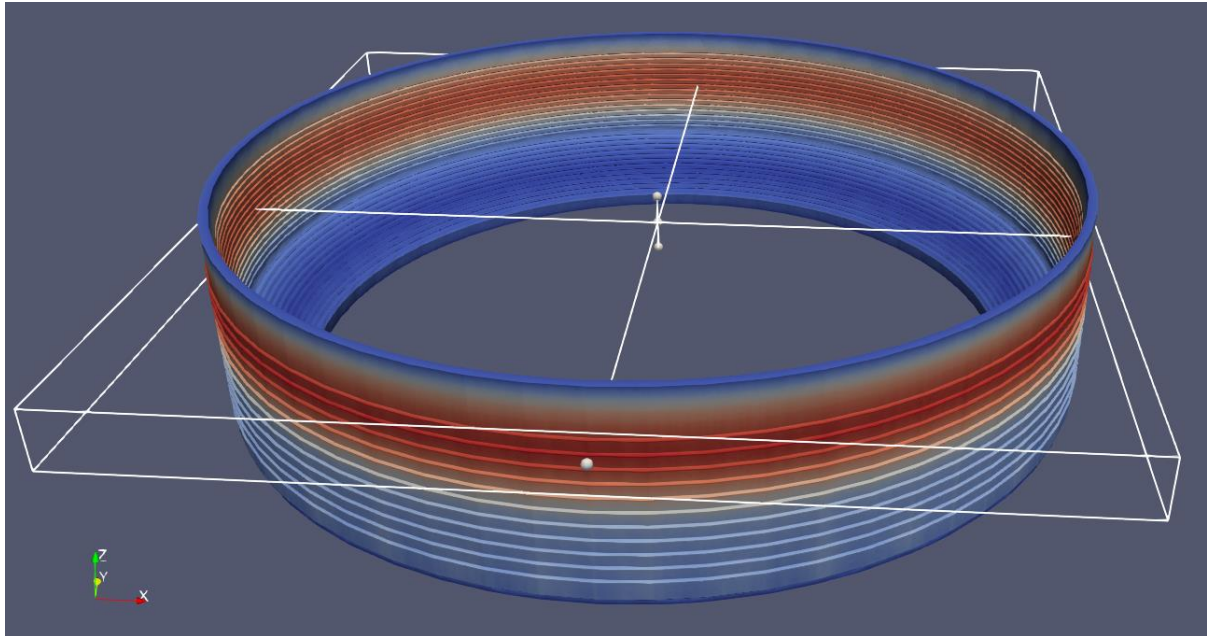


Figure 5-4: Small-scale injector flange build simulation showing distortion results and region (white box selection) employed for calibration measurements.

Table 5-3: Ansys Additive distortion calibration results.

MicroCT derived average distortion (mm)		0.0320989		
Iterative Build Simulation Results				
Simulation iteration	Average distortion (mm)	Simulation Settings	New Setting	Error (%)
		SSF	SSF	
1st	0.0310853	1.000	1.033	3.2%
2nd	0.0318682	1.058	1.066	0.7%
3rd	0.0320825	1.066	1.067	0.1%

As indicated, an SSF parameter of 1.066 yielded a 0.1% error in distortion measurement when compared to the MicroCT derived distortion. According to the Ansys Calibration Guide, this error value is below the threshold for acceptable iterations, and the final “new” SSF setting of 1.067 was taken to be appropriate for this specific AM process, material and geometry combination.

Calibrated Build Simulation and Geometry Pre-Distortion

Using the above-defined build simulation calibration parameters, an updated material model was saved in Ansys Additive to include the SSF value of 1.067. This material model was then used for a build simulation of the full-scale injector flange component. The build simulation parameters used here were identical to those presented in

Table 5-1, with the exception of the updated SSF value. After processing the distortion data from this updated build simulation model, the maximum deviation, and averaged cumulative distributions of distortion are obtained, which are shown in Table 5-4.

Table 5-4: SSF 1.067 build simulation distortion data.

	<i>Maximum Deviation (mm)</i>	<i>50% Cumulative Distribution (mm)</i>	<i>90% Cumulative Distribution (mm)</i>	<i>95% Cumulative Distribution (mm)</i>	<i>98% Cumulative Distribution (mm)</i>
<i>SSF 1.067 Build Simulation</i>	0.413	0.0888	0.342	0.357	0.381

During build simulations, Ansys Additive has an optional output parameter for a “Distortion Compensated STL file”. Once a build simulation has been completed, this output generates a mesh geometry file of the component in pre-distorted form. This pre-distorted geometry represents the geometric model which, when simulated with the same build simulation parameters, would result in a final geometry identical to the nominal CAD geometry. This pre-distorted geometry can then be manufactured on an AM system with the intention of mitigating the distortion or thermal warpage expected to occur during the AM process.

5.2.3. Full-Scale Injector Flange NDT Process

Due to the scale and density of the full-scale injector flange component, standard, lower-power MicroCT systems would not provide adequate X-ray penetration to achieve accurate results. Therefore, high-power MicroCT systems were used to obtain the relevant scan information. The systems used were the YXLON UX50 and the YXLON FF85, which are both capable of analysing large, dense components at X-ray voltages up to 450 kV. Achieving high scan quality on high-power systems such as these can be challenging, especially due to the large bounding box of the injector flange (200x200x100 mm). To achieve good scan quality, multiple scans were performed using different CT parameters and the scan quality was assessed. Furthermore, higher resolution scans were performed on specific regions of interest to investigate specific defects with higher quality. These higher resolution scans were performed using a different scan orientation, as shown in Figure 5-5 (right).

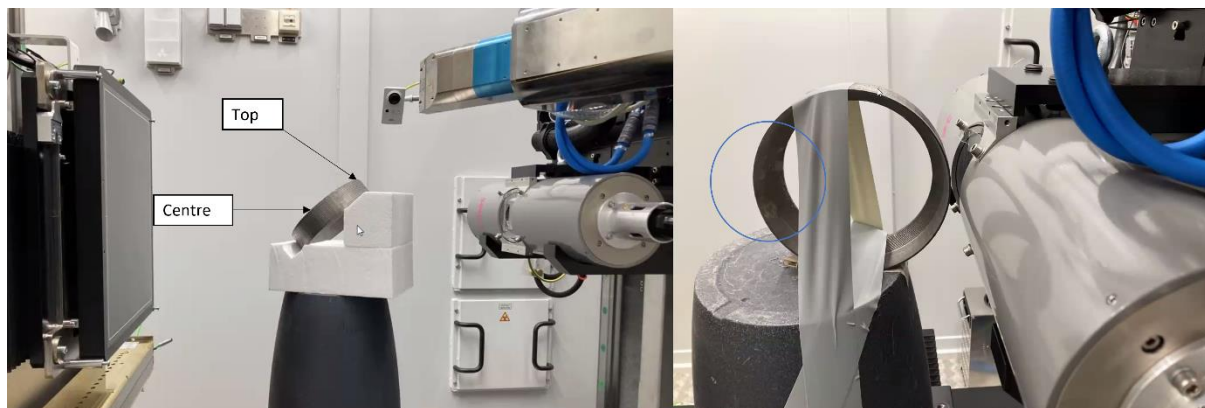


Figure 5-5: Full-scale injector flange mounted in YXLON UX50 MicroCT system. Full scan of whole component (left) with high-quality (centre) and low-quality (top/bottom) regions annotated. Higher resolution scan of region of interest (right) annotated by blue marking.

The CT parameters used for all scans performed are listed in Table 5-5. The specific quality achieved using these different CT parameters varied highly; scan number 2 achieved the best overall quality, and scan number 1 displayed specific defect regions at even higher resolutions. The scan quality achieved for scan 2 was dependant on the orientation of the component mounting in the MicroCT system (Figure 5-5 (left)). This scan's quality was significantly higher in the centre section.

Table 5-5: Full-scale injector flange MicroCT parameters for both the FF85 and UX50 systems.

<i>MicroCT system used</i>	FF85		UX50	
<i>Scan Number</i>	1	2	3	4
<i>Voltage (kV)</i>	450	450	450	450
<i>Current (μA)</i>	222.22	555.55	1400	1400
<i>Voxel Resolution (μm)</i>	150	150	150	150
<i>Sensitivity</i>	12	6	6	6
<i>Additional Parameters</i>	Continuous Cone Beam, 60 μm focal spot	Continuous Cone Beam, 250 μm focal spot	Stop and go Cone Beam, Scatter fix method, 400 μm focal spot	Stop and go Cone Beam, 400 μm focal spot

The difference in scan quality is due to the X-rays needing to pass through thicker sections of material in the top/bottom regions compared to the centre region, at all angles of orientation. Figure 5-6 shows the relative scan quality of scan 2 in it's best (a) and worst case (b) locations, as well as the relative quality of scan 1 (c). Scan 1 shows much more detail and less X-ray scattering artifacts, and this scan was therefore used where higher detail was required. While its scan quality was found to be much poorer in the thicker regions of the injector flange, scan 2 displays acceptable levels of quality in the thin-wall sections throughout, and was consequently used for overall surface analysis in these thin regions.

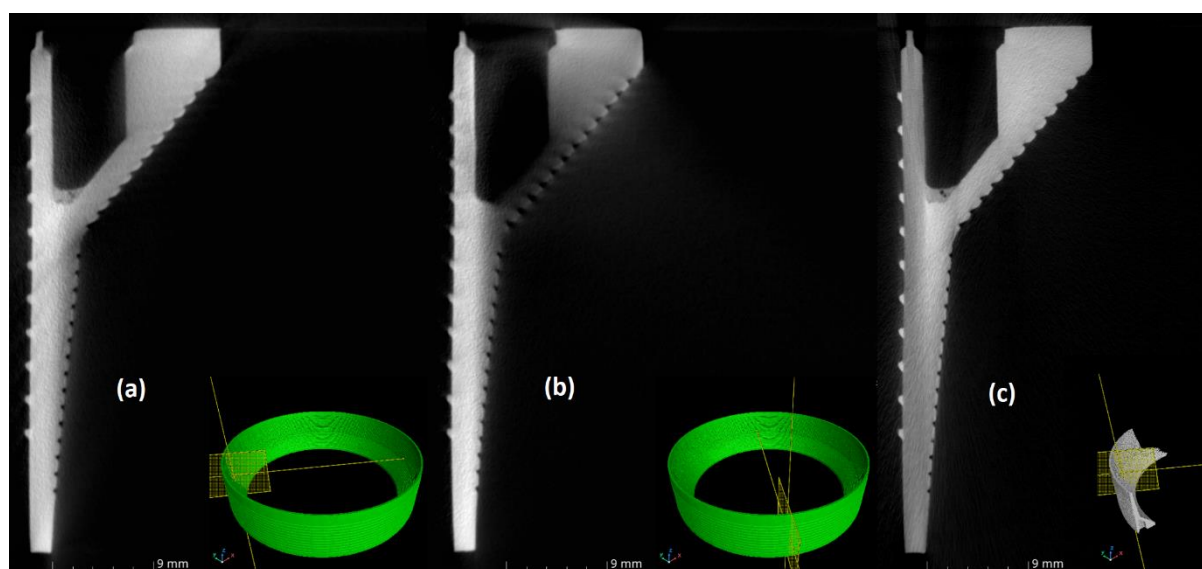


Figure 5-6: MicroCT derived slice views of injector flange scan quality for (a) scan 2 best case quality, (b) scan 2 worst case quality and (c) scan 1 quality.

Following MicroCT processes, the scan data sets were imported into Volume Graphics VGStudioMax 3.5 and processed using filtering methods to achieve the highest quality, dependant on the scan type. Similarly to the methods described in Section 3.6, accurate surface determinations were acquired using the CAD model surface determination method. This method specifically uses the imported nominal CAD model as a baseline for surface determination processes, and the surface is further optimized based on manual image analysis methods dependant on the quality of each scan. Once an accurate surface was determined, both qualitative and quantitative data were obtained for specific defect analyses.

5.3. Results

After manufacturing and initial heat treatment processes, the injector flange component was found to contain significant surface defects quite visible to the naked eye. These defects, shown in Figure 5-7, were not present directly after the AM process, and were only observed after the stress-relieving heat treatment process. Physical measurements of these pitting defects showed that the worst-case defects were roughly 0.2 mm in depth.

These defects were likely to be the result of pitting caused by excessive oxidation during the stress relieving heat treatment process and need further investigation. The stress relieving process used an inert gas (nitrogen) purging process to remove the oxygen from the furnace before the heat cycle, with this inert environment subsequently being maintained throughout the cycle. It is possible that during the heat treatment of the full-scale injector flange, the inert gas flowrate was insufficient, resulting in a non-negligible amount of oxygen being present in the environment and subsequent part oxidation.

Of interest, these defects were not evenly distributed on all surfaces of the component (Figure 5-7 (left)), suggesting that the oxidation which seemed to have occurred was somehow localized, and that further observations were required to verify the integrity of these localized regions.



Figure 5-7: Full-scale injector flange heat treatment pitting defects showing whole component (left) and close-up view of surface defects (right).

5.3.1. Surface Defects

The pitting defects found above were characterised further using MicroCT techniques. Since sub-surface porosity can be a potential cause of these surface defects, ensuring no excessive sub-surface porosity was present in the injector flange was critical. Through qualitative analyses, no sub-surface or bulk porosity was observed in the component on any of the scans. However, due to the scan qualities achieved, and the resolutions of the scans, definitive quantitative results for porosities are not possible unless smaller sections of the component are analysed. The deviation magnitude associated with these pitting defects is represented in Figure 5-8, with the localization of defects annotated.

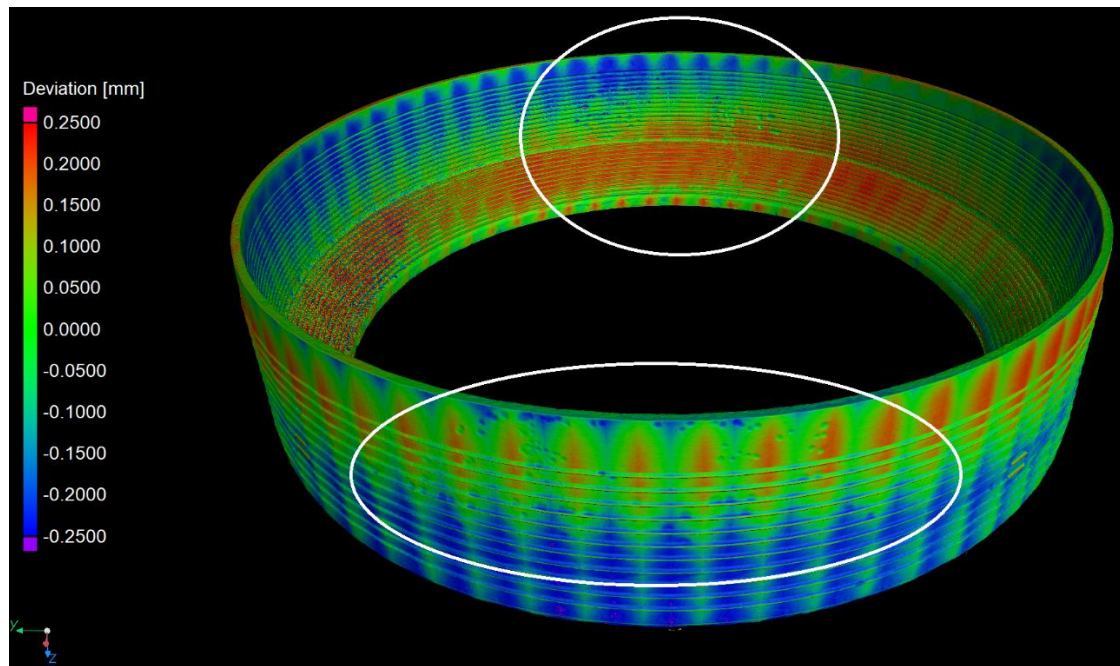


Figure 5-8: Full-scale injector flange MicroCT deviation analysis showing pitting surface defects and magnitude of deviation. The defect locations are annotated by white circles showing the localization of defects.

Further manual analyses were performed to fully characterise these defects using surface measurement tools. Figure 5-9 shows MicroCT derived slice views of pitting defect locations, with defect depths annotated. The average depth of these defects was determined to be 0.134 mm, while the worst-case defect was observed to be 0.272 mm in depth on the inner surface of the component. These results roughly correlate with the physical measurements made. While it would be good practice to mitigate them, the pitting defects are not expected to instigate part failure during operation due to their small size and the lack of observed sub-surface porosity.

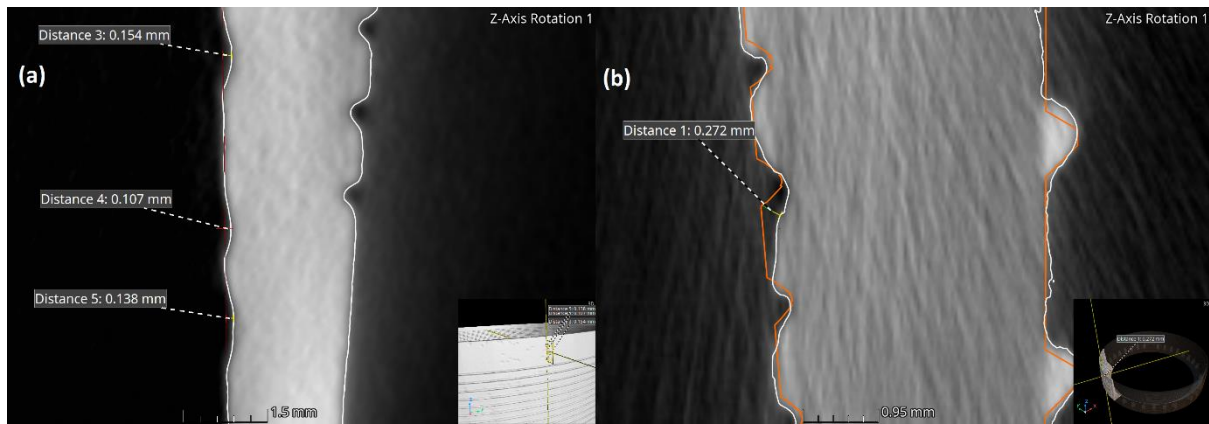


Figure 5-9: Injector flange MicroCT slice views showing pitting defect sizes quantified on both (a) outer surface (scan 2) and (b) inner surface (scan 1).

5.3.2. Trapped Powder

Trapped powder was detected in the injector flange during qualitative inspection through MicroCT techniques. As shown in Figure 5-10, this trapped, partially-consolidated powder was observed to exist at the bottom of the recessed holes. The trapped powder varied in volume across the multiple hole locations, but was found in the majority of the recesses. Since these holes would remain open after final machining, the trapped powder and/or partially consolidated metal could easily be removed using hand tools. In more complex components, where internal channels are present, removal of trapped powder is more difficult and design mitigation techniques may be required to prevent trapped powder and/or partially consolidated powder build up. Figure 5-10 shows a MicroCT slice view of the trapped and/or consolidated powder from the high-quality, local region scan (scan 1). The density observed on this slice view suggests that the powder was only partially consolidated in this recess location. It is possible that the powder was unconsolidated after the initial AM process, and that the heat treatment applied thereafter (stress-relief) resulted in subsequent partial consolidation.

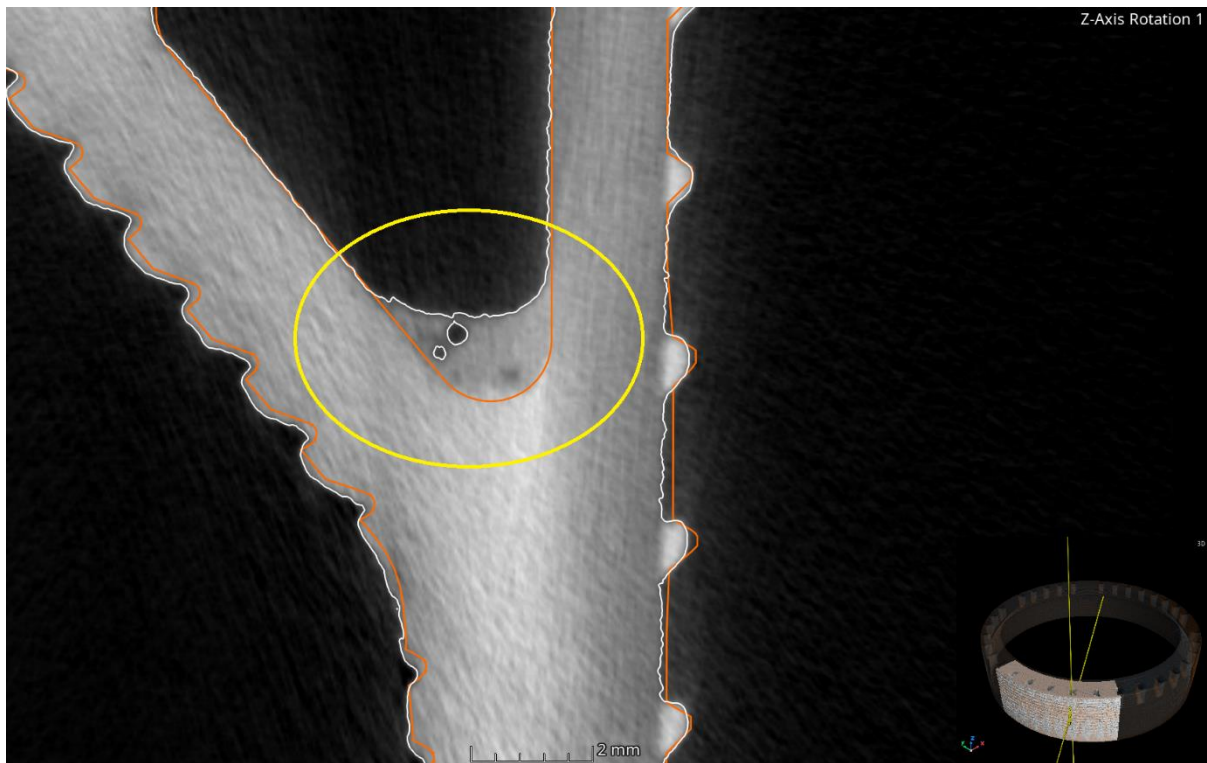


Figure 5-10: MicroCT slice view showing trapped/consolidated powder annotated by yellow ellipse. Orange lines represent nominal CAD geometry and white lines represent MicroCT scan surface.

5.3.3. Geometric Deviation

The primary defect of interest on the injector flange component was the magnitude of thermal warpage or deviation on the side surfaces. When comparing the surface deviation of the MicroCT data to the nominal CAD geometry, good quality results were found in all regions of the component. For the purpose of deviation evaluation, three regions were considered for different observational analyses. These three regions were the whole component using scan 2, the isolated region of interest (ROI) from scan 1, and the same ROI from scan 2. “Nominal/actual deviation” analyses were performed on these three regions, and both quantitative and qualitative results were obtained. The qualitative results of these analyses are illustrated in Figure 5-11, showing the scan 1 isolated region results (a), and scan 2 whole component region results (b).

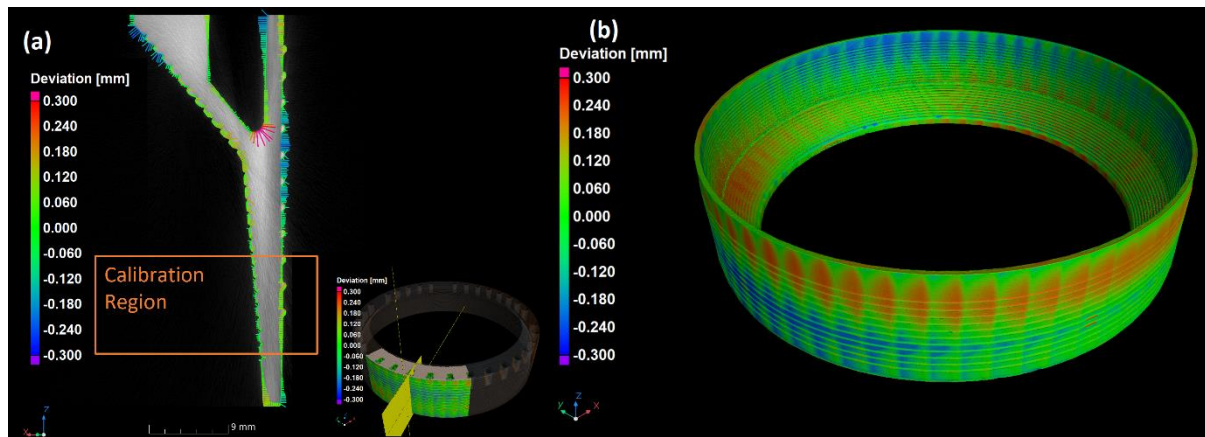


Figure 5-11: Nominal/actual deviation analyses results for both (a) scan 1 isolated region and (b) scan 2 whole component region. Equivalent region used in build simulation calibration annotated by orange box (a).

The geometric deviation observed, as shown in Figure 5-11 (a), was found to be higher in the thicker sections of the component, and lower in the thin section. This differs from the warpage location results from initial build simulations as shown in Figure 5-3, where the highest warpage was observed in the thin sections. This suggests that the build simulation pre-distortion DfAM techniques applied were successful, and that the overall distortion was reduced and evenly distributed. Furthermore, these deviation results clearly show that the distortion in the part is lowest in the region annotated in Figure 5-11 (a). This region closely correlates with the region used for average distortion measurements for build simulation calibration, which is shown in Figure 5-4. This suggests that while the build simulation calibration and pre-distortion procedure greatly reduced the distortion in the calibration region, other regions which are not in the calibration region do not have as much reduction in distortion.

Further quantitative analyses were performed on the deviations of all surfaces of the component, indicating the achievement of high quality results, with the minimum, maximum, and the averaged cumulative distributions of deviations shown in Table 5-6. Also shown is the initially predicted deviation results from the calibrated build simulation. In comparison to the build simulation results, which are not pre-distorted, the actual scanned results show much lower measured deviation, particularly in the 90%, 95% and 98% cumulative distributions.

The scan 1 ROI shows the highest and lowest deviation magnitudes observed for the part. This is primarily caused by the high-quality characterisation of the trapped/consolidated powder, and the deviation of the “surfaces” of this partially consolidated powder. In contrast, due to the lower scan quality and the surface determination techniques used, scan 2 does not correctly characterise these partially consolidated powder surfaces, and resultant deviations are lower. This, combined with the lower scan quality of scan 2, may show more accurate results from the lower bound average cumulative distributions ($\leq 50\%$) for scan 1, and the upper bound average cumulative distributions ($\geq 50\%$) for scan 2. For a relatively large AM component, these results are of very high quality with very minimal deviation occurring.

Table 5-6: Quantitative deviation results from each scan in the respective regions of interest (ROI), and predicted deviation results from calibrated build simulation.

	<i>Minimum Deviation (mm)</i>	<i>Maximum Deviation (mm)</i>	<i>50% Cumulative Distribution (mm)</i>	<i>90% Cumulative Distribution (mm)</i>	<i>95% Cumulative Distribution (mm)</i>	<i>98% Cumulative Distribution (mm)</i>
<i>Scan 2 Whole Component</i>	-0.439	0.391	0.098	0.201	0.219	0.232
<i>Scan 2 ROI</i>	-0.295	0.286	0.057	0.133	0.156	0.18
<i>Scan 1 ROI</i>	<-0.5	>0.5	0.066	0.17	0.207	0.254
<i>SSF 1.067 Build Simulation</i>	N/A	0.413	0.0888	0.342	0.357	0.381

The deviation results reflected above are subject to errors associated with scan quality and the surface determination method employed. As shown in Figure 5-6, the grey value distributions in the solid sections of the component are highly irregular, which is an indication of poor scan quality. The surface determination method used for scan 2, surface determination from CAD body baseline, can mitigate the effects of uneven grey values by searching for local grey value gradients in the vicinity, but various factors can influence the accuracy of this method. When regions have small grey value gradients between the air and material interface, this CAD body baseline surface determination method is less accurate. The CAD geometry alignment procedure also effects the surface determination accuracy and, furthermore, the accuracy of this alignment heavily effects the deviation analysis results.

5.4. Conclusions and Recommendations

Component-specific DfAM techniques were applied to the SAFFIRE injector flange design. AM build simulation techniques were applied to predict the possible defects which could arise during the AM process, and excessive thermal warpage was observed to be a potential build process flaw. To mitigate these potential flaws and ensure first-time-right manufacturing quality, small-scale samples were additively manufactured and, using MicroCT techniques, tested to calibrate the accuracy of build simulation procedures. Once calibrated, high-fidelity build simulations were performed on the final component geometry for the purpose of creating a pre-distorted geometry model which, when manufactured, should contain reduced thermal warpage characteristics. This geometry was then additively manufactured and analysed using specific high-power MicroCT techniques. The distortion characteristics observed correlated with the mitigation strategies applied, which provided insight into DfAM techniques applied to larger scale AM-built components. Further defects were observed in the form of trapped or partially consolidated powder, as well as pitting defects, presumed to have arisen from excessive oxidation. These pitting defects were characterised through MicroCT techniques, and the magnitude of these defects was found to be minimal, especially when compared to the expected thermal warpage.

Recommendations for further work:

The build simulation and geometry pre-distortion technique used in this chapter showed promising results. However, for the full-scale AM-built component, no baseline component was manufactured to verify the performance of the method. For verification purposes, the pre-distortion method could be applied to a small-scale injector flange component and analysed through MicroCT techniques to evaluate the accuracy of the method. Furthermore, the build simulation calibration method used was applied only to a specific region of interest, which effectively calibrates the results of the pre-distortion to this region alone. This could be improved by manufacturing and analysing the effects of different calibration regions and pre-distortion methods. There are also higher fidelity AM build simulation methods such as Thermal Strain and Scan Pattern simulations which could yield improved results.

There remain unanswered research questions concerning the definitive cause of the pitting surface defects found on the full-scale component. This is further obscured by the localization of the pitting defects. The suspected cause of these defects is excessive oxidation occurring during the stress-relieving heat treatment cycle. Another possible cause of these pitting defects could be the presence of porosity near the surface (sub-surface porosity) of the component after the AM process. These sub-surface pores could have been closed during heat treatment processes, which results in the pores “collapsing” and deforming the surface in the form of a pit. The veracity of these hypotheses could be assessed by testing different inert gas flow rates during the heat treatment process, and analysing resulting part samples using MicroCT techniques. Another verification process could be to analyse AM-built samples which have not undergone stress-relief to check for the presence of sub-surface porosity. These pitting defects could also be characterised further through the use of superior physical measurement methods such as tactile probe techniques, which could also further verify the MicroCT results.

6. CONCLUSION AND RECOMENDATIONS

The development of a liquid propellant fuelled rocket engine is the primary goal of the South African First Integrated Rocket Engine (SAFFIRE) programme at the Aerospace Systems Research Group (ASReG). Additive Manufacturing (AM) technologies were identified to potentially enable higher performance and more cost-effective components for the SAFFIRE engine. These manufacturing techniques are relatively new and require in-depth knowledge. The primary aim of this study was to gain insight and experience in the AM-based production of liquid rocket engine components according to specified requirements.

6.1. Achievement of Objectives

The achievement of each of the research objectives set out in Section 1.4 is addressed as follows:

Objective 1: Perform extensive research into the advantages and practical use-cases for AM in aerospace components.

The first objective of the study was achieved by means of the literature review presented in Chapter 2, which was subsequently published in the Elsevier Materials & Design Journal under the title *Metal additive manufacturing in aerospace: a review* [5].

Of the AM processes under investigation, Laser Powder Bed Fusion (L-PBF) was identified as the most widely used for applications in aerospace, with its capabilities to produce fully dense components at high resolution with complex geometries, and with small to medium build volumes. Application of this manufacturing technology has resulted in cost and schedule reductions, and in certain instances, mass reduction improvements and significant levels of part consolidation.

Distinct engineering alloys for production using AM techniques were identified and categorised according to AM process type. Aluminium alloys, nickel-based alloys, and titanium alloys are just a few of the many materials available on AM platforms. Titanium alloys, in particular, are most prevalently used in the AM fabrication of aerospace components, and also constitute the most widely researched alloy group in AM. While most AM alloys have traceability to common alloys manufactured by traditional processes, many new and custom alloys are under development primarily for aerospace applications. These alloys are being developed for specific applications, and out-perform other commonly available materials in their specific applications. Besides monolithic alloys, AM is capable of creating custom bimetallic or multimetallic material solutions, adding material to a design locally to optimize thermal or structural performance.

The use of advanced Design for Additive Manufacturing (DfAM) techniques such as Topology Optimization (TO) and lattice structures was observed in some AM aerospace applications. As a result of the manufacturing freedoms provided by AM technologies, TO is particularly suited to highly efficient designs with internal components and intricate features that would not be feasible otherwise. In the near future, lattice structures are expected to be an increasingly popular topic of research for aerospace applications due to their unique characteristics, such as energy absorption, lightweighting, and heat transfer. While there are many potential advantages to using TO and lattice structures, many different mechanical, thermal, and integrated loads need to be considered, which are not always well defined in associated applications.

Objective 2: Investigate material characteristics and manufacturing processes of various AM engineering alloys

The second objective of the study was achieved by manufacturing and evaluating various additively manufactured alloys which are under consideration for components associated with the SAFFIRE programme. The manufacturing, testing and evaluation methods for this objective are presented in Chapter 3.

Porosity characteristics were measured for the Ti-6Al-4V, MS1, 316L and 6061-T6 alloys, both quantitatively and qualitatively. Percentage porosity quantities and the conditions of the porosity were measured using the non-destructive testing (NDT) technique, micro-computed tomography (MicroCT). For Ti-6Al-4V, MS1, 316L and 6061-T6, percentage porosities were measured to be 0.01%, 0.02%, 0.02% and 0.25% respectively. These measurements were in agreement with associated parameters provided by AM material suppliers. The specific conditions for porosity, such as number of pores, largest pore size and distribution of pores, were also measured. For the Ti-6Al-4V, MS1 and 316L samples, no excessively large pores were found, and pore distributions were mostly uniform. For the 6061-T6 sample, large pores were observed and the distributions of pores were concentrated at roughly 0.25 mm from all surfaces, which indicated high contour porosity.

A specific material characteristic optimization was performed with the primary aim of reducing the contour porosity in the 6061-T6 material. AM processing parameters were varied and the percentage porosity was reduced from 0.25% to 0.03% in the best case. The characteristics of these pores were also improved, showing smaller maximum pore size and more evenly distributed pores, in turn indicating lower contour porosity. This method of varying AM processing parameters and measuring porosity through MicroCT techniques was found to provide an effective means of improving material characteristics of AM-built materials where poor quality is observed. Powder quality control methods were also investigated, and a framework for reducing the effects of poor powder quality was proposed.

For the Ti-6Al-4V, MS1, 316L and 6061-T6 materials, surface roughness measurements were obtained through MicroCT on surfaces at varying inclination angles relative to the AM build plate. The surface roughness was measured on unsupported surfaces at an inclination angle of 90° (vertical surface), as well as inclination angles ranging from 55° to 25° at 5° increments. For all materials, the measured surface roughness was higher than data provided by the AM material supplier. The study showed that surface roughness increases as inclination angle decreases. The surface roughness sensitivity to inclination angles was therefore discussed for each material. Pore sizes and distributions were also measured qualitatively on the samples at lower inclination angles. The characteristics of the porosity (sizes and distributions) were similar at all inclination angles for all materials, showing that porosity is independent of inclination angle. Surface deviation results were also obtained for all inclination angles, which, for all materials, showed an increase in surface deviation as inclination angle decreased.

Objective 3: Design, manufacture and test iterations of pump components in order to assess the practicality of using AM for liquid rocket engine component fabrication

The third objective of the study was achieved through the iterative design of volute and impeller pump components, and subsequent testing methods applied as presented in Chapter 4.

Iterative prototyping and DfAM techniques were applied to improve the practicality and quality of impeller and volute pump components. Three iterations of impeller and volute sets were manufactured using AM techniques,

and defects were quantified using MicroCT techniques. Specific defects such as wall thickness thinning, swarf generation, excessive surface roughness and surface geometric inaccuracy were measured. For each progressive iteration, the defects observed in previous iterations were mitigated successfully using DfAM techniques such as build orientation optimization, support structure optimization, internal surface orientation optimization and AM build simulations, amongst other methods.

While most of the observed defects were measured qualitatively, quantitative results were also obtained for all impeller and volute component iterations. MicroCT techniques were used to obtain maximum and minimum surface deviation results, and averaged surface deviation results were calculated using cumulative distributions. For the volute components, which had a large amount of DfAM modifications applied, these surface deviation results generally improved across the progressive iterations. For the impeller component iterations, larger surface deviations were measured on the 3rd iteration component in the impeller blade regions. This phenomenon is believed to be due to the presence of lower inclination angles on the 3rd iteration impeller, and mitigation strategies were proposed to alleviate these deviations.

Besides the defect mitigation and surface deviation reduction techniques used, AM specific post-processing methods were employed for the impeller and volute components. These methods were predominantly used to improve the AM process output, reduce post-processing requirements and improve final component quality after post-processing. TO techniques were used on the 3rd iteration volute to improve design aspects of the volute, while leveraging the design complexity permitted by using AM methods. The generated TO structure, when observed through MicroCT techniques, was found to add no further defects to the component.

Objective 4: Investigate and optimize the production of an injector flange component within the framework of AM process characteristics

The last objective of the study was achieved through the design, manufacture and evaluation of the injector flange component as presented in Chapter 5.

Due to the scale and cost of the injector flange component, extensive DfAM methods were applied to the design before manufacture to improve part quality. In particular, AM build simulations were employed to reduce the geometric distortion in the as-built component. A small-scale, geometrically similar injector flange was manufactured and analyzed using MicroCT techniques. These analyses provided insight into the nature of the distortion, and the distortion data was used to calibrate the full-scale injector flange build simulation. Using the calibrated build simulation method, an accurate pre-distorted geometry model was built.

After manufacturing the pre-distorted part, high-power MicroCT methods were used to assess the geometric accuracy of the component, validate the build simulation results, and identify defects. In comparison to the calibrated build simulation deviation results, the measured deviation of the full-scale component was much lower, with 98% of deviation at ≤ 0.232 mm compared to ≤ 0.381 mm predicted by the build simulation. For an AM-built component of this scale, these distortion results are of very high quality. Defects in the component were investigated both qualitatively and quantitatively, and no excessive pores or cracks were observed. Both pitting and trapped/consolidated powder defects were identified, analysed and discussed. The magnitude of these defects was not of concern in terms of the integrity of the component, and defect mitigation strategies were presented to reduce the occurrence of such defects in future components, if necessary.

6.2. Constraints and Limitations

Various constraints and limitations were imposed on this study. MicroCT scan quality is heavily influenced by X-ray absorption, which is dependant on material density and thickness. For all samples and components, the achievable scan quality influences defect identification, where smaller defects may be present but are not detected. This presents potential errors for all porosity conclusions, and small defects such as cracks could also be undetected.

For MicroCT surface-based analyses, the accuracy of measured surfaces is influenced by both the surface determination method used, and the scan quality. For the surface determination by CAD body baseline method, the accuracy of the measured surfaces depends heavily on the alignment procedure for the CAD geometry, which presents potential measurement errors. The surface deviation results from the larger components presented in Chapters 4 and 5 are heavily influenced by scan quality, and higher quality data are achievable through higher power MicroCT systems, which were not available.

The AM build simulation techniques used in Chapters 4 and 5 were simplified due to licensing constraints and computing power available, and higher fidelity build simulations are possible. The calibration procedure used for the build simulations is limited to using maximum predicted and measured values. Due to the cost of the injector flange component, validating the effects of the pre-distortion method was not directly possible, since it would require manufacturing a second test component without the pre-distortion method.

6.3. Recommendations

Extensive specific recommendations are presented in each chapters' conclusion. In the general context of the SAFFIRE programme, however, overall recommendations are presented here.

It is clear that AM techniques provide various advantages compared to traditional manufacturing techniques, especially for aerospace applications. AM-built components should be considered for all future SAFFIRE engine components, where feasible. While the clear advantages that AM technologies present are evidenced by numerous successful application instances, material characteristics and responses are not always well-known. As such, the material testing methods described and discussed in Chapter 3 should be applied for any materials under consideration in the SAFFIRE programme. Developing an understanding of other AM material characteristics would provide higher degrees of certainty concerning the use of AM in the fabrication of SAFFIRE engine components.

Due to the relative novelty and complexity achievable through AM methods, the DfAM methods available are vastly underdeveloped. New design techniques such as Topology Optimization and lattice structures are expected to increase in use as these methods are further refined, and future component applications could feature performance benefits using these methods. The DfAM methods used throughout this thesis show clear benefits in production quality, and further application of these methods is recommended. While the DfAM methods used in Chpaters 4 and 5 may be applied in similar future components, most likely, different methods should be tested due to DfAM being highly dependant on the specific component design.

7. REFERENCES

- [1] J.R. Behrens, B. Lal, Exploring Trends in the Global Small Satellite Ecosystem, *New Space*. 7 (2019) 126–136. <https://doi.org/10.1089/space.2018.0017>.
- [2] E. Sacco, S.K. Moon, Additive manufacturing for space: status and promises, *The International Journal of Advanced Manufacturing Technology*. 105 (2019) 4123–4146. <https://doi.org/10.1007/s00170-019-03786-z>.
- [3] P. Kit, Jupiter Orbit Insertion - Press Kit, 2016. https://www.jpl.nasa.gov/news/press_kits/juno/.
- [4] RocketLab, Rutherford Engine, (n.d.). <https://amfg.ai/2019/08/28/application-spotlight-3d-printed-rockets-and-the-future-of-spacecraft-manufacturing/> (accessed February 2, 2021).
- [5] B. Blakey-Milner, P. Gradl, G. Snedden, M. Brooks, J. Pitot, E. Lopez, M. Leary, F. Berto, A. du Plessis, Metal additive manufacturing in aerospace: a review, *Mater Des*. 209 (2021) 110008. <https://doi.org/10.1016/j.matdes.2021.110008>.
- [6] The Business Research Company, Aerospace Global Market Report 2021: COVID 19 Impact and Recovery to 2030, 2021. https://www.reportlinker.com/p06018925/Aerospace-Global-Market-Report-COVID-19-Impact-and-Recovery-to.html?utm_source=GNW.
- [7] B.N. Bhat, ed., *Aerospace Materials and Applications*, American Institute of Aeronautics and Astronautics, Inc., Reston ,VA, 2018. <https://doi.org/10.2514/4.104893>.
- [8] R.R. Boyer, J.D. Cotton, M. Mohaghegh, R.E. Schafrik, Materials considerations for aerospace applications, *MRS Bull*. 40 (2015) 1055–1066. <https://doi.org/10.1557/mrs.2015.278>.
- [9] Market Research Future, Global Aerospace Additive Manufacturing Market Research Report – Forecast 2016-2021, 2019. <https://www.marketresearchfuture.com/reports/aerospace-additive-manufacturing-market-1551>.
- [10] D. Herzog, V. Seyda, E. Wycisk, C. Emmelmann, Additive manufacturing of metals, *Acta Mater*. 117 (2016) 371–392. <https://doi.org/10.1016/j.actamat.2016.07.019>.
- [11] A. Angrish, A critical analysis of additive manufacturing technologies for aerospace applications, in: 2014 IEEE Aerospace Conference, IEEE, 2014: pp. 1–6. <https://doi.org/10.1109/AERO.2014.6836456>.
- [12] A.A. Shapiro, J.P. Borgonia, Q.N. Chen, R.P. Dillon, B. McEnerney, R. Polit-Casillas, L. Soloway, Additive Manufacturing for Aerospace Flight Applications, *J Spacecr Rockets*. 53 (2016) 952–959. <https://doi.org/10.2514/1.A33544>.
- [13] J.C. Snyder, K.A. Thole, Effect of Additive Manufacturing Process Parameters on Turbine Cooling, in: Volume 5B: Heat Transfer, American Society of Mechanical Engineers, 2019. <https://doi.org/10.1115/GT2019-90862>.

- [14] F. Kerstens, A. Cervone, P. Gradl, End to end process evaluation for additively manufactured liquid rocket engine thrust chambers, *Acta Astronaut.* 182 (2021) 454–465. <https://doi.org/10.1016/j.actaastro.2021.02.034>.
- [15] M.E. Orme, M. Gschweitl, M. Ferrari, R. Vernon, I.J. Madera, R. Yancey, F. Mouriaux, Additive manufacturing of lightweight, optimized, metallic components suitable for space flight, *J Spacecr Rockets.* 54 (2017) 1050–1059. <https://doi.org/10.2514/1.A33749>.
- [16] General Electric - Additive, LEAP engine Fuel nozzle, (n.d.). <https://www.ge.com/additive/stories/new-manufacturing-milestone-30000-additive-fuel-nozzles> (accessed February 2, 2021).
- [17] Fraunhofer IWS, Annual report 2019/2020 Fraunhofer IWS, (2020). https://www.iws.fraunhofer.de/en/newsandmedia/mediacenter/annual_reports.html (accessed March 16, 2021).
- [18] S. Cellcore, Rocket Engine, (n.d.). <https://www.etmm-online.com/3d-printing-a-rocket-engine-a-886960/> (accessed February 2, 2021).
- [19] I. Yadroitsev, I. Yadroitsava, A. du Plessis, E. MacDonald, *Fundamentals of Laser Powder Bed Fusion of Metals*, 1st ed., 2021. <https://www.elsevier.com/books/fundamentals-of-laser-powder-bed-fusion-of-metals/yadroitsev/978-0-12-824090-8>.
- [20] M. Leary, D. Downing, B. Lozanovski, J. Harris, Design principles, in: I. Yadroitsev, I. Yadroitsava, A. du Plessis, E. MacDonald (Eds.), *Fundamentals of Laser Powder Bed Fusion of Metals*, 1st ed., 2021. <https://www.elsevier.com/books/fundamentals-of-laser-powder-bed-fusion-of-metals/yadroitsev/978-0-12-824090-8> (accessed March 19, 2021).
- [21] M. Gebler, A.J.M. Schoot Uiterkamp, C. Visser, A global sustainability perspective on 3D printing technologies, *Energy Policy.* 74 (2014) 158–167. <https://doi.org/10.1016/j.enpol.2014.08.033>.
- [22] B. Dutta, F.H. (Sam) Froes, The additive manufacturing (AM) of titanium alloys, in: *Titanium Powder Metallurgy*, Elsevier, 2015: pp. 447–468. <https://doi.org/10.1016/B978-0-12-800054-0.00024-1>.
- [23] A. Barz, T. Buer, H.-D. Haasis, A Study on the Effects of Additive Manufacturing on the Structure of Supply Networks, *IFAC-PapersOnLine.* 49 (2016) 72–77. <https://doi.org/10.1016/j.ifacol.2016.03.013>.
- [24] J. Allen, An Investigation into the Comparative Costs of Additive Manufacture vs. Machine from Solid for Aero Engine Parts, *Cost Effective Manufacture via Net-Shape Processing.* (2006) 17-1-17–10. <http://oai.dtic.mil/oai/oai?verb=getRecord&metadataPrefix=html&identifier=ADA521730>.
- [25] S.C. Joshi, A.A. Sheikh, 3D printing in aerospace and its long-term sustainability, *Virtual Phys Prototyp.* 10 (2015) 175–185. <https://doi.org/10.1080/17452759.2015.1111519>.
- [26] T.A. Campbell, O.S. Ivanova, Additive Manufacturing As a Disruptive Technology: Implications of Three-Dimensional Printing, *Technol Innov.* 15 (2013) 67–79. <https://doi.org/10.3727/194982413x13608676060655>.

- [27] R. Russell, D. Wells, J. Waller, B. Poorganji, E. Ott, T. Nakagawa, H. Sandoval, N. Shamsaei, M. Seifi, Qualification and certification of metal additive manufactured hardware for aerospace applications, in: *Additive Manufacturing for the Aerospace Industry*, Elsevier, 2019: pp. 33–66. <https://doi.org/10.1016/B978-0-12-814062-8.00003-0>.
- [28] I. Gibson, D. Rosen, B. Stucker, *Additive manufacturing technologies: 3D printing, rapid prototyping, and direct digital manufacturing*, second edition, Springer New York, New York, NY, 2015. <https://doi.org/10.1007/978-1-4939-2113-3>.
- [29] L.E. Murr, *Handbook of Materials Structures, Properties, Processing and Performance*, Springer International Publishing, Cham, 2015. <https://doi.org/10.1007/978-3-319-01815-7>.
- [30] H. Bikas, P. Stavropoulos, G. Chryssolouris, Additive manufacturing methods and modelling approaches: a critical review, *The International Journal of Advanced Manufacturing Technology*. 83 (2016) 389–405. <https://doi.org/10.1007/s00170-015-7576-2>.
- [31] SAE International, AS9100 - Quality Systems - Aerospace - Model for Quality Assurance in Design, Development, Production, Installation and Servicing, (1999). <https://www.sae.org/standards/content/as9100/>.
- [32] NASA, MSFC-SPEC-3717 - Specification for Control and Qualification of Laser Powder Bed Fusion Metallurgical Processes, (2017) 58. <https://repository.msfc.nasa.gov/docs/multiprogram/MSFC-SPEC-3717.pdf>.
- [33] ISO/ASTM, Additive manufacturing — System performance and reliability (ISO/ASTM 52941-20), (2020) 12. <https://www.iso.org/standard/74948.html>.
- [34] ASTM, Guide for Additive manufacturing – Installation/Operation and Performance Qualification (IQ/OQ/PQ) of Laser-Beam Powder Bed Fusion Equipment for Production Manufacturing (ASTM F3434 - 20), (2020) 7. <https://doi.org/10.1520/F3434-20>.
- [35] ISO/ASTM, Additive manufacturing — Qualification principles (ISO/ASTM 52942-20), (2020). <https://www.iso.org/standard/74949.html>.
- [36] NASA, Additive Manufacturing Requirements for Crewed Spaceflight Systems (NASA-STD-6030), (n.d.). <https://standards.nasa.gov/nasa-technical-standards>.
- [37] J.J. Lewandowski, M. Seifi, Metal Additive Manufacturing: A Review of Mechanical Properties, *Annu Rev Mater Res*. 46 (2016) 151–186. <https://doi.org/10.1146/annurev-matsci-070115-032024>.
- [38] M. Seifi, A. Salem, J. Beuth, O. Harrysson, J.J. Lewandowski, Overview of Materials Qualification Needs for Metal Additive Manufacturing, *JOM*. 68 (2016) 747–764. <https://doi.org/10.1007/s11837-015-1810-0>.
- [39] M. Seifi, M. Gorelik, J. Waller, N. Hrabe, N. Shamsaei, S. Daniewicz, J.J. Lewandowski, Progress Towards Metal Additive Manufacturing Standardization to Support Qualification and Certification, *JOM*. 69 (2017) 439–455. <https://doi.org/10.1007/s11837-017-2265-2>.

- [40] P. Li, D.H. Warner, A. Fatemi, N. Phan, Critical assessment of the fatigue performance of additively manufactured Ti-6Al-4V and perspective for future research, *Int J Fatigue*. 85 (2016) 130–143. <https://doi.org/10.1016/j.ijfatigue.2015.12.003>.
- [41] A. Fatemi, R. Molaei, S. Sharifimehr, N. Phan, N. Shamsaei, Multiaxial fatigue behavior of wrought and additive manufactured Ti-6Al-4V including surface finish effect, *Int J Fatigue*. 100 (2017) 347–366. <https://doi.org/10.1016/j.ijfatigue.2017.03.044>.
- [42] A. Fatemi, R. Molaei, S. Sharifimehr, N. Shamsaei, N. Phan, Torsional fatigue behavior of wrought and additive manufactured Ti-6Al-4V by powder bed fusion including surface finish effect, *Int J Fatigue*. 99 (2017) 187–201. <https://doi.org/10.1016/j.ijfatigue.2017.03.002>.
- [43] I. Yadroitsev, P. Krakhmalev, I. Yadroitsava, A. du Plessis, Qualification of Ti6Al4V ELI Alloy Produced by Laser Powder Bed Fusion for Biomedical Applications, *JOM*. 70 (2018) 372–377. <https://doi.org/10.1007/s11837-017-2655-5>.
- [44] ISO/ASTM, Additive Manufacturing - General Principles Terminology (ASTM52900), (2015). <https://www.iso.org/standard/69669.html>.
- [45] T. DebRoy, H.L.L. Wei, J.S.S. Zuback, T. Mukherjee, J.W.W. Elmer, J.O.O. Milewski, A.M.M. Beese, A. Wilson-Heid, A. De, W. Zhang, Additive manufacturing of metallic components – Process, structure and properties, *Prog Mater Sci*. 92 (2018) 112–224. <https://doi.org/10.1016/j.pmatsci.2017.10.001>.
- [46] A. Uriondo, M. Esperon-Miguez, S. Perinpanayagam, The present and future of additive manufacturing in the aerospace sector: A review of important aspects, *Proc Inst Mech Eng G J Aerosp Eng*. 229 (2015) 2132–2147. <https://doi.org/10.1177/0954410014568797>.
- [47] C.Y. Yap, C.K. Chua, Z.L. Dong, Z.H. Liu, D.Q. Zhang, L.E. Loh, S.L. Sing, Review of selective laser melting: Materials and applications, *Appl Phys Rev*. 2 (2015) 041101. <https://doi.org/10.1063/1.4935926>.
- [48] V. Bhavar, P. Kattire, V. Patil, S. Khot, K. Gujar, R. Singh, A review on powder bed fusion technology of metal additive manufacturing, in: 4th International Conference and Exhibition on Additive Manufacturing Technologies-AM-2014, 2014: pp. 1–2.
- [49] I. Yadroitsev, I. Yadroitsava, A. du Plessis, Basics of laser powder bed fusion, in: *Fundamentals of Laser Powder Bed Fusion of Metals*, Elsevier, 2021: pp. 15–38. <https://doi.org/10.1016/B978-0-12-824090-8.00024-X>.
- [50] J. Parthasarathy, B. Starly, S. Raman, A. Christensen, Mechanical evaluation of porous titanium (Ti6Al4V) structures with electron beam melting (EBM), *J Mech Behav Biomed Mater*. 3 (2010) 249–259. <https://doi.org/10.1016/j.jmbbm.2009.10.006>.
- [51] J.C. Pereira, H. Borovkov, F. Zubiri, M.C. Guerra, J. Caminos, Optimization of Thin Walls with Sharp Corners in SS316L and IN718 Alloys Manufactured with Laser Metal Deposition, *Journal of Manufacturing and Materials Processing*. 5 (2021) 5. <https://doi.org/10.3390/jmmp5010005>.

- [52] P.R. Gradl, C.S. Protz, T. Wammen, Additive Manufacturing and Hot-fire Testing of Liquid Rocket Channel Wall Nozzles Using Blown Powder Directed Energy Deposition Inconel 625 and JBK-75 Alloys, in: AIAA Propulsion and Energy 2019 Forum, American Institute of Aeronautics and Astronautics, Reston, Virginia, 2019. <https://doi.org/10.2514/6.2019-4362>.
- [53] J. Yu, M. Rombouts, G. Maes, F. Motmans, Material Properties of Ti6Al4V Parts Produced by Laser Metal Deposition, *Phys Procedia*. 39 (2012) 416–424. <https://doi.org/10.1016/j.phpro.2012.10.056>.
- [54] S.M. Kelly, S.L. Kampe, Microstructural evolution in laser-deposited multilayer Ti-6Al-4V builds: Part II. Thermal Modeling, *Metall Mater Trans A Phys Metall Mater Sci*. 35 A (2004) 1869–1879. <https://doi.org/10.1007/s11661-004-0095-7>.
- [55] S.M. Kelly, S.L. Kampe, Microstructural evolution in laser-deposited multilayer Ti-6Al-4V builds: Part 1. Microstructural characterization, *Metall Mater Trans A Phys Metall Mater Sci*. 35 A (2004) 1861–1867. <https://doi.org/10.1007/s11661-004-0094-8>.
- [56] A. Antonysamy, Microstructure, Texture and Mechanical Property Evolution during Additive Manufacturing of Ti6Al4V Alloy for Aerospace Applications, 2012.
- [57] F. Wang, S. Williams, P. Colegrove, A.A. Antonysamy, Microstructure and Mechanical Properties of Wire and Arc Additive Manufactured Ti-6Al-4V, *Metallurgical and Materials Transactions A*. 44 (2013) 968–977. <https://doi.org/10.1007/s11661-012-1444-6>.
- [58] P. Rambabu, N. Eswara Prasad, V. v. Kutumbarao, R.J.H. Wanhil, Aluminium Alloys for Aerospace Applications, in: 2017: pp. 29–52. https://doi.org/10.1007/978-981-10-2134-3_2.
- [59] A.G. Evans, J.W. Hutchinson, N.A. Fleck, M.F. Ashby, H.N.G. Wadley, The topological design of multifunctional cellular metals, *Prog Mater Sci*. 46 (2001) 309–327. [https://doi.org/10.1016/S0079-6425\(00\)00016-5](https://doi.org/10.1016/S0079-6425(00)00016-5).
- [60] N. Haghdadi, M. Laleh, M. Moyle, S. Primig, Additive manufacturing of steels: a review of achievements and challenges, *J Mater Sci*. 56 (2021) 64–107. <https://doi.org/10.1007/s10853-020-05109-0>.
- [61] A. Adeyemi, E.T. Akinlabi, R.M. Mahamood, Powder Bed Based Laser Additive Manufacturing Process of Stainless Steel: A Review, *Mater Today Proc*. 5 (2018) 18510–18517. <https://doi.org/10.1016/j.matpr.2018.06.193>.
- [62] X. Zhang, Y. Chen, J. Hu, Recent advances in the development of aerospace materials, *Progress in Aerospace Sciences*. 97 (2018) 22–34. <https://doi.org/10.1016/j.paerosci.2018.01.001>.
- [63] M. Peters, J. Kumpfert, C.H. Ward, C. Leyens, Titanium Alloys for Aerospace Applications, *Adv Eng Mater*. 5 (2003) 419–427. <https://doi.org/10.1002/adem.200310095>.
- [64] E. Uhlmann, R. Kersting, T.B. Klein, M.F. Cruz, A.V. Borille, Additive Manufacturing of Titanium Alloy for Aircraft Components, *Procedia CIRP*. 35 (2015) 55–60. <https://doi.org/10.1016/j.procir.2015.08.061>.

- [65] J.C. Williams, R.R. Boyer, Opportunities and Issues in the Application of Titanium Alloys for Aerospace Components, *Metals* (Basel). 10 (2020) 705. <https://doi.org/10.3390/met10060705>.
- [66] K. Minet, A. Saharan, A. Loesser, N. Raitanen, Superalloys, powders, process monitoring in additive manufacturing, in: *Additive Manufacturing for the Aerospace Industry*, Elsevier Inc., 2019; pp. 163–185. <https://doi.org/10.1016/B978-0-12-814062-8.00009-1>.
- [67] P.R. Gradl, S.E. Greene, C. Protz, B. Bullard, J. Buzzell, C. Garcia, J. Wood, R. Osborne, J. Hulka, K.G. Cooper, J. Hulka, R. Osborne, Additive Manufacturing of Liquid Rocket Engine Combustion Devices: A Summary of Process Developments and Hot-Fire Testing Results, in: *2018 Joint Propulsion Conference, American Institute of Aeronautics and Astronautics*, Reston, Virginia, 2018. <https://doi.org/10.2514/6.2018-4625>.
- [68] M. Perrut, P. Caron, M. Thomas, A. Couret, High temperature materials for aerospace applications: Ni-based superalloys and γ -TiAl alloys, *C R Phys.* 19 (2018) 657–671. <https://doi.org/10.1016/j.crhy.2018.10.002>.
- [69] E. Akca, A. Gürsel, A Review on Superalloys and IN718 Nickel-Based INCONEL Superalloy, *Periodicals of Engineering and Natural Sciences (PEN)*. 3 (2015). <https://doi.org/10.21533/pen.v3i1.43>.
- [70] J.A. Lee, Hydrogen embrittlement of nickel, cobalt and iron-based superalloys, in: *Gaseous Hydrogen Embrittlement of Materials in Energy Technologies: The Problem, Its Characterisation and Effects on Particular Alloy Classes*, Elsevier Ltd., 2012; pp. 624–667. <https://doi.org/10.1533/9780857093899.3.624>.
- [71] P.R. Gradl, C.S. Protz, D.L. Ellis, S.E. Greene, Progress in additively manufactured copper-alloy GRCOP-84, GRCOP-42, and bimetallic combustion chambers for liquid rocket engines, in: *Proceedings of the International Astronautical Congress, IAC*, 2019.
- [72] P.R. Gradl, C. Protz, S.E. Greene, D. Ellis, B. Lerch, I. Locci, Development and Hot-fire Testing of Additively Manufactured Copper Combustion Chambers for Liquid Rocket Engine Applications, in: *53rd AIAA/SAE/ASEE Joint Propulsion Conference, American Institute of Aeronautics and Astronautics*, Reston, Virginia, 2017. <https://doi.org/10.2514/6.2017-4670>.
- [73] S. MORIYA, T. INOUE, M. SASAKI, T. NAKAMOTO, T. KIMURA, N. NOMURA, K. KIKUCHI, A. KAWASAKI, T. KATO, I. MASUDA, Feasibility Study on Additive Manufacturing of Liquid Rocket Combustion Chamber, *TRANSACTIONS OF THE JAPAN SOCIETY FOR AERONAUTICAL AND SPACE SCIENCES, AEROSPACE TECHNOLOGY JAPAN*. 16 (2018) 261–266. <https://doi.org/10.2322/tastj.16.261>.
- [74] H.C. de Groh, D.L. Ellis, W.S. Loewenthal, Comparison of GRCop-84 to Other Cu Alloys with High Thermal Conductivities, *J Mater Eng Perform*. 17 (2008) 594–606. <https://doi.org/10.1007/s11665-007-9175-3>.

- [75] C. Zeng, H. Wen, B.C. Bernard, J.R. Raush, P.R. Gradl, M. Khonsari, S.M. Guo, Effect of Temperature History on Thermal Properties of Additively Manufactured C-18150 Alloy Samples, *Manuf Lett.* (2021). <https://doi.org/10.1016/j.mfglet.2021.02.002>.
- [76] T.J. Horn, D. Gamzina, Additive Manufacturing of Copper and Copper Alloys, in: *Additive Manufacturing Processes*, ASM International, 2020: pp. 388–418. <https://doi.org/10.31399/asm.hb.v24.a0006579>.
- [77] N.R. Philips, M. Carl, N.J. Cunningham, New Opportunities in Refractory Alloys, *Metallurgical and Materials Transactions A*. 51 (2020) 3299–3310. <https://doi.org/10.1007/s11661-020-05803-3>.
- [78] G.J. Schiller, Additive manufacturing for Aerospace, in: *2015 IEEE Aerospace Conference*, IEEE, 2015: pp. 1–8. <https://doi.org/10.1109/AERO.2015.7118958>.
- [79] J.O. Milewski, Additive Manufacturing Metal, the Art of the Possible, in: 2017: pp. 7–33. https://doi.org/10.1007/978-3-319-58205-4_2.
- [80] Zitelli, Folgarait, di Schino, Laser Powder Bed Fusion of Stainless Steel Grades: A Review, *Metals* (Basel). 9 (2019) 731. <https://doi.org/10.3390/met9070731>.
- [81] D. Manfredi, R. Bidulský, Laser powder bed fusion of aluminum alloys, *Acta Metallurgica Slovaca*. 23 (2017) 276. <https://doi.org/10.12776/ams.v23i3.988>.
- [82] H. Eskandari Sabzi, Powder bed fusion additive layer manufacturing of titanium alloys, *Materials Science and Technology*. 35 (2019) 875–890. <https://doi.org/10.1080/02670836.2019.1602974>.
- [83] Z. Tian, C. Zhang, D. Wang, W. Liu, X. Fang, D. Wellmann, Y. Zhao, Y. Tian, A Review on Laser Powder Bed Fusion of Inconel 625 Nickel-Based Alloy, *Applied Sciences*. 10 (2019) 81. <https://doi.org/10.3390/app10010081>.
- [84] C. Gradl, P., Tinker, D., Park, A., Mireles, P., Garcia, M., Wilkerson, R., Mckinney, C. Gradl, T. P., P. D., M. A., G. P., W. M., M. R., Robust Metal Additive Manufacturing Process Selection and Development for Aerospace Components, *Journal of Materials Engineering and Performance*, Springer. (2021).
- [85] K. Amato, J. Hernandez, L.E. Murr, E. Martinez, S.M. Gaytan, P.W. Shindo, S. Collins, Comparison of Microstructures and Properties for a Ni-Base Superalloy (Alloy 625) Fabricated by Electron Beam Melting, *Journal of Materials Science Research*. 1 (2012) p3. <https://doi.org/10.5539/JMSR.V1N2P3>.
- [86] L.E. Murr, S.M. Gaytan, D.A. Ramirez, E. Martinez, J. Hernandez, K.N. Amato, P.W. Shindo, F.R. Medina, R.B. Wicker, Metal Fabrication by Additive Manufacturing Using Laser and Electron Beam Melting Technologies, *J Mater Sci Technol*. 28 (2012) 1–14. [https://doi.org/10.1016/S1005-0302\(12\)60016-4](https://doi.org/10.1016/S1005-0302(12)60016-4).
- [87] A.N. Jinoop, C.P. Paul, K.S. Bindra, Laser-assisted directed energy deposition of nickel super alloys: A review, *Proceedings of the Institution of Mechanical Engineers, Part L: Journal of Materials: Design and Applications*. 233 (2019) 2376–2400. <https://doi.org/10.1177/1464420719852658>.

- [88] O.G. Rivera, P.G. Allison, J.B. Jordon, O.L. Rodriguez, L.N. Brewer, Z. McClelland, W.R. Whittington, D. Francis, J. Su, R.L. Martens, N. Hardwick, Microstructures and mechanical behavior of Inconel 625 fabricated by solid-state additive manufacturing, *Materials Science and Engineering A*. 694 (2017) 1–9. <https://doi.org/10.1016/j.msea.2017.03.105>.
- [89] G. Marchese, X. Garmendia, F. Calignano, D. Manfredi, Characterization and Comparison of Inconel 625 Processed by Selective Laser Melting and Laser Metal Deposition, *ResearchGate*. (2017).
- [90] N. Tuncer, A. Bose, Solid-State Metal Additive Manufacturing: A Review, *Jom*. 72 (2020) 3090–3111. <https://doi.org/10.1007/s11837-020-04260-y>.
- [91] G. Mladenov, E. Koleva, L. Koleva, V. Dzharov, State of the arts of additive manufacturing by selective electron beam melting, *Electrotechnica & Electronica*. 51 (2016) 14–24.
- [92] Z. Tian, C. Zhang, D. Wang, W. Liu, X. Fang, W. Daniel, Y. Zhao, T. Tian, A Review on Laser Powder Bed Fusion of Inconel 625, *Applied Scien*. 10 (2020). <https://doi.org/10.3390/app10010081>.
- [93] J.-R. Zhao, F.-Y. Hung, C.-S. Lu, I.-C. Lai, Comparison of Laser Powder Bed Fusion and Cast Inconel 713 Alloy in Terms of Their Microstructure, Mechanical Properties, and Fatigue Life, *Adv Eng Mater*. 23 (2021) 2001366. <https://doi.org/10.1002/ADEM.202001366>.
- [94] R. Udriou, Additive Manufacturing Technologies Used for Superalloys Processing, *MATEC Web of Conferences*. 94 (2014) 35–41.
- [95] W. Wong, E. Irissou, P. Vo, M. Sone, F. Bernier, J.-G. Legoux, H. Fukanuma, S. Yue, Cold Spray Forming of Inconel 718, *Journal of Thermal Spray Technology* 2012 22:2. 22 (2012) 413–421. <https://doi.org/10.1007/S11666-012-9827-1>.
- [96] P. Kumar, J. Farah, J. Akram, C. Teng, J. Ginn, M. Misra, Influence of processing parameters on properties of Inconel 718, *The International Journal of Advanced Manufacturing Technology*. 103 (2019) 1497–1507.
- [97] M.M. Kirka, F. Medina, R. Dehoff, A. Okello, Mechanical behavior of post-processed Inconel 718 manufactured through the electron beam melting process, *Materials Science and Engineering: A*. 680 (2017) 338–346. <https://doi.org/10.1016/J.MSEA.2016.10.069>.
- [98] A.N. Jinoop, C.P. Paul, S.K. Mishra, K.S. Bindra, Laser Additive Manufacturing using directed energy deposition of Inconel-718 wall structures with tailored characteristics, *Vacuum*. 166 (2019) 270–278. <https://doi.org/10.1016/j.vacuum.2019.05.027>.
- [99] X. Wang, X. Gong, K. Chou, Review on powder-bed laser additive manufacturing of Inconel 718 parts, *Proc Inst Mech Eng B J Eng Manuf*. 231 (2017) 1890–1903. <https://doi.org/10.1177/0954405415619883>.
- [100] P. Karimi, E. Sadeghi, J. Ålgårdh, P. Harlin, J. Andersson, Effect of build location on microstructural characteristics and corrosion behavior of EB-PBF built Alloy 718, *The International Journal of Advanced Manufacturing Technology* 2020 106:7. 106 (2020) 3597–3607. <https://doi.org/10.1007/S00170-019-04859-9>.

- [101] S. Bagherifard, S. Monti, M.V. Zuccoli, M. Riccio, J. Kondás, M. Guagliano, Cold spray deposition for additive manufacturing of freeform structural components compared to selective laser melting, *Materials Science and Engineering: A*. 721 (2018) 339–350. <https://doi.org/10.1016/J.MSEA.2018.02.094>.
- [102] Y.-N. Zhang, X. Cao, P. Wanjara, M. Medraj, Tensile properties of laser additive manufactured Inconel 718 using filler wire, *Journal of Materials Research* 29:17. 29 (2014) 2006–2020. <https://doi.org/10.1557/JMR.2014.199>.
- [103] C.R. Cunningham, J.M. Flynn, A. Shokrani, V. Dhokia, S.T. Newman, Invited review article: Strategies and processes for high quality wire arc additive manufacturing, *Addit Manuf.* 22 (2018) 672–686. <https://doi.org/10.1016/J.ADDMA.2018.06.020>.
- [104] J. Schneider, Comparison of Microstructural Response to Heat Treatment of Inconel 718 Prepared by Three Different Metal Additive Manufacturing Processes, *JOM* 2020 72:3. 72 (2020) 1085–1091. <https://doi.org/10.1007/S11837-020-04021-X>.
- [105] B. Lim, H. Chen, Z. Chen, N. Haghdadi, X. Liao, S. Primig, S.S. Babu, A.J. Breen, S.P. Ringer, Microstructure-property gradients in Ni-based superalloy (Inconel 738) additively manufactured via electron beam powder bed fusion, *Addit Manuf.* 46 (2021) 102121. <https://doi.org/10.1016/J.ADDMA.2021.102121>.
- [106] A. Ramakrishnan, G.P. Dinda, Direct laser metal deposition of Inconel 738, *Materials Science and Engineering: A*. 740–741 (2019) 1–13. <https://doi.org/10.1016/J.MSEA.2018.10.020>.
- [107] K. Kunze, T. Etter, J. Grässlin, V. Shklover, Texture, anisotropy in microstructure and mechanical properties of IN738LC alloy processed by selective laser melting (SLM), *Materials Science and Engineering A*. 620 (2015) 213–222.
- [108] N. Menon, T.H. Mahdi, A. Basak, Microstructure of In738Lc Fabricated Using Laser Powder Bed Fusion Additive Manufacturing, *J Turbomach.* (2021) 1–19. <https://doi.org/10.1115/1.4052404>.
- [109] B. V. Padmini, D.G. Bhosale, H.B. Niranjana, A study of T11 boiler steel protection by cold sprayed Inconel 738 coating against high temperature erosion, *Surfaces and Interfaces*. 23 (2021) 101002. <https://doi.org/10.1016/J.SURFIN.2021.101002>.
- [110] J. Norrish, J. Polden, I. Richardson, A review of wire arc additive manufacturing: development, principles, process physics, implementation and current status, *J Phys D Appl Phys*. 54 (2021) 473001. <https://doi.org/10.1088/1361-6463/ac1e4a>.
- [111] P. Kanagarajah, F. Brenne, T. Niendorf, H.J. Maier, Inconel 939 processed by selective laser melting: Effect of microstructure and temperature on the mechanical properties under static and cyclic loading, *Materials Science and Engineering: A*. 588 (2013) 188–195. <https://doi.org/10.1016/J.MSEA.2013.09.025>.

- [112] Y. Li, X. Liang, Y. Yu, H. Li, W. Kan, F. Lin, Microstructures and mechanical properties evolution of IN939 alloy during electron beam selective melting process, *J Alloys Compd.* 883 (2021) 160934. <https://doi.org/10.1016/J.JALLCOM.2021.160934>.
- [113] M.D. Barath Kumar, M. Manikandan, Assessment of Process, Parameters, Residual Stress Mitigation, Post Treatments and Finite Element Analysis Simulations of Wire Arc Additive Manufacturing Technique, The Korean Institute of Metals and Materials, 2021. <https://doi.org/10.1007/s12540-021-01015-5>.
- [114] M. Romedenne, R. Pillai, M. Kirka, S. Dryepondt, High temperature air oxidation behavior of Hastelloy X processed by Electron Beam Melting (EBM) and Selective Laser Melting (SLM), *Corros Sci.* 171 (2020) 108647. <https://doi.org/10.1016/J.CORSCI.2020.108647>.
- [115] D. Tomus, Y. Tian, P.A. Rometsch, M. Heilmaier, X. Wu, Influence of post heat treatments on anisotropy of mechanical behaviour and microstructure of Hastelloy-X parts produced by selective laser melting, *Materials Science and Engineering: A.* 667 (2016) 42–53. <https://doi.org/10.1016/J.MSEA.2016.04.086>.
- [116] R. Ghiaasiaan, M. Muhammad, P.R. Gradl, S. Shao, N. Shamsaei, Superior tensile properties of Hastelloy X enabled by additive manufacturing, *Mater Res Lett.* 9 (2021) 308–314. <https://doi.org/10.1080/21663831.2021.1911870>.
- [117] M. Dinovitzer, X. Chen, J. Laliberte, X. Huang, H. Frei, Effect of wire and arc additive manufacturing (WAAM) process parameters on bead geometry and microstructure, *Addit Manuf.* 26 (2019) 138–146. <https://doi.org/10.1016/J.ADDMA.2018.12.013>.
- [118] C. Tan, F. Weng, S. Sui, Y. Chew, G. Bi, Progress and Perspectives in Laser Additive Manufacturing of Key Aeroengine Materials, *Int J Mach Tools Manuf.* (2021) 135907. <https://doi.org/10.1016/j.ijmachtools.2021.103804>.
- [119] and N.A. Gradl, Paul, Omar Mireles, Introduction to additive manufacturing for propulsion and energy systems, in: *AIAA Propulsion and Energy Forum*, 2020.
- [120] K. Bauer, Thomas; Dawson, K; Spierings, Adriaan B.; Wegener, Microstructure and mechanical characterisation of SLM processed Haynes® 230, in: *Proceedings of the 26th Annual International Solid Freeform Fabrication Symposium*, 2015: pp. 813–822. <https://doi.org/10.3929/ethz-a-010584903>.
- [121] N. Shamsaei, A. Yadollahi, L. Bian, S.M. Thompson, An overview of Direct Laser Deposition for additive manufacturing; Part II: Mechanical behavior, process parameter optimization and control, *Addit Manuf.* 8 (2015) 12–35. <https://doi.org/10.1016/j.addma.2015.07.002>.
- [122] M. Haack, M. Kuczyk, A. Seidel, E. López, F. Brueckner, C. Leyens, Comprehensive study on the formation of grain boundary serrations in additively manufactured Haynes 230 alloy, *Mater Charact.* 160 (2020) 110092. <https://doi.org/10.1016/J.MATCHAR.2019.110092>.
- [123] Z. GaoLe, J. Yun, H. XiaoAn, H. Jia, W. JinWu, W. Yun, High-temperature mechanical properties of nickel-based superalloys manufactured by additive manufacturing,

- <https://doi.org/10.1080/02670836.2020.1799137>. 36 (2020) 1523–1533.
<https://doi.org/10.1080/02670836.2020.1799137>.
- [124] M.A. Alvin, Materials and Component Development for Advanced Turbine Systems: Project Summary, Proceedings of the ASME Turbo Expo. 1 (2010) 1005–1016. <https://doi.org/10.1115/GT2010-23173>.
- [125] P.R. Gradl, C.S. Protz, Technology advancements for channel wall nozzle manufacturing in liquid rocket engines, *Acta Astronaut.* 174 (2020). <https://doi.org/10.1016/j.actaastro.2020.04.067>.
- [126] S. Rathee, M. Srivastava, P.M. Pandey, A. Mahawar, S. Shukla, Metal additive manufacturing using friction stir engineering: A review on microstructural evolution, tooling and design strategies, *CIRP J Manuf Sci Technol.* 35 (2021) 560–588. <https://doi.org/10.1016/j.cirpj.2021.08.003>.
- [127] K.A. Unocic, M.M. Kirka, E. Cakmak, D. Greeley, A.O. Okello, S. Dryepondt, Evaluation of additive electron beam melting of haynes 282 alloy, *Materials Science and Engineering: A.* 772 (2020) 138607. <https://doi.org/10.1016/J.MSEA.2019.138607>.
- [128] A. Kumar Srivastava, N. Kumar, A. Rai Dixit, Friction stir additive manufacturing – An innovative tool to enhance mechanical and microstructural properties, *Materials Science and Engineering: B.* 263 (2021) 114832. <https://doi.org/10.1016/J.MSEB.2020.114832>.
- [129] T.A. Rodrigues, V. Duarte, R.M. Miranda, T.G. Santos, J.P. Oliveira, Current status and perspectives on wire and arc additive manufacturing (WAAM), *Materials.* 12 (2019). <https://doi.org/10.3390/ma12071121>.
- [130] A.W. Gebisa, H.G. Lemu, Additive Manufacturing for the Manufacture of Gas Turbine Engine Components: Literature Review and Future Perspectives, in: *ASME Turbo Expo 2018: Turbomachinery Technical Conference and Exposition*, Oslo, Norway, 2018. <https://doi.org/10.1115/GT2018-76686>.
- [131] O. Marenych, A. Kostryzhev, C. Shen, Z. Pan, H. Li, S. van Duin, Precipitation strengthening in Ni–Cu alloys fabricated using wire arc additive manufacturing technology, *Metals (Basel).* 9 (2019). <https://doi.org/10.3390/met9010105>.
- [132] A.R. Kannan, S.M. Kumar, R. Pramod, N.P. Kumar, N.S. Shanmugam, Y. Palguna, Microstructure and mechanical properties of wire arc additive manufactured bi-metallic structure, <https://doi.org/10.1080/13621718.2020.1833140>. 26 (2020) 47–57.
<https://doi.org/10.1080/13621718.2020.1833140>.
- [133] E. Strandh, P. Mellin, S. Dubiez-Le Goff, J. Gårdstam, M. Ahlfors, A. Ströndl, Surface pick-up of argon during hot isostatic pressing of material built by laser powder bed fusion, *Addit Manuf.* 38 (2021) 101763. <https://doi.org/10.1016/j.addma.2020.101763>.
- [134] L.E. Murr, E. Martinez, X.M. Pan, S.M. Gaytan, J.A. Castro, C.A. Terrazas, F. Medina, R.B. Wicker, D.H. Abbott, Microstructures of Rene 142 nickel-based superalloy fabricated by electron beam melting, *Acta Mater.* 61 (2013) 4289–4296. <https://doi.org/10.1016/J.ACTAMAT.2013.04.002>.

- [135] L. Sexton, S. Lavin, G. Byrne, A. Kennedy, Laser cladding of aerospace materials, *J Mater Process Technol.* 122 (2002) 63–68. [https://doi.org/10.1016/S0924-0136\(01\)01121-9](https://doi.org/10.1016/S0924-0136(01)01121-9).
- [136] A. Basak, Additive manufacturing of high-gamma prime nickel-based superalloys through selective laser melting (SLM), *Solid Freeform Fabrication 2019: Proceedings of the 30th Annual International Solid Freeform Fabrication Symposium - An Additive Manufacturing Conference, SFF 2019.* (2019) 554–575.
- [137] S. Vezzù, S. Rech, E. Vedelago, G.P. Zanon, G. Alfeo, A. Scialpi, R. Huang, On deposition of Waspaloy coatings by cold spray, <https://doi.org/10.1179/1743294413Y.0000000199>. 30 (2014) 342–351. <https://doi.org/10.1179/1743294413Y.0000000199>.
- [138] N.I.S. Hussein, J. Segal, D.G. McCartney, I.R. Pashby, Microstructure formation in Waspaloy multilayer builds following direct metal deposition with laser and wire, *Materials Science and Engineering: A.* 497 (2008) 260–269. <https://doi.org/10.1016/J.MSEA.2008.07.021>.
- [139] K.A. Mumtaz, P. Erasenthiran, N. Hopkinson, High density selective laser melting of Waspaloy®, in: *J Mater Process Technol, 2006 International Solid Freeform Fabrication Symposium, 2008:* pp. 77–87. <https://doi.org/10.1016/j.jmatprotec.2007.04.117>.
- [140] N. Haghdadi, M. Laleh, M. Moyle, S. Primig, Additive manufacturing of steels: a review of achievements and challenges, *J Mater Sci.* 56 (2021) 64–107. <https://doi.org/10.1007/s10853-020-05109-0>.
- [141] A. Zadi-Maad, R. Rohib, A. Irawan, Additive manufacturing for steels: A review, *IOP Conf Ser Mater Sci Eng.* 285 (2018). <https://doi.org/10.1088/1757-899X/285/1/012028>.
- [142] S.M. Thompson, L. Bian, N. Shamsaei, A. Yadollahi, An overview of Direct Laser Deposition for additive manufacturing; Part I: Transport phenomena, modeling and diagnostics, *Addit Manuf.* 8 (2015) 36–62. <https://doi.org/10.1016/j.addma.2015.07.001>.
- [143] J. Wanwan, Z. Chaoqun, J. Shuoya, T. Yingtao, W. Daniel, L. Wen, Wire Arc Additive Manufacturing of Stainless Steels: A Review, *Applied Sciences (Switzerland).* (2020).
- [144] M. Galati, P. Minetola, G. Rizza, Surface roughness characterisation and analysis of the Electron Beam Melting (EBM) process, *Materials.* 12 (2019). <https://doi.org/10.3390/ma12132211>.
- [145] M. Marya, V. Singh, J.Y. Hascoet, S. Marya, A Metallurgical Investigation of the Direct Energy Deposition Surface Repair of Ferrous Alloys, *J Mater Eng Perform.* 27 (2018) 813–824. <https://doi.org/10.1007/s11665-017-3117-5>.
- [146] M.Yu. Panchenko, G.G. Maier, V.A. Moskvina, E. V. Melnikov, S. V. Astafurov, V.E. Rubtsov, E.A. Kolubaev, E.G. Astafurova, Microstructural inhomogeneity of phase composition and grain structure in electron beam wire-feed additive manufactured AISI 304 stainless steel, *AIP Conf Proc.* 2167 (2019) 020261. <https://doi.org/10.1063/1.5132128>.
- [147] V. Gopan, K. Leo Dev Wins, A. Surendran, Innovative potential of additive friction stir deposition among current laser based metal additive manufacturing processes: A review, *CIRP J Manuf Sci Technol.* 32 (2021) 228–248. <https://doi.org/10.1016/J.CIRPJ.2020.12.004>.

- [148] J. Gordon, J. Hochhalter, C. Haden, D.G. Harlow, Enhancement in fatigue performance of metastable austenitic stainless steel through directed energy deposition additive manufacturing, *Mater Des.* 168 (2019) 107630. <https://doi.org/10.1016/j.matdes.2019.107630>.
- [149] H. Yeom, T. Dabney, N. Pocquette, K. Ross, F.E. Pfefferkorn, K. Sridharan, Cold spray deposition of 304L stainless steel to mitigate chloride-induced stress corrosion cracking in canisters for used nuclear fuel storage, *Journal of Nuclear Materials.* 538 (2020) 152254. <https://doi.org/10.1016/J.JNUCMAT.2020.152254>.
- [150] M.O. Shaikh, C.C. Chen, H.C. Chiang, J.R. Chen, Y.C. Chou, T.Y. Kuo, K. Ameyama, C.H. Chuang, Additive manufacturing using fine wire-based laser metal deposition, *Rapid Prototyp J.* 26 (2020) 473–483. <https://doi.org/10.1108/RPJ-04-2019-0110>.
- [151] V.R. Utyaganova, A. V. Filippov, N.N. Shamarin, A. V. Vorontsov, N.L. Savchenko, S. V. Fortuna, D.A. Gurianov, A. V. Chumaevskii, V.E. Rubtsov, S.Y. Tarasov, Controlling the porosity using exponential decay heat input regimes during electron beam wire-feed additive manufacturing of Al-Mg alloy, *International Journal of Advanced Manufacturing Technology.* 108 (2020) 2823–2838. <https://doi.org/10.1007/s00170-020-05539-9>.
- [152] A. Hehr, M. Norfolk, A comprehensive review of ultrasonic additive manufacturing, *Rapid Prototyp J.* 26 (2020) 445–458. <https://doi.org/10.1108/RPJ-03-2019-0056>.
- [153] A. Sova, S. Grigoriev, A. Okunkova, I. Smurov, Cold spray deposition of 316L stainless steel coatings on aluminium surface with following laser post-treatment, *Surf Coat Technol.* 235 (2013) 283–289. <https://doi.org/10.1016/J.SURFCOAT.2013.07.052>.
- [154] T.A. Kalashnikova, E.S. Khoroshko, A. V. Chumaevskii, A. V. Filippov, Surface morphology of 321 stainless steel obtained by electron-beam wire-feed additive manufacturing technology, *AIP Conf Proc.* 2051 (2018) 020114. <https://doi.org/10.1063/1.5083357>.
- [155] P.A. Kuznetsov, I. V Shakirov, A.S. Zukov, V. V Bobyr', M. V Starytsin, Effect of particle size distribution on the structure and mechanical properties in the process of laser powder bed fusion, *J. Phys.* (2021). <https://doi.org/10.1088/1742-6596/1758/1/012021>.
- [156] S.M. Kumar, R. Sasikumar, A.R. Kannan, R. Pramod, N.P. Kumar, N.S. Shanmugam, M. Vishnukumar, Microstructural administered mechanical properties and corrosion behaviour of wire plus arc additive manufactured SS 321 plate:, <https://doi.org/10.1177/09544062211042039>. (2021) 095440622110420. <https://doi.org/10.1177/09544062211042039>.
- [157] R. Duraisamy, S.M. Kumar, A.R. Kannan, N.S. Shanmugam, K. Sankaranarayanan, Fatigue Behavior of Austenitic Stainless Steel 347 Fabricated via Wire Arc Additive Manufacturing, *Journal of Materials Engineering and Performance* 2021. (2021) 1–7. <https://doi.org/10.1007/S11665-021-06033-3>.
- [158] P. Wanjara, M. Brochu, M. Jahazi, Electron beam freeforming of stainless steel using solid wire feed, *Mater Des.* 28 (2007) 2278–2286. <https://doi.org/10.1016/J.MATDES.2006.08.008>.

- [159] T. Han, C.H. Kuo, N. Sridharan, L.M. Headings, S.S. Babu, M.J. Dapino, Effect of preheat temperature and post-process treatment on the microstructure and mechanical properties of stainless steel 410 made via ultrasonic additive manufacturing, *Materials Science and Engineering: A*. 769 (2020) 138457. <https://doi.org/10.1016/J.MSEA.2019.138457>.
- [160] M. Kottman, S. Zhang, J. McGuffin-Cawley, P. Denney, B.K. Narayanan, Laser Hot Wire Process: A Novel Process for Near-Net Shape Fabrication for High-Throughput Applications, (n.d.). <https://doi.org/10.1007/s11837-014-1288-1>.
- [161] B. Zhu, J. Lin, Y. Lei, Y. Zhang, Q. Sun, S. Cheng, Additively manufactured δ -ferrite-free 410 stainless steel with desirable performance, *Mater Lett.* 293 (2021) 129579. <https://doi.org/10.1016/J.MATLET.2021.129579>.
- [162] M.K. Alam, M. Mehdi, R.J. Urbanic, A. Edrisy, Mechanical behavior of additive manufactured AISI 420 martensitic stainless steel, *Materials Science and Engineering: A*. 773 (2020) 138815. <https://doi.org/10.1016/J.MSEA.2019.138815>.
- [163] I. Zetková, L. Kučerová, M. Zetek, J. Česánek, P. Hanzl, M. Dana, M. Nozar, J. Kána, Evaluation of metal powder for additive manufacturing of maraging steel, *Annals of DAAAM and Proceedings of the International DAAAM Symposium*. (2017) 410–416. <https://doi.org/10.2507/28th.daaam.proceedings.057>.
- [164] A. Dass, A. Moridi, State of the Art in Directed Energy Deposition: From Additive Manufacturing to Materials Design, *Coatings*. 9 (2019) 418. <https://doi.org/10.3390/coatings9070418>.
- [165] A. Levy, A. Miriyev, N. Sridharan, T. Han, E. Tuval, S.S. Babu, M.J. Dapino, N. Frage, Ultrasonic additive manufacturing of steel: Method, post-processing treatments and properties, *J Mater Process Technol.* 256 (2018) 183–189. <https://doi.org/10.1016/J.JMATPROTEC.2018.02.001>.
- [166] A.A. Hassen, M. Noakes, P. Nandwana, S. Kim, V. Kunc, U. Vaidya, L. Love, A. Nycz, Scaling Up metal additive manufacturing process to fabricate molds for composite manufacturing, *Addit Manuf.* 32 (2020) 101093. <https://doi.org/10.1016/J.ADDMA.2020.101093>.
- [167] C. Qiu, N.J.E. Adkins, M.M. Attallah, Selective laser melting of Invar 36: Microstructure and properties, *Acta Mater.* 103 (2016) 382–395. <https://doi.org/10.1016/J.ACTAMAT.2015.10.020>.
- [168] M. Yakout, M.A. Elbestawi, S.C. Veldhuis, A review of metal additive manufacturing technologies, *Solid State Phenomena*. 278 SSP (2018) 1–14. <https://doi.org/10.4028/www.scientific.net/SSP.278.1>.
- [169] C. Chen, Y. Xie, L. Liu, R. Zhao, X. Jin, S. Li, R. Huang, J. Wang, H. Liao, Z. Ren, Cold spray additive manufacturing of Invar 36 alloy: microstructure, thermal expansion and mechanical properties, *J Mater Sci Technol.* 72 (2021) 39–51. <https://doi.org/10.1016/J.JMST.2020.07.038>.
- [170] P.R. Gradl, C. Protz, T. Wammen, Additive manufacturing development and hot-fire testing of liquid rocket channel wall nozzles using blown powder directed energy deposition inconel 625 and jbk-75 alloys,

- in: AIAA Propulsion and Energy Forum and Exposition, 2019, American Institute of Aeronautics and Astronautics Inc, AIAA, 2019. <https://doi.org/10.2514/6.2019-4362>.
- [171] P. Bhaskar, Investigation of LPBF built JBK-75 stainless steel, University West, 2021.
 - [172] Z.C. Jones, P.R. Gradl, Powder Bed Fusion Additive Manufacturing of JBK-75, in: JANNAF Joint Propulsion Conference, NASA, 2019: pp. 148–162.
 - [173] C. Katsarelis, P. Chen, P.R. Gradl, C. Protz, Z. Jones, N. Marshall, S. Flight, D. Ellis, L. Evans, Additive Manufacturing of NASA HR-1 Material for Liquid Rocket Engine Component Applications, JANNAF Joint Propulsion Conference. (2019) <https://ntrs.nasa.gov/search.jsp?R=20200001007>.
 - [174] P.R. Gradl, T.W. Teasley, C.S. Protz, C. Katsarelis, P. Chen, Process Development and Hot-fire Testing of Additively Manufactured NASA HR-1 for Liquid Rocket Engine Applications, in: AIAA Propulsion and Energy 2021, 2021: pp. 1–23. <https://doi.org/10.2514/6.2021-3236>.
 - [175] P.S. Chen, C.C. Katsarelis, W.M. Medders, P.R. Gradl, Segregation Evolution and Diffusion of Titanium in Directed Energy Deposited, 2021.
 - [176] P.R. Gradl, C. Protz, J. Fikes, A. Clark, L. Evans, S. Miller, D. Ellis, T. Hudson, Lightweight thrust chamber assemblies using multi-alloy additive manufacturing and composite overwrap, in: AIAA Propulsion and Energy 2020 Forum, American Institute of Aeronautics and Astronautics Inc, AIAA, 2020: pp. 1–25. <https://doi.org/10.2514/6.2020-3787>.
 - [177] P. Tarafder, C. Rock, T. Horn, Quasi-static tensile properties of unalloyed copper produced by electron beam powder bed fusion additive manufacturing, *Materials*. 14 (2021). <https://doi.org/10.3390/ma14112932>.
 - [178] J. Huang, X. Yan, C. Chang, Y. Xie, W. Ma, R. Huang, R. Zhao, S. Li, M. Liu, H. Liao, Pure copper components fabricated by cold spray (CS) and selective laser melting (SLM) technology, *Surf Coat Technol*. 395 (2020) 125936. <https://doi.org/10.1016/j.surfcoat.2020.125936>.
 - [179] Q. Jiang, P. Zhang, Z. Yu, H. Shi, D. Wu, H. Yan, X. Ye, Q. Lu, Y. Tian, A review on additive manufacturing of pure copper, *Coatings*. 11 (2021). <https://doi.org/10.3390/coatings11060740>.
 - [180] J.D. Martin, Exploring Additive Manufacturing Processes for Direct 3D Printing of Copper Induction Coils, ASME International Mechanical Engineering Congress and Exposition, Proceedings (IMECE). 2 (2018). <https://doi.org/10.1115/IMECE2017-71685>.
 - [181] T.J. Horn, D. Gamzina, Additive Manufacturing of Copper and Copper Alloys, in: Additive Manufacturing Processes, ASM International, 2020: pp. 388–418. <https://doi.org/10.31399/asm.hb.v24.a0006579>.
 - [182] A. Panfilov, E. Knyazhev, T. Kalashnikova, K. Kalashnikov, S. Nikonov, A. Gusarova, Deformation uniformity of additively manufactured materials on the example of austenitic stainless steel 321 and copper C11000, *J Phys Conf Ser*. 1611 (2020) 12005. <https://doi.org/10.1088/1742-6596/1611/1/012005>.

- [183] P.R. Gradl, C. Protz, K. Cooper, C. Garcia, D. Ellis, L. Evans, GRCop-42 development and hot-fire testing using additive manufacturing powder bed fusion for channel-cooled combustion chambers, in: AIAA Propulsion and Energy Forum and Exposition, 2019, American Institute of Aeronautics and Astronautics Inc, AIAA, 2019. <https://doi.org/10.2514/6.2019-4228>.
- [184] P.R. Gradl, T.W. Teasley, C.S. Protz, M.B. Garcia, D. Ellis, C. Kantzos, Advancing GRCop-based Bimetallic Additive Manufacturing to Optimize Component Design and Applications for Liquid Rocket Engines, in: AIAA Propulsion and Energy 2021, 2021: pp. 1–28. <https://doi.org/10.2514/6.2021-3231>.
- [185] K.G. Cooper, J.L. Lydon, M.D. Lecorre, Z.C. Jones, Three-Dimensional Printing GRCop-42, 2019.
- [186] T. Suresh, S. Landes, T. Letcher, A. Prasad, P. Gradl, D. Ellis, Nanomechanical Characterization of Additively Manufactured GRCop-42 Alloy Developed by Directed Energy Deposition Methods, in: International Mechanical Engineering Congress and Exposition (IMECE2020), 2020: pp. 1–7. <https://doi.org/10.1115/IMECE2020-24400>.
- [187] P.R. Gradl, S. Elam Greene, C.S. Protz, D.L. Ellis, B.A. Lerch, S.E. Greene, C.S. Protz, D.L. Ellis, B.A. Lerch, I.E. Locci, Development and Hot-fire Testing of Additively Manufactured Copper Combustion Chambers for Liquid Rocket Engine Applications, 53rd AIAA/SAE/ASEE Joint Propulsion Conference. (2017) 1–27. <https://doi.org/10.2514/6.2017-4670>.
- [188] A.H. Seltzman, S.J. Wukitch, Surface roughness and finishing techniques in selective laser melted GRCop-84 copper for an additive manufactured lower hybrid current drive launcher, Fusion Engineering and Design. 160 (2020) 111801. <https://doi.org/10.1016/j.fusengdes.2020.111801>.
- [189] P.R. Gradl, C. Protz, K. Zagorski, V. Doshi, H. McCallum, Additive manufacturing and hot-fire testing of bimetallic grcop-84 and c-18150 channel-cooled combustion chambers using powder bed fusion and inconel 625 hybrid directed energy deposition, in: AIAA Propulsion and Energy Forum and Exposition, 2019, American Institute of Aeronautics and Astronautics Inc, AIAA, 2019. <https://doi.org/10.2514/6.2019-4390>.
- [190] T.W. Teasley, P.R. Gradl, M.B. Garcia, B.B. Williams, C.S. Protz, Extreme Environment Hot Fire Durability of Post Processed Additively Manufactured GRCop-Alloy Combustion Chambers, in: AIAA Propulsion and Energy 2021, 2021: pp. 1–20. <https://doi.org/10.2514/6.2021-3233>.
- [191] A.H. Seltzman, S.J. Wukitch, Resolution and geometric limitations in laser powder bed fusion additively manufactured GRCop-84 structures for a lower hybrid current drive launcher, Fusion Engineering and Design. 173 (2021) 112847. <https://doi.org/10.1016/j.fusengdes.2021.112847>.
- [192] R.P. Minneci, E.A. Lass, J.R. Bunn, H. Choo, C.J. Rawn, Copper-based alloys for structural high-heat-flux applications: a review of development, properties, and performance of Cu-rich Cu–Cr–Nb alloys, International Materials Reviews. (2020). <https://doi.org/10.1080/09506608.2020.1821485>.
- [193] F. Kerstens, A. Cervone, P. Gradl, End to end process evaluation for additively manufactured liquid rocket engine thrust chambers, Acta Astronaut. 182 (2021) 454–465. <https://doi.org/10.1016/j.actaastro.2021.02.034>.

- [194] C. Zeng, H. Wen, B.C. Bernard, J.R. Raush, P.R. Gradl, M. Khonsari, S.M. Guo, Effect of temperature history on thermal properties of additively manufactured C-18150 alloy samples, *Manuf Lett.* 28 (2021) 25–29. <https://doi.org/10.1016/j.mfglet.2021.02.002>.
- [195] P.R. Gradl, Rapid fabrication techniques for liquid rocket channel wall nozzles, in: 52nd AIAA/SAE/ASEE Joint Propulsion Conference, 2016, American Institute of Aeronautics and Astronautics Inc, AIAA, 2016. <https://doi.org/10.2514/6.2016-4771>.
- [196] T. Wegener, J. Koopmann, J. Richter, P. Krooß, T. Niendorf, CuCrZr processed by laser powder bed fusion—Processability and influence of heat treatment on electrical conductivity, microstructure and mechanical properties, *Fatigue Fract Eng Mater Struct.* 44 (2021) 2570–2590. <https://doi.org/10.1111/ffe.13527>.
- [197] C. Salvan, L. Briottet, T. Baffie, L. Guetaz, C. Flament, CuCrZr alloy produced by laser powder bed fusion: Microstructure, nanoscale strengthening mechanisms, electrical and mechanical properties, *Materials Science and Engineering: A.* 826 (2021) 141915. <https://doi.org/10.1016/j.msea.2021.141915>.
- [198] S. MORIYA, T. INOUE, M. SASAKI, T. NAKAMOTO, T. KIMURA, N. NOMURA, K. KIKUCHI, A. KAWASAKI, T. KATO, I. MASUDA, Feasibility Study on Additive Manufacturing of Liquid Rocket Combustion Chamber, *Transactions of the Japan Society for Aeronautical and Space Sciences, Aerospace Technology Japan.* 16 (2018) 261–266. <https://doi.org/10.2322/tastj.16.261>.
- [199] Y. Bai, C. Zhao, Y. Zhang, J. Chen, H. Wang, Additively manufactured CuCrZr alloy: Microstructure, mechanical properties and machinability, *Materials Science and Engineering A.* 819 (2021) 141528. <https://doi.org/10.1016/j.msea.2021.141528>.
- [200] C. Zeng, H. Wen, B.C. Bernard, H. Ding, J.R. Raush, P.R. Gradl, M. Khonsari, S. Guo, Tensile Properties of Additively Manufactured C-18150 Copper Alloys, *Metals and Materials International* 2021. (2021) 1–13. <https://doi.org/10.1007/S12540-021-01052-0>.
- [201] M.A. Melnikova, A.U. Shishov, A.A. Kholopov, A.I. Misyurov, A.V. Perestoronin, A.R. Smirnov, Application features of laser coaxial melting of powder materials to produce objects from copper alloys, *AIP Conf Proc.* 2318 (2021). <https://doi.org/10.1063/5.0035860>.
- [202] S.D. Jadhav, P.P. Dhekne, S. Dadbakhsh, J.P. Kruth, J. Van Humbeeck, K. Vanmeensel, Surface Modified Copper Alloy Powder for Reliable Laser-based Additive Manufacturing, *Addit Manuf.* 35 (2020) 101418. <https://doi.org/10.1016/J.ADDMA.2020.101418>.
- [203] P.R. Gradl, S.E. Greene, C. Protz, B. Bullard, J. Buzzell, C. Garcia, J. Wood, R. Osborne, J. Hulka, K.G. Cooper, J. Hulka, R. Osborne, Additive Manufacturing of Liquid Rocket Engine Combustion Devices: A Summary of Process Developments and Hot-Fire Testing Results, in: 2018 Joint Propulsion Conference, American Institute of Aeronautics and Astronautics, Reston, Virginia, 2018. <https://doi.org/10.2514/6.2018-4625>.

- [204] P. Sudharshan Phani, V. Vishnukanthan, G. Sundararajan, Effect of heat treatment on properties of cold sprayed nanocrystalline copper alumina coatings, *Acta Mater.* 55 (2007) 4741–4751. <https://doi.org/10.1016/J.ACTAMAT.2007.04.044>.
- [205] Y. Li, Y. Liu, H. Peng, Laser net shape manufacturing of Haynes 188 alloy, *International Congress on Applications of Lasers & Electro-Optics*. 2008 (2018) 506. <https://doi.org/10.2351/1.5061320>.
- [206] P.K. Gokuldoss, S. Kolla, J. Eckert, Additive manufacturing processes: Selective laser melting, electron beam melting and binder jetting-selection guidelines, *Materials*. 10 (2017). <https://doi.org/10.3390/ma10060672>.
- [207] Z. Zhang, F. Kong, R. Kovacevic, Laser hot-wire cladding of Co-Cr-W metal cored wire, *Opt Lasers Eng.* 128 (2020) 105998. <https://doi.org/10.1016/J.OPTLASENG.2019.105998>.
- [208] N. Cinca, E. López, S. Dosta, J.M. Guilemany, Study of stellite-6 deposition by cold gas spraying, *Surf Coat Technol.* 232 (2013) 891–898. <https://doi.org/10.1016/J.SURFCOAT.2013.06.120>.
- [209] Z. Li, Y. Cui, J. Wang, C. Liu, J. Wang, T. Xu, T. Lu, H. Zhang, J. Lu, S. Ma, H. Fan, S. Tang, Characterization of Microstructure and Mechanical Properties of Stellite 6 Part Fabricated by Wire Arc Additive Manufacturing, *Metals* 2019, Vol. 9, Page 474. 9 (2019) 474. <https://doi.org/10.3390/MET9040474>.
- [210] B. Ren, M. Zhang, C. Chen, X. Wang, T. Zou, Z. Hu, Effect of Heat Treatment on Microstructure and Mechanical Properties of Stellite 12 Fabricated by Laser Additive Manufacturing, *Journal of Materials Engineering and Performance* 2017 26:11. 26 (2017) 5404–5413. <https://doi.org/10.1007/S11665-017-2984-0>.
- [211] P. Magarò, A.L. Marino, A. Di Schino, F. Furgiuele, C. Maletta, R. Pileggi, E. Sgambitterra, C. Testani, M. Tului, Effect of process parameters on the properties of Stellite-6 coatings deposited by Cold Gas Dynamic Spray, *Surf Coat Technol.* 377 (2019) 124934. <https://doi.org/10.1016/J.SURFCOAT.2019.124934>.
- [212] P. Agrawal, R.S. Haridas, S. Yadav, S. Thapliyal, S. Gaddam, R. Verma, R.S. Mishra, Processing-structure-property correlation in additive friction stir deposited Ti-6Al-4V alloy from recycled metal chips, *Addit Manuf.* 47 (2021) 102259. <https://doi.org/10.1016/j.addma.2021.102259>.
- [213] S.W. Williams, F. Martina, A.C. Addison, J. Ding, G. Pardal, P. Colegrove, Wire + Arc Additive Manufacturing, *Materials Science and Technology*. 32 (2016) 641–647. <https://doi.org/10.1179/1743284715Y.0000000073>.
- [214] M. Vaezi, P. Drescher, H. Seitz, Beamless Metal Additive Manufacturing, *Materials*. 13 (2020). <https://doi.org/10.3390/ma13040922>.
- [215] Z. Lin, K. Song, X. Yu, A review on wire and arc additive manufacturing of titanium alloy, *J Manuf Process.* 70 (2021) 24–45. <https://doi.org/10.1016/j.jmapro.2021.08.018>.

- [216] D. Herzog, V. Seyda, E. Wycisk, C. Emmelmann, Additive manufacturing of metals, *Acta Mater.* 117 (2016) 371–392. <https://doi.org/10.1016/J.ACTAMAT.2016.07.019>.
- [217] K.F. Graff, M. Short, M. Norfolk, Very high power ultrasonic additive manufacturing (VHP UAM) for advanced materials, in: *Solid Freeform Fabrication Symposium*, Austin, TX, 2010.
- [218] Y. Chen, S.J. Clark, L. Sinclair, C.L.A. Leung, S. Marussi, T. Connolley, R.C. Atwood, G.J. Baxter, M.A. Jones, I. Todd, P.D. Lee, Synchrotron X-ray imaging of directed energy deposition additive manufacturing of titanium alloy Ti-6242, *Addit Manuf.* 41 (2021) 101969. <https://doi.org/10.1016/J.ADDMA.2021.101969>.
- [219] H. Fan, S. Yang, Effects of direct aging on near-alpha Ti-6Al-2Sn-4Zr-2Mo (Ti-6242) titanium alloy fabricated by selective laser melting (SLM), *Materials Science and Engineering: A.* 788 (2020) 139533. <https://doi.org/10.1016/J.MSEA.2020.139533>.
- [220] E. Lannunziata, Process parameters optimisation of Ti6242 produced by Electron Beam Melting, (2021).
- [221] P. Harish, Understanding Effects of Isothermal Heat Treatments on Microstructure of LMD-w Titanium Alloy (Ti-6242), University West, 2020.
- [222] A. Mohammad, A.M. Al-Ahmari, A. AlFaify, M.K. Mohammed, Effect of melt parameters on density and surface roughness in electron beam melting of gamma titanium aluminide alloy, *Rapid Prototyp J.* 23 (2017) 474–485. <https://doi.org/10.1108/RPJ-12-2014-0187>.
- [223] M. Balichakra, S. Bontha, P. Krishna, V.K. Balla, Prediction and validation of residual stresses generated during laser metal deposition of γ titanium aluminide thin wall structures, *Mater Res Express.* 6 (2019) 106550. <https://doi.org/10.1088/2053-1591/ab38ee>.
- [224] W. Chen, Z. Li, Additive manufacturing of titanium aluminides, *Additive Manufacturing for the Aerospace Industry.* (2019) 235–263. <https://doi.org/10.1016/B978-0-12-814062-8.00013-3>.
- [225] G. Baudana, S. Biamino, D. Ugues, M. Lombardi, P. Fino, M. Pavese, C. Badini, Titanium aluminides for aerospace and automotive applications processed by Electron Beam Melting: Contribution of Politecnico di Torino, *Metal Powder Report.* 71 (2016) 193–199. <https://doi.org/10.1016/J.MPRP.2016.02.058>.
- [226] C. Leyens, F. Brueckner, A. Seidel, A. Straubel, A. Marquardt, R. Willner, Laser Additive Manufacturing of Titanium Alloys and Titanium Aluminides, *Conference: World Conference on Titanium, Symposium 5. Intermetallics and MMCs.* (2015) 1–6.
- [227] P. Cao, B. Gabbitas, A. Salman, D.L. Zhang, Z.H. Han, Consolidation of TiAl Powder by Thermal Spray Processes, *Adv Mat Res.* 29–30 (2007) 159–162. <https://doi.org/10.4028/www.scientific.net/amr.29-30.159>.
- [228] T. Novoselova, P. Fox, R. Morgan, W. O'Neill, Experimental study of titanium/aluminium deposits produced by cold gas dynamic spray, *Surf Coat Technol.* 200 (2006) 2775–2783. <https://doi.org/10.1016/J.SURFCOAT.2004.10.133>.

- [229] G.M. Karthik, G.D.J. Ram, R.S. Kottada, Friction deposition of titanium particle reinforced aluminum matrix composites, *Materials Science and Engineering: A*. 653 (2016) 71–83. <https://doi.org/10.1016/J.MSEA.2015.12.005>.
- [230] M. Muhammad, P.D. Nezhadfar, S. Thompson, A. Saharan, N. Phan, N. Shamsaei, A comparative investigation on the microstructure and mechanical properties of additively manufactured aluminum alloys, *Int J Fatigue*. 146 (2021) 106165. <https://doi.org/10.1016/j.ijfatigue.2021.106165>.
- [231] D. Svetlizky, M. Das, B. Zheng, A.L. Vyatskikh, S. Bose, A. Bandyopadhyay, J.M. Schoenung, E.J. Lavernia, N. Eliaz, Directed energy deposition (DED) additive manufacturing: Physical characteristics, defects, challenges and applications, *Materials Today*. (2021). <https://doi.org/10.1016/J.MATTOD.2021.03.020>.
- [232] A. Wang, H. Wang, Y. Wu, H. Wang, 3D printing of aluminum alloys using laser powder deposition: a review, *International Journal of Advanced Manufacturing Technology*. 116 (2021). <https://doi.org/10.1007/s00170-021-07440-5>.
- [233] S. Kumar, A. Kar, A Review of Solid-State Additive Manufacturing Processes, *Transactions of the Indian National Academy of Engineering*. (2021). <https://doi.org/10.1007/s41403-021-00270-7>.
- [234] K. Petráčková, J. Kondás, M. Guagliano, Mechanical Performance of Cold-Sprayed A357 Aluminum Alloy Coatings for Repair and Additive Manufacturing, *Journal of Thermal Spray Technology* 2017 26:8. 26 (2017) 1888–1897. <https://doi.org/10.1007/S11666-017-0643-5>.
- [235] H.R. Kotadia, G. Gibbons, A. Das, P.D. Howes, A review of Laser Powder Bed Fusion Additive Manufacturing of aluminium alloys: Microstructure and properties, *Addit Manuf.* 46 (2021) 102155. <https://doi.org/10.1016/j.addma.2021.102155>.
- [236] A.K. Gujba, M. Medraj, Power Ultrasonic Additive Manufacturing: Process Parameters, Microstructure, and Mechanical Properties, *Advances in Materials Science and Engineering*. 2020 (2020). <https://doi.org/10.1155/2020/1064870>.
- [237] H. Wu, X. Xie, S. Liu, S. Xie, R. Huang, C. Verdy, M. Liu, H. Liao, S. Deng, Y. Xie, Bonding behavior of Bi-metal-deposits produced by hybrid cold spray additive manufacturing, *J Mater Process Technol.* 299 (2021) 117375. <https://doi.org/10.1016/j.jmatprotec.2021.117375>.
- [238] E. Elfishawy, M.M.Z. Ahmed, M.M. El-Sayed Seleman, Additive Manufacturing of Aluminum Using Friction Stir Deposition, *Minerals, Metals and Materials Series*. (2020) 227–238. https://doi.org/10.1007/978-3-030-36296-6_21.
- [239] Z. Wang, Y. Zhang, A review of aluminum alloy fabricated by different processes of wire arc additive manufacturing, *Medziagotyra*. 27 (2021) 18–26. <https://doi.org/10.5755/j02.ms.22772>.
- [240] S. Cooke, K. Ahmadi, S. Willerth, R. Herring, Metal additive manufacturing: Technology, metallurgy and modelling, *J Manuf Process*. 57 (2020) 978–1003. <https://doi.org/10.1016/j.jmapro.2020.07.025>.

- [241] M. Kumar, G.J. Gibbons, A. Das, I. Manna, D. Tanner, H.R. Kotadia, Additive manufacturing of aluminium alloy 2024 by laser powder bed fusion: microstructural evolution, defects and mechanical properties, *Rapid Prototyp J.* (2021) 1388–1397. <https://doi.org/10.1108/RPJ-10-2020-0241>.
- [242] F. Caiazzo, V. Alfieri, P. Argenio, V. Sergi, Additive manufacturing by means of laser-aided directed metal deposition of 2024 aluminium powder: Investigation and optimization, *9* (2017) 1–12. <Http://Dx.Doi.Org/10.1177/1687814017714982>. <https://doi.org/10.1177/1687814017714982>.
- [243] Z. Qi, B. Qi, B. Cong, H. Sun, G. Zhao, J. Ding, Microstructure and mechanical properties of wire + arc additively manufactured 2024 aluminum alloy components: As-deposited and post heat-treated, *J Manuf Process.* 40 (2019) 27–36. <https://doi.org/10.1016/J.JMAPRO.2019.03.003>.
- [244] T. Mahale, D. Cormier, O. Harrysson, K. Ervin, E.P. Fitts, Advances in Electron Beam Melting of aluminum alloys, in: *Solid Freeform Fabrication Symposium Proceedings, 2007*: pp. 312–323.
- [245] L. Ajdelsztajn, A. Zúñiga, B. Jodoin, E.J. Lavernia, Cold-spray processing of a nanocrystalline Al-Cu-Mg-Fe-Ni alloy with Sc, *Journal of Thermal Spray Technology.* 15 (2006) 184–190. <https://doi.org/10.1361/105996306X107995>.
- [246] Y. Zhou, X. Lin, N. Kang, W. Huang, Z. Wang, Mechanical properties and precipitation behavior of the heat-treated wire + arc additively manufactured 2219 aluminum alloy, *Mater Charact.* 171 (2021) 110735. <https://doi.org/10.1016/J.MATCHAR.2020.110735>.
- [247] O.G. Rivera, P.G. Allison, L.N. Brewer, O.L. Rodriguez, J.B. Jordon, T. Liu, W.R. Whittington, R.L. Martens, Z. McClelland, C.J.T. Mason, L. Garcia, J.Q. Su, N. Hardwick, Influence of texture and grain refinement on the mechanical behavior of AA2219 fabricated by high shear solid state material deposition, *Materials Science and Engineering A.* 724 (2018) 547–558. <https://doi.org/10.1016/j.msea.2018.03.088>.
- [248] Y. Zhou, X. Lin, N. Kang, W. Huang, J. Wang, Z. Wang, Influence of travel speed on microstructure and mechanical properties of wire + arc additively manufactured 2219 aluminum alloy, *J Mater Sci Technol.* 37 (2020) 143–153. <https://doi.org/10.1016/j.jmst.2019.06.016>.
- [249] Q. Miao, D. Wu, D. Chai, Y. Zhan, G. Bi, F. Niu, G. Ma, Comparative study of microstructure evaluation and mechanical properties of 4043 aluminum alloy fabricated by wire-based additive manufacturing, *Mater Des.* 186 (2020) 108205. <https://doi.org/10.1016/J.MATDES.2019.108205>.
- [250] G. Langelandsvik, O.M. Akselsen, T. Furu, T. Furu, Review of Aluminum Alloy Development for Wire Arc, *Materials.* 14 (2021) 1–26. <https://doi.org/10.3390/ma14185370>.
- [251] D. Waller, A.J. Polizzi, J.J. Iten, Feasibility study of additively manufactured al-6061 ram2 parts for aerospace applications, *AIAA Scitech 2019 Forum.* (2019). <https://doi.org/10.2514/6.2019-0409>.
- [252] Y.K. Wei, X.T. Luo, X. Chu, G.S. Huang, C.J. Li, Solid-state additive manufacturing high performance aluminum alloy 6061 enabled by an in-situ micro-forging assisted cold spray, *Materials Science and Engineering: A.* 776 (2020) 139024. <https://doi.org/10.1016/J.MSEA.2020.139024>.

- [253] D. Gonzales, S. Liu, M. Domack, R. Hafley, Using powder cored tubular wire technology to enhance electron beam freeform fabricated structures, TMS Annual Meeting. 0 (2016) 183–189. https://doi.org/10.1007/978-3-319-48254-5_23.
- [254] K.E.K. Vimal, M. Naveen Srinivas, S. Rajak, Wire arc additive manufacturing of aluminium alloys: A review, Mater Today Proc. 41 (2019) 1139–1145. <https://doi.org/10.1016/j.matpr.2020.09.153>.
- [255] D. Gu, X. Shi, R. Poprawe, D.L. Bourell, R. Setchi, J. Zhu, Material-structure-performance integrated laser-metal additive manufacturing, Science (1979). 1487 (2021). <https://doi.org/10.1126/science.abg1487>.
- [256] A. Singh, A. Ramakrishnan, G. Dinda, Direct Laser Metal Deposition of Al 7050 Alloy, in: WCX™ 17: SAE World Congress Experience, SAE, 2017. <https://doi.org/10.4271/2017-01-0286>.
- [257] J.P. Isaac, S. Lee, N. Shamsaei, H. V. Tippur, Dynamic fracture behavior of additively manufactured Scalmlloy®: Effects of build orientation, heat-treatment and loading-rate, Materials Science and Engineering: A. 826 (2021) 141978. <https://doi.org/10.1016/j.msea.2021.141978>.
- [258] I. Karakurt, K.Y. Ho, C. Ledford, D. Gamzina, T. Horn, N.C. Luhmann, L. Lin, Development of a magnetically driven abrasive polishing process for additively manufactured copper structures, Procedia Manuf. 26 (2018) 798–805. <https://doi.org/10.1016/j.promfg.2018.07.097>.
- [259] I. Gibson, D. Rosen, B. Stucker, Directed Energy Deposition Processes, Additive Manufacturing Technologies. (2015) 245–268. https://doi.org/10.1007/978-1-4939-2113-3_10.
- [260] A. Bose, C.A. Schuh, J.C. Tobia, N. Tuncer, N.M. Mykulowycz, A. Preston, A.C. Barbati, B. Kernan, M.A. Gibson, D. Krause, T. Brzezinski, J. Schroers, R. Fulop, J.S. Myerberg, M. Sowerbutts, Y.M. Chiang, A. John Hart, E.M. Sachs, E.E. Lomeli, A.C. Lund, Traditional and additive manufacturing of a new Tungsten heavy alloy alternative, Int J Refract Metals Hard Mater. 73 (2018) 22–28. <https://doi.org/10.1016/J.IJRMHM.2018.01.019>.
- [261] F. Calignano, D. Manfredi, E.P. Ambrosio, L. Iuliano, P. Fino, Influence of process parameters on surface roughness of aluminum parts produced by DMLS, The International Journal of Advanced Manufacturing Technology. 67 (2012) 2743–2751. <https://doi.org/10.1007/s00170-012-4688-9>.
- [262] Q. Li, H. Zhang, D. Li, Z. Chen, S. Huang, Z. Lu, H. Yan, WxNbMoTa Refractory High-Entropy Alloys Fabricated by Laser Cladding Deposition, Materials 2019, Vol. 12, Page 533. 12 (2019) 533. <https://doi.org/10.3390/MA12030533>.
- [263] N.R. Philips, M. Carl, N.J. Cunningham, New Opportunities in Refractory Alloys, Metall Mater Trans A Phys Metall Mater Sci. 51 (2020) 3299–3310. <https://doi.org/10.1007/s11661-020-05803-3>.
- [264] G. Munday, J. Hogan, A. McDonald, On the microstructure-dependency of mechanical properties and failure of low-pressure cold-sprayed tungsten carbide-nickel metal matrix composite coatings, Surf Coat Technol. 396 (2020) 125947. <https://doi.org/10.1016/J.SURFCOAT.2020.125947>.

- [265] G. Wang, X. Sun, M. Huang, Y. Qin, Y. Yao, S. Yang, Influence of processing parameters on the microstructure and tensile property of 85 W-15Ni produced by laser direct deposition, *Int J Refract Metals Hard Mater.* 82 (2019) 227–233. <https://doi.org/10.1016/j.ijrmhm.2019.04.016>.
- [266] P. Rebesan, M. Bonesso, C. Gennari, R. Dima, A. Pepato, M. Vedani, Tungsten Fabricated by Laser Powder Bed Fusion, *BHM Berg- Und Hüttenmännische Monatshefte.* 166 (2021) 263–269. <https://doi.org/10.1007/s00501-021-01109-y>.
- [267] O. Mireles, Additive Manufacture of Refractory Metals for Aerospace Applications, in: *AIAA Propulsion and Energy 2020 Forum*, 2021: pp. 1–9. <https://doi.org/10.2514/6.2021-3234>.
- [268] A. V. Müller, G. Schlick, R. Neu, C. Anstatt, T. Klimkait, J. Lee, B. Pascher, M. Schmitt, C. Seidel, Additive manufacturing of pure tungsten by means of selective laser beam melting with substrate preheating temperatures up to 1000 °C, *Nuclear Materials and Energy.* 19 (2019) 184–188. <https://doi.org/10.1016/J.NME.2019.02.034>.
- [269] C.C. Eckley, R.A. Kemnitz, C.P. Fassio, C.R. Hartsfield, T.A. Leonhardt, Selective Laser Melting of Tungsten-Rhenium Alloys, *Jom.* (2021). <https://doi.org/10.1007/s11837-021-04776-x>.
- [270] G. Marinelli, F. Martina, S. Ganguly, S. Williams, H. Lewtas, D. Hancock, S. Mehraban, N. Lavery, Microstructure and thermal properties of unalloyed tungsten deposited by Wire + Arc Additive Manufacture, *Journal of Nuclear Materials.* 522 (2019) 45–53. <https://doi.org/10.1016/J.JNUCMAT.2019.04.049>.
- [271] M. Moorehead, K. Bertsch, M. Niezgoda, C. Parkin, M. Elbakhshwan, K. Sridharan, C. Zhang, D. Thoma, A. Couet, High-throughput synthesis of Mo-Nb-Ta-W high-entropy alloys via additive manufacturing, *Mater Des.* 187 (2020) 108358. <https://doi.org/10.1016/j.matdes.2019.108358>.
- [272] H. Dobbelsstein, M. Thiele, E.L. Gurevich, E.P. George, A. Ostendorf, Direct Metal Deposition of Refractory High Entropy Alloy MoNbTaW, *Phys Procedia.* 83 (2016) 624–633. <https://doi.org/10.1016/J.PHPRO.2016.08.065>.
- [273] O. Mireles, J. Caffrey, T. Godfroy, O. Rodriguez, C. Cox, Additive Manufacture of U-8Mo for LEU Space Nuclear Power Systems, in: *Proceedings of the JANNAF Additive Manufacture TIM*, 2020: p. 7523.
- [274] M.T. Stawovy, Comparison of LCAC and PM Mo deposited using Sciaky EBAM™, *Int J Refract Metals Hard Mater.* 73 (2018) 162–167. <https://doi.org/10.1016/j.ijrmhm.2018.02.009>.
- [275] J.L. Johnson, T. Palmer, Directed energy deposition of molybdenum, *Int J Refract Metals Hard Mater.* 84 (2019) 105029. <https://doi.org/10.1016/j.ijrmhm.2019.105029>.
- [276] P. Rebesan, M. Ballan, M. Bonesso, A. Campagnolo, S. Corradetti, R. Dima, C. Gennari, G.A. Longo, S. Mancin, M. Manzolaro, G. Meneghetti, A. Pepato, E. Visconti, M. Vedani, Pure molybdenum manufactured by Laser Powder Bed Fusion: thermal and mechanical characterization at room and high temperature, *Addit Manuf.* (2021) 102277. <https://doi.org/10.1016/J.ADDMA.2021.102277>.

- [277] P. Fernandez-Zelaia, C. Ledford, E.A.I. Ellis, Q. Campbell, A.M. Rossy, D.N. Leonard, M.M. Kirka, Crystallographic texture evolution in electron beam melting additive manufacturing of pure Molybdenum, *Mater Des.* 207 (2021) 109809. <https://doi.org/10.1016/j.matdes.2021.109809>.
- [278] G. Marinelli, F. Martina, H. Lewtas, D. Hancock, S. Ganguly, S. Williams, Functionally graded structures of refractory metals by wire arc additive manufacturing, *https://Doi.Org/10.1080/13621718.2019.1586162.* 24 (2019) 495–503. <https://doi.org/10.1080/13621718.2019.1586162>.
- [279] W.E. Frazier, Metal Additive Manufacturing: A Review, *J Mater Eng Perform.* 23 (2014) 1917–1928. <https://doi.org/10.1007/s11665-014-0958-z>.
- [280] C. Sungail, A. Abid, Spherical tantalum feed powder for metal additive manufacturing, *Metal Powder Report.* 73 (2018) 316–318. <https://doi.org/10.1016/J.MPRP.2018.03.046>.
- [281] D.R. Jones, S.J. Fensin, B.G. Ndefru, D.T. Martinez, C.P. Trujillo, G.T. Gray, Spall fracture in additive manufactured tantalum, *J Appl Phys.* 124 (2018) 225902. <https://doi.org/10.1063/1.5063930>.
- [282] H. Assadi, H. Kreye, F. Gärtner, T. Klassen, Cold spraying – A materials perspective, *Acta Mater.* 116 (2016) 382–407. <https://doi.org/10.1016/j.actamat.2016.06.034>.
- [283] G. Marinelli, F. Martina, S. Ganguly, S. Williams, Microstructure, hardness and mechanical properties of two different unalloyed tantalum wires deposited via wire + arc additive manufacture, *Int J Refract Metals Hard Mater.* 83 (2019) 1–20. <https://doi.org/10.1016/j.ijrmhm.2019.104974>.
- [284] D.M. Jafarlou, B.C. Sousa, M.A. Gleason, G. Ferguson, A.T. Nardi, D.L. Cote, I.R. Grosse, Solid-state additive manufacturing of tantalum using high-pressure cold gas-dynamic spray, *Addit Manuf.* 47 (2021) 102243. <https://doi.org/10.1016/J.ADDMA.2021.102243>.
- [285] O.R. Mireles, O. Rodriguez, Y. Gao, N. Philips, Additive manufacture of refractory alloy c103 for propulsion applications, in: *AIAA Propulsion and Energy 2020 Forum*, American Institute of Aeronautics and Astronautics Inc, AIAA, 2020: pp. 1–13. <https://doi.org/10.2514/6.2020-3500>.
- [286] P. Bangalore Venkatesh, L. Osborne, M. Fitzpatrick, M. Hollinbeck, D. Barnes, Pulse Performance Analysis of a 45 Newton Additively Manufactured Bipropellant Thruster, *J Propuls Power.* (2021) 1–10. <https://doi.org/10.2514/1.b38326>.
- [287] M. Hedges, N. Calder, Near net shape rapid manufacture & repair by LENS, in: *Cost Eff. Manuf. via Net-Shape Process*, Meet. Proc. RTO-MP-AVT-139. 2 (2006) 13.
- [288] A. Essop, NASA becomes first customer of HRL high strength aluminum 3d printing powder, (2019). <https://3dprintingindustry.com/news/nasa-becomes-first-customer-of-hrl-high-strength-aluminum-3d-printing-powder-163530/> (accessed March 16, 2021).
- [289] M. Muhammad, P.D. Nezhadfar, S. Thompson, A. Saharan, N. Phan, N. Shamsaei, A comparative investigation on the microstructure and mechanical properties of additively manufactured aluminum alloys, *Int J Fatigue.* 146 (2021) 106165. <https://doi.org/10.1016/j.ijfatigue.2021.106165>.

- [290] N.T. Aboulkhair, M. Simonelli, L. Parry, I. Ashcroft, C. Tuck, R. Hague, 3D printing of Aluminium alloys: Additive Manufacturing of Aluminium alloys using selective laser melting, *Prog Mater Sci.* 106 (2019) 100578. <https://doi.org/10.1016/j.pmatsci.2019.100578>.
- [291] S.C. Altıparmak, V.A. Yardley, Z. Shi, J. Lin, Challenges in additive manufacturing of high-strength aluminium alloys and current developments in hybrid additive manufacturing, *International Journal of Lightweight Materials and Manufacture.* 4 (2021) 246–261. <https://doi.org/10.1016/j.ijlmm.2020.12.004>.
- [292] D. Waller, A. Polizzi, J. Iten, Feasibility Study of Additively Manufactured Al-6061 RAM2 Parts for Aerospace Applications, in: *AIAA Scitech 2019 Forum*, American Institute of Aeronautics and Astronautics, Reston, Virginia, 2019. <https://doi.org/10.2514/6.2019-0409>.
- [293] C. Katsarelis, P. Chen, P. Gradl, C. Protz, Z. Jones, D. Ellis, L. Evans, Additive Manufacturing of NASA HR-1 Material for Liquid Rocket Engine Component Applications, *JANNAF Dec.* (2019) <https://ntrs.nasa.gov/search.jsp?R=20200001007>. <https://ntrs.nasa.gov/search.jsp?R=20200001007>.
- [294] T.M. Pollock, A.J. Clarke, S.S. Babu, Design and Tailoring of Alloys for Additive Manufacturing, *Metallurgical and Materials Transactions A.* 51 (2020) 6000–6019. <https://doi.org/10.1007/s11661-020-06009-3>.
- [295] Y.T. Tang, J.N. Ghoussoub, C. Panwisawas, D.M. Collins, S. Amirkhanlou, J.W.G. Clark, A.A.N. Németh, D. Graham McCartney, R.C. Reed, The Effect of Heat Treatment on Tensile Yielding Response of the New Superalloy ABD-900AM for Additive Manufacturing, in: 2020: pp. 1055–1065. https://doi.org/10.1007/978-3-030-51834-9_103.
- [296] Y.T. Tang, C. Panwisawas, J.N. Ghoussoub, Y. Gong, J.W.G. Clark, A.A.N. Németh, D.G. McCartney, R.C. Reed, Alloys-by-design: Application to new superalloys for additive manufacturing, *Acta Mater.* 202 (2021) 417–436. <https://doi.org/10.1016/j.actamat.2020.09.023>.
- [297] S. Ewald, F. Kies, S. Hermsen, M. Voshage, C. Haase, J.H. Schleifenbaum, Rapid Alloy Development of Extremely High-Alloyed Metals Using Powder Blends in Laser Powder Bed Fusion, *Materials.* 12 (2019) 1706. <https://doi.org/10.3390/ma12101706>.
- [298] P.R. Gradl, S. Greene, T. Wammen, Bimetallic Channel Wall Nozzle Development and Hot-fire Testing Using Additively Manufactured Laser Wire Direct Closeout Technology, in: *AIAA Propulsion and Energy 2019 Forum*, American Institute of Aeronautics and Astronautics, Reston, Virginia, 2019. <https://doi.org/10.2514/6.2019-4361>.
- [299] P.R. Gradl, C. Protz, J. Fikes, D. Ellis, L. Evans, A. Clark, S. Miller, T. Hudson, Lightweight Thrust Chamber Assemblies using Multi-Alloy Additive Manufacturing and Composite Overwrap, in: *AIAA Propulsion and Energy 2020 Forum*, American Institute of Aeronautics and Astronautics, Reston, Virginia, 2020. <https://doi.org/10.2514/6.2020-3787>.
- [300] C. Zhang, F. Chen, Z. Huang, M. Jia, G. Chen, Y. Ye, Y. Lin, W. Liu, B. Chen, Q. Shen, L. Zhang, E.J. Lavernia, Additive manufacturing of functionally graded materials: A review, *Materials Science and Engineering A.* 764 (2019) 138209. <https://doi.org/10.1016/j.msea.2019.138209>.

- [301] Y. Li, Z. Feng, L. Hao, L. Huang, C. Xin, Y. Wang, E. Bilotti, K. Essa, H. Zhang, Z. Li, F. Yan, T. Peijs, A Review on Functionally Graded Materials and Structures via Additive Manufacturing: From Multi-Scale Design to Versatile Functional Properties, *Adv Mater Technol.* 5 (2020) 1900981. <https://doi.org/10.1002/admt.201900981>.
- [302] A. Reichardt, A.A. Shapiro, R. Otis, R.P. Dillon, J.P. Borgonia, B.W. McEnerney, P. Hosemann, A.M. Beese, Advances in additive manufacturing of metal-based functionally graded materials, *International Materials Reviews.* 66 (2021) 1–29. <https://doi.org/10.1080/09506608.2019.1709354>.
- [303] J.C. Najmon, S. Raeisi, A. Tovar, Review of additive manufacturing technologies and applications in the aerospace industry, in: *Additive Manufacturing for the Aerospace Industry*, Elsevier, 2019: pp. 7–31. <https://doi.org/10.1016/B978-0-12-814062-8.00002-9>.
- [304] B. Westerweel, R.J.I. Basten, G.J. van Houtum, Traditional or Additive Manufacturing? Assessing Component Design Options through Lifecycle Cost Analysis, *Eur J Oper Res.* 270 (2018) 570–585. <https://doi.org/10.1016/j.ejor.2018.04.015>.
- [305] S. Singamneni, Y. LV, A. Hewitt, R. Chalk, W. Thomas, D. Jordison, Additive Manufacturing for the Aircraft Industry: A Review, *Journal of Aeronautics & Aerospace Engineering.* 08 (2019). <https://doi.org/10.35248/2168-9792.19.8.215>.
- [306] A. du Plessis, S. Beretta, Killer notches: The effect of as-built surface roughness on fatigue failure in AlSi10Mg produced by laser powder bed fusion, *Addit Manuf.* 35 (2020) 101424. <https://doi.org/10.1016/j.addma.2020.101424>.
- [307] A. du Plessis, E. Macdonald, Hot isostatic pressing in metal additive manufacturing: X-ray tomography reveals details of pore closure, *Addit Manuf.* 34 (2020) 101191. <https://doi.org/10.1016/j.addma.2020.101191>.
- [308] N. Sanaei, A. Fatemi, Defect-based fatigue life prediction of L-PBF additive manufactured metals, *Eng Fract Mech.* 244 (2021) 107541. <https://doi.org/10.1016/j.engfracmech.2021.107541>.
- [309] A. du Plessis, I. Yadroitsev, I. Yadroitsava, S.G. le Roux, X-Ray Microcomputed Tomography in Additive Manufacturing: A Review of the Current Technology and Applications, *3D Print Addit Manuf.* 5 (2018) 227–247. <https://doi.org/10.1089/3dp.2018.0060>.
- [310] O. Sigmund, K. Maute, Topology optimization approaches: A comparative review, *Structural and Multidisciplinary Optimization.* 48 (2013) 1031–1055. <https://doi.org/10.1007/s00158-013-0978-6>.
- [311] M.P. Bendsøe, N. Kikuchi, Generating optimal topologies in structural design using a homogenization method, *Comput Methods Appl Mech Eng.* 71 (1988) 197–224. [https://doi.org/10.1016/0045-7825\(88\)90086-2](https://doi.org/10.1016/0045-7825(88)90086-2).
- [312] F. Cucinotta, E. Guglielmino, G. Longo, G. Risitano, D. Santonocito, F. Sfravara, Topology optimization additive manufacturing-oriented for a biomedical application, in: *Lecture Notes in Mechanical Engineering*, 2019: pp. 184–193. https://doi.org/10.1007/978-3-030-12346-8_18.

- [313] J. Liu, A.T. Gaynor, S. Chen, Z. Kang, K. Suresh, A. Takezawa, L. Li, J. Kato, J. Tang, C.C.L. Wang, L. Cheng, X. Liang, A.C. To, Current and future trends in topology optimization for additive manufacturing, *Structural and Multidisciplinary Optimization*. 57 (2018) 2457–2483. <https://doi.org/10.1007/s00158-018-1994-3>.
- [314] M. Benedetti, A. du Plessis, R.O. Ritchie, M. Dallago, S.M.J. Razavi, F. Berto, Architected cellular materials: A review on their mechanical properties towards fatigue-tolerant design and fabrication, *Materials Science and Engineering R: Reports*. 144 (2021) 100606. <https://doi.org/10.1016/j.mser.2021.100606>.
- [315] A. Panesar, M. Abdi, D. Hickman, I. Ashcroft, Strategies for functionally graded lattice structures derived using topology optimisation for Additive Manufacturing, *Addit Manuf.* 19 (2018) 81–94. <https://doi.org/10.1016/j.addma.2017.11.008>.
- [316] H. NAKAJIMA, Fabrication, properties and application of porous metals with directional pores, *Prog Mater Sci.* 52 (2007) 1091–1173. <https://doi.org/10.1016/j.pmatsci.2006.09.001>.
- [317] T. Maconachie, M. Leary, B. Lozanovski, X. Zhang, M. Qian, O. Faruque, M. Brandt, SLM lattice structures: Properties, performance, applications and challenges, *Mater Des.* 183 (2019) 108137. <https://doi.org/10.1016/j.matdes.2019.108137>.
- [318] A. du Plessis, C. Broeckhoven, I. Yadroitsava, I. Yadroitsev, C.H. Hands, R. Kunju, D. Bhate, Beautiful and Functional: A Review of Biomimetic Design in Additive Manufacturing, *Addit Manuf.* 27 (2019) 408–427. <https://doi.org/10.1016/j.addma.2019.03.033>.
- [319] C. Yan, L. Hao, A. Hussein, P. Young, D. Raymont, Advanced lightweight 316L stainless steel cellular lattice structures fabricated via selective laser melting, *Mater Des.* 55 (2014) 533–541. <https://doi.org/10.1016/j.matdes.2013.10.027>.
- [320] C. Yan, L. Hao, A. Hussein, D. Raymont, Evaluations of cellular lattice structures manufactured using selective laser melting, *Int J Mach Tools Manuf.* 62 (2012) 32–38. <https://doi.org/10.1016/j.ijmachtools.2012.06.002>.
- [321] J.-H. Zhu, W.-H. Zhang, L. Xia, Topology Optimization in Aircraft and Aerospace Structures Design, *Archives of Computational Methods in Engineering*. 23 (2016) 595–622. <https://doi.org/10.1007/s11831-015-9151-2>.
- [322] M. Tomlin, J. Meyer, Topology Optimization of an Additive Layer Manufactured (ALM) Aerospace Part, *The 7th Altair CAE Technology Conference 2011*. (2011) 1–9.
- [323] L. Meng, W. Zhang, D. Quan, G. Shi, L. Tang, Y. Hou, P. Breitkopf, J. Zhu, T. Gao, From Topology Optimization Design to Additive Manufacturing: Today’s Success and Tomorrow’s Roadmap, *Archives of Computational Methods in Engineering*. 27 (2020) 805–830. <https://doi.org/10.1007/s11831-019-09331-1>.

- [324] D. Jia, F. Li, Y. ZHANG, 3D-printing process design of lattice compressor impeller based on residual stress and deformation, *Sci Rep.* 10 (2020) 600. <https://doi.org/10.1038/s41598-019-57131-1>.
- [325] P. Rochus, J.-Y. Plessier, M. van Elsen, J.-P. Kruth, R. Carrus, T. Dormal, New applications of rapid prototyping and rapid manufacturing (RP/RM) technologies for space instrumentation, *Acta Astronaut.* 61 (2007) 352–359. <https://doi.org/10.1016/j.actaastro.2007.01.004>.
- [326] EOS GmbH, Certified for Universal Success: Additive Manufacturing of Satellite Components, 2018. https://cdn.eos.info/b63671baf552082c/88339183a70a/CS_M_Aerospace_RUAG_Citim_en_WEB.pdf.
- [327] EOS, Future Ariane propulsion module: Simplified by additive manufacturing, (2018). https://www.eos.info/01_parts-and-applications/case_studies_applications_parts/_case_studies_pdf/en_cases/cs_m_aerospace_arianegroup_en.pdf.
- [328] NASA, NASA Tests Limits of 3-D Printing with Powerful Rocket Engine Check, (2013). <https://www.nasa.gov/exploration/systems/sls/3d-printed-rocket-injector.html> (accessed March 16, 2021).
- [329] NASA, Hot-Fire Tests Show 3-D Printed Rocket Parts Rival Traditionally Manufactured Parts, (2013). <https://www.nasa.gov/exploration/systems/sls/3dprinting.html> (accessed March 15, 2021).
- [330] S. Kim, S.K. Moon, A Part Consolidation Design Method for Additive Manufacturing based on Product Disassembly Complexity, *Applied Sciences.* 10 (2020) 1100. <https://doi.org/10.3390/app10031100>.
- [331] T.D. Ngo, A. Kashani, G. Imbalzano, K.T.Q. Nguyen, D. Hui, Additive manufacturing (3D printing): A review of materials, methods, applications and challenges, *Compos B Eng.* 143 (2018) 172–196. <https://doi.org/10.1016/j.compositesb.2018.02.012>.
- [332] ManufacturingTalkRadio, Pratt & Whitney uses 3D printing for Aero Engine Parts, (2019). <https://optics.org/news/6/4/7>.
- [333] P. Gradl, O. Mireles, N. Andrews, Intro to Additive Manufacturing for Propulsion Systems, AIAA Joint Propulsion Conference. (2018) 83. <https://ntrs.nasa.gov/api/citations/20180006357/downloads/20180006357.pdf>.
- [334] J. Loy, P. Tatham, Redesigning production systems, Environmental Footprints and Eco-Design of Products and Processes. (2016) 145–168. https://doi.org/10.1007/978-981-10-0549-7_7.
- [335] General Electric - Additive, GE9X Engine Additive Parts, (n.d.). [https://www.ge.com/additive/sites/default/files/2020-08/GE9X Additive parts.pdf](https://www.ge.com/additive/sites/default/files/2020-08/GE9X%20Additive%20parts.pdf) (accessed February 2, 2021).
- [336] D. Saltzman, M. Bichnevicius, S. Lynch, T.W. Simpson, E.W. Reutzel, C. Dickman, R. Martukanitz, Design and evaluation of an additively manufactured aircraft heat exchanger, Elsevier Ltd, 2018. <https://doi.org/10.1016/j.applthermaleng.2018.04.032>.

- [337] M. Bici, S. Brischetto, F. Campana, C.G. Ferro, C. Seclì, S. Varetto, P. Maggiore, A. Mazza, Development of a multifunctional panel for aerospace use through SLM additive manufacturing, *Procedia CIRP*. 67 (2018) 215–220. <https://doi.org/10.1016/j.procir.2017.12.202>.
- [338] S. Donath, 3D printing a rocket engine, (2019). <https://www.etmm-online.com/3d-printing-a-rocket-engine-a-886960/>.
- [339] P.R. Gradl, C.S. Protz, Technology advancements for channel wall nozzle manufacturing in liquid rocket engines, *Acta Astronaut.* 174 (2020) 148–158. <https://doi.org/10.1016/j.actaastro.2020.04.067>.
- [340] P.R. Gradl, C.S. Protz, K. Cooper, D. Ellis, L.J. Evans, C. Garcia, GRCop-42 Development and Hot-fire Testing Using Additive Manufacturing Powder Bed Fusion for Channel-cooled Combustion Chambers, in: *AIAA Propulsion and Energy 2019 Forum*, American Institute of Aeronautics and Astronautics, Reston, Virginia, 2019. <https://doi.org/10.2514/6.2019-4228>.
- [341] S. Woolley, Hot Stuff: To Build More Affordable Rocket Engines, NASA Researchers Are Using The Latest 3D Printers — And 1 Ancient Metal, (2019). <https://www.ge.com/news/reports/hot-stuff-to-build-more-affordable-rocket-engines-nasa-researchers-are-using-the-latest-3d-printers-and-1-ancient-metal> (accessed March 16, 2021).
- [342] Team NASA, NASA Advances New Alloys and Scale of Metal Additive Manufacturing, (2021). <https://3dprint.com/278670/nasa-advances-new-alloys-and-scale-of-metal-additive-manufacturing/> (accessed March 16, 2021).
- [343] NASA, 3D Printed Rocket Engine Parts Survive 23 Hot-Fire Tests, (2020). <https://www.nasa.gov/centers/marshall/news/releases/2020/3d-printed-rocket-engine-parts-survive-23-hot-fire-tests.html> (accessed March 16, 2021).
- [344] R. Liu, Z. Wang, T. Sparks, F. Liou, J. Newkirk, Aerospace applications of laser additive manufacturing, in: *Laser Additive Manufacturing: Materials, Design, Technologies, and Applications*, Elsevier, 2017: pp. 351–371. <https://doi.org/10.1016/B978-0-08-100433-3.00013-0>.
- [345] E. RUAG Altair, Sentinel Antenna Bracket, (n.d.). https://www.eos.info/01_parts-and-applications/case_studies_applications_parts/_case_studies_pdf/en_cases/cs_m_aerospace_ruag_en.pdf (accessed February 2, 2021).
- [346] Altair, Generative Design and Topology Optimization, 2019. https://web.altair.com/generative-design-report?product__c=Inspire&msdcampaignid=COMP-06217-C9P7Y&detailed_lead_source=AdWords_Generative_Design_ad&campaign_source=AdWords&utm_campaign=Generative+Design&utm_term=%2Bgenerative%2Bdesign&utm_medium=ppc&utm.
- [347] Optomec, LENS® Component Repair Solution, (2016). https://optomec.com/wp-content/uploads/2014/04/Optomec_LENS_Blisk_Repair_Datasheet.pdf.
- [348] P.R. Gradl, C.S. Protz, K. Zagorski, V. Doshi, H. McCallum, Additive Manufacturing and Hot-fire Testing of Bimetallic GRCop-84 and C-18150 Channel-Cooled Combustion Chambers Using Powder

- Bed Fusion and Inconel 625 Hybrid Directed Energy Deposition, in: AIAA Propulsion and Energy 2019 Forum, American Institute of Aeronautics and Astronautics, Reston, Virginia, 2019. <https://doi.org/10.2514/6.2019-4390>.
- [349] M. Leary, Design for additive manufacturing, 2019. <https://doi.org/10.1016/C2017-0-04238-6>.
- [350] J.-H. Zhu, Y. Li, W.-H. Zhang, J. Hou, Shape preserving design with structural topology optimization, *Structural and Multidisciplinary Optimization*. 53 (2016) 893–906. <https://doi.org/10.1007/s00158-015-1364-3>.
- [351] L.J. Kumar, C.G. Krishnadas Nair, Current Trends of Additive Manufacturing in the Aerospace Industry, in: *Advances in 3D Printing & Additive Manufacturing Technologies*, Springer Singapore, Singapore, 2017: pp. 39–54. https://doi.org/10.1007/978-981-10-0812-2_4.
- [352] M.E. Orme, M. Gschweidl, M. Ferrari, I. Madera, F. Mouriaux, Designing for additive manufacturing: Lightweighting through topology optimization enables lunar spacecraft, *Journal of Mechanical Design, Transactions of the ASME*. 139 (2017). <https://doi.org/10.1115/1.4037304>.
- [353] N. Titanium, Norsk Titanium to Deliver the World’s First FAA-Approved, 3D-Printed, Structural Titanium Components to Boeing | Additive Manufacturing (AM), (n.d.). <http://additivemanufacturing.com/2017/04/10/norsk-titanium-to-deliver-the-worlds-first-faa-approved-3d-printed-structural-titanium-components-to-boeing/>.
- [354] Marie Caujolle, First titanium 3D-printed part installed into serial production aircraft, (2017) 1. <https://www.airbus.com/newsroom/press-releases/en/2017/09/first-titanium-3d-printed-part-installed-into-serial-production-.html> (accessed February 2, 2021).
- [355] D.A. Pisano, J.L. Bromberg, NASA and the Space Industry, 2000. <https://doi.org/10.2307/2652132>.
- [356] S.M. Yusuf, S. Cutler, N. Gao, S. Mohd Yusuf, S. Cutler, N. Gao, Review: The impact of metal additive manufacturing on the aerospace industry, *Metals* (Basel). 9 (2019) 1286. <https://doi.org/10.3390/met9121286>.
- [357] L. Valdevit, A.J. Jacobsen, J.R. Greer, W.B. Carter, Protocols for the Optimal Design of Multi-Functional Cellular Structures: From Hypersonics to Micro-Architected Materials, *Journal of the American Ceramic Society*. 94 (2011) s15–s34. <https://doi.org/10.1111/j.1551-2916.2011.04599.x>.
- [358] Y. Wang, L. Zhang, S. Daynes, H. Zhang, S. Feih, M.Y. Wang, Design of graded lattice structure with optimized mesostructures for additive manufacturing, *Mater Des.* 142 (2018) 114–123. <https://doi.org/10.1016/j.matdes.2018.01.011>.
- [359] A. Hussein, L. Hao, C. Yan, R. Everson, P. Young, Advanced lattice support structures for metal additive manufacturing, *J Mater Process Technol.* 213 (2013) 1019–1026. <https://doi.org/10.1016/j.jmatprotec.2013.01.020>.
- [360] nTopology, Air Force optimizes cubesat using Architected Materials, (n.d.). <https://ntopology.com/case-studies/air-force-optimizes-cubesat-using-architected-materials/> (accessed February 15, 2021).

- [361] L. Chougrani, J.-P. Pernot, P. Véron, S. Abed, Parts internal structure definition using non-uniform patterned lattice optimization for mass reduction in additive manufacturing, *Eng Comput.* 35 (2019) 277–289. <https://doi.org/10.1007/s00366-018-0598-2>.
- [362] Thales Alenia Space, Diphasic heat spreader, (2017). <https://www.metal-am.com/large-scale-metal-lattice-structure-complex-satellite-part/> (accessed March 22, 2021).
- [363] D. Nagy, D. Zhao, D. Benjamin, Nature-Based Hybrid Computational Geometry System for Optimizing Component Structure, in: *Humanizing Digital Reality*, Springer Singapore, Singapore, 2018: pp. 167–176. https://doi.org/10.1007/978-981-10-6611-5_15.
- [364] Airbus Group, Pioneering bionic 3D printing - Learning from nature, (2016). <https://www.airbus.com/newsroom/news/en/2016/03/Pioneering-bionic-3D-printing.html>.
- [365] B. Danon, How this light-weight airplane seat can save airlines \$200,000,000 (and dramatically reduce carbon emissions), (2017). <https://adsknews.autodesk.com/stories/how-this-light-weight-airplane-seat-can-save-airlines-200000000-and-dramatically-reduce-carbon-emissions>.
- [366] F. Froes, Combining additive manufacturing with conventional casting and reduced density materials to greatly reduce the weight of airplane components such as passenger seat frames, in: *Additive Manufacturing for the Aerospace Industry*, Elsevier, 2019: pp. 419–425. <https://doi.org/10.1016/B978-0-12-814062-8.00021-2>.
- [367] W. Tao, M.C. Leu, Design of lattice structure for additive manufacturing, in: *2016 International Symposium on Flexible Automation (ISFA)*, IEEE, 2016: pp. 325–332. <https://doi.org/10.1109/ISFA.2016.7790182>.
- [368] T. Allison, A. Rimple, R. Pelton, J. Moore, J. Wilkes, K. Wygant, Manufacturing and Testing Experience with Direct Metal Laser Sintering for Closed Centrifugal Compressor Impellers, in: *43rd Turbomachinery & 30th Pump Users Symposia*, 2014. <https://doi.org/https://doi.org/10.21423/R1F64W>.
- [369] C. Noon, Taking Wing: Boeing’s 777X, Powered By GE’s Record-Breaking Engines, Soars Above Seattle, (2020). <https://www.ge.com/news/reports/taking-wing-boeings-777x-powered-by-ges-record-breaking-engines-soars-above-seattle>.
- [370] S. Gautam Chhibber, and S.T. Jahrmarcht, Stack Up the Layers WP, 2020. <https://assets.new.siemens.com/siemens/assets/api/uuid:a4659ee4-06f3-4089-91cb-399854611dbb/version:1588178306/wp-march2020-stackupthelayers-article.pdf>.
- [371] D. Godfrey, Additive Manufacturing (AM) and the Honeywell Global Initiative, *International Society of Air Breathing Engines Conference*, Phoenix, AZ. (2015). [https://drc.libraries.uc.edu/bitstream/handle/2374.UC/745770/ISABE2015_CS%26A_Donald Godfrey_227_MANUSCRIPT_20193.pdf?sequence=2](https://drc.libraries.uc.edu/bitstream/handle/2374.UC/745770/ISABE2015_CS%26A_Donald%20Godfrey_227_MANUSCRIPT_20193.pdf?sequence=2).
- [372] General Electric, New manufacturing milestone: 30,000 additive fuel nozzles, *GE Additive*. (2018). <https://www.ge.com/additive/stories/new-manufacturing-milestone-30000-additive-fuel->

- nozzles%0Ahttps://www.ge.com/additive/blog/new-manufacturing-milestone-30000-additive-fuel-nozzles.
- [373] C. Hauser, Case Study: Laser Powder Metal Deposition Manufacturing of Complex Real Parts, *Materials Joining and Engineering Technologies*. (2014). <https://docplayer.net/48458850-Case-study-laser-powder-metal-deposition-manufacturing-of-complex-real-parts.html>.
 - [374] W.E. Frazier, Metal additive manufacturing: A review, *J Mater Eng Perform*. 23 (2014) 1917–1928. <https://doi.org/10.1007/s11665-014-0958-z>.
 - [375] F. Walachowicz, I. Bernsdorf, U. Papenfuss, C. Zeller, A. Graichen, V. Navrotsky, N. Rajvanshi, C. Kiener, Comparative Energy, Resource and Recycling Lifecycle Analysis of the Industrial Repair Process of Gas Turbine Burners Using Conventional Machining and Additive Manufacturing, *J Ind Ecol*. 21 (2017) S203–S215. <https://doi.org/10.1111/jiec.12637>.
 - [376] Optomec, LENS® Component Repair Solution, (2016). https://optomec.com/wp-content/uploads/2014/04/Optomec_LENS_Blisk_Repair_Datasheet.pdf (accessed February 2, 2021).
 - [377] F. Froes, R. Boyer, Additive manufacturing for the aerospace industry, *Additive Manufacturing for the Aerospace Industry*. (2019) 1–482. <https://doi.org/10.1201/9780429466236-10>.
 - [378] L. Xue, Y. Li, J. Chen, S. Wang, Laser Consolidation: A Novel Additive Manufacturing Process for Making Net-Shape Functional Metallic Components for Gas Turbine Applications, in: *Proceedings of the ASME Turbo Expo, American Society of Mechanical Engineers*, 2015. <https://doi.org/10.1115/GT2015-43971>.
 - [379] NASA, NASA Rocket Fuel Pump Tests Pave Way for Methane-Fueled Mars Lander, (2016). <https://www.nasa.gov/centers/marshall/news/news/releases/2016/nasa-rocket-fuel-pump-tests-pave-way-for-methane-fueled-mars-lander.html> (accessed March 16, 2021).
 - [380] NASA, Successful NASA Rocket Fuel Pump Tests Pave Way for 3-D Printed Demonstrator Engine, (2015). <https://www.nasa.gov/centers/marshall/news/news/releases/2015/successful-nasa-rocket-fuel-pump-tests-pave-way-for-3-d-printed-demonstrator-engine.html> (accessed March 16, 2021).
 - [381] P. Gradl, W. Brandsmeier, M. Calvert, S. Greene, D.O. Neal, J. Richard, K. Morgan, Additive Manufacturing Overview : Propulsion Applications , Design for and Lessons Learned, in: 2018. <https://core.ac.uk/download/pdf/154738243.pdf>.
 - [382] B. West, E. Robertson, R. Osborne, M. Calvert, Additive Manufacturing for Affordable Rocket Engines, (2016). <http://hdl.handle.net/2060/20160012051>.
 - [383] D. O’Neal, Evaluation of Additively Manufactured Demonstration Hardware for a Turbopump Application, in: *Liquid Propulsion Subcommittee and Advanced Materials Panel Additive Manufacturing for Propulsion Applications Technical Interchange Meeting*, 2014. <https://ntrs.nasa.gov/api/citations/20140016517/downloads/20140016517.pdf>.

- [384] D. Jafari, W.W. Wits, The utilization of selective laser melting technology on heat transfer devices for thermal energy conversion applications: A review, *Renewable and Sustainable Energy Reviews*. 91 (2018) 420–442. <https://doi.org/10.1016/j.rser.2018.03.109>.
- [385] P.R. Gradl, Rapid Fabrication Techniques for Liquid Rocket Channel Wall Nozzles, in: 52nd AIAA/SAE/ASEE Joint Propulsion Conference, American Institute of Aeronautics and Astronautics, Reston, Virginia, 2016. <https://doi.org/10.2514/6.2016-4771>.
- [386] Z. Gobetz, A. Rowen, C. Dickman, K. Meinert, R. Martukanitz, Utilization of Additive Manufacturing for Aerospace Heat Exchangers, (2016). <https://apps.dtic.mil/sti/citations/AD1010878>.
- [387] NASA, NASA's Perseverance Rover Bringing 3D-Printed Metal Parts to Mars, (2020). <http://www.nasa.gov/feature/jpl/nasas-perseverance-rover-bringing-3d-printed-metal-parts-to-mars> (accessed February 19, 2021).
- [388] H.P. de Bock, Exploration of a Hybrid Analytical Thermal Topology Optimization Method for an Additively Manufactured Heat Sink, in: 2018 17th IEEE Intersociety Conference on Thermal and Thermomechanical Phenomena in Electronic Systems (ITherm), IEEE, 2018: pp. 761–767. <https://doi.org/10.1109/ITHERM.2018.8419585>.
- [389] Conflux Technologies, Conflux sectioned F1 heat exchanger, (n.d.). https://www.eos.info/01_parts-and-applications/case_studies_applications_parts/_case_studies_pdf/en_cases/cs_m_industry_conflux_en.pdf (accessed February 2, 2021).
- [390] nTopology, Cobra Aero uses multiphysics simulation to optimize UAV engine for thermal mitigation, (n.d.). <https://ntopology.com/case-studies/cobra-aero/> (accessed February 16, 2021).
- [391] Hieta, High and Low temperature heat exchangers, (n.d.). <https://www.hieta.biz/wp-content/uploads/2018/09/HiETA-Brochure.pdf> (accessed February 2, 2021).
- [392] A. Orlova, R. Nogueira, P. Chimenti, The Present and Future of the Space Sector: A Business Ecosystem Approach, *Space Policy*. 52 (2020) 101374. <https://doi.org/10.1016/j.spacepol.2020.101374>.
- [393] R. Pyle, Space 2.0: How Private Spaceflight, a Resurgent NASA, and International Partners are Creating a New Space Age, 2019. <https://books.google.com/books?hl=en&lr=&id=IxRbDwAAQBAJ&oi=fnd&pg=PT295&dq=cosmic+dust+nasa&ots=MfLHyB6N1y&sig=DuM9nZMGtN1w6uUSgUWCF7OgEWE>.
- [394] K. Whealan George, The Economic Impacts of the Commercial Space Industry, *Space Policy*. 47 (2019) 181–186. <https://doi.org/10.1016/j.spacepol.2018.12.003>.
- [395] J.E. Catchpole, The International Space Station, Praxis, New York, NY, 2008. <https://doi.org/10.1007/978-0-387-78145-7>.
- [396] C. Marchesan, LinkedIn Post, (n.d.). <https://www.linkedin.com/feed/update/urn:li:activity:6742612928245198848/> (accessed December 10, 2020).

- [397] C. Suborbitals, Coaxial Swirl Injectors, (n.d.). <https://www.3dprintingmedia.network/copenhagen-suborbitals-3d-prints-coaxial-swirl-injector-by-metal-binder-jetting/> (accessed February 2, 2021).
- [398] M.K. Ilja Müller, I. Petkov, S. Bletsch, K. Huybrechts, and P. van Cauwenbergh, 3D-PRINTED COAXIAL INJECTOR FOR A LOX/KEROSENE ROCKET ENGINE, (n.d.).
- [399] NASA, Rocket Injector, (n.d.). <https://www.designworldonline.com/building-a-3d-printed-rocket-engine/> (accessed February 2, 2021).
- [400] NASA, RS-25 Pogo z-baffle, (n.d.). https://www.nasa.gov/exploration/systems/sls/j2x/3d_print.html (accessed February 2, 2021).
- [401] Aerojet Rocketdyne, AR1 Preburner, (n.d.). <https://3dprintingindustry.com/news/aerojet-rocketdyne-successfully-tests-3d-printed-ar1-engine-bid-replace-current-rd-180-112732/> (accessed February 2, 2021).
- [402] Aerojet Rocketdyne, Thrust Chamber, (n.d.). <https://www.i3dmfg.com/aerojet-rocketdyne-tests-3d-printed-thrust-chamber-success/> (accessed February 2, 2021).
- [403] E. Ariane Group, Rocket Combustion Chamber, (n.d.). <https://3dprintingindustry.com/news/arianegroup-successfully-tests-fully-3d-printed-combustion-chamber-for-esa-rocket-engine-172357/> (accessed February 2, 2021).
- [404] L. Sorrentino, M. Marchetti, C. Bellini, A. Delfini, M. Albano, Design and manufacturing of an isogrid structure in composite material: Numerical and experimental results, *Compos Struct.* 143 (2016) 189–201. <https://doi.org/10.1016/j.compstruct.2016.02.043>.
- [405] Lockheed Martin, Satalite Fuel Tank, (n.d.). https://news.lockheedmartin.com/2018-07-11-Giant-Satellite-Fuel-Tank-Sets-New-Record-for-3-D-Printed-Space-Parts#assets_115 (accessed February 2, 2021).
- [406] Relativity Space, Terran 1 Fuel Tank, (n.d.). <https://spectrum.ieee.org/aerospace/space-flight/the-worlds-largest-3d-metal-printer-is-churning-out-rockets> (accessed February 2, 2021).
- [407] SpaceX, SuperDraco Engine Chamber, (n.d.). <https://www.designnews.com/design-hardware-software/spacex-reveals-3d-printed-rocket-engine-parts> (accessed February 2, 2021).
- [408] S. Clark, Astronauts fly with SpaceX in landmark launch for commercial spaceflight, n.d. <https://spaceflightnow.com/2020/11/16/astronauts-ride-spacex-crew-capsule-in-landmark-launch-for-commercial-spaceflight/>.
- [409] SpaceX, SuperDraco Test Fire Mosaic, (2015). <https://www.flickr.com/photos/spacex/22925132396/in/photolist-rzAxfp-AVPiao/> (accessed March 16, 2021).
- [410] SpaceX, SuperDracos, (2014). <https://www.flickr.com/photos/spacex/16789102495/in/photolist-rzAxfp-AVPiao/> (accessed March 16, 2021).

- [411] Relativity Space, Aeon Engine, (n.d.). <https://spectrum.ieee.org/aerospace/space-flight/the-worlds-largest-3d-metal-printer-is-churning-out-rockets> (accessed February 2, 2021).
- [412] L. EOS, E-2 Rocket Engine, (n.d.). <https://launcherspace.com/engine-2> (accessed February 2, 2021).
- [413] Team NASA, NASA Advances New Alloys and Scale of Metal Additive Manufacturing, (2021). <https://3dprint.com/278670/nasa-advances-new-alloys-and-scale-of-metal-additive-manufacturing/amp/> (accessed February 21, 2021).
- [414] P. Gradl, Principles of Directed Energy Deposition for Aerospace Applications, in: Drive AM Industry Connect W.M. Keck Center for 3D Innovation, 2021. <https://doi.org/https://doi.org/10.13140>.
- [415] W. Bryan, Future Rocket Engines May Include Large-Scale 3D Printing, (2020). <https://www.nasa.gov/centers/marshall/news/releases/2020/future-rocket-engines-may-include-large-scale-3d-printing.html> (accessed March 16, 2021).
- [416] Team NASA, NASA Advances New Alloys and Scale of Metal Additive Manufacturing, (2021). <https://3dprint.com/278670/nasa-advances-new-alloys-and-scale-of-metal-additive-manufacturing/> (accessed February 21, 2021).
- [417] J. Hahn, Hyperganic uses AI to design 3D-printed rocket engine, (2020). <https://www.dezeen.com/2020/03/30/hyperganic-ai-rocket-engine-3d-printed/> (accessed February 2, 2021).
- [418] A. Sola, A. Nouri, Microstructural porosity in additive manufacturing: The formation and detection of pores in metal parts fabricated by powder bed fusion, *J Adv Manuf Process.* 1 (2019). <https://doi.org/10.1002/amp2.10021>.
- [419] N. Sanaei, A. Fatemi, N. Phan, Defect characteristics and analysis of their variability in metal L-PBF additive manufacturing, *Mater Des.* 182 (2019) 108091. <https://doi.org/10.1016/j.matdes.2019.108091>.
- [420] A. du Plessis, Effects of process parameters on porosity in laser powder bed fusion revealed by X-ray tomography, *Addit Manuf.* 30 (2019) 100871. <https://doi.org/10.1016/j.addma.2019.100871>.
- [421] S.A. Khairallah, A.T. Anderson, A. Rubenchik, W.E. King, Laser powder-bed fusion additive manufacturing: Physics of complex melt flow and formation mechanisms of pores, spatter, and denudation zones, *Acta Mater.* 108 (2016) 36–45. <https://doi.org/10.1016/j.actamat.2016.02.014>.
- [422] Y.Y. Sun, S.L. Lu, S. Gulizia, C.H. Oh, D. Fraser, M. Leary, M. Qian, Fatigue Performance of Additively Manufactured Ti-6Al-4V: Surface Condition vs. Internal Defects, *JOM.* 72 (2020) 1022–1030. <https://doi.org/10.1007/s11837-020-04025-7>.
- [423] N. Sanaei, A. Fatemi, Defects in additive manufactured metals and their effect on fatigue performance: A state-of-the-art review, *Prog Mater Sci.* 117 (2021) 100724. <https://doi.org/10.1016/j.pmatsci.2020.100724>.

- [424] A. du Plessis, I. Yadroitsava, I. Yadroitsev, Effects of defects on mechanical properties in metal additive manufacturing: A review focusing on X-ray tomography insights, *Mater Des.* 187 (2020) 108385. <https://doi.org/10.1016/j.matdes.2019.108385>.
- [425] N.E. Uzan, R. Shneck, O. Yeheskel, N. Frage, Fatigue of AlSi10Mg specimens fabricated by additive manufacturing selective laser melting (AM-SLM), *Materials Science and Engineering: A.* 704 (2017) 229–237. <https://doi.org/10.1016/j.msea.2017.08.027>.
- [426] H. Masuo, Y. Tanaka, S. Morokoshi, H. Yagura, T. Uchida, Y. Yamamoto, Y. Murakami, Influence of defects, surface roughness and HIP on the fatigue strength of Ti-6Al-4V manufactured by additive manufacturing, *Int J Fatigue.* 117 (2018) 163–179. <https://doi.org/10.1016/j.ijfatigue.2018.07.020>.
- [427] W.E. King, A.T. Anderson, R.M. Ferencz, N.E. Hodge, C. Kamath, S.A. Khairallah, A.M. Rubenchik, Laser powder bed fusion additive manufacturing of metals; physics, computational, and materials challenges, *Appl Phys Rev.* 2 (2015) 041304. <https://doi.org/10.1063/1.4937809>.
- [428] H.L. Wei, T. Mukherjee, W. Zhang, J.S. Zuback, G.L. Knapp, A. De, T. DebRoy, Mechanistic models for additive manufacturing of metallic components, *Prog Mater Sci.* 116 (2021) 100703. <https://doi.org/10.1016/j.pmatsci.2020.100703>.
- [429] C. Meier, R.W. Penny, Y. Zou, J.S. Gibbs, A.J. Hart, Thermophysical phenomena in metal additive manufacturing by selective laser melting: fundamentals, modeling, simulation, and experimentation, *Annual Review of Heat Transfer.* 20 (2017) 241–316. <https://doi.org/10.1615/AnnualRevHeatTransfer.2018019042>.
- [430] A. du Plessis, E. MacDonald, J.M. Waller, F. Berto, Non-destructive testing of parts produced by laser powder bed fusion, in: *Fundamentals of Laser Powder Bed Fusion of Metals*, Elsevier, 2021: pp. 277–300. <https://doi.org/10.1016/B978-0-12-824090-8.00016-0>.
- [431] A. du Plessis, J.M. Waller, S.G. le Roux, I. Yadroitsava, I. Yadroitsev, J. Els, J. Prinsloo, X-Ray Computed Tomography Inspection in Metal Additive Manufacturing: The Role of Witness Specimens, in: *Structural Integrity of Additive Manufactured Materials & Parts*, ASTM International, 100 Barr Harbor Drive, PO Box C700, West Conshohocken, PA 19428-2959, 2020: pp. 139–156. <https://doi.org/10.1520/STP163120190109>.
- [432] F.H. Kim, S.P. Moylan, Literature review of metal additive manufacturing defects, Gaithersburg, MD, 2018. <https://doi.org/10.6028/NIST.AMS.100-16>.
- [433] A. du Plessis, S.G. le Roux, Standardized X-ray tomography testing of additively manufactured parts: A round robin test, *Addit Manuf.* 24 (2018) 125–136. <https://doi.org/10.1016/j.addma.2018.09.014>.
- [434] J.C. Williams, R.R. Boyer, Opportunities and Issues in the Application of Titanium Alloys for Aerospace Components, *Metals (Basel).* 10 (2020) 705. <https://doi.org/10.3390/met10060705>.
- [435] D. Gu, X. Shi, R. Poprawe, D.L. Bourell, R. Setchi, J. Zhu, Material-structure-performance integrated laser-metal additive manufacturing, *Science*, (n.d.). <https://doi.org/10.1126/science.abg1487>.

- [436] I. Yadroitsev, I. Yadroitsava, A. du Plessis, E. MacDonald, *Fundamentals of Laser Powder Bed Fusion of Metals*, 1st ed., Elsevier, 2021. <https://www.elsevier.com/books/fundamentals-of-laser-powder-bed-fusion-of-metals/yadroitsev/978-0-12-824090-8>.
- [437] A. du Plessis, I. Yadroitsava, I. Yadroitsev, Effects of defects on mechanical properties in metal additive manufacturing: A review focusing on X-ray tomography insights, *Mater Des.* 187 (2020) 108385. <https://doi.org/10.1016/j.matdes.2019.108385>.
- [438] M.C. Brennan, J.S. Keist, T.A. Palmer, Defects in Metal Additive Manufacturing Processes, *J Mater Eng Perform.* 30 (2021) 4808–4818. <https://doi.org/10.1007/s11665-021-05919-6>.
- [439] N. Sanaei, A. Fatemi, N. Phan, Defect characteristics and analysis of their variability in metal L-PBF additive manufacturing, *Mater Des.* 182 (2019) 108091. <https://doi.org/10.1016/j.matdes.2019.108091>.
- [440] J.A. Slotwinski, E.J. Garboczi, K.M. Hebenstreit, Porosity Measurements and Analysis for Metal Additive Manufacturing Process Control, *J Res Natl Inst Stand Technol.* 119 (2014) 494. <https://doi.org/10.6028/jres.119.019>.
- [441] A. Sola, A. Nouri, Microstructural porosity in additive manufacturing: The formation and detection of pores in metal parts fabricated by powder bed fusion, *J Adv Manuf Process.* 1 (2019). <https://doi.org/10.1002/amp2.10021>.
- [442] J.A. Slotwinski, E.J. Garboczi, Porosity of additive manufacturing parts for process monitoring, in: 2014: pp. 1197–1204. <https://doi.org/10.1063/1.4864957>.
- [443] J.A. Slotwinski, E.J. Garboczi, Porosity of additive manufacturing parts for process monitoring, in: 2014: pp. 1197–1204. <https://doi.org/10.1063/1.4864957>.
- [444] R. Morgan, C.J. Sutcliffe, W. O'Neill, Density analysis of direct metal laser re-melted 316L stainless steel cubic primitives, *J Mater Sci.* 39 (2004) 1195–1205. <https://doi.org/10.1023/B:JMSC.0000013875.62536.fa>.
- [445] S. Coeck, M. Bisht, J. Plas, F. Verbist, Prediction of lack of fusion porosity in selective laser melting based on melt pool monitoring data, *Addit Manuf.* 25 (2019) 347–356. <https://doi.org/10.1016/j.addma.2018.11.015>.
- [446] I. Watanabe, J.H. Watkins, H. Nakajima, M. Atsuta, T. Okabe, Effect of Pressure Difference on the Quality of Titanium Casting, *J Dent Res.* 76 (1997) 773–779. <https://doi.org/10.1177/00220345970760031001>.
- [447] A. du Plessis, I. Yadroitsava, I. Yadroitsev, Effects of defects on mechanical properties in metal additive manufacturing: A review focusing on X-ray tomography insights, *Mater Des.* 187 (2020) 108385. <https://doi.org/10.1016/j.matdes.2019.108385>.
- [448] H. Gong, K. Rafi, H. Gu, G.D. Janaki Ram, T. Starr, B. Stucker, Influence of defects on mechanical properties of Ti–6Al–4V components produced by selective laser melting and electron beam melting, *Mater Des.* 86 (2015) 545–554. <https://doi.org/10.1016/j.matdes.2015.07.147>.

- [449] U. Zerbst, M. Madia, C. Klinger, D. Bettge, Y. Murakami, Defects as a root cause of fatigue failure of metallic components. III: Cavities, dents, corrosion pits, scratches, *Eng Fail Anal.* 97 (2019) 759–776. <https://doi.org/10.1016/j.engfailanal.2019.01.034>.
- [450] N.E. Uzan, S. Ramati, R. Shneck, N. Frage, O. Yeheskel, On the effect of shot-peening on fatigue resistance of AlSi10Mg specimens fabricated by additive manufacturing using selective laser melting (AM-SLM), *Addit Manuf.* 21 (2018) 458–464. <https://doi.org/10.1016/j.addma.2018.03.030>.
- [451] J. Damon, S. Dietrich, F. Vollert, J. Gibmeier, V. Schulze, Process dependent porosity and the influence of shot peening on porosity morphology regarding selective laser melted AlSi10Mg parts, *Addit Manuf.* 20 (2018) 77–89. <https://doi.org/10.1016/j.addma.2018.01.001>.
- [452] L. Hackel, J.R. Rankin, A. Rubenchik, W.E. King, M. Matthews, Laser peening: A tool for additive manufacturing post-processing, *Addit Manuf.* 24 (2018) 67–75. <https://doi.org/10.1016/j.addma.2018.09.013>.
- [453] Y. Kajima, A. Takaichi, T. Nakamoto, T. Kimura, N. Kittikundecha, Y. Tsutsumi, N. Nomura, A. Kawasaki, H. Takahashi, T. Hanawa, N. Wakabayashi, Effect of adding support structures for overhanging part on fatigue strength in selective laser melting, *J Mech Behav Biomed Mater.* 78 (2018) 1–9. <https://doi.org/10.1016/j.jmbbm.2017.11.009>.
- [454] L. Nickels, Free standing 3D printing, *Metal Powder Report.* 73 (2018) 307–309. <https://doi.org/10.1016/j.mprp.2018.09.003>.
- [455] P.J. Withers, C. Bouman, S. Carmignato, V. Cnudde, D. Grimaldi, C.K. Hagen, E. Maire, M. Manley, A. du Plessis, S.R. Stock, X-ray computed tomography, *Nature Reviews Methods Primers.* 1 (2021) 18. <https://doi.org/10.1038/s43586-021-00015-4>.
- [456] A.C. Kak, M. Slaney, Principles of Computerized Tomographic Imaging, *Principles of Computerized Tomographic Imaging.* (2001). <https://doi.org/10.1137/1.9780898719277>.
- [457] A. du Plessis, I. Yadroitsev, I. Yadroitsava, S.G. le Roux, X-Ray Microcomputed Tomography in Additive Manufacturing: A Review of the Current Technology and Applications, *3D Print Addit Manuf.* 5 (2018) 227–247. <https://doi.org/10.1089/3dp.2018.0060>.
- [458] A. du Plessis, S.G. le Roux, A. Guelpa, The CT Scanner Facility at Stellenbosch University: An open access X-ray computed tomography laboratory, *Nucl Instrum Methods Phys Res B.* 384 (2016) 42–49. <https://doi.org/10.1016/j.nimb.2016.08.005>.
- [459] A. du Plessis, C. Broeckhoven, A. Guelpa, S.G. le Roux, Laboratory x-ray micro-computed tomography: a user guideline for biological samples, *Gigascience.* 6 (2017). <https://doi.org/10.1093/gigascience/gix027>.
- [460] A. du Plessis, P. Sperling, A. Beerlink, L. Tshabalala, S. Hoosain, N. Mathe, S.G. le Roux, Standard method for microCT-based additive manufacturing quality control 2: Density measurement, *MethodsX.* 5 (2018) 1117–1123. <https://doi.org/10.1016/j.mex.2018.09.006>.

- [461] A. du Plessis, P. Sperling, A. Beerlink, L. Tshabalala, S. Hoosain, N. Mathe, S.G. le Roux, Standard method for microCT-based additive manufacturing quality control 1: Porosity analysis, *MethodsX*. 5 (2018) 1102–1110. <https://doi.org/10.1016/j.mex.2018.09.005>.
- [462] A. Thompson, N. Senin, I. Maskery, L. Körner, S. Lawes, R. Leach, Internal surface measurement of metal powder bed fusion parts, *Addit Manuf.* 20 (2018) 126–133. <https://doi.org/10.1016/j.addma.2018.01.003>.
- [463] G. Pyka, G. Kerckhofs, I. Papantoniou, M. Speirs, J. Schrooten, M. Wevers, Surface Roughness and Morphology Customization of Additive Manufactured Open Porous Ti6Al4V Structures, *Materials*. 6 (2013) 4737–4757. <https://doi.org/10.3390/ma6104737>.
- [464] A. Thompson, N. Senin, C. Giusca, R. Leach, Topography of selectively laser melted surfaces: A comparison of different measurement methods, *CIRP Annals*. 66 (2017) 543–546. <https://doi.org/10.1016/j.cirp.2017.04.075>.
- [465] A. du Plessis, P. Sperling, A. Beerlink, O. Kruger, L. Tshabalala, S. Hoosain, S.G. le Roux, Standard method for microCT-based additive manufacturing quality control 3: Surface roughness, *MethodsX*. 5 (2018) 1111–1116. <https://doi.org/10.1016/j.mex.2018.09.004>.
- [466] D. Ramos, F. Belblidia, J. Sienz, New scanning strategy to reduce warpage in additive manufacturing, *Addit Manuf.* 28 (2019) 554–564. <https://doi.org/10.1016/j.addma.2019.05.016>.
- [467] M. Shange, I. Yadroitsava, A. du Plessis, I. Yadroitsev, Roughness and Near-Surface Porosity of Unsupported Overhangs Produced by High-Speed Laser Powder Bed Fusion, *3D Print Addit Manuf.* (2021) 3dp.2020.0097. <https://doi.org/10.1089/3dp.2020.0097>.
- [468] ASTM, B822-17, (2020). <https://doi.org/https://www.doi.org/10.1520/B0822-17>.
- [469] R. Cunningham, A. Nicolas, J. Madsen, E. Fodran, E. Anagnostou, M.D. Sangid, A.D. Rollett, Analyzing the effects of powder and post-processing on porosity and properties of electron beam melted Ti-6Al-4V, *Mater Res Lett*. 5 (2017) 516–525. <https://doi.org/10.1080/21663831.2017.1340911>.
- [470] A. du Plessis, P. Sperling, A. Beerlink, W.B. du Preez, S.G. le Roux, Standard method for microCT-based additive manufacturing quality control 4: Metal powder analysis, *MethodsX*. 5 (2018) 1336–1345. <https://doi.org/10.1016/j.mex.2018.10.021>.
- [471] J.A. Slotwinski, E.J. Garboczi, P.E. Stutzman, C.F. Ferraris, S.S. Watson, M.A. Peltz, Characterization of Metal Powders Used for Additive Manufacturing, *J Res Natl Inst Stand Technol*. 119 (2014) 460. <https://doi.org/10.6028/jres.119.018>.
- [472] H.P. Tang, M. Qian, N. Liu, X.Z. Zhang, G.Y. Yang, J. Wang, Effect of Powder Reuse Times on Additive Manufacturing of Ti-6Al-4V by Selective Electron Beam Melting, *JOM*. 67 (2015) 555–563. <https://doi.org/10.1007/s11837-015-1300-4>.

- [473] T. DebRoy, T. Mukherjee, J.O. Milewski, J.W. Elmer, B. Ribic, J.J. Blecher, W. Zhang, Scientific, technological and economic issues in metal printing and their solutions, *Nat Mater.* 18 (2019) 1026–1032. <https://doi.org/10.1038/s41563-019-0408-2>.
- [474] J.M. Waller, B.H. Parker, K.L. Hodges, E.R. Burke, J.L. Walker, E.R. Generazio, *Nondestructive Evaluation of Additive Manufacturing State-of-the-Discipline Report*, 2014.
- [475] N. Wunderlin, Design of an Ablatively Cooled Combustion Chamber for the SAFFIRE engine, in: *Annual Conference of the Aeronautical Society of South Africa (AESSA)*, Pretoria, 2016.
- [476] ASReG, ASReG SAFFIRE Rocket Engine, (n.d.). <https://aerospace.ukzn.ac.za/rocket-projects/saffire/> (accessed June 10, 2022).
- [477] N. Singh, *The Hydrodynamic Design and Analysis of a Liquid Oxygen Pump Impeller for a Rocket Engine*, University of KwaZulu-Natal, 2018.
- [478] C. Chetty, *The Hydrodynamic Design and Analysis of an RP-1 Pump for a Liquid Rocket Engine*, University of KwaZulu-Natal, 2018.
- [479] Y. Singh, *Design, Manufacture and Testing of a Liquid Rocket Engine Electropump*, University of KwaZulu-Natal, 2022.
- [480] Spaceflight101, Electron, Spaceflight, (2017). <https://spaceflight101.com/spacerockets/electron/> (accessed June 10, 2022).
- [481] RocketLab, RocketLab Electron, (n.d.). <https://www.rocketlabusa.com/about/gallery/> (accessed June 10, 2022).
- [482] Sulzer Pumps Ltd., *Centrifugal Pump Handbook*, 3rd ed., 2010. <https://doi.org/10.1016/c2009-0-15015-8>.
- [483] Autodesk, Netfabb Knowledge Network, (n.d.). <https://knowledge.autodesk.com/support/netfabb?sort=score> (accessed September 11, 2022).
- [484] A. du Plessis, M. Tshibalanganda, S.G. le Roux, Not all scans are equal: X-ray tomography image quality evaluation, *Mater Today Commun.* 22 (2020) 100792. <https://doi.org/10.1016/j.mtcomm.2019.100792>.
- [485] ANSYS, ANSYS Additive 2021 R1 User Guide, (2021). https://ansyshelp.ansys.com/account/secured?returnurl=/Views/Secured/prod_page.html?pn=AdditiveManufacturing&prodver=21.1&lang=en (accessed September 11, 2022).
- [486] nTopology, nTopology Knowledge Base, (n.d.). <https://support.ntopology.com/hc/en-us/categories/360003678133-Knowledge-Base> (accessed September 11, 2022).
- [487] A.J. Dunbar, E.R. Denlinger, M.F. Gouge, T.W. Simpson, P. Michaleris, Comparisons of laser powder bed fusion additive manufacturing builds through experimental in situ distortion and temperature measurements, *Addit Manuf.* 15 (2017) 57–65. <https://doi.org/10.1016/j.addma.2017.03.003>.

- [488] T. Mukherjee, V. Manvatkar, A. De, T. DebRoy, Mitigation of thermal distortion during additive manufacturing, *Scr Mater.* 127 (2017) 79–83. <https://doi.org/10.1016/j.scriptamat.2016.09.001>.
- [489] A. du Plessis, S.G. le Roux, Standardized X-ray tomography testing of additively manufactured parts: A round robin test, *Addit Manuf.* 24 (2018) 125–136. <https://doi.org/10.1016/j.addma.2018.09.014>.
- [490] B. Bhattacharyya, B. Doloi, Advanced finishing processes, in: *Modern Machining Technology*, Elsevier, 2020: pp. 675–743. <https://doi.org/10.1016/B978-0-12-812894-7.00008-6>.
- [491] A. Nycz, A.I. Adediran, M.W. Noakes, L.J. Love, Large scale metal additive techniques review, *Solid Freeform Fabrication 2016: Proceedings of the 27th Annual International Solid Freeform Fabrication Symposium - An Additive Manufacturing Conference, SFF 2016*. (2016) 2001–2006.
- [492] W.J. Sames, F.A. List, S. Pannala, R.R. Dehoff, S.S. Babu, The metallurgy and processing science of metal additive manufacturing, *International Materials Reviews.* 61 (2016) 315–360. <https://doi.org/10.1080/09506608.2015.1116649>.
- [493] AMCM, AMCM M 4K, (n.d.). <https://amcm.com/machines/amcm-m4k> (accessed June 9, 2022).
- [494] A. du Plessis, P. Rossouw, X-ray computed tomography of a titanium aerospace investment casting, *Case Studies in Nondestructive Testing and Evaluation.* 3 (2015) 21–26. <https://doi.org/10.1016/j.csndt.2015.03.001>.
- [495] S. Ucsnik, M. Scheerer, S. Zaremba, D.H. Pahr, Experimental investigation of a novel hybrid metal–composite joining technology, *Compos Part A Appl Sci Manuf.* 41 (2010) 369–374. <https://doi.org/10.1016/j.compositesa.2009.11.003>.
- [496] R. Falck, S.M. Goushegir, J.F. dos Santos, S.T. Amancio-Filho, AddJoining: A novel additive manufacturing approach for layered metal-polymer hybrid structures, *Mater Lett.* 217 (2018) 211–214. <https://doi.org/10.1016/j.matlet.2018.01.021>.
- [497] ANSYS, Ansys Additive Calibration Guide 2021 R1, (2021). https://storage.ansys.com/mbu-assets/additive/Calibration/211/ANSYS_Additive_Calibration_Guide_2021_R1.pdf (accessed November 5, 2021).
- [498] Airborne Engineering, 3D printing technology provides another breakthrough at Westcott, (2019). <https://www.westcottvp.com/news-and-insights/news/3d-printing-technology-provides-another-breakthrough-at-westcott> (accessed March 26, 2021).
- [499] Airbus, Reflector Bracket, (n.d.). https://www.eos.info/01_parts-and-applications/case_studies_applications_parts/_case_studies_pdf/en_cases/cs_m_aerospace_airbus_en.pdf (accessed February 2, 2021).
- [500] Airbus, A350 Cabin Bracket Connector, (n.d.). <https://www.makepartsfast.com/additive-manufacturing-enables-bionic-aircraft-designs/> (accessed February 2, 2021).

- [501] Airbus, A350 XWB Pylon Bracket, (n.d.). <https://www.airbus.com/newsroom/press-releases/en/2017/09/first-titanium-3d-printed-part-installed-into-serial-production-.html> (accessed February 2, 2021).
- [502] A. 3D Systems, RF Filter, (n.d.). <https://www.3dsystems.com/customer-stories/first-air-worthy-metal-printed-rf-filter-ready-to-take> (accessed February 2, 2021).
- [503] A. EOS, Aircraft Door Locking Shaft, (n.d.). <https://www.eos.info/en/3d-printing-examples-applications/aerospace-3d-printing/airbus-case-study> (accessed February 2, 2021).
- [504] Autodesk, Bionic Partition Project, (2016). <https://www.archdaily.com/780661/the-livings-parametric-3d-printed-airplane-partition-is-designed-to-mimic-bone-structure> (accessed February 2, 2021).
- [505] S. Amaero Engineering, Aero-Engine, (n.d.). <https://sief.org.au/wp-content/uploads/2019/02/RP04-153AeroFinalReportSummary.pdf> (accessed February 2, 2021).
- [506] E. Ariane Group, Ariane 6 Injector head, (n.d.). https://www.eos.info/01_parts-and-applications/case_studies_applications_parts/_case_studies_pdf/en_cases/cs_m_aerospace_arianegroup_en.pdf (accessed February 2, 2021).
- [507] Autodesk, Lattice Aircraft Seat, (n.d.). <https://adsknews.autodesk.com/stories/how-this-light-weight-airplane-seat-can-save-airlines-200000000-and-dramatically-reduce-carbon-emissions> (accessed February 2, 2021).
- [508] Betatype Ltd, Rocket Nozzle - Betatype, (2020). <https://www.betaty.pe/case-studies/rocket-nozzle/> (accessed February 2, 2021).
- [509] Blue Origin, BE-7 Engine, (n.d.). <https://www.amchronicle.com/news/blue-origin-3d-printed-be-7-engine-confirms-capability-to-land-on-moon/> (accessed February 2, 2021).
- [510] Blue Origin, Jeff Bezos gives a geeky boost to Blue Origin's BE-4 rocket engine boost pump, (2017). <https://www.geekwire.com/2017/jeff-bezos-blue-origin-be4-ox-boost-pump/> (accessed March 17, 2021).
- [511] D. 3D systems, Liquid Rocket Engine Injector, (n.d.). https://www.small-launcher.eu/wp-content/uploads/Müller-3d-printed-coaxial-injector-for-a-LOX-kerosene-rocket-engine-SP2018_176-Sevilla-2018.pdf (accessed February 2, 2021).
- [512] E. Airbus, Borescope Bosses, (n.d.). https://www.eos.info/01_parts-and-applications/case_studies_applications_parts/_case_studies_pdf/en_cases/cs_m_aerospace_mtuaeroengines_en.pdf (accessed February 2, 2021).
- [513] E. Airbus, Hydraulic spoiler manifold, (n.d.). https://www.eos.info/01_parts-and-applications/case_studies_applications_parts/_case_studies_pdf/en_cases/cs_m_aerospace_liebherr_en.pdf (accessed February 2, 2021).

- [514] E. Vectoflow, Flow Measurement Probe, (n.d.). https://www.eos.info/01_parts-and-applications/case_studies_applications_parts/_case_studies_pdf/en_cases/cs_m_aerospace_vectoflow_en.pdf (accessed February 2, 2021).
- [515] S. and A. EOS, Vertical tail plane bracket, (n.d.). https://www.eos.info/01_parts-and-applications/case_studies_applications_parts/_case_studies_pdf/en_cases/cs_m_aerospace_sogeti_en.pdf (accessed February 2, 2021).
- [516] F. 3D Systems, GE Aircraft Bracket, (n.d.). <https://www.3dsystems.com/learning-center/case-studies/topology-optimization-and-dmp-combine-meet-ge-aircraft-engine-bracket> (accessed February 2, 2021).
- [517] General Electric - Additive, Sump Cover, (n.d.). <https://www.ge.com/additive/us-air-force> (accessed February 2, 2021).
- [518] General Electric - Additive, NACA inlet, (n.d.). https://www.ge.com/additive/sites/default/files/2020-06/GE_Addworks_Brochure_US_EN_Digital.pdf (accessed February 2, 2021).
- [519] GKN Aerospace, ArianeGroup, GKN Aerospace Wins ArianeGroup Contract for Ground-breaking Additively Manufactured Rocket Engine Turbines, (2018).
- [520] GKN Aerospace, Turbine exhaust casting H sectors, (2020). <https://www.gknaerospace.com/en/our-technology/2020/gkn-aerospace-ansys-and-additive-industries-have-collaborated-to-demonstrate-their-shared-capabilities-in-the-production-of-a-large-aero-engine-component/> (accessed June 10, 2021).
- [521] GKN Aerospace, GKN Aerospace: The development of Additive Manufacturing at a global Tier 1 aerospace supplier, (2016). <https://www.metal-am.com/articles/gkn-aerospace-3d-printing-at-a-tier-1-aerospace-supplier/> (accessed June 10, 2021).
- [522] GKN Aerospace, GKN Aerospace engine systems journey to introduce additive manufacturing, (2017). <https://www.gknaerospace.com/en/our-technology/2017/gkn-aerospace-engine-systems-journey-to-introduce-additive-manufacturing/> (accessed June 10, 2021).
- [523] GKN Aerospace, GKN Aerospace has successful first flight of additive manufactured optical ice detection probe, (2017). <https://www.gknaerospace.com/en/newsroom/news-releases/2017/gkn-aerospace-has-successful-first-flight-of-additive-manufactured-optical-ice-detection-probe/> (accessed June 10, 2021).
- [524] L.S. Oerlikon, Propulsion system Impeller, (n.d.). <http://lenaspace.com/2017/06/16/lena-space-developing-turbopumps/> (accessed February 2, 2021).
- [525] L.-A. Airbus, Landing Gear Sensor Bracket, (n.d.). <https://amfg.ai/2020/01/23/applications-spotlight-3d-printed-brackets/> (accessed February 2, 2021).
- [526] NASA, Rocket Combustion Chamber, (n.d.). <https://3dprint.com/59881/nasa-3d-prints-copper-rocket/> (accessed February 2, 2021).

- [527] NASA, Rocket Engine, (n.d.). <https://3dprint.com/111799/nasa-3d-printed-rocket-engine/> (accessed February 2, 2021).
- [528] NASA, SLS B2 Turbopump, (n.d.). <https://www.nasa.gov/centers/marshall/news/news/releases/2015/successful-nasa-rocket-fuel-pump-tests-pave-way-for-3-d-printed-demonstrator-engine.html> (accessed February 2, 2021).
- [529] NASA, Pogo Accumulator Assembly, (n.d.). <https://3dprint.com/205157/nasa-tests-rs-25-engines/> (accessed February 2, 2021).
- [530] NASA, Maintenance Port Cover, (n.d.). https://www.nasa.gov/exploration/systems/sls/j2x/3d_print.html (accessed February 2, 2021).
- [531] Oerlikon, Turbine Blade, (n.d.). <https://www.oerlikon.com/stories/2018/08/03/whats-being-added-to-additive-manufacturing/> (accessed February 2, 2021).
- [532] Oerlikon, Double Nozzle, (n.d.). <https://www.metal-am.com/articles/oerlikon-a-leading-developer-of-3d-printed-components-and-materials/> (accessed February 2, 2021).
- [533] Orbex, Rocket Engine, (n.d.). <https://www.aero-mag.com/orbex-single-piece-rocket-engine-3d-printing-slm-800/> (accessed February 2, 2021).
- [534] Orbital ATK, Orbital ATK Successfully Tests Hypersonic Engine Combustor Produced Through Additive Manufacturing, (2016). <http://www.parabolicarc.com/2016/01/19/orbital-atk-successfully-tests-hypersonic-engine-combustor-produced-additive-manufacturing/> (accessed March 17, 2021).
- [535] Parabilis Space Technologies, Multi-Material Nozzle, (n.d.). <https://parabilis-space.com/december-8-2017-parabilis-demonstrates-multi-material-printing-technology/> (accessed February 2, 2021).
- [536] P.S. Technologies, RCS Thruster, (n.d.). <https://parabilis-space.com/october-30-2017-parabilis-test-fires-innovative-3d-printed-thruster/> (accessed February 2, 2021).
- [537] E. RUAG Altair, Star Tracker Bracket, (n.d.). https://cdn2.hubspot.net/hubfs/47251/Altair_Generative_Design_Report.pdf (accessed February 2, 2021).
- [538] E. RUAG Altair, LEROS Engine support bracket, (n.d.). https://cdn2.hubspot.net/hubfs/47251/Altair_Generative_Design_Report.pdf (accessed February 2, 2021).
- [539] Siemens, High Pressure Hydraulic Manifold, (n.d.). <https://www.siemens-energy.com/global/en/news/key-topics/additive-manufacturing.html#Download> (accessed February 2, 2021).
- [540] SpaceX, Main Oxidiser Valve Body, (n.d.). <https://www.designnews.com/design-hardware-software/spacex-reveals-3d-printed-rocket-engine-parts> (accessed February 2, 2021).

- [541] SpaceX, Raptor Engine Components, (n.d.). <https://spaceflight101.com/spx/spacex-raptor/> (accessed February 2, 2021).
- [542] S.A. Boeing, Access Door Latch, (n.d.). <https://www.additivemanufacturing.media/articles/what-is-the-role-for-additive-manufacturing-in-aircraft-structural-components> (accessed February 2, 2021).
- [543] T.A.S. 3D Systems, Antenna Bracket, (n.d.). <https://www.3dsystems.com/learning-center/case-studies/tipping-point> (accessed February 2, 2021).
- [544] The Welding Institute, Helicopter combustion chamber, (n.d.). <https://docplayer.net/48458850-Case-study-laser-powder-metal-deposition-manufacturing-of-complex-real-parts.html> (accessed February 2, 2021).
- [545] ULA, Vulcan Bellows Feedline Housing, (n.d.). <https://www.ge.com/additive/sites/default/files/2019-05/GE Additive Print Services - Large Format Prints - ULA Case Study.pdf> (accessed February 2, 2021).
- [546] NASA, Virgin Orbit, NASA and Virgin Orbit 3D Print, Test Rocket Combustion Chamber, (2019). <https://www.nasa.gov/centers/marshall/news/news/releases/2019/nasa-and-virgin-orbit-3d-print-test-rocket-combustion-chamber.html> (accessed March 17, 2021).
- [547] M. Godec, S. Malej, D. Feizpour, Donik, M. Balažic, D. Klobčar, L. Pambaguian, M. Conradi, A. Kocijan, Hybrid additive manufacturing of Inconel 718 for future space applications, *Mater Charact.* 172 (2021). <https://doi.org/10.1016/j.matchar.2020.110842>.
- [548] Z. Zhan, H. Li, A novel approach based on the elastoplastic fatigue damage and machine learning models for life prediction of aerospace alloy parts fabricated by additive manufacturing, 2021. <https://doi.org/10.1016/j.ijfatigue.2020.106089>.
- [549] P. van den Berg, B. Jyoti, R. Hermesen, Investigation of Thermal Behaviour of Additively Manufactured Green Bi-Propellant Thrusters in CubeSat Applications Using Transient Thermal Modelling, in: *Space Propulsion Conference 2020+12*, 2021.
- [550] I. Waugh, E. Moore, A. Greig, J. Macfarlane, Additive manufacture of rocket engine combustion chambers from CuCrZr (C-18150) using the DMLS process, in: *Space Propulsion Conference 2020+12*, 2021.
- [551] I. Waugh, Moore, A. Greig, J. Macfarlane, W.D. Cleland, Additive manufacture of rocket engine combustion chambers using the ABD R -900AM nickel Superalloy, in: *Space Propulsion Conference 2020+1*, 2021.
- [552] G. D'Emilia, A. di Ilio, A. Gaspari, E. Natale, A.G. Stamopoulos, Uncertainty assessment for measurement and simulation in selective laser melting: A case study of an aerospace part, *Acta IMEKO.* 9 (2020) 96–105. https://doi.org/10.21014/acta_imeko.v9i4.720.
- [553] C. Dordlofva, P. Törlind, Qualification challenges with additive manufacturing in space applications, *Solid Freeform Fabrication 2017: Proceedings of the 28th Annual International Solid Freeform Fabrication Symposium - An Additive Manufacturing Conference, SFF 2017.* (2020) 2699–2712.

- [554] E. Fereiduni, A. Ghasemi, M. Elbestawi, Selective laser melting of aluminum and titanium matrix composites: Recent progress and potential applications in the aerospace industry, *Aerospace*. 7 (2020). <https://doi.org/10.3390/AEROSPACE7060077>.
- [555] L. Gao, S. Yu, R. Yu, T. He, Y. Shi, Study on Arc Additive Manufacturing Process and Properties of 5356 Aluminum Alloy Rocket Booster Module Transition End Frame, *Jixie Gongcheng Xuebao/Journal of Mechanical Engineering*. 56 (2020) 28–36. <https://doi.org/10.3901/JME.2020.08.028>.
- [556] D. Gu, H. Zhang, H. Chen, H. Zhang, L. Xi, Laser Additive Manufacturing of High-Performance Metallic Aerospace Components, *Zhongguo Jiguang/Chinese Journal of Lasers*. 47 (2020). <https://doi.org/10.3788/CJL202047.0500002>.
- [557] A. Hehr, M. Norfolk, J. Wenning, J. Sheridan, M. Ahlfors, A.J. Mastropietro, E. Maghsoudi, S.N. Roberts, Hot Isostatic Pressing of Ultrasonic Additive Manufacturing Liquid Cold Plate Heat Exchangers, *J Spacecr Rockets*. (2020) 1–5. <https://doi.org/10.2514/1.a34888>.
- [558] L.L. Jing, Y.H. Li, K.W. Cheng, F. Wang, J.W. Shi, Application of Selective Laser Melting Technology Based on Titanium Alloy in Aerospace Products, *IOP Conf Ser Mater Sci Eng*. 740 (2020). <https://doi.org/10.1088/1757-899X/740/1/012056>.
- [559] S. Kuntanapreeda, D. Hess, Opening access to space by maximizing utilization of 3D printing in launch vehicle design and production, *Applied Science and Engineering Progress*. 14 (2020) 1–3. <https://doi.org/10.14416/J.ASEP.2020.12.002>.
- [560] F. Liu, W. Jingchao, T. Zhi, L. Bin, G. Yue, D.Z. Zhang, An aerospace integrated component application based on selective laser melting: Design, fabrication and FE simulation, *Solid Freeform Fabrication 2018: Proceedings of the 29th Annual International Solid Freeform Fabrication Symposium - An Additive Manufacturing Conference, SFF 2018*. (2020) 488–499.
- [561] P. Manil, Y. Jan, F. Nunio, F. Lomello, A. Arhancet, M. Lacroix, D. Bachet, J. Lapresle, Structural optimization, additive manufacturing and vibration testing of titanium alloy supports based on the space detector SVOM-MXT, (2021) 119. <https://doi.org/10.1117/12.2562351>.
- [562] M.O. Oyesola, K. Mpofu, N.R. Mathe, I.A. Daniyan, Hybrid-additive manufacturing cost model: A sustainable through-life engineering support for maintenance repair overhaul in the aerospace, *Procedia Manuf*. 49 (2020) 199–205. <https://doi.org/10.1016/j.promfg.2020.07.019>.
- [563] G. SHI, C. GUAN, D. QUAN, D. WU, L. TANG, T. GAO, An aerospace bracket designed by thermo-elastic topology optimization and manufactured by additive manufacturing, *Chinese Journal of Aeronautics*. 33 (2020) 1252–1259. <https://doi.org/10.1016/j.cja.2019.09.006>.
- [564] R. Stolt, F. Elgh, Introducing design for selective laser melting in aerospace industry, *J Comput Des Eng*. 7 (2020) 489–497. <https://doi.org/10.1093/jcde/qwaa042>.

- [565] R. Willner, S. Lender, A. Ihl, C. Wilsnack, S. Gruber, A. Brandão, L. Pambaguian, M. Riede, E. López, F. Brueckner, C. Leyens, Potential and challenges of additive manufacturing for topology optimized spacecraft structures, *J Laser Appl.* 32 (2020) 032012. <https://doi.org/10.2351/7.0000111>.
- [566] O. Mireles, O. Rodriguez, Y. Gao, N. Philips, Additive Manufacture of Refractory Alloy C103 for Propulsion Applications, in: *AIAA Propulsion and Energy 2020 Forum*, American Institute of Aeronautics and Astronautics, Reston, Virginia, 2020. <https://doi.org/10.2514/6.2020-3500>.
- [567] M. Abi-Fadel, M. al Harbi, M. Chafen, T. Chen, A. Farias, S. Halpin, Z. Jiang, S. Khan, J. Liu, M. McGrath, A. Nawal, K. Patel, A. Rahamimoff, S. Rocha, I. Romanov-Chernigovsky, D. Rotko, E. Tuohy, H. Vertadier, Leveraging additive manufacturing to enable deep space crewed missions, *Proceedings of the International Astronautical Congress, IAC.* 2019-Octob (2019).
- [568] B. Barroqueiro, A. Andrade-Campos, R.A.F. Valente, V. Neto, Metal additive manufacturing cycle in aerospace industry: A comprehensive review, *Journal of Manufacturing and Materials Processing.* 3 (2019). <https://doi.org/10.3390/jmmp3030052>.
- [569] F. Belfi, F. Iorizzo, C. Galbiati, F. Lepore, Space structures with embedded flat plate pulsating heat pipe built by additive manufacturing technology: Development, test and performance analysis, *J Heat Transfer.* 141 (2019). <https://doi.org/10.1115/1.4042082>.
- [570] L. Berrocal, R. Fernández, S. González, A. Perrián, S. Tudela, J. Vilanova, L. Rubio, J.M. Martín Márquez, J. Guerrero, F. Lasagni, Topology optimization and additive manufacturing for aerospace components, *Progress in Additive Manufacturing.* 4 (2019) 83–95. <https://doi.org/10.1007/s40964-018-0061-3>.
- [571] N. Chekir, J.J. Sixsmith, R. Tollett, M. Brochu, Laser wire deposition of a large Ti-6Al-4V space component the methodology for creating a functional Ti-6Al-4V satellite part using LWD additive manufacturing is detailed, *Weld J.* 98 (2019) 172S-180S. <https://doi.org/10.29391/2019.98.014>.
- [572] A. Diaz, Surface texture characterization and optimization of metal additive manufacturing-produced components for aerospace applications, *Additive Manufacturing for the Aerospace Industry.* (2019) 341–374. <https://doi.org/10.1016/B978-0-12-814062-8.00018-2>.
- [573] F. Facchini, A. de Chirico, G. Mummolo, Comparative cost evaluation of material removal process and additive manufacturing in aerospace industry, *Springer Proceedings in Mathematics and Statistics.* 280 (2019) 47–59. https://doi.org/10.1007/978-3-030-14969-7_5.
- [574] F. Froes, R. Boyer, B. Dutta, Introduction to aerospace materials requirements and the role of additive manufacturing, *Additive Manufacturing for the Aerospace Industry.* (2019) 1–6. <https://doi.org/10.1016/B978-0-12-814062-8.00001-7>.
- [575] C. Fuchs, D. Baier, D. Elitzer, R. Kleinwort, A. Bachmann, M.F. Zäh, Additive Manufacturing for Structural Components in Aerospace Engineering, *ZWF Zeitschrift Fuer Wirtschaftlichen Fabrikbetrieb.* 114 (2019) 431–434. <https://doi.org/10.3139/104.112124>.

- [576] M. Gas, A. Rene-Corail, S. Eyrygnoux, G. Iker, S. Sudre, Improving Competitiveness of Additive Manufacturing Aerospace Serial Parts, SAE Technical Papers. 2019-Septe (2019). <https://doi.org/10.4271/2019-01-1900>.
- [577] N. Gloria, W. Bassett, I. Vorbach, J. Kolano, A. Gupta, A. Kaplan, I. Rall, N. Folz, S. Dirks, Performance of a small-orifice rocket injector utilizing additive manufacturing, AIAA Propulsion and Energy Forum and Exposition, 2019. (2019). <https://doi.org/10.2514/6.2019-4231>.
- [578] A. Hehr, J. Wenning, M. Norfolk, J. Sheridan, J.A. Newman, M. Domack, Selective Reinforcement of Aerospace Structures Using Ultrasonic Additive Manufacturing, J Mater Eng Perform. 28 (2019) 633–640. <https://doi.org/10.1007/s11665-018-3614-1>.
- [579] E. Hilpert, J. Hartung, H. von Lukowicz, T. Herffurth, N. Heidler, Design, additive manufacturing, processing, and characterization of metal mirror made of aluminum silicon alloy for space applications, Optical Engineering. 58 (2019) 1. <https://doi.org/10.1117/1.oe.58.9.092613>.
- [580] M. Kalender, S.E. Kilic, S. Ersoy, Y. Bozkurt, S. Salman, Additive manufacturing and 3D printer technology in aerospace industry, Proceedings of 9th International Conference on Recent Advances in Space Technologies, RAST 2019. (2019) 689–695. <https://doi.org/10.1109/RAST.2019.8767881>.
- [581] M. Kamal, G. Rizza, Design for metal additive manufacturing for aerospace applications, Additive Manufacturing for the Aerospace Industry. (2019) 67–86. <https://doi.org/10.1016/B978-0-12-814062-8.00005-4>.
- [582] P. Kohl, M. Kilian, A. Schinagl-Weis, C. Hartwanger, Additive Manufacturing Developments for Satellite Antenna Applications from C- to Ka-Band, GeMiC 2019 - 2019 German Microwave Conference. (2019) 222–225. <https://doi.org/10.23919/GEMIC.2019.8698174>.
- [583] T. Kozmel, J. Grabowski, C. Kantner, J. Sebastian, B. Nez, J. Fetty, Additive manufacturing of ferrium® c64® for aerospace gear applications, The Vertical Flight Society - Forum 75: The Future of Vertical Flight - Proceedings of the 75th Annual Forum and Technology Display. (2019).
- [584] S.B. Liu, H. Chai, J. Ni, H.Y. Liu, J. Zhao, K. Shi, H.W. Qu, A Summary of Research on 3D Printing Technology of Titanium Alloys for Aerospace Field, Zhuzao/Foundry. 68 (2019) 965–970.
- [585] B.S. Mekki, J. Langer, S. Lynch, Genetic Algorithm Based Topology Optimization of Heat Exchanger Fins Used in Aerospace Applications, in: ASME International Mechanical Engineering Congress and Exposition, Proceedings (IMECE), American Society of Mechanical Engineers, 2019. <https://doi.org/10.1115/IMECE2019-10617>.
- [586] T.F. Murphy, C.T. Schade, Measurement of powder characteristics and quality for additive manufacturing in aerospace alloys, Additive Manufacturing for the Aerospace Industry. (2019) 99–142. <https://doi.org/10.1016/B978-0-12-814062-8.00007-8>.
- [587] X.D. Niu, 3D printing in China's aerospace industries, 59th Israel Annual Conference on Aerospace Sciences, IACAS 2019. (2019) 667–674.

- [588] M.M.J. Opgenoord, K.E. Willcox, Design for additive manufacturing: cellular structures in early-stage aerospace design, *Structural and Multidisciplinary Optimization*. 60 (2019) 411–428. <https://doi.org/10.1007/s00158-019-02305-8>.
- [589] M. Oyesola, K. Mpofu, N. Mathe, A techno-economic analytical approach of laser-based additive manufacturing processes for aerospace application, *Procedia Manuf.* 35 (2019) 155–163. <https://doi.org/10.1016/j.promfg.2019.05.019>.
- [590] N. Patel, S. Standbridge, M. van den Berghe, V. Devalaraju, Design and additive manufacturing considerations for liquid rocket engine development, *AIAA Propulsion and Energy Forum and Exposition*, 2019. (2019). <https://doi.org/10.2514/6.2019-4392>.
- [591] R. Piticescu, I. Vlaicu, A. Katz-Demyanetz, V. v. Popov, A. Kovalevsky, D. Safranchik, A. Koptuyug, Powder-bed additive manufacturing for aerospace application: Techniques, metallic and metal/ceramic composite materials and trends, *Manuf Rev (Les Ulis)*. 6 (2019). <https://doi.org/10.1051/mfreview/2019003>.
- [592] A. Riou, Metal powders for additive manufacturing in aerospace & land turbines. Trends and challenges, *Metallurgia Italiana*. 111 (2019) 56–61.
- [593] E. Sacco, S.K. Moon, Additive manufacturing for space: status and promises, *International Journal of Advanced Manufacturing Technology*. 105 (2019) 4123–4146. <https://doi.org/10.1007/s00170-019-03786-z>.
- [594] T.E. Shelton, C.R. Hartsfield, G.R. Cobb, C.A. Sheffield, Additive manufacturing: Designing to withstand space launch, *AIAA Scitech 2019 Forum*. (2019). <https://doi.org/10.2514/6.2019-0314>.
- [595] N. Tepylo, X. Huang, P.C. Patnaik, Laser-Based Additive Manufacturing Technologies for Aerospace Applications, *Adv Eng Mater*. 21 (2019). <https://doi.org/10.1002/adem.201900617>.
- [596] Vetrivel, P. Varun Teja, R. Sivaprakash, V. Vinay Krishnan, Vasanth, Potential and scope of additive manufacturing in aerospace industry with reference to India, *International Journal of Innovative Technology and Exploring Engineering*. 8 (2019) 1085–1089. <https://doi.org/10.35940/ijitee.F1225.0486S419>.
- [597] P.A. Vityaz, M.L. Kheifetz, S.A. Chizhik, Synergetic technologies of direct layer deposition in aerospace additive manufacturing, *Additive Manufacturing for the Aerospace Industry*. (2019) 427–447. <https://doi.org/10.1016/B978-0-12-814062-8.00022-4>.
- [598] J.C. Withers, Fusion and/or solid state additive manufacturing for aerospace applications, *Additive Manufacturing for the Aerospace Industry*. (2019) 187–211. <https://doi.org/10.1016/B978-0-12-814062-8.00010-8>.
- [599] G. YongXin, Z. Lihua, L. Zhijia, G. Yimin, M. Lingxi, Optimization Design of Star Tracker Bracket of Small Satellite for 3D Printing, *2019 5th International Conference on Control, Automation and Robotics, ICCAR 2019*. (2019) 808–811. <https://doi.org/10.1109/ICCAR.2019.8813432>.

- [600] X. Zhang, E. Liang, Metal additive manufacturing in aircraft: Current application, opportunities and challenges, *IOP Conf Ser Mater Sci Eng.* 493 (2019). <https://doi.org/10.1088/1757-899X/493/1/012032>.
- [601] G. Allevi, M. Cibeca, R. Fioretti, R. Marsili, R. Montanini, G. Rossi, Non-Contact Measurement Techniques for Qualification of Aerospace Brackets Made by Additive Manufacturing Technologies, *J Phys Conf Ser.* 1110 (2018). <https://doi.org/10.1088/1742-6596/1110/1/012002>.
- [602] Y. Cailloce, P. Hourlay, F. Lebrun, B. Palacin, Additive manufacturing of Ku band horn antennas for telecommunications space applications, *IET Conference Publications.* 2018 (2018). <https://doi.org/10.1049/cp.2018.0602>.
- [603] S. Eberle, A. Reutlinger, B. Curzadd, M. Mueller, M. Riede, C. Wilsnack, A. Brandão, L. Pambaguian, A. Seidel, E. López, F. Brueckner, E. Beyer, C. Leyens, Additive manufacturing of an AlSi40 mirror coated with electroless nickel for cryogenic space applications, (2019) 40. <https://doi.org/10.1117/12.2535960>.
- [604] J. Fessl, H. Shen, N. Patel, T.W. Chen, S. Ghirnkar, M. van den Berghe, Liquid rocket engine design for additive manufacturing, *Proceedings of the International Astronautical Congress, IAC.* 2018-Octob (2018).
- [605] K. v. Fetisov, P. v. Maksimov, Topology optimization and laser additive manufacturing in design process of efficiency lightweight aerospace parts, *J Phys Conf Ser.* 1015 (2018). <https://doi.org/10.1088/1742-6596/1015/5/052006>.
- [606] P. Gaudenzi, S. Atek, V. Cardini, M. Eugeni, G. Graterol Nisi, L. Lampani, M. Pasquali, L. Pollice, Revisiting the configuration of small satellites structures in the framework of 3D Additive Manufacturing, *Acta Astronaut.* 146 (2018) 249–258. <https://doi.org/10.1016/j.actaastro.2018.01.036>.
- [607] P.R. Gradl, W. Brandsmeier, S.E. Greene, Channel Wall Nozzle Manufacturing and Hot-Fire Testing using a Laser Wire Direct Closeout Technique for Liquid Rocket Engines, in: 2018 Joint Propulsion Conference, American Institute of Aeronautics and Astronautics, Reston, Virginia, 2018. <https://doi.org/10.2514/6.2018-4860>.
- [608] T. Hettesheimer, S. Hirzel, H.B. Roß, Energy savings through additive manufacturing: an analysis of selective laser sintering for automotive and aircraft components, *Energy Effic.* 11 (2018) 1227–1245. <https://doi.org/10.1007/s12053-018-9620-1>.
- [609] P. Maksimov, O. Smetannikov, A. Dubrovskaya, K. Dongauzer, L. Bushuev, Numeric simulation of aircraft engine parts additive manufacturing process, *MATEC Web of Conferences.* 224 (2018). <https://doi.org/10.1051/mateconf/201822401065>.
- [610] M. Oyesola, N. Mathe, K. Mpofu, S. Fatoba, Sustainability of Additive Manufacturing for the South African aerospace industry: A business model for laser technology production, commercialization and market prospects, *Procedia CIRP.* 72 (2018) 1530–1535. <https://doi.org/10.1016/j.procir.2018.03.072>.

- [611] R. Stolt, T. Heikkinen, F. Elgh, Integrating additive manufacturing in the design of aerospace components, *Advances in Transdisciplinary Engineering*. 7 (2018) 145–154. <https://doi.org/10.3233/978-1-61499-898-3-145>.
- [612] K. Vishnu Prashant Reddy, I. Meera Mirzana, A. Koti Reddy, Application of Additive Manufacturing technology to an Aerospace component for better trade-off's, *Mater Today Proc.* 5 (2018) 3895–3902. <https://doi.org/10.1016/j.matpr.2017.11.644>.
- [613] D. Vogel, L. Geismayr, M. Langer, P. Leslabay, G. Schlick, Combining additive manufacturing and biomimetics for the optimization of satellite structures, *Proceedings of the International Astronautical Congress, IAC. 2018-Octob* (2018).
- [614] A.M. Cismilianu, C.E. Munteanu, I.C. Oncescu, R.P. Bibire, V. Stoenescu, M.V. Pricop, E. Rey, S. Eyrygnoux, End-to-end process of hollow spacecraft structures with high frequency and low mass obtained with in-house structural optimization tool and additive manufacturing, *INCAS Bulletin*. 9 (2017) 13–20. <https://doi.org/10.13111/2066-8201.2017.9.3.2>.
- [615] R. Clinton, NASA additive manufacturing initiatives: In space manufacturing and rocket engines, *Proceedings of the International Astronautical Congress, IAC.* 12 (2017) 8226–8246.
- [616] M.F. Cruz, A.V. Borille, Decision methods application to compare conventional manufacturing process with metal additive manufacturing process in the aerospace industry, *Journal of the Brazilian Society of Mechanical Sciences and Engineering*. 39 (2017) 177–193. <https://doi.org/10.1007/s40430-016-0532-8>.
- [617] D.B. Froes F.H., Boyer R., Additive manufacturing for aerospace applications-part I, *Advanced Materials and Processes*. (2017). <https://www.scopus.com/inward/record.uri?eid=2-s2.0-85026852703&partnerID=40&md5=6dedfc6ba21e7823cfee6191ba0591b5%0A>.
- [618] F.H. Froes, R. Boyer, B. Dutta, Additive manufacturing for aerospace applications-part II, *Advanced Materials and Processes*. 175 (2017) 18–22.
- [619] S.S. Gill, H. Arora, Jidesh, V. Sheth, On the development of Antenna feed array for space applications by additive manufacturing technique, *Addit Manuf.* 17 (2017) 39–46. <https://doi.org/10.1016/j.addma.2017.06.010>.
- [620] N. Judalet, A. Kazakçi, E. le Gouguec, A. Balvay, Performance monitoring & control for an additive manufacturing factory - A case study in the aerospace industry, *21st International Conference on Engineering Design, ICED 2017*. 5 (2017) 249–258. <https://www.scopus.com/inward/record.uri?eid=2-s2.0-85029815782&partnerID=40&md5=bf943f31ccb7f7f666cbc5a96328b3d9>.
- [621] K. Kasim, A. Muley, M. Stoia, F. Ladeinde, Advanced Heat Transfer Devices for Aerospace Applications, in: *ASME International Mechanical Engineering Congress and Exposition, Proceedings (IMECE)*, American Society of Mechanical Engineers, 2017. <https://doi.org/10.1115/IMECE2017-72382>.

- [622] A. Lindwall, C. Dordlofva, A. Öhrwall Rönnbäck, Additive manufacturing & the product development process: Insights from the space industry, *Proceedings of the International Conference on Engineering Design, ICED*. 5 (2017) 345–354.
- [623] C.E. Munteanu, A.M. Cismilianu, A.I. Chira, D. Baran, Structural optimization of space components adapted for 3d printing, *Annals of DAAAM and Proceedings of the International DAAAM Symposium*. (2017) 821–825. <https://doi.org/10.2507/28th.daaam.proceedings.115>.
- [624] A. Nagasaki, D. Watanabe, T. Tamura, T. Onga, A. Ogawara, Suyalatu, N. Higuchi, H. Sakai, Y. Nunome, H. Kawashima, Additive manufacturing development for rocket engine in Japan, *Proceedings of the International Astronautical Congress, IAC*. 13 (2017) 8713–8719.
- [625] D. Obilanade, J. Kingston, Spacecraft designers' guide to using additive manufacturing processes for large metallic spacecraft structures., *Proceedings of the International Astronautical Congress, IAC*. 13 (2017) 8280–8304.
- [626] R. Pacurar, A. Pacurar, N. Bâlc, Research on the mechanical behaviour of an airplane component made by selective laser melting technology, *MATEC Web of Conferences*. 94 (2017). <https://doi.org/10.1051/mateconf/20179403012>.
- [627] M. Yakout, A. Cadamuro, M.A. Elbestawi, S.C. Veldhuis, The selection of process parameters in additive manufacturing for aerospace alloys, *International Journal of Advanced Manufacturing Technology*. 92 (2017) 2081–2098. <https://doi.org/10.1007/s00170-017-0280-7>.

Appendix A: Prominent examples of AM applications in non-technical literature

Non-Technical Literature					
Manufacturer or Author	Application	Design Approach	Material used	Technologies used	Reference
Aerojet Rocketdyne	Thrust Chamber	AM	Copper Alloy	L-PBF	[402]
Aerojet Rocketdyne	AR1 Preburner	AM	Mondaloy 200™	L-PBF	[401]
Airborne Engineering	C-18150 copper chamber	AM	Copper Alloy	Not Stated	[498]
Airbus	Reflector Bracket	AM	Titanium	L-PBF	[499]
Airbus	A350 Cabin Bracket Connector	AM & TO	Ti-6Al-4V	L-PBF	[500]
Airbus	A350 XWB Pylon Bracket	AM	Titanium Alloy	L-PBF	[501]
Airbus & 3D Systems	RF Filter	AM	AlSi10Mg	L-PBF	[502]
Airbus & EOS	Aircraft Door Locking Shaft	AM	Ti6Al4V	L-PBF	[503]
Airbus, APWorks, Autodesk	Bionic Partition wall Project	AM, TO, Lattice	AlMgSc	L-PBF	[504]
Amaero Engineering, Safran	Aero-Engine	AM	Al7SiMg, Hastelloy X, Ti6Al4V	L-PBF, EBM, CS	[505]
Ariane Group, EOS	Ariane 6 Injector head	AM	Inconel 718	L-PBF	[506]
Ariane Group, ESA	Rocket Combustion Chamber	AM	Copper Alloy	L-PBF	[403]
Autodesk	Lattice Aircraft Seat	AM, TO, Lattice and “lost wax” casting	Magnesium	N/A	[507]
Betatype	Rocket Nozzle	AM, Lattice	Stainless Steel 316L	L-PBF	[508]
Blue Origin	BE-7 Engine	AM	Various	Not Stated	[509]
Blue Origin	BE-4 rocket engine boost pump	AM	Aluminium alloy, Monel	L-PBF	[510]
Copenhagen Suborbitals	Coaxial Swirl Injectors	AM	Not Stated	Binder Jetting	[397]
DLR & 3D systems	Liquid Rocket Engine Injector	AM	Ni718	L-PBF	[511]
EOS & Airbus	Borescope Bosses	AM	Not Stated	L-PBF	[512]
EOS & Airbus	Hydraulic spoiler manifold	AM	Titanium	L-PBF	[513]
EOS & Vectoflow	Flow Measurement Probe	AM	Nickel-Chromium Alloy	L-PBF	[514]
EOS, Sogeti and Airbus	Vertical tailplane bracket	AM & TO	AlSi10Mg	L-PBF	[515]
Frustum & 3D Systems	GE Aircraft Bracket	AM & TO	Titanium Alloy	L-PBF	[516]
General Electric	LEAP engine Fuel nozzle	AM	Cobalt-Chrome Alloy	L-PBF	[16]
General Electric	GE9X Engine Additive Components	AM	Various	EBM, L-PBF	[335]
General Electric	Sump Cover	AM	Cobalt-Chrome Alloy	L-PBF	[517]
General Electric	NACA inlet	AM	Ti-6Al-4V	L-PBF	[518]
GKN Aerospace	Rocket Engine Turbines	AM	IN718	DED	[519]

Non-Technical Literature					
Manufacturer or Author	Application	Design Approach	Material used	Technologies used	Reference
GKN Aerospace	Turbine exhaust casing	AM	Not stated	L-PBF	[520]
GKN Aerospace	Bracket	AM	Ti6Al4V	EBM	[521]
GKN Aerospace	Intermediate compressor case bosses	AM	Ti-6Al-4V	DED	[522]
GKN Aerospace	Optical ice protection probe	AM	Not stated	Not stated	[523]
Hietia	High and Low temperature heat exchangers	AM	Inconel 718/625 or CM247 LC	L-PBF	[391]
Hyperganic	Rocket Nozzle	AM, Lattice	Inconel 718	L-PBF	[417]
Launcher & EOS	E-2 Rocket Engine	AM	C-18150 Copper-alloy	L-PBF	[412]
Lena Space and Oerlikon	Propulsion system Impeller	AM	Not Stated	Not Stated	[524]
Liebherr-Aerospace & Airbus	Landing Gear Sensor Bracket	AM	Titanium Alloy	L-PBF	[525]
Lockheed Martin	Satellite Fuel Tank	AM	Titanium Alloy	EB-DED	[405]
NASA	Rocket Combustion Chamber	AM	GCop-84	L-PBF	[526]
NASA	Perseverance Heat Exchangers and structures	AM	Titanium Alloy, Nickel-based superalloy	L-PBF	[387]
NASA	7k and 40k coupled chamber and nozzle	AM	GRCop-84, GRCop-42, NASA HR-1, Inconel 625	L-PBF, EB-DED, DED	[413]
NASA	Rocket Injector	AM	Inconel 625	L-PBF	[399]
NASA	Rocket Engine	AM	Various	Not Stated	[527]
NASA	LPS Fuel Turbopump	AM	Various	L-PBF	[528]
NASA	Pogo Accumulator Assembly	AM	Inconel 718	Not Stated	[529]
NASA	Maintenance Port Cover	AM	Not Stated	L-PBF	[530]
NASA	RS-25 Pogo z-baffle	AM	Inconel 718	L-PBF	[400]
nTopology & Cobra Aero	Cobra Aero uses multiphysics simulation to optimize UAV engine	AM, Lattice	AlSi10MG	L-PBF	[390]
nTopology & USAF	Air Force optimizes cubesat using Architected Materials	AM, Lattice	Inconel 718	L-PBF	[360]
Oerlikon	Turbine Blade	AM	Not Stated	Not Stated	[531]
Oerlikon	Double Nozzle	AM	Inconel 625	L-PBF	[532]
Optomec	T700 Blisk repair	AM	Stellite® 21	LENS	[347]
Orbex	Rocket Engine	AM	Not Stated	L-PBF	[533]
Orbital ATK	Hypersonic Engine Combustor	AM	Not Stated	L-PBF	[534]
Parabilis Space Technologies	Multi-Material Nozzle	AM	Copper Alloy, Isomolded Graphite, Inconel 718	Various	[535]

Non-Technical Literature					
Manufacturer or Author	Application	Design Approach	Material used	Technologies used	Reference
Parabilis Space Technologies	RCS Thruster	AM	Not Stated	Not Stated	[536]
Pratt & Whitney	Compressor Stators	AM	Ti-6Al-4V	L-PBF	[332]
Relativity Space	Terran 1 Fuel Tank	AM	Not Stated	DED	[406]
Relativity Space	Aeon Engine	AM	Not Stated	L-PBF	[411]
RocketLab	Rutherford Engine	AM & Lattice	Not Stated	EBM	[4]
RUAG, EOS, Altair	Sentinel Antenna Bracket	AM & TO	AlSi10Mg	L-PBF	[345]
RUAG, EOS, Altair	Star Tracker Bracket	AM & TO	AlSi10Mg	L-PBF	[537]
RUAG, EOS, Altair	LEROS Engine support bracket	AM & TO	AlSi10Mg	L-PBF	[538]
Siemens	High-Pressure Hydraulic Manifold	AM	Titanium Alloy	L-PBF	[539]
SLM & Cellcore	Rocket Engine	AM & Lattice	Inconel 718	L-PBF	[18]
SpaceX	SuperDraco Engine Chamber	AM, Cert	Inconel Alloy	L-PBF	[407]
SpaceX	Main Oxidiser Valve Body	AM	Not Stated	Not Stated	[540]
SpaceX	Raptor Engine Components	AM	Various	Various	[541]
Spirit AeroSystems & Boeing	Access Door Latch	AM	Titanium Alloy	DED	[542]
Thales Alenia Space & 3D Systems	Antenna Bracket	AM & TO	Ti Gr5	L-PBF	[543]
The Welding Institute	Helicopter combustion chamber	AM	Inconel 718	DED	[544]
ULA	Vulcan Bellows Feedline Housing	AM	Inconel 718	L-PBF	[545]
Virgin Orbit & NASA	Rocket Combustion Chamber	AM	GRCop-84	L-PBF	[546]

Appendix B: Prominent examples of AM applications in technical literature

Technical Literature				
Authors	Title	Application Areas	Year	Reference
Godec M.,et al.	Hybrid additive manufacturing of Inconel 718 for future space applications	Spacecraft	2021	[547]
Zhan Z., Li H.	A novel approach based on the elastoplastic fatigue damage and machine learning models for life prediction of aerospace alloy parts fabricated by additive manufacturing	Aerospace, Fatigue	2021	[548]
Van Den Berg P, Jyoti B, Hermesen R	Investigation of Thermal Behaviour of Additively Manufactured Green Bi-Propellant Thrusters in CubeSat Applications Using Transient Thermal Modelling	Spacecraft, Engines	2021	[549]
Waugh I., et al.	Additive manufacture of rocket engine combustion chambers from CuCrZr (C-18150) using the DMLS process	Rocket, Engines	2021	[550]
Waugh I., et al.	Additive manufacture of rocket engine combustion chambers using the ABD R -900AM nickel Superalloy	Rocket, Engines	2021	[551]
D'Emilia G.,et al.	Uncertainty assessment for measurement and simulation in selective laser melting: A case study of an aerospace part	Aerospace, Case Study	2020	[552]
Dordlofva C., Törlind P.	Qualification challenges with additive manufacturing in space applications	Qualification, Spacecraft	2020	[553]
Fereiduni E., Ghasemi A., Elbestawi M.	Selective laser melting of aluminum and titanium matrix composites: Recent progress and potential applications in the aerospace industry	Aerospace	2020	[554]
Gao L.,et al.	Study on Arc Additive Manufacturing Process and Properties of 5356 Aluminum Alloy Rocket Booster Module Transition End Frame	Rocket	2020	[555]
Gu D.,et al.	Laser Additive Manufacturing of High-Performance Metallic Aerospace Components	Aerospace	2020	[556]
Hehr, A.,et al.	Hot Isostatic Pressing of Ultrasonic Additive Manufacturing Liquid Cold Plate Heat Exchangers.	Case Study, Heat Transfer	2020	[557]
Jia, D., Li, F., & Zhang, Y.	3D-printing process design of lattice compressor impeller based on residual stress and deformation.	Case Study, Design, Lattice	2020	[324]
Jing L.-L.,et al.	Application of Selective Laser Melting Technology Based on Titanium Alloy in Aerospace Products	Aerospace	2020	[558]
Kuntanapreeda S., Hess D.	Opening access to space by maximizing utilization of 3D printing in launch vehicle design and production	Spacecraft, Design	2020	[559]
Liu F.,et al.	An aerospace integrated component application based on selective laser melting: Design, fabrication and FE simulation	Aerospace, Design	2020	[560]
Manil P.,et al.	Structural optimization, additive manufacturing and vibration testing of titanium alloy supports based on the space detector SVOM-MXT	Spacecraft, Optimization Techniques	2020	[561]
Oyesola M.O.,et al.	Hybrid-additive manufacturing cost model: A sustainable through-life engineering support for maintenance repair overhaul in the aerospace	Aerospace, Repair	2020	[562]
SHI G.,et al.	An aerospace bracket designed by thermo-elastic topology optimization and manufactured by additive manufacturing	Aerospace, Design, Topology Optimization	2020	[563]
Stolt R., Elgh F.	Introducing design for selective laser melting in aerospace industry	Aerospace, Design	2020	[564]
Willner R.,et al.	Potential and challenges of additive manufacturing for topology optimized spacecraft structures	Spacecraft, Structures, Topology Optimization	2020	[565]
Gradl P.,et al.	Lightweight Thrust Chamber Assemblies using Multi-Alloy Additive Manufacturing and Composite Overwrap	Rocket, Engines	2020	[299]

Technical Literature				
Authors	Title	Application Areas	Year	Reference
Mireles O., et al.	Additive Manufacture of Refractory Alloy C103 for Propulsion Applications	Rocket, Engines	2020	[566]
Abi-Fadel M.,et al.	Leveraging additive manufacturing to enable deep space crewed missions	Spacecraft	2019	[567]
Barroqueiro B.,et al.	Metal additive manufacturing cycle in aerospace industry: A comprehensive review	Review, Aerospace	2019	[568]
Belfi F.,et al.	Space structures with embedded flat plate pulsating heat pipe built by additive manufacturing technology: Development, test and performance analysis	Spacecraft, Structures, Heat Transfer	2019	[569]
Berrocal L.,et al.	Topology optimization and additive manufacturing for aerospace components	Aerospace, Topology Optimization	2019	[570]
Chekir N.,et al.	Laser wire deposition of a large Ti-6Al-4V space component the methodology for creating a functional Ti-6Al-4V satellite part using LWD additive manufacturing is detailed	Spacecraft, Structures	2019	[571]
Chougrani L., et al.	Parts internal structure definition using non-uniform patterned lattice optimization for mass reduction in additive manufacturing	Spacecraft, Structures, Lattice	2019	[361]
Diaz A	Surface texture characterization and optimization of metal additive manufacturing-produced components for aerospace applications	Aerospace, Optimization Techniques	2019	[572]
Facchini F., De Chirico A., Mummolo G.	Comparative cost evaluation of material removal process and additive manufacturing in aerospace industry	Aerospace	2019	[573]
Froes F., Boyer R.	Additive manufacturing for the aerospace industry	Aerospace	2019	[377]
Froes F., Boyer R., Dutta B.	Introduction to aerospace materials requirements and the role of additive manufacturing	Aerospace	2019	[574]
Fuchs C.,et al.	Additive Manufacturing for Structural Components in Aerospace Engineering	Aerospace	2019	[575]
Gas M.,et al.	Improving Competitiveness of Additive Manufacturing Aerospace Serial Parts	Aerospace	2019	[576]
Gloria N.,et al.	Performance of a small-orifice rocket injector utilizing additive manufacturing	Rocket	2019	[577]
Gradl P.R., Protz C., Wammen T.	Additive manufacturing development and hot-fire testing of liquid rocket channel wall nozzles using blown powder directed energy deposition Inconel 625 and JBK-75 alloys	Rocket	2019	[52]
Hehr A.,et al.	Selective Reinforcement of Aerospace Structures Using Ultrasonic Additive Manufacturing	Aerospace, Structures	2019	[578]
Hilpert E.,et al.	Design, additive manufacturing, processing, and characterization of metal mirror made of aluminum silicon alloy for space applications	Spacecraft, Design	2019	[579]
Kalender M.,et al.	Additive manufacturing and 3D printer technology in aerospace industry	Aerospace	2019	[580]
Kamal M., Rizza G.	Design for metal additive manufacturing for aerospace applications	Aerospace, Design	2019	[581]
Kohl P.,et al.	Additive Manufacturing Developments for Satellite Antenna Applications from C- to Ka-Band	Spacecraft, Structures	2019	[582]
Kozmel T.,et al.	Additive manufacturing of Ferrium® c64® for aerospace gear applications	Aerospace	2019	[583]
Liu S.-B.,et al.	A Summary of Research on 3D Printing Technology of Titanium Alloys for Aerospace Field	Aerospace	2019	[584]
Mekki, B. S., Langer, J., & Lynch, S.	Genetic Algorithm Based Topology Optimization of Heat Exchanger Fins Used in Aerospace Applications.	Aerospace, Topology Optimization, Heat Transfer	2019	[585]
Murphy T.F., Schade C.T.	Measurement of powder characteristics and quality for additive manufacturing in aerospace alloys	Qualification, Aerospace	2019	[586]

Technical Literature				
Authors	Title	Application Areas	Year	Reference
Najmon J.C., Raeisi S., Tovar A.	Review of additive manufacturing technologies and applications in the aerospace industry	Review, Aerospace	2019	[303]
Niu X.-D	3D printing in China's aerospace industries	Aerospace	2019	[587]
Opgenoord M.M.J., Willcox K.E.	Design for additive manufacturing: cellular structures in early-stage aerospace design	Aerospace, Structures, Design	2019	[588]
Oyesola M., Mpofu K., Mathe N.	A techno-economic analytical approach of laser-based additive manufacturing processes for aerospace application	Aerospace	2019	[589]
Patel N.,et al.	Design and additive manufacturing considerations for liquid rocket engine development	Rocket, Design	2019	[590]
Piticescu R.,et al.	Powder-bed additive manufacturing for aerospace application: Techniques, metallic and metal/ceramic composite materials and trends	Aerospace	2019	[591]
Riou A	Metal powders for additive manufacturing in aerospace & land turbines. Trends and challenges	Aerospace, Turbomachinery	2019	[592]
Sacco E., Moon S.K.	Additive manufacturing for space: status and promises	Spacecraft	2019	[593]
Shelton T.E.,et al.	Additive manufacturing: Designing to withstand space launch	Spacecraft, Design	2019	[594]
Teplyo N., Huang X., Patnaik P.C.	Laser-Based Additive Manufacturing Technologies for Aerospace Applications	Aerospace	2019	[595]
Vetrivel, Varun Teja P.,et al.	Potential and scope of additive manufacturing in aerospace industry with reference to India	Aerospace	2019	[596]
Vityaz P.A., Kheifetz M.L., Chizhik S.A.	Synergetic technologies of direct layer deposition in aerospace additive manufacturing	Aerospace	2019	[597]
Withers J.C.	Fusion and/or solid state additive manufacturing for aerospace applications	Aerospace	2019	[598]
YongXin G.,et al.	Optimization Design of Star Tracker Bracket of Small Satellite for 3D Printing	Spacecraft, Structures, Design, Optimization Techniques	2019	[599]
Yusuf S.M., Cutler S., Gao N.	Review: The impact of metal additive manufacturing on the aerospace industry	Review, Aerospace	2019	[356]
Zhang X., Liang E.	Metal additive manufacturing in aircraft: Current application, opportunities and challenges	Aircraft	2019	[600]
Gradl P.,et al.	Bimetallic Channel Wall Nozzle Development and Hot-fire Testing Using Additively Manufactured Laser Wire Direct Closeout Technology	Rocket, Engines	2019	[298]
Gradl P.,et al.	GRCop-42 Development and Hot-fire Testing Using Additive Manufacturing Powder Bed Fusion for Channel-cooled Combustion Chambers	Rocket, Engines	2019	[340]
Gradl P.,et al.	Additive Manufacturing and Hot-fire Testing of Bimetallic GRCop-84 and C-18150 Channel-Cooled Combustion Chambers Using Powder Bed Fusion and Inconel 625 Hybrid Directed Energy Deposition	Rocket, Engines	2019	[348]
Gradl P.,et al.	Progress in additively manufactured copper-alloy GRCOP-84, GRCOP-42, and bimetallic combustion chambers for liquid rocket engines	Rocket, Engines	2019	[71]
Allevi G.,et al.	Non-Contact Measurement Techniques for Qualification of Aerospace Brackets Made by Additive Manufacturing Technologies	Qualification, Aerospace	2018	[601]
Bici M.,et al.	Development of a multifunctional panel for aerospace use through SLM additive manufacturing	Aerospace	2018	[337]
Cailloce Y.,et al.	Additive manufacturing of Ku band horn antennas for telecommunications space applications	Spacecraft	2018	[602]

Technical Literature				
Authors	Title	Application Areas	Year	Reference
Eberle S.,et al.	Additive manufacturing of an ALSI40 mirror coated with electroless nickel for cryogenic space applications	Spacecraft	2018	[603]
Fessler J.,et al.	Liquid rocket engine design for additive manufacturing	Rocket, Design	2018	[604]
Fetisov K.V., Maksimov P.V.	Topology optimization and laser additive manufacturing in design process of efficiency lightweight aerospace parts	Aerospace, Design, Topology Optimization	2018	[605]
Gaudenzi P.,et al.	Revisiting the configuration of small satellites structures in the framework of 3D Additive Manufacturing	Spacecraft, Structures	2018	[606]
Gradl P.R.,et al.	Additive manufacturing of liquid rocket engine combustion devices: A summary of process developments and hot-fire testing results	Rocket	2018	[67]
Gradl P.,et al.	Channel Wall Nozzle Manufacturing and Hot-Fire Testing using a Laser Wire Direct Closeout Technique for Liquid Rocket Engines	Rocket, Engines	2018	[607]
Hettesheimer T., Hirzel S., Roß H.B.	Energy savings through additive manufacturing: an analysis of selective laser sintering for automotive and aircraft components	Aircraft	2018	[608]
Maksimov P.,et al.	Numeric simulation of aircraft engine parts additive manufacturing process	Aircraft	2018	[609]
Oyesola M.,et al.	Sustainability of Additive Manufacturing for the South African aerospace industry: A business model for laser technology production, commercialization and market prospects	Aerospace	2018	[610]
Saltzman, D.,et al.	Design and evaluation of an additively manufactured aircraft heat exchanger.	Aircraft, Design, Heat Transfer	2018	[336]
Stolt R., Heikkinen T., Elgh F.	Integrating additive manufacturing in the design of aerospace components	Aerospace, Design	2018	[611]
Vishnu Prashant Reddy K., Meera Mirzana I., Koti Reddy A.	Application of Additive Manufacturing technology to an Aerospace component for better trade-off's	Aerospace	2018	[612]
Vogel D.,et al.	Combining additive manufacturing and biomimetics for the optimization of satellite structures	Spacecraft, Structures, Optimization Techniques	2018	[613]
Gradl P.,et al.	Development and Hot-fire Testing of Additively Manufactured Copper Combustion Chambers for Liquid Rocket Engine Applications	Rocket, Engines	2017	[72]
Cismilianu A.-M.,et al.	End-to-end process of hollow spacecraft structures with high frequency and low mass obtained with in-house structural optimization tool and additive manufacturing	Spacecraft, Structures, Optimization Techniques	2017	[614]
Clinton R.	NASA additive manufacturing initiatives: In space manufacturing and rocket engines	Spacecraft, Rocket	2017	[615]
Cruz M.F., Borille A.V.	Decision methods application to compare conventional manufacturing process with metal additive manufacturing process in the aerospace industry	Aerospace	2017	[616]
Froes F.H., Boyer R., Dutta B.	Additive manufacturing for aerospace applications-part I	Aerospace	2017	[617]
Froes F.H., Boyer R., Dutta B.	Additive manufacturing for aerospace applications-part II	Aerospace	2017	[618]

Technical Literature				
Authors	Title	Application Areas	Year	Reference
Gill S.S., Arora H., Jidesh, Sheth V.	On the development of Antenna feed array for space applications by additive manufacturing technique	Spacecraft	2017	[619]
Judalet N.,et al.	Performance monitoring & control for an additive manufacturing factory - A case study in the aerospace industry	Aerospace, Case Study	2017	[620]
Kasim, K.,et al.	Advanced Heat Transfer Devices for Aerospace Applications.	Aerospace, Heat Transfer	2017	[621]
Lindwall A., Dordlofva C., Öhrwall Rönnbäck A.	Additive manufacturing & the product development process: Insights from the space industry	Spacecraft	2017	[622]
Liu R.,et al.	Aerospace applications of laser additive manufacturing	Aerospace	2017	[344]
Munteanu C.E.,et al.	Structural optimization of space components adapted for 3d printing	Spacecraft, Optimization Techniques	2017	[623]
Nagasaki A.,et al.	Additive manufacturing development for rocket engine in Japan	Rocket	2017	[624]
Obilanade D., Kingston J.	Spacecraft designers' guide to using additive manufacturing processes for large metallic spacecraft structures.	Spacecraft, Structures, Design	2017	[625]
Orme M.E.,et al.	Designing for additive manufacturing: Lightweighting through topology optimization enables lunar spacecraft	Spacecraft, Structures, Design, Topology Optimization	2017	[352]
Orme M.E.,et al.	Additive manufacturing of lightweight, optimized, metallic components suitable for space flight	Spacecraft	2017	[15]
Pacurar R., Pacurar A., Bâlc N.	Research on the mechanical behaviour of an airplane component made by selective laser melting technology	Aircraft	2017	[626]
Yakout M.,et al.	The selection of process parameters in additive manufacturing for aerospace alloys	Aerospace	2017	[627]
Gradl P.,et al.	Rapid Fabrication Techniques for Liquid Rocket Channel Wall Nozzles	Rocket, Engines	2016	[385]
Godfrey, D	Additive Manufacturing (AM) and the Honeywell Global Initiative.	Case Study	2015	[371]
Xue, L.,et al.	Laser Consolidation: A Novel Additive Manufacturing Process for Making Net-Shape Functional Metallic Components for Gas Turbine Applications.	Case Study, Turbomachinery, Repair	2015	[378]
Allison, T.,et al.	Manufacturing and Testing Experience with Direct Metal Laser Sintering for Closed Centrifugal Compressor Impellers.	Case Study	2014	[368]
Tomlin M., Meyer J.	Topology Optimization of an Additive Layer Manufactured (ALM) Aerospace Part	Aerospace, Topology Optimization	2011	[322]

Appendix C: EOS M280 Additive Manufacturing Machine Specifications

EOSINT M 280



Laser-Sintering for the production of tooling inserts, prototype parts and end products directly in metal

The technology:

Laser-sintering – the key to e-Manufacturing

Laser-sintering is well known as the technology of choice for ensuring the quickest route from product idea to market launch. Innovative companies from a broad range of industries are using this technology for e-Manufacturing – the fast, flexible and cost effective production directly from electronic data for every phase of the product life cycle.

The system:

e-Manufacturing for the industrial sector

The EOSINT M 280 is the updated and further improved version of the EOSINT M 270, the leading system on the market for the additive layer manufacturing of metal components. It directly produces top-quality metal parts on the basis of three-dimensional

CAD data – fully automatically, in only a few hours, and with no need for tools. The Direct Metal Laser-Sintering (DMLS) process builds the parts up layer by layer by melting fine metal powder with a laser beam, which enables the creation of extremely complex geometries such as free form surfaces, deep grooves and three-dimensional cooling channels.

The system is optionally equipped with a solid state laser of either 200 or 400 watt. This laser provides an exceptionally high quality radiation and stable performance. The Laser Power Monitoring (LPM) makes it possible to control all this during the building process. Together with an optimized Gas Management System this guarantees optimal and consistent processing conditions for highest and constant part building qualities. The system operates in both

protective nitrogen and argon atmospheres. This allows the system to process a wide range of materials: from light metals to stainless and tooling steel to super alloys.

The process software has been further developed over a period of many years and contains an array of intelligent exposure strategies and features; these enable the optimization and adaption of the build process for a variety of material types and applications. For the EOSINT M 280 EOS offers a number of powdered metal materials with corresponding parameter sets that have been optimized according to the application. They produce parts with standardized property profiles (PPPs). In addition, EOS ensures maximum reliability by subjecting all the relevant products to an intensive process development

procedure and constant quality assurance.

The system's capacity can be adapted to different customer requirements with a variety of options and additional equipment. The Integrated Process Chain Management (IPCM) modules enable greater productivity, higher quality and increased user-friendliness, and can also be added to at any time.

The distinctive features of the EOSINT M 280 system are the quality of the parts it produces and the ergonomically designed peripherals. These features are what make the system the ideal production tool for the economical batch-size optimized manufacture of parts at all stages of the product life cycle. The system is therefore perfectly suited for an industrial environment.

EOSINT M 280

Knee Implant
Built in bio-compatible
Cobalt-Chrome alloy
using EOSINT M 280.
(Project: Stryker Orthopaedics)



The software:

Achieve maximum productivity automatically

EOS offers various software packages for processing CAD

data and tracking production flows. EOSTATE was developed to provide users with an overview of all production-related data at any desired point in time. The soft-

ware processes production data for freely definable timeframes and displays it clearly. The user's requirements are accommodated within the integrated Basic,

Quality Assurance, Controlling and Machine Park Management (MPM) modules. They ensure that production flows are easy to track and to manage.

Building volume (including building platform)	250 mm x 250 mm x 325 mm (9.85 x 9.85 x 12.8 in)
Laser type	Yb-fibre laser, 200 W or 400 W (optional)
Precision optics	F-theta-lens, high-speed scanner
Scan speed	up to 7.0 m/s (23 ft/sec)
Variable focus diameter	100 - 500 µm (0.004 - 0.02 in)
Power supply	32 A
Power consumption	maximum 8.5 kW
Nitrogen generator	integrated
Compressed air supply	7,000 hPa; 20 m³/h (102 psi; 26.2 yd³/h)
Argon supply	4,000 hPa; 100 l/min
Dimensions (B x D x H)	
System	2,200 mm x 1,070 mm x 2,290 mm (86.6 x 42.1 x 90.1 in)
Recommended installation space	min. 4.8 m x 3.6 m x 2.9 m (189 x 142 x 114 in)
Weight	approx. 1,250 kg (2,756 lb)
Data preparation	
Software	EOS RP Tools; EOSTATE Magics RP (Materialise)
CAD interface	STL. Optional: converter for all standard formats
Network	Ethernet

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Appendix D: EOS Ti-6Al-4V Powder/Material Specifications



Material data sheet

EOS Titanium Ti64

EOS Titanium Ti64 is a titanium alloy powder which has been optimized especially for processing on EOSINT M systems.

This document provides information and data for parts built using EOS Titanium Ti64 powder (EOS art.-no. 9011-0014) on the following system specifications:

- EOSINT M 280 with PSW 3.6 and Original EOS Parameter Set Ti64_Speed 1.0
- EOS M 290 400W with EOSPRINT 1.0 and Original EOS Parameter Set Ti64_Performance 1.0 und Ti64_Speed 1.0

Description

Parts built in EOS Titanium Ti64 have a chemical composition corresponding to ISO 5832-3, ASTM F1472 and ASTM B348.

This well-known light alloy is characterized by having excellent mechanical properties and corrosion resistance combined with low specific weight and biocompatibility.

This material is ideal for many high-performance engineering applications, for example in aerospace and motor racing, and also for the production of biomedical implants (note: subject to fulfilment of statutory validation requirements where appropriate).

Due to the layerwise building method, the parts have a certain anisotropy, which can be reduced or removed by appropriate heat treatment – see Technical Data for examples.



Material data sheet

Technical data

General process and geometric data

Typical achievable part accuracy [1], [8]	$\pm 50 \mu\text{m}$
Min. wall thickness [2], [8]	approx. 0.3 – 0.4 mm approx. 0.012 – 0.016 inch
Surface roughness, as built [3], [8]	
Ti64 Performance (30 μm)	R_a 9 – 12 μm , R_z 40 – 80 μm R_a 0.36 – 0.47 $\times 10^{-3}$ inch, R_z 1.6 – 3.2 $\times 10^{-3}$ inch
Ti64 Speed (60 μm)	R_a 6 – 10 μm , R_z 35 – 40 μm R_a 0.23 – 0.39 $\times 10^{-3}$ inch, R_z 1.37 – 1.57 $\times 10^{-3}$ inch
Volume rate [4]	
Ti64 Performance (30 μm)	5 mm ³ /s (18 cm ³ /h) 0.82 in ³ /h
Ti64 Speed (60 μm)	9 mm ³ /s (32.4 cm ³ /h) 1.98 in ³ /h

- [1] Based on users' experience of dimensional accuracy for typical geometries. Part accuracy is subject to appropriate data preparation and post-processing, in accordance with EOS training.
- [2] Mechanical stability is dependent on geometry (wall height etc.) and application
- [3] Due to the layerwise building, the surface structure depends strongly on the orientation of the surface, for example sloping and curved surfaces exhibit a stair-step effect. The values also depend on the measurement method used. The values quoted here given an indication of what can be expected for horizontal (up-facing) or vertical surfaces.
- [4] Volume rate is a measure of build speed during laser exposure. The total build speed depends on the average volume rate, the recoating time (related to the number of layers) and other factors such as DMLS-Start settings.



Material data sheet

Physical and chemical properties of parts

Material composition	Ti (balance) Al (5.5 – 6.75 wt.-%) V (3.5 – 4.5 wt.-%) O (< 2000 ppm) N (< 500 ppm) C (< 800 ppm) H (< 150 ppm) Fe (< 3000 ppm)
Relative density	approx. 100 %
Density	4.41 g/cm ³ 0.159 lb/in ³

Material data sheet

Mechanical properties of parts [8]

	As built	Heat treated [6]
Tensile strength [5]		
- in horizontal direction (XY)	typ. 1290 ± 50 MPa typ. 187 ± 7 ksi	min. 930 MPa (134.8 ksi) typ. 1100 ± 40 MPa (160 ± 6 ksi)
- in vertical direction (Z)	typ. 1240 ± 50 MPa typ. 187 ± 7 ksi	min. 930 MPa (134.8 ksi) typ. 1100 ± 40 MPa (160 ± 6 ksi)
Yield strength (R _{p0.2}) [5]		
- in horizontal direction (XY)	typ. 1140 ± 50 MPa typ. 165 ± 7 ksi	min. 860 MPa (124.7 ksi) typ. 1000 ± 50 MPa (145 ± 7 ksi)
- in vertical direction (Z)	typ. 1120 ± 80 MPa typ. 162 ± 12 ksi	min. 860 MPa (124.7 ksi) typ. 1000 ± 60 MPa (145 ± 9 ksi)
Elongation at break [5]		
- in horizontal direction (XY)	typ. (7 ± 3) %	min. 10 % typ. (13.5 ± 2 %)
- in vertical direction (Z)	typ. (10 ± 3) %	min. 10 % typ. (14.5 ± 2 %)
Modulus of elasticity [5]		
- in horizontal direction (XY)	typ. 110 ± 15 GPa typ. 16 ± 2 Msi	typ. 110 ± 15 GPa typ. 16 ± 2 Msi
- in vertical direction (Z)	typ. 110 ± 15 GPa typ. 16 ± 2 Msi	typ. 110 ± 15 GPa typ. 16 ± 2 Msi
Hardness [7]		
	typ. 320 ± 12 HV5	

[5] Tensile testing according to ISO 6892-1:2009 (B) Annex D, proportional test pieces, diameter of the neck area 5 mm (0.2 inch), original gauge length 25 mm (1 inch).

[6] Specimens were treated at 800 °C (1470 °F) for 4 hours in argon inert atmosphere. Mechanical properties are expressed as minimum values to indicate that mechanical properties exceed the minimum requirements of material specification standards. ASTM F1472-08. By fulfilling these minimum values, also the specifications of standards ASTM B348-09 and ISO 5832-3:2000 are met.

[7] Vickers hardness measurement (HV) according to EN ISO 6507-1 on polished surface. Note that measured hardness can vary significantly depending on how the specimen has been prepared.

[8] Hint: these properties were determined for Ti64_Performance 1.0 on an EOSINT M 280-400W and EOSINT M 290-400W. Test parts from Ti64_Speed 1.0 were determined on machine types EOSINT M 280-400W and correspond with data from an EOS M 290-400W.



Material data sheet

Thermal properties of parts

Maximum long-term operating temperature	approx. 350 °C approx. 660 °F
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Abbreviations

typ.	typical
min.	minimum
wt.	weight
approx.	approximately

Notes

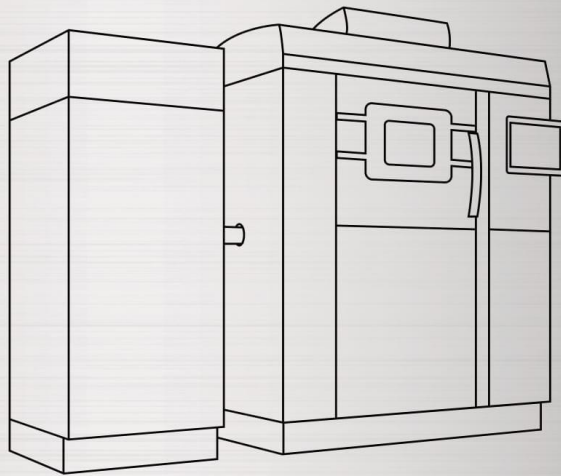
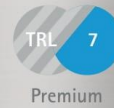
The data are valid for the combinations of powder material, machine and parameter sets referred to on page 1, when used in accordance with the relevant Operating Instructions (including Installation Requirements and Maintenance) and Parameter Sheet. Part properties are measured using defined test procedures. Further details of the test procedures used by EOS are available on request.

The data correspond to our knowledge and experience at the time of publication. They do not on their own provide a sufficient basis for designing parts. Neither do they provide any agreement or guarantee about the specific properties of a part or the suitability of a part for a specific application. The producer or the purchaser of a part is responsible for checking the properties and the suitability of a part for a particular application. This also applies regarding any rights of protection as well as laws and regulations. The data are subject to change without notice as part of EOS' continuous development and improvement processes.

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Appendix E: EOS 316L Powder/Material Specifications



EOS StainlessSteel 316L for EOS M 290 | 40 μm

Process Information
Chemical and Physical Part Properties
Heat Treatment
Mechanical Properties
Additional Data

CR365 v03/12.2016

EOS StainlessSteel 316L for EOS M 290 | 40 µm

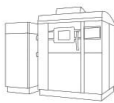
Process Information

This process product is optimized for building high quality parts with EOS M 290 system reliably using EOS StainlessSteel 316L. Mechanical properties have been validated to TRL7 level.

System set-up	EOS M 290
EOS ParameterSet	316L 40µm FlexLine
EOSPAR name	316L_040_FlexM291_1.X
Software requirements	EOSPRINT 2.7 or newer EOSYSTEM 2.11 or newer
Powder part no.	9011-0032
Recoater blade	EOS HSS blade
Nozzle	EOS grid nozzle
Inert gas	Argon
Sieve	63 µm

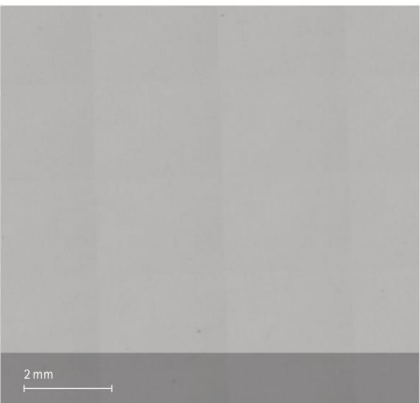
Additional information	
Layer thickness	40 µm
Min. wall thickness	0.1 mm
Typical dimensional change after HT	+0.2 %
Volume rate	3.7 mm ³ /s

Chemical and Physical Properties of Parts



Chemical composition of built parts
is compliant to EOS StainlessSteel
316L powder chemical composition.

Micrograph of polished surface



Microstructure solution annealed
Etched with etchant Kallings 2



Defects	Result	Number of samples
Average defect percentage	0.015 %	20
Density, ISO3369	Result	Number of samples
Average density	$\geq 7.97 \text{ g/cm}^3$	20

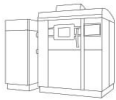
Heat Treatment

Heat treatment according to AMS 2759 is optional.

Stress relief: Hold temperature 900 °C, hold time minimum 2 h when thoroughly heated, water quenching

Solution annealing: Hold temperature 1 150 °C, hold time minimum 1.5 h when thoroughly heated, water quenching

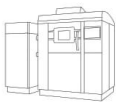
Mechanical Properties as manufactured



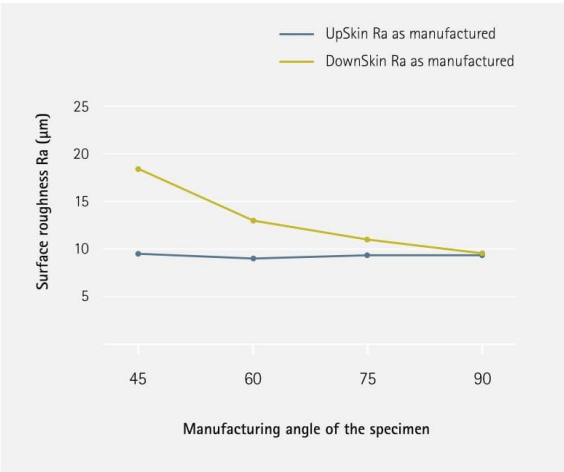
Mechanical properties ISO6892-1				
	Yield strength $R_{p0.2}$ [MPa]	Tensile strength R_m [MPa]	Elongation at break A [%]	Number of samples
Vertical	480	570	51	105
Horizontal	540	640	40	90



Additional Data



Surface Roughness



Coefficient of Thermal Expansion ASTM E228

Temperature	25-100 °C	25-200 °C	25-300 °C	25-400 °C
CTE	15.72 *10 ⁻⁶ /K	16.75 *10 ⁻⁶ /K	17.27 *10 ⁻⁶ /K	17.70 *10 ⁻⁶ /K

Appendix F: EOS MS1 Powder/Material Specifications



Material data sheet

EOS MaragingSteel MS1

EOS MaragingSteel MS1 is a steel powder which has been optimized especially for processing on EOSINT M systems.

This document provides information and data for parts built using EOS MaragingSteel MS1 powder (EOS art.-no. 9011-0016) on the following system specifications:

- EOSINT M 280 400W
with PSW 3.6 and EOS Original Parameter Set MS1_Performance 1.0 or MS1_Speed 1.0
- EOS M290 400W
with EOSPRINT 1.0 and Parameter Set MS1_Performance 1.0 or MS1_Speed 1.0

Description

Parts built in EOS MaragingSteel MS1 have a chemical composition corresponding to US classification 18% Ni Maraging 300, European 1.2709 and German X3NiCoMoTi 18-9-5. This kind of steel is characterized by having very good mechanical properties, and being easily heat-treatable using a simple thermal age-hardening process to obtain excellent hardness and strength.

Parts built from EOS MaragingSteel MS1 are easily machinable after the building process and can be easily post-hardened to more than 50 HRC by age-hardening at 490 °C (914 °F) for 6 hours. In both as-built and age-hardened states the parts can be machined, spark-eroded, welded, micro shot-peened, polished and coated if required. Due to the layerwise building method, the parts have a certain anisotropy, which can be reduced or removed by appropriate heat treatment – see Technical Data for examples.

Material data sheet

Technical data

General process data

Typical achievable part accuracy [1], [8]	
- small parts (< 80 x 80 mm)	approx. $\pm 20 \mu\text{m}$ approx. $\pm 0.8 \times 10^{-3}$ inch
- large parts	approx. $\pm 50 \mu\text{m}$ approx. ± 0.002 inch
Age hardening shrinkage [2], [8]	
approx. 0.08 %	
Min. wall thickness [3], [8]	
approx. 0.3 – 0.4 mm approx. 0.012 – 0.016 inch	
Surface roughness (approx.) [4]	
- as manufactured	
MS1 Performance (40 μm)	R_a 5 μm ; R_z 28 μm R_a 0.19 $\times 10^{-3}$ inch, R_z 1.10 $\times 10^{-3}$ inch
MS1 Speed (50 μm)	R_a 9 μm ; R_z 50 μm R_a 0.47 $\times 10^{-3}$ inch, R_z 2.36 $\times 10^{-3}$ inch
- after shot-peening	R_a 4 – 6.5 μm ; R_z 20 – 50 μm R_a 0.16 – 0.26 $\times 10^{-3}$ inch R_z 0.78 – 1.97 $\times 10^{-3}$ inch
- after polishing	R_z up to < 0.5 μm R_z up to < 0.02 $\times 10^{-3}$ inch (can be very finely polished)
Volume rate [5]	
- Parameter set MS1_Performance (40 μm)	4.2 mm ³ /s (15.1 cm ³ /h) 0.92 in ³ /h
- Parameter set MS1_Speed 1.0 (50 μm)	5.5 mm ³ /s (19.8 cm ³ /h) 1.21 in ³ /h

[1] Based on users' experience of dimensional accuracy for typical geometries, as built. Part accuracy is subject to appropriate data preparation and post-processing, in accordance with EOS training.

[2] Ageing temperature 490 °C (914 °F), 6 hours, air cooling

[3] Mechanical stability is dependent on geometry (wall height etc.) and application



Material data sheet

- [4] Due to the layerwise building, the surface structure depends strongly on the orientation of the surface, for example sloping and curved surfaces exhibit a stair-step effect. The values also depend on the measurement method used. The values quoted here given an indication of what can be expected for horizontal (up-facing) or vertical surfaces.
- [5] Volume rate is a measure of build speed during laser exposure of hatched areas. The total build speed depends on the average volume rate, the recoating time (related to the number of layers) and other geometry- and machine setting-related factors.

Physical and chemical properties of parts

Material composition	Fe (balance) Ni (17 - 19 wt-%) Co (8.5 - 9.5 wt-%) Mo (4.5 - 5.2 wt-%) Ti (0.6 - 0.8 wt-%) Al (0.05 - 0.15 wt-%) Cr, Cu (each $\leq$ 0.5 wt-%) C ($\leq$ 0.03 wt-%) Mn, Si (each $\leq$ 0.1 wt-%) P, S (each $\leq$ 0.01 wt-%)
Relative density	approx. 100 %
Density	8.0 - 8.1 g/cm ³ 0.289 - 0.293 lb/in ³



Material data sheet

Mechanical properties of parts at 20 °C (68°F) [8]

As built	
Tensile strength [6]	
- in horizontal direction (XY)	typ. 1200 ± 100 MPa typ. 160 ± 15 ksi
- in vertical direction (Z)	typ. 1100 ± 150 MPa typ. 160 ± 22 ksi
Yield strength (Rp 0.2 %) [6]	
- in horizontal direction (XY)	typ. 1100 ± 100 MPa typ. 1xx ± 22 ksi
- in vertical direction (Z)	typ. 930 ± 150 MPa typ. 145 ± 22 ksi
Elongation at break [6]	
- in horizontal direction (XY)	typ. (12 ± 4) %
- in vertical direction (Z)	x
Modulus of elasticity [6]	
- in horizontal direction (XY)	typ. 150 ± 25 GPa typ. 22 ± 4 Msi
- in vertical direction (Z)	typ. 140 ± 25 GPa typ. 20 ± 4 Msi
Hardness [7]	typ. 33 – 37 HRC

[6] Tensile testing according to ISO 6892-1:2009 (B) Annex D, proportional test pieces, diameter of the neck area 5mm (0.2 inch), original gauge length 25mm (1 inch).

[7] Rockwell C (HRC) hardness measurement according to EN ISO 6508-1 on polished surface. Note that measured hardness can vary significantly depending on how the specimen has been prepared.

[8] Hint: these properties were determined on an EOSINT M 280-400W. Test parts from machine type EOS M 290-400W correspond with these data.



Material data sheet

Thermal properties of parts

	As built	After age hardening [2]
Thermal conductivity	typ. $15 \pm 0.8 \text{ W/m}^\circ\text{C}$ typ. $104 \pm 6 \text{ Btu in/(h ft}^2 \text{ }^\circ\text{F)}$	typ. $20 \pm 1 \text{ W/m}^\circ\text{C}$ typ. $139 \pm 7 \text{ Btu in/(h ft}^2 \text{ }^\circ\text{F)}$
Specific heat capacity	typ. $450 \pm 20 \text{ J/kg}^\circ\text{C}$ typ. $0.108 \pm 0.005 \text{ Btu/(lb }^\circ\text{F)}$	typ. $450 \pm 20 \text{ J/kg}^\circ\text{C}$ typ. $0.108 \pm 0.005 \text{ Btu/(lb }^\circ\text{F)}$
Maximum operating temperature		approx. $400 \text{ }^\circ\text{C}$ approx. $750 \text{ }^\circ\text{F}$

Abbreviations

typ.	typical
min.	minimum
approx.	approximately
wt	weight

Notes

The data are valid for the combinations of powder material, machine and parameter sets referred to on page 1, when used in accordance with the relevant Operating Instructions (including Installation Requirements and Maintenance) and Parameter Sheet. Part properties are measured using defined test procedures. Further details of the test procedures used by EOS are available on request. Unless otherwise specified, the data refer to the default job MS1_040_default.job or the equivalent parameter set MS1_Performance 2.0. The corresponding data for the default job MS1_020_default.job or the equivalent parameter set MS1_Surface 1.0 are approximately the same except where otherwise specified.

The data correspond to our knowledge and experience at the time of publication. They do not on their own provide a sufficient basis for designing parts. Neither do they provide any agreement or guarantee about the specific properties of a part or the suitability of a part for a specific application. The producer or the purchaser of a part is responsible for checking the properties and the suitability of a part for a particular application. This also applies regarding any rights of protection as well as laws and regulations. The data are subject to change without notice as part of EOS' continuous development and improvement processes.

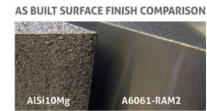
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Appendix G: Elementum3D 6061-T6 RAM2 Powder/Material Specifications

ELEMENTUM®

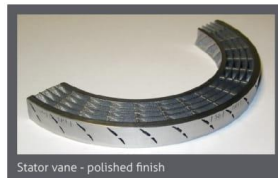
A6061-RAM2
Materials Data Sheet



► A6061-RAM2 (Highly Versatile and Cost Effective)

Product Information

Elementum 3D's A6061-RAM2 is a general-purpose AM aluminum alloy that features a good combination of excellent ductility, high strength, and good thermal conductivity. An additive printed AL6061 is of particular interest for many applications due to its unique balance of properties and efficient build speed. A6061-RAM2 builds more than 50% faster than ALSi10Mg on an EOS M290 and has a better "as built" surface finish.



Physical and Chemical Properties

Material composition: Proprietary A6061 w/2% ceramic (E3D-T6 Condition)

Theoretical maximum density: 2.74 g/cm³

Printed relative density: > 99.7%

Ultimate tensile strength^[1]: 48 ± 3.0 ksi (331 MPa)

Yield strength^[1]: 43 ± 2.0 ksi (297 MPa)

Elongation^[1]: 12 ± 1.5 %

Hardness^[2]: 60 ± 2.0 HRB

Modulus of elasticity^[3]: 11.0 ± 0.10 Msi (76 GPa)

Deposition rate^[4]: 2.3 in³/hr (10.4 mm³/s)

Wear volume loss^[5]: 5.1x10⁻³ in³ (84 mm³) (Note: Lower volume loss is better)

Comparison: ^[6]17-4 Stainless Steel 300mm³, ^[7]A380 Cast Aluminum 304 mm³

Thermal conductivity^[8]: 162±3 W/m·K (measured in z)

Coefficient of Thermal Expansion (CTE)^[9]: 22.4ppm/°C

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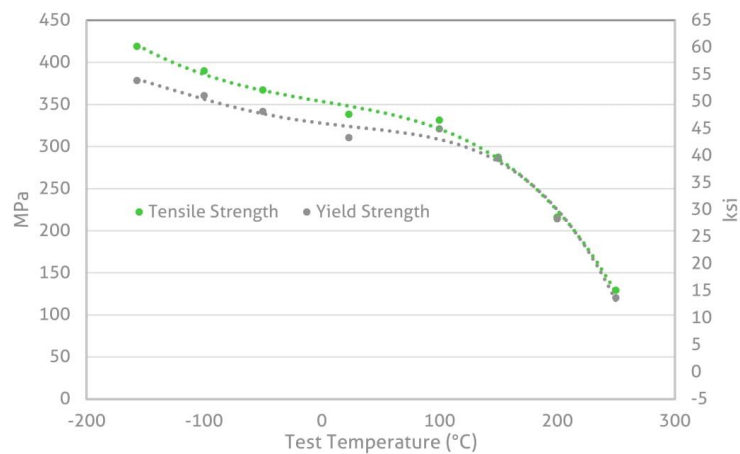


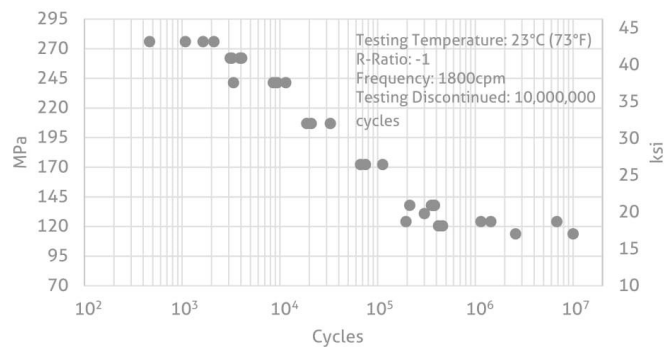
Surface roughness as built^[10]:

Angle Deg. °	Upskin		Downskin	
	Ra μm	Ra μin	Ra μm	Ra μin
0 (top)	2.27±0.31	89.4±12		
40	7.35±1.72	289±68	18.91±1.0	744±39
45	6.32±1.73	249±68	18.14±2.29	714±90
50	6.74±1.13	265±44	16.97±3.86	668±152
90 (vertical)	7.28±0.31	287±12		

Elevated temperature tensile^[11]:

Test temperature °C °F		Ultimate tensile strength		Yield strength		Elongation %
		MPa	ksi	MPa	ksi	
-157	-251	419±11	60.7±2	378±11	54.8±2	15.1±1
-100	-148	390±11	56.5±2	360±12	52.2±2	13.0±1
-50	-58	367±17	53.2±3	342±17	49.5±3	13.1±1
23	73	338±20	49.0±3	311±14	45.0±2	12.9±2
100	212	331±5	48.0±1	320±7	46.5±1	14.5±1
150	302	287±6	41.6±1	286±5	41.4±0.7	18.4±3
200	392	215±9	31.3±1	214±8	31.0±1	22.0±4
250	482	129±15	18.7±2	120±6	17.4±1	29.6±5



Fatigue^[12]:

All stated values are from heat treated samples.

^[1]ASTM E8, ^[2]ASTM E18, ^[3]ASTM E494-15, ^[4]Deposition rate calculation is for comparison purposes on an EOS M290 and does not include recoating time, laser migration time, contour exposures, etc., ^[5]ASTM G65, Procedure E, ^[6]Suthar et al. (2015). Comparative evaluation of abrasive wear resistance of various stainless steel grades. GE- International Journal of Engineering Research, 3(7), ^[7]Lall and Williamson. Wear Resistance and Mechanical Properties of Selected PM Aluminum Alloys and Composites, Metal Powder Products Company, ^[8]ISO/DIS 22007-2.2 (Transient Place Source, TPS), ^[9]ASTM E228, ^[10]Surface roughness determined by stylus profilometry, ^[11]ASTM E21, ^[12]ASTM E466.

All stated values are approximate values. All details given above are our current knowledge and experience, and are dependent on the equipment, parameters and operating conditions. The data provided in this document is subject to change and only intended as general information on a material set that is continually improving and developing. The data does not provide a sufficient basis for engineering parts. All samples were produced on an EOS M290. All tensile tests were performed at third party certified test labs such as Westmoreland Mechanical Testing & Research.

Please contact us at sales@elementum3d.com for additional information.